



EUROPEAN CLIMATE, INFRASTRUCTURE AND ENVIRONMENT EXECUTIVE AGENCY (CINEA)

CINEA.C — Green research and innovation
C.2 — Horizon Europe Energy

GRANT AGREEMENT

Project 101103998 — DEDALUS

PREAMBLE

This **Agreement** ('the Agreement') is **between** the following parties:

on the one part,

the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and

on the other part,

1. 'the coordinator':

ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG), PIC 999960488, established in PIAZZALE DELL'AGRICOLTURA 24, ROMA 00144, Italy,

and the following other beneficiaries, if they sign their 'accession form' (see Annex 3 and Article 40):

2. **ETHNICON METSOVION POLYTECHNION (NTUA)**, PIC 999978142, established in HEROON POLYTECHNIOU 9 ZOGRAPHOU CAMPUS, ATHINA 157 80, Greece,

3. **FUNDACION CARTIF (CARTIF)**, PIC 999929836, established in PQ TECNOLOGICO BOECILLO 205, BOECILLO 47151, Spain,

4. **UNIVERSITA POLITECNICA DELLE MARCHE (UNIVPM)**, PIC 999866689, established in PIAZZA ROMA 22, ANCONA 60121, Italy,

5. **ARCELIK A.S. (ARC)**, PIC 998821708, established in KARAAGAC, SUTLUCE BEYOGLU CAD. NO:2-6, ISTANBUL 34445, Türkiye,

6. **COMSENSUS, KOMUNIKACIJE IN SENZORIKA, DOO (COMS)**, PIC 937154831, established in BREZJE PRI DOBU 8A, DOB 1233, Slovenia,

7. **OURPOWER ENERGIEGENOSSENSCHAFT SCE MIT BESCHRANKTER HAFTUNG (OURPOWER)**, PIC 898281208, established in MARIAHILFER STRASSE 1D/13, WIEN 1060, Austria,

8. **GREEN-POINT 62 GMBH (GREENPOINT)**, PIC 885248773, established in POTTENDORFER STRASSE 62/EG04, WIEN 2700, Austria,
9. **BLUEPRINT ENERGY SOLUTIONS GMBH (BLUEPRINT)**, PIC 904096067, established in AM GESTADE 3, WIEN 1010, Austria,
10. **FAELLESBO (FAELLESBO)**, PIC 892008024, established in NYGADE 20, HERNING 7400, Denmark,
11. **NEOGRID TECHNOLOGIES APS (NEOGRID)**, PIC 953024419, established in NIELS JERNES VEJ 10, AALBORG OST 9220, Denmark,
12. **EUROPEAN GREEN CITIES APS (GREEN CITIES)**, PIC 950585936, established in NORSGADE 17, AARHUS C 8000, Denmark,
13. **HERON SINGLE MEMBER S.A. ENERGY SERVICES (HERON)**, PIC 954579911, established in LEOFOROS MESOGEION 85, ATHINA 115 26, Greece,
14. **DOMX IDIOTIKI KEFALAIOUCHIKI ETAIREIA (DOM-X)**, PIC 898645637, established in STR. SARAFI 48E, THESSALONIKI 551 32, Greece,
15. **ISTITUTO PER SERVIZI DI RICOVERO E ASSISTENZA AGLI ANZIANI (ISRAA)**, PIC 930262690, established in BORGO MAZZINI 48, TREVISO 31100, Italy,
16. **DYNAME SRL (DYNAME)**, PIC 885390102, established in VICOLO CADORE 45, SAN VENDEMIANO 31020, Italy,
17. **CENTRE INTERNACIONAL DE METODES NUMERICS EN ENGINYERIA (CIMNE)**, PIC 999658721, established in C GRAN CAPITAN, EDIFICI C1, CAMPUS NORD UPC SN, BARCELONA 08034, Spain,
18. **PRODUCTORA ELECTRICA URGELLENCA SA (PEUSA)**, PIC 899979775, established in PASEO JOAN BRUDIEU, 17, LA SEU D'URGELL 25700, Spain,
19. **ROMUR RENOVALBES SL (ROMUR)**, PIC 885732512, established in URB SAN PATRICIO 97, ALGINET 46230, Spain,
20. **CENTRICA BUSINESS SOLUTIONS BELGIUM (CENTRICA)**, PIC 927493825, established in RODERVELDLAAN 2, ANTWERPEN 2600, Belgium,
21. **UNIVERSITATEA TEHNICA CLUJ-NAPOCA (TUC)**, PIC 999897244, established in STR MEMORANDUMULUI 28, CLUJ NAPOCA 400114, Romania,
22. **SMART INNOVATION NORWAY AS (SIN)**, PIC 950648016, established in HAKON MELBERGS VEI 16, HALDEN 1783, Norway,
23. **FONDAZIONE ICONS (ICONS)**, PIC 917939810, established in PIAZZA DELLA VITTORIA 1, LODI 26900, Italy,

Unless otherwise specified, references to ‘beneficiary’ or ‘beneficiaries’ include the coordinator and affiliated entities (if any).

If only one beneficiary signs the grant agreement (‘mono-beneficiary grant’), all provisions referring

to the ‘coordinator’ or the ‘beneficiaries’ will be considered — mutatis mutandis — as referring to the beneficiary.

The parties referred to above have agreed to enter into the Agreement.

By signing the Agreement and the accession forms, the beneficiaries accept the grant and agree to implement the action under their own responsibility and in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

The Agreement is composed of:

Preamble

Terms and Conditions (including Data Sheet)

- Annex 1 Description of the action¹
- Annex 2 Estimated budget for the action
- Annex 2a Additional information on unit costs and contributions (if applicable)
- Annex 3 Accession forms (if applicable)²
- Annex 3a Declaration on joint and several liability of affiliated entities (if applicable)³
- Annex 4 Model for the financial statements
- Annex 5 Specific rules (if applicable)

¹ Template published on [Portal Reference Documents](#).

² Template published on [Portal Reference Documents](#).

³ Template published on [Portal Reference Documents](#).

TERMS AND CONDITIONS

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DATA SHEET

1. General data

Project summary:

Project summary
DEDALUS will design, develop and demonstrate SSH-driven multi-value energy carrier-agnostic micro (home/apartment)-to macro(building & district-scale) participatory Demand Response (DR) ecosystem, aimed to: (a) facilitate and scale up residential energy consumers massive participation to DR; (b) adapt to a variety of different mono-carrier (electricity, heat) or multi-carrier synergetic scenarios (electricity vs heat and natural gas) at building/district scale, while strengthening social interactions within respective communities. To this aim, we will deploy: 1) a Social Science Framework for multi-dimensional incentives and nudging interventions tools; 2) AI-based individual/building/districts consumers clustering and segmentation algorithms; 3) Open APIs for DR-ready smart appliances; 4) Energy DataSpace adaptation for extended DR interoperability and privacy-preserving DLT/Blockchain Data Governance and flexibility coordination; 5) Digital Twins for consumers-aware DR flexibility planning; 6) Comfort-based flexibility models and tools for serviced apartments for elderly people; 7) flexibility management tools for building and physical/virtual districts energy communities, based on pre-aggregation and shared DR assets; 8) Optimal aggregation of flexible energy assets for district heating and decentralized power2heat 9) Business Sandbox with novel sharing economy social innovation-based business models. DEDALUS solutions will be validated by 5 Front Runners full-scale pilots and replicated by 2 Multipliers in 7 countries, deployed at the interplay of energy, and non-energy (comfort/healthcare, ageing, social innovation) sectors, and operated by different stakeholders (social housing operators, building operators, aggregators, energy communities, retailers/suppliers, DSOs/District Heating Network operators). Capacity building and blueprints will support policy makers and regulators to enable market take-up and EU-wide replication of residential DR.

Keywords:

- Energy efficient buildings
- Social science multi-dimensional incentives, Cross-Energy commodity Residential Buildings Demand Response, DR-ready smart residential appliances, Digital Twins & data-driven services for Comfort-based DR

Project number: 101103998

Project name: Data-driven Residential Energy Carrier-agnostic Demand Response Tools and Multi-value Services

Project acronym: DEDALUS

Call: HORIZON-CL5-2022-D4-01

Topic: HORIZON-CL5-2022-D4-01-01

Type of action: HORIZON Innovation Actions

Granting authority: European Climate, Infrastructure and Environment Executive Agency

Grant managed through EU Funding & Tenders Portal: Yes (eGrants)

Project starting date: fixed date: 1 May 2023

Project end date: 30 April 2026

Project duration: 36 months

Consortium agreement: Yes

2. Participants

List of participants:

Nº	Role	Short name	Legal name	Ctry	PIC	Total eligible costs (BEN and AE)	Max grant amount
1	COO	ENG	ENGINEERING - INGEGNERIA INFORMATICA SPA	IT	999960488	785 875.00	550 112.50
2	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142	494 500.00	494 500.00

N°	Role	Short name	Legal name	Ctry	PIC	Total eligible costs (BEN and AE)	Max grant amount
3	BEN	CARTIF	FUNDACION CARTIF	ES	999929836	251 875.00	251 875.00
4	BEN	UNIVPM	UNIVERSITA POLITECNICA DELLE MARCHE	IT	999866689	283 125.00	283 125.00
5	BEN	ARC	ARCELIK A.S.	TR	998821708	238 375.00	166 862.50
6	BEN	COMS	COMSENSUS, KOMUNIKACIJE IN SENZORIKA, DOO	SI	937154831	280 875.00	196 612.50
7	BEN	OURPOWER	OURPOWER ENERGIEGENOSSENSCHAFT SCE MIT BESCHRANKTER HAFTUNG	AT	898281208	438 650.00	307 055.00
8	BEN	GREENPOINT	GREEN-POINT 62 GMBH	AT	885248773	175 625.00	122 937.50
9	BEN	BLUEPRINT	BLUEPRINT ENERGY SOLUTIONS GMBH	AT	904096067	370 625.00	259 437.50
10	BEN	FAELLESBO	FAELLESBO	DK	892008024	250 037.50	250 037.50
11	BEN	NEOGRID	NEOGRID TECHNOLOGIES APS	DK	953024419	371 750.00	260 225.00
12	BEN	GREEN CITIES	EUROPEAN GREEN CITIES APS	DK	950585936	214 125.00	214 125.00
13	BEN	HERON	HERON SINGLE MEMBER S.A. ENERGY SERVICES	EL	954579911	430 250.00	301 175.00
14	BEN	DOM-X	DOMX IDIOTIKI KEFALAIOUCHIKI ETAIREIA	EL	898645637	421 375.00	294 962.50
15	BEN	ISRAA	ISTITUTO PER SERVIZI DI RICOVERO E ASSISTENZA AGLI ANZIANI	IT	930262690	238 750.00	238 750.00
16	BEN	DYNAME	DYNAME SRL	IT	885390102	160 500.00	112 350.00
17	BEN	CIMNE	CENTRE INTERNACIONAL DE METODES NUMERICS EN ENGINYERIA	ES	999658721	278 375.00	278 375.00
18	BEN	PEUSA	PRODUCTORA ELECTRICA URGELLENCA SA	ES	899979775	145 762.50	102 033.75
19	BEN	ROMUR	ROMUR RENOVALBES SL	ES	885732512	207 500.00	145 250.00
20	BEN	CENTRICA	CENTRICA BUSINESS SOLUTIONS BELGIUM	BE	927493825	503 750.00	352 625.00
21	BEN	TUC	UNIVERSITATEA TEHNICA CLUJ-NAPOCA	RO	999897244	330 000.00	330 000.00
22	BEN	SIN	SMART INNOVATION NORWAY AS	NO	950648016	194 875.00	194 875.00
23	BEN	ICONS	FONDAZIONE ICONS	IT	917939810	292 500.00	292 500.00
Total						7 359 075.00	5 999 801.25

Coordinator:

- ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG)

3. Grant**Maximum grant amount, total estimated eligible costs and contributions and funding rate:**

Total eligible costs (BEN and AE)	Funding rate (%)	Maximum grant amount (Annex 2)	Maximum grant amount (award decision)
7 359 075.00	100, 70	5 999 801.25	5 999 801.25

Grant form: Budget-based**Grant mode:** Action grant**Budget categories/activity types:**

- A. Personnel costs
 - A.1 Employees, A.2 Natural persons under direct contract, A.3 Seconded persons
 - A.4 SME owners and natural person beneficiaries
- B. Subcontracting costs

- C. Purchase costs
 - C.1 Travel and subsistence
 - C.2 Equipment
 - C.3 Other goods, works and services
- D. Other cost categories
 - D.2 Internally invoiced goods and services
- E. Indirect costs

Cost eligibility options:

- In-kind contributions eligible costs
- Parental leave
- Project-based supplementary payments
- Average personnel costs (unit cost according to usual cost accounting practices)
- Limitation for subcontracting
- Travel and subsistence:
 - Travel: Actual costs
 - Accommodation: Actual costs
 - Subsistence: Actual costs
- Equipment: depreciation only
- Indirect cost flat-rate: 25% of the eligible direct costs (categories A-D, except volunteers costs, subcontracting costs, financial support to third parties and exempted specific cost categories, if any)
- VAT: Yes
- Other ineligible costs

Budget flexibility: Yes (no flexibility cap)**4. Reporting, payments and recoveries****4.1 Continuous reporting** (art 21)**Deliverables:** see Funding & Tenders Portal Continuous Reporting tool**4.2 Periodic reporting and payments****Reporting and payment schedule** (art 21, 22):

Reporting					Payments	
Reporting periods			Type	Deadline	Type	Deadline (time to pay)
RP No	Month from	Month to				
					Initial prefinancing	30 days from entry into force/10 days before starting date – whichever is the latest

Reporting					Payments	
Reporting periods			Type	Deadline	Type	Deadline (time to pay)
RP No	Month from	Month to				
1	1	18	Periodic report	60 days after end of reporting period	Interim payment	90 days from receiving periodic report
2	19	36	Periodic report	60 days after end of reporting period	Final payment	90 days from receiving periodic report

Prefinancing payments and guarantees:

Prefinancing payment	
Type	Amount
Prefinancing 1 (initial)	4 799 841.00

Reporting and payment modalities (art 21, 22):

Mutual Insurance Mechanism (MIM): Yes

MIM contribution: 5% of the maximum grant amount (299 990.06), retained from the initial prefinancing

Restrictions on distribution of initial prefinancing: The prefinancing may be distributed only if the minimum number of beneficiaries set out in the call conditions (if any) have acceded to the Agreement and only to beneficiaries that have acceded.

Interim payment ceiling (if any): 90% of the maximum grant amount

Exception for revenues: Yes

No-profit rule: Yes

Late payment interest: ECB + 3.5%

Bank account for payments:

IT60P0306903201100000112526

Conversion into euros: Double conversion

Reporting language: Language of the Agreement

4.3 Certificates (art 24):

Certificates on the financial statements (CFS):

Conditions:

Schedule: only at final payment, if threshold is reached

Standard threshold (beneficiary-level):

- financial statement: requested EU contribution to costs $\geq$ EUR 430 000.00

Special threshold for beneficiaries with a systems and process audit(see Article 24): financial statement: requested EU contribution to costs $\geq$ EUR 725 000.00

4.4 Recoveries (art 22)

First-line liability for recoveries:

Beneficiary termination: Beneficiary concerned

Final payment: Each beneficiary for their own debt

After final payment: Beneficiary concerned

Joint and several liability for enforced recoveries (in case of non-payment):

Individual financial responsibility: Each beneficiary is liable only for its own debts (and those of its affiliated entities, if any)

5. Consequences of non-compliance, applicable law & dispute settlement forum

Suspension and termination:

Additional suspension grounds (art 31)

Additional termination grounds (art 32)

Applicable law (art 43):

Standard applicable law regime: EU law + law of Belgium

Dispute settlement forum (art 43):

Standard dispute settlement forum:

EU beneficiaries: EU General Court + EU Court of Justice (on appeal)

Non-EU beneficiaries: Courts of Brussels, Belgium (unless an international agreement provides for the enforceability of EU court judgements)

6. Other

Specific rules (Annex 5): Yes

Standard time-limits after project end:

Confidentiality (for X years after final payment): 5

Record-keeping (for X years after final payment): 5 (or 3 for grants of not more than EUR 60 000)

Reviews (up to X years after final payment): 2

Audits (up to X years after final payment): 2

Extension of findings from other grants to this grant (no later than X years after final payment): 2

Impact evaluation (up to X years after final payment): 5 (or 3 for grants of not more than EUR 60 000)

CHAPTER 1 GENERAL

ARTICLE 1 — SUBJECT OF THE AGREEMENT

This Agreement sets out the rights and obligations and terms and conditions applicable to the grant awarded for the implementation of the action set out in Chapter 2.

ARTICLE 2 — DEFINITIONS

For the purpose of this Agreement, the following definitions apply:

Actions — The project which is being funded in the context of this Agreement.

Grant — The grant awarded in the context of this Agreement.

EU grants — Grants awarded by EU institutions, bodies, offices or agencies (including EU executive agencies, EU regulatory agencies, EDA, joint undertakings, etc.).

Participants — Entities participating in the action as beneficiaries, affiliated entities, associated partners, third parties giving in-kind contributions, subcontractors or recipients of financial support to third parties.

Beneficiaries (BEN) — The signatories of this Agreement (either directly or through an accession form).

Affiliated entities (AE) — Entities affiliated to a beneficiary within the meaning of Article 187 of EU Financial Regulation 2018/1046⁴ which participate in the action with similar rights and obligations as the beneficiaries (obligation to implement action tasks and right to charge costs and claim contributions).

Associated partners (AP) — Entities which participate in the action, but without the right to charge costs or claim contributions.

Purchases — Contracts for goods, works or services needed to carry out the action (e.g. equipment, consumables and supplies) but which are not part of the action tasks (see Annex 1).

Subcontracting — Contracts for goods, works or services that are part of the action tasks (see Annex 1).

In-kind contributions — In-kind contributions within the meaning of Article 2(36) of EU Financial

⁴ For the definition, see Article 187 Regulation (EU, Euratom) 2018/1046 of the European Parliament and of the Council of 18 July 2018 on the financial rules applicable to the general budget of the Union, amending Regulations (EU) No 1296/2013, (EU) No 1301/2013, (EU) No 1303/2013, (EU) No 1304/2013, (EU) No 1309/2013, (EU) No 1316/2013, (EU) No 223/2014, (EU) No 283/2014, and Decision No 541/2014/EU and repealing Regulation (EU, Euratom) No 966/2012 ('EU Financial Regulation') (OJ L 193, 30.7.2018, p. 1): "**affiliated entities** [are]:

- (a) entities that form a sole beneficiary [(i.e. where an entity is formed of several entities that satisfy the criteria for being awarded a grant, including where the entity is specifically established for the purpose of implementing an action to be financed by a grant)];
- (b) entities that satisfy the eligibility criteria and that do not fall within one of the situations referred to in Article 136(1) and 141(1) and that have a link with the beneficiary, in particular a legal or capital link, which is neither limited to the action nor established for the sole purpose of its implementation".

Regulation 2018/1046, i.e. non-financial resources made available free of charge by third parties.

Fraud — Fraud within the meaning of Article 3 of EU Directive 2017/1371⁵ and Article 1 of the Convention on the protection of the European Communities' financial interests, drawn up by the Council Act of 26 July 1995⁶, as well as any other wrongful or criminal deception intended to result in financial or personal gain.

Irregularities — Any type of breach (regulatory or contractual) which could impact the EU financial interests, including irregularities within the meaning of Article 1(2) of EU Regulation 2988/95⁷.

Grave professional misconduct — Any type of unacceptable or improper behaviour in exercising one's profession, especially by employees, including grave professional misconduct within the meaning of Article 136(1)(c) of EU Financial Regulation 2018/1046.

Applicable EU, international and national law — Any legal acts or other (binding or non-binding) rules and guidance in the area concerned.

Portal — EU Funding & Tenders Portal; electronic portal and exchange system managed by the European Commission and used by itself and other EU institutions, bodies, offices or agencies for the management of their funding programmes (grants, procurements, prizes, etc.).

CHAPTER 2 ACTION

ARTICLE 3 — ACTION

The grant is awarded for the action **101103998 — DEDALUS** ('action'), as described in Annex 1.

ARTICLE 4 — DURATION AND STARTING DATE

The duration and the starting date of the action are set out in the Data Sheet (see Point 1).

CHAPTER 3 GRANT

ARTICLE 5 — GRANT

5.1 Form of grant

The grant is an action grant⁸ which takes the form of a budget-based mixed actual cost grant (i.e. a

⁵ Directive (EU) 2017/1371 of the European Parliament and of the Council of 5 July 2017 on the fight against fraud to the Union's financial interests by means of criminal law (OJ L 198, 28.7.2017, p. 29).

⁶ OJ C 316, 27.11.1995, p. 48.

⁷ Council Regulation (EC, Euratom) No 2988/95 of 18 December 1995 on the protection of the European Communities financial interests (OJ L 312, 23.12.1995, p. 1).

⁸ For the definition, see Article 180(2)(a) EU Financial Regulation 2018/1046: '**action grant**' means an EU grant to finance "an action intended to help achieve a Union policy objective".

grant based on actual costs incurred, but which may also include other forms of funding, such as unit costs or contributions, flat-rate costs or contributions, lump sum costs or contributions or financing not linked to costs).

5.2 Maximum grant amount

The maximum grant amount is set out in the Data Sheet (see Point 3) and in the estimated budget (Annex 2).

5.3 Funding rate

The funding rate for costs is 100% of the eligible costs for beneficiaries that are non-profit legal entities⁹ and 70% of the eligible costs for beneficiaries that are profit legal entities.

Contributions are not subject to any funding rate.

5.4 Estimated budget, budget categories and forms of funding

The estimated budget for the action is set out in Annex 2.

It contains the estimated eligible costs and contributions for the action, broken down by participant and budget category.

Annex 2 also shows the types of costs and contributions (forms of funding)¹⁰ to be used for each budget category.

If unit costs or contributions are used, the details on the calculation will be explained in Annex 2a.

5.5 Budget flexibility

The budget breakdown may be adjusted — without an amendment (see Article 39) — by transfers (between participants and budget categories), as long as this does not imply any substantive or important change to the description of the action in Annex 1.

However:

- changes to the budget category for volunteers (if used) always require an amendment
- changes to budget categories with lump sums costs or contributions (if used; including financing not linked to costs) always require an amendment
- changes to budget categories with higher funding rates or budget ceilings (if used) always require an amendment
- addition of amounts for subcontracts not provided for in Annex 1 either require an amendment or simplified approval in accordance with Article 6.2

⁹ For the definition, see Article XX of the Horizon Europe Framework Programme and Rules for Participation Regulation (EU) XXX: ‘**non-profit legal entity**’ means a legal entity which by its legal form is non-profit-making or which has a legal or statutory obligation not to distribute profits to its shareholders or individual members.

¹⁰ See Article 125 EU Financial Regulation 2018/1046.

- other changes require an amendment or simplified approval, if specifically provided for in Article 6.2
- flexibility caps: not applicable.

ARTICLE 6 — ELIGIBLE AND INELIGIBLE COSTS AND CONTRIBUTIONS

In order to be eligible, costs and contributions must meet the **eligibility** conditions set out in this Article.

6.1 General eligibility conditions

The **general eligibility conditions** are the following:

- (a) for actual costs:
 - (i) they must be actually incurred by the beneficiary
 - (ii) they must be incurred in the period set out in Article 4 (with the exception of costs relating to the submission of the final periodic report, which may be incurred afterwards; see Article 21)
 - (iii) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2
 - (iv) they must be incurred in connection with the action as described in Annex 1 and necessary for its implementation
 - (v) they must be identifiable and verifiable, in particular recorded in the beneficiary's accounts in accordance with the accounting standards applicable in the country where the beneficiary is established and with the beneficiary's usual cost accounting practices
 - (vi) they must comply with the applicable national law on taxes, labour and social security and
 - (vii) they must be reasonable, justified and must comply with the principle of sound financial management, in particular regarding economy and efficiency
- (b) for unit costs or contributions (if any):
 - (i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2
 - (ii) the units must:
 - be actually used or produced by the beneficiary in the period set out in Article 4 (with the exception of units relating to the submission of the final periodic report, which may be used or produced afterwards; see Article 21)
 - be necessary for the implementation of the action and
 - (iii) the number of units must be identifiable and verifiable, in particular supported by records and documentation (see Article 20)

(c) for flat-rate costs or contributions (if any):

- (i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2
- (ii) the costs or contributions to which the flat-rate is applied must:
 - be eligible
 - relate to the period set out in Article 4 (with the exception of costs or contributions relating to the submission of the final periodic report, which may be incurred afterwards; see Article 21)

(d) for lump sum costs or contributions (if any):

- (i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2
- (ii) the work must be properly implemented by the beneficiary in accordance with Annex 1
- (iii) the deliverables/outputs must be achieved in the period set out in Article 4 (with the exception of deliverables/outputs relating to the submission of the final periodic report, which may be achieved afterwards; see Article 21)

(e) for unit, flat-rate or lump sum costs or contributions according to usual cost accounting practices (if any):

- (i) they must fulfil the general eligibility conditions for the type of cost concerned
- (ii) the cost accounting practices must be applied in a consistent manner, based on objective criteria, regardless of the source of funding

(f) for financing not linked to costs (if any): the results must be achieved or the conditions must be fulfilled as described in Annex 1.

In addition, for direct cost categories (e.g. personnel, travel & subsistence, subcontracting and other direct costs) only costs that are directly linked to the action implementation and can therefore be attributed to it directly are eligible. They must not include any indirect costs (i.e. costs that are only indirectly linked to the action, e.g. via cost drivers).

In-kind contributions provided by third parties free of charge may be declared as eligible direct costs by the beneficiaries which use them (under the same conditions as if they were their own, provided that they concern only direct costs and that the third parties and their in-kind contributions are set out in Annex 1 (or approved ex post in the periodic report, if their use does not entail changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants; ‘simplified approval procedure’).

6.2 Specific eligibility conditions for each budget category

For each budget category, the **specific eligibility conditions** are as follows:

Direct costs

A. Personnel costs

A.1 Costs for employees (or equivalent) are eligible as personnel costs if they fulfil the general eligibility conditions and are related to personnel working for the beneficiary under an employment contract (or equivalent appointing act) and assigned to the action.

They must be limited to salaries (including net payments during parental leave), social security contributions, taxes and other costs linked to the remuneration, if they arise from national law or the employment contract (or equivalent appointing act) and be calculated on the basis of the costs actually incurred, in accordance with the following method:

{daily rate for the person
multiplied by
number of day-equivalents worked on the action (rounded up or down to the nearest half-day)}.

The daily rate must be calculated as:

{annual personnel costs for the person
divided by
215}.

The number of day-equivalents declared for a person must be identifiable and verifiable (see Article 20).

The actual time spent on parental leave by a person assigned to the action may be deducted from the 215 days indicated in the above formula.

The total number of day-equivalents declared in EU grants, for a person for a year, cannot be higher than 215, minus time spent on parental leave (if any).

For personnel which receives supplementary payments for work in projects (project-based remuneration), the personnel costs must be calculated at a rate which:

- corresponds to the actual remuneration costs paid by the beneficiary for the time worked by the person in the action over the reporting period
- does not exceed the remuneration costs paid by the beneficiary for work in similar projects funded by national schemes ('national projects reference')
- is defined based on objective criteria allowing to determine the amount to which the person is entitled

and

- reflects the usual practice of the beneficiary to pay consistently bonuses or supplementary payments for work in projects funded by national schemes.

The national projects reference is the remuneration defined in national law, collective labour agreement or written internal rules of the beneficiary applicable to work in projects funded by national schemes.

If there is no such national law, collective labour agreement or written internal rules or if the project-

based remuneration is not based on objective criteria, the national project reference will be the average remuneration of the person in the last full calendar year covered by the reporting period, excluding remuneration paid for work in EU actions.

If the beneficiary uses average personnel costs (unit cost according to usual cost accounting practices), the personnel costs must fulfil the general eligibility conditions for such unit costs and the daily rate must be calculated:

- using the actual personnel costs recorded in the beneficiary's accounts and excluding any costs which are ineligible or already included in other budget categories; the actual personnel costs may be adjusted on the basis of budgeted or estimated elements, if they are relevant for calculating the personnel costs, reasonable and correspond to objective and verifiable information

and

- according to usual cost accounting practices which are applied in a consistent manner, based on objective criteria, regardless of the source of funding.

A.2 and A.3 Costs for natural persons working under a direct contract other than an employment contract and costs for **seconded persons by a third party against payment** are also eligible as personnel costs, if they are assigned to the action, fulfil the general eligibility conditions and:

- (a) work under conditions similar to those of an employee (in particular regarding the way the work is organised, the tasks that are performed and the premises where they are performed) and
- (b) the result of the work belongs to the beneficiary (unless agreed otherwise).

They must be calculated on the basis of a rate which corresponds to the costs actually incurred for the direct contract or secondment and must not be significantly different from those for personnel performing similar tasks under an employment contract with the beneficiary.

A.4 The work of **SME owners** for the action (i.e. owners of beneficiaries that are small and medium-sized enterprises¹¹ not receiving a salary) or **natural person beneficiaries** (i.e. beneficiaries that are natural persons not receiving a salary) may be declared as personnel costs, if they fulfil the general eligibility conditions and are calculated as unit costs in accordance with the method set out in Annex 2a.

B. Subcontracting costs

Subcontracting costs for the action (including related duties, taxes and charges, such as non-deductible or non-refundable value added tax (VAT)) are eligible, if they are calculated on the basis

¹¹ For the definition, see Commission Recommendation 2003/361/EC: micro, small or medium-sized enterprise (SME) are enterprises

- engaged in an economic activity, irrespective of their legal form (including, in particular, self-employed persons and family businesses engaged in craft or other activities, and partnerships or associations regularly engaged in an economic activity) and
- employing fewer than 250 persons (expressed in 'annual working units' as defined in Article 5 of the Recommendation) and which have an annual turnover not exceeding EUR 50 million, and/or an annual balance sheet total not exceeding EUR 43 million.

of the costs actually incurred, fulfil the general eligibility conditions and are awarded using the beneficiary's usual purchasing practices — provided these ensure subcontracts with best value for money (or if appropriate the lowest price) and that there is no conflict of interests (see Article 12).

Beneficiaries that are 'contracting authorities/entities' within the meaning of the EU Directives on public procurement must also comply with the applicable national law on public procurement.

Subcontracting may cover only a limited part of the action.

The tasks to be subcontracted and the estimated cost for each subcontract must be set out in Annex 1 and the total estimated costs of subcontracting per beneficiary must be set out in Annex 2 (or may be approved ex post in the periodic report, if the use of subcontracting does not entail changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants; 'simplified approval procedure').

C. Purchase costs

Purchase costs for the action (including related duties, taxes and charges, such as non-deductible or non-refundable value added tax (VAT)) are eligible if they fulfil the general eligibility conditions and are bought using the beneficiary's usual purchasing practices — provided these ensure purchases with best value for money (or if appropriate the lowest price) and that there is no conflict of interests (see Article 12).

Beneficiaries that are 'contracting authorities/entities' within the meaning of the EU Directives on public procurement must also comply with the applicable national law on public procurement.

C.1 Travel and subsistence

Purchases for **travel, accommodation and subsistence** must be calculated as follows:

- travel: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel
- accommodation: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel
- subsistence: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel .

C.2 Equipment

Purchases of **equipment, infrastructure or other assets** used for the action must be declared as depreciation costs, calculated on the basis of the costs actually incurred and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.

Only the portion of the costs that corresponds to the rate of actual use for the action during the action duration can be taken into account.

Costs for **renting or leasing** equipment, infrastructure or other assets are also eligible, if they do not exceed the depreciation costs of similar equipment, infrastructure or assets and do not include any financing fees.

C.3 Other goods, works and services

Purchases of **other goods, works and services** must be calculated on the basis of the costs actually incurred.

Such goods, works and services include, for instance, consumables and supplies, promotion, dissemination, protection of results, translations, publications, certificates and financial guarantees, if required under the Agreement.

D. Other cost categories

D.2 Internally invoiced goods and services

Costs for internally invoiced goods and services directly used for the action may be declared as unit cost according to usual cost accounting practices, if and as declared eligible in the call conditions, if they fulfil the general eligibility conditions for such unit costs and the amount per unit is calculated:

- using the actual costs for the good or service recorded in the beneficiary's accounts, attributed either by direct measurement or on the basis of cost drivers, and excluding any cost which are ineligible or already included in other budget categories; the actual costs may be adjusted on the basis of budgeted or estimated elements, if they are relevant for calculating the costs, reasonable and correspond to objective and verifiable information

and

- according to usual cost accounting practices which are applied in a consistent manner, based on objective criteria, regardless of the source of funding.

'Internally invoiced goods and services' means goods or services which are provided within the beneficiary's organisation directly for the action and which the beneficiary values on the basis of its usual cost accounting practices.

This cost will not be taken into account for the indirect cost flat-rate.

Indirect costs

E. Indirect costs

Indirect costs will be reimbursed at the flat-rate of 25% of the eligible direct costs (categories A-D, except volunteers costs, subcontracting costs, financial support to third parties and exempted specific cost categories, if any).

Contributions

Not applicable

6.3 Ineligible costs and contributions

The following costs or contributions are **ineligible**:

- (a) costs or contributions that do not comply with the conditions set out above (Article 6.1 and 6.2), in particular:

- (i) costs related to return on capital and dividends paid by a beneficiary
 - (ii) debt and debt service charges
 - (iii) provisions for future losses or debts
 - (iv) interest owed
 - (v) currency exchange losses
 - (vi) bank costs charged by the beneficiary's bank for transfers from the granting authority
 - (vii) excessive or reckless expenditure
 - (viii) deductible or refundable VAT (including VAT paid by public bodies acting as public authority)
 - (ix) costs incurred or contributions for activities implemented during grant agreement suspension (see Article 31)
 - (x) in-kind contributions by third parties: not applicable
- (b) costs or contributions declared under other EU grants (or grants awarded by an EU Member State, non-EU country or other body implementing the EU budget), except for the following cases:
- (i) Synergy actions: not applicable
 - (ii) if the action grant is combined with an operating grant¹² running during the same period and the beneficiary can demonstrate that the operating grant does not cover any (direct or indirect) costs of the action grant
- (c) costs or contributions for staff of a national (or regional/local) administration, for activities that are part of the administration's normal activities (i.e. not undertaken only because of the grant)
- (d) costs or contributions (especially travel and subsistence) for staff or representatives of EU institutions, bodies or agencies
- (e) other :
- (i) country restrictions for eligible costs: not applicable
 - (ii) costs or contributions declared specifically ineligible in the call conditions.

6.4 Consequences of non-compliance

If a beneficiary declares costs or contributions that are ineligible, they will be rejected (see Article 27).

This may also lead to other measures described in Chapter 5.

¹² For the definition, see Article 180(2)(b) of EU Financial Regulation 2018/1046: '**operating grant**' means an EU grant to finance "the functioning of a body which has an objective forming part of and supporting an EU policy".

CHAPTER 4 GRANT IMPLEMENTATION

SECTION 1 CONSORTIUM: BENEFICIARIES, AFFILIATED ENTITIES AND OTHER PARTICIPANTS

ARTICLE 7 — BENEFICIARIES

The beneficiaries, as signatories of the Agreement, are fully responsible towards the granting authority for implementing it and for complying with all its obligations.

They must implement the Agreement to their best abilities, in good faith and in accordance with all the obligations and terms and conditions it sets out.

They must have the appropriate resources to implement the action and implement the action under their own responsibility and in accordance with Article 11. If they rely on affiliated entities or other participants (see Articles 8 and 9), they retain sole responsibility towards the granting authority and the other beneficiaries.

They are jointly responsible for the *technical* implementation of the action. If one of the beneficiaries fails to implement their part of the action, the other beneficiaries must ensure that this part is implemented by someone else (without being entitled to an increase of the maximum grant amount and subject to an amendment; see Article 39). The *financial* responsibility of each beneficiary in case of recoveries is governed by Article 22.

The beneficiaries (and their action) must remain eligible under the EU programme funding the grant for the entire duration of the action. Costs and contributions will be eligible only as long as the beneficiary and the action are eligible.

The **internal roles and responsibilities** of the beneficiaries are divided as follows:

(a) Each beneficiary must:

- (i) keep information stored in the Portal Participant Register up to date (see Article 19)
- (ii) inform the granting authority (and the other beneficiaries) immediately of any events or circumstances likely to affect significantly or delay the implementation of the action (see Article 19)
- (iii) submit to the coordinator in good time:
 - the prefinancing guarantees (if required; see Article 23)
 - the financial statements and certificates on the financial statements (CFS) (if required; see Articles 21 and 24.2 and Data Sheet, Point 4.3)
 - the contribution to the deliverables and technical reports (see Article 21)
 - any other documents or information required by the granting authority under the Agreement
- (iv) submit via the Portal data and information related to the participation of their affiliated entities.

(b) The coordinator must:

- (i) monitor that the action is implemented properly (see Article 11)
- (ii) act as the intermediary for all communications between the consortium and the granting authority, unless the Agreement or granting authority specifies otherwise, and in particular:
 - submit the prefinancing guarantees to the granting authority (if any)
 - request and review any documents or information required and verify their quality and completeness before passing them on to the granting authority
 - submit the deliverables and reports to the granting authority
 - inform the granting authority about the payments made to the other beneficiaries (report on the distribution of payments; if required, see Articles 22 and 32)
- (iii) distribute the payments received from the granting authority to the other beneficiaries without unjustified delay (see Article 22).

The coordinator may not delegate or subcontract the above-mentioned tasks to any other beneficiary or third party (including affiliated entities).

However, coordinators which are public bodies may delegate the tasks set out in Point (b)(ii) last indent and (iii) above to entities with ‘authorisation to administer’ which they have created or which are controlled by or affiliated to them. In this case, the coordinator retains sole responsibility for the payments and for compliance with the obligations under the Agreement.

Moreover, coordinators which are ‘sole beneficiaries’¹³ (or similar, such as European research infrastructure consortia (ERICs)) may delegate the tasks set out in Point (b)(i) to (iii) above to one of their members. The coordinator retains sole responsibility for compliance with the obligations under the Agreement.

The beneficiaries must have **internal arrangements** regarding their operation and co-ordination, to ensure that the action is implemented properly.

If required by the granting authority (see Data Sheet, Point 1), these arrangements must be set out in a written **consortium agreement** between the beneficiaries, covering for instance:

- the internal organisation of the consortium
- the management of access to the Portal
- different distribution keys for the payments and financial responsibilities in case of recoveries (if any)
- additional rules on rights and obligations related to background and results (see Article 16)

¹³ For the definition, see Article 187(2) EU Financial Regulation 2018/1046: “Where several entities satisfy the criteria for being awarded a grant and together form one entity, that entity may be treated as the **sole beneficiary**, including where it is specifically established for the purpose of implementing the action financed by the grant.”

- settlement of internal disputes
- liability, indemnification and confidentiality arrangements between the beneficiaries.

The internal arrangements must not contain any provision contrary to this Agreement.

ARTICLE 8 — AFFILIATED ENTITIES

Not applicable

ARTICLE 9 — OTHER PARTICIPANTS INVOLVED IN THE ACTION

9.1 Associated partners

Not applicable

9.2 Third parties giving in-kind contributions to the action

Other third parties may give in-kind contributions to the action (i.e. personnel, equipment, other goods, works and services, etc. which are free-of-charge) if necessary for the implementation.

Third parties giving in-kind contributions do not implement any action tasks. They may not charge costs or contributions to the action, but the costs for the in-kind contributions are eligible and may be charged by the beneficiaries which use them, under the conditions set out in Article 6. The costs will be included in Annex 2 as part of the beneficiaries' costs.

The third parties and their in-kind contributions should be set out in Annex 1.

The beneficiaries must ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the third parties giving in-kind contributions.

9.3 Subcontractors

Subcontractors may participate in the action, if necessary for the implementation.

Subcontractors must implement their action tasks in accordance with Article 11. The costs for the subcontracted tasks (invoiced price from the subcontractor) are eligible and may be charged by the beneficiaries, under the conditions set out in Article 6. The costs will be included in Annex 2 as part of the beneficiaries' costs.

The beneficiaries must ensure that their contractual obligations under Articles 11 (proper implementation), 12 (conflict of interest), 13 (confidentiality and security), 14 (ethics), 17.2 (visibility), 18 (specific rules for carrying out action), 19 (information) and 20 (record-keeping) also apply to the subcontractors.

The beneficiaries must ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the subcontractors.

9.4 Recipients of financial support to third parties

If the action includes providing financial support to third parties (e.g. grants, prizes or similar forms of

support), the beneficiaries must ensure that their contractual obligations under Articles 12 (conflict of interest), 13 (confidentiality and security), 14 (ethics), 17.2 (visibility), 18 (specific rules for carrying out action), 19 (information) and 20 (record-keeping) also apply to the third parties receiving the support (recipients).

The beneficiaries must also ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the recipients.

ARTICLE 10 — PARTICIPANTS WITH SPECIAL STATUS

10.1 Non-EU participants

Participants which are established in a non-EU country (if any) undertake to comply with their obligations under the Agreement and:

- to respect general principles (including fundamental rights, values and ethical principles, environmental and labour standards, rules on classified information, intellectual property rights, visibility of funding and protection of personal data)
- for the submission of certificates under Article 24: to use qualified external auditors which are independent and comply with comparable standards as those set out in EU Directive 2006/43/EC¹⁴
- for the controls under Article 25: to allow for checks, reviews, audits and investigations (including on-the-spot checks, visits and inspections) by the bodies mentioned in that Article (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.).

Special rules on dispute settlement apply (see Data Sheet, Point 5).

10.2 Participants which are international organisations

Participants which are international organisations (IOs; if any) undertake to comply with their obligations under the Agreement and:

- to respect general principles (including fundamental rights, values and ethical principles, environmental and labour standards, rules on classified information, intellectual property rights, visibility of funding and protection of personal data)
- for the submission of certificates under Article 24: to use either independent public officers or external auditors which comply with comparable standards as those set out in EU Directive 2006/43/EC
- for the controls under Article 25: to allow for the checks, reviews, audits and investigations by the bodies mentioned in that Article, taking into account the specific agreements concluded by them and the EU (if any).

For such participants, nothing in the Agreement will be interpreted as a waiver of their privileges or immunities, as accorded by their constituent documents or international law.

¹⁴ Directive 2006/43/EC of the European Parliament and of the Council of 17 May 2006 on statutory audits of annual accounts and consolidated accounts or similar national regulations (OJ L 157, 9.6.2006, p. 87).

Special rules on applicable law and dispute settlement apply (see Article 43 and Data Sheet, Point 5).

10.3 Pillar-assessed participants

Pillar-assessed participants (if any) may rely on their own systems, rules and procedures, in so far as they have been positively assessed and do not call into question the decision awarding the grant or breach the principle of equal treatment of applicants or beneficiaries.

‘Pillar-assessment’ means a review by the European Commission on the systems, rules and procedures which participants use for managing EU grants (in particular internal control system, accounting system, external audits, financing of third parties, rules on recovery and exclusion, information on recipients and protection of personal data; see Article 154 EU Financial Regulation 2018/1046).

Participants with a positive pillar assessment may rely on their own systems, rules and procedures, in particular for:

- record-keeping (Article 20): may be done in accordance with internal standards, rules and procedures
- currency conversion for financial statements (Article 21): may be done in accordance with usual accounting practices
- guarantees (Article 23): for public law bodies, prefinancing guarantees are not needed
- certificates (Article 24):
 - certificates on the financial statements (CFS): may be provided by their regular internal or external auditors and in accordance with their internal financial regulations and procedures
 - certificates on usual accounting practices (CoMUC): are not needed if those practices are covered by an ex-ante assessment

and use the following specific rules, for:

- recoveries (Article 22): in case of financial support to third parties, there will be no recovery if the participant has done everything possible to retrieve the undue amounts from the third party receiving the support (including legal proceedings) and non-recovery is not due to an error or negligence on its part
- checks, reviews, audits and investigations by the EU (Article 25): will be conducted taking into account the rules and procedures specifically agreed between them and the framework agreement (if any)
- impact evaluation (Article 26): will be conducted in accordance with the participant’s internal rules and procedures and the framework agreement (if any)
- grant agreement suspension (Article 31): certain costs incurred during grant suspension are eligible (notably, minimum costs necessary for a possible resumption of the action and costs relating to contracts which were entered into before the pre-information letter was received and which could not reasonably be suspended, reallocated or terminated on legal grounds)

- grant agreement termination (Article 32): the final grant amount and final payment will be calculated taking into account also costs relating to contracts due for execution only after termination takes effect, if the contract was entered into before the pre-information letter was received and could not reasonably be terminated on legal grounds
- liability for damages (Article 33.2): the granting authority must be compensated for damage it sustains as a result of the implementation of the action or because the action was not implemented in full compliance with the Agreement only if the damage is due to an infringement of the participant's internal rules and procedures or due to a violation of third parties' rights by the participant or one of its employees or individual for whom the employees are responsible.

Participants whose pillar assessment covers procurement and granting procedures may also do purchases, subcontracting and financial support to third parties (Article 6.2) in accordance with their internal rules and procedures for purchases, subcontracting and financial support.

Participants whose pillar assessment covers data protection rules may rely on their internal standards, rules and procedures for data protection (Article 15).

The participants may however not rely on provisions which would breach the principle of equal treatment of applicants or beneficiaries or call into question the decision awarding the grant, such as in particular:

- eligibility (Article 6)
- consortium roles and set-up (Articles 7-9)
- security and ethics (Articles 13, 14)
- IPR (including background and results, access rights and rights of use), communication, dissemination and visibility (Articles 16 and 17)
- information obligation (Article 19)
- payment, reporting and amendments (Articles 21, 22 and 39)
- rejections, reductions, suspensions and terminations (Articles 27, 28, 29-32)

If the pillar assessment was subject to remedial measures, reliance on the internal systems, rules and procedures is subject to compliance with those remedial measures.

Participants whose assessment has not yet been updated to cover (the new rules on) data protection may rely on their internal systems, rules and procedures, provided that they ensure that personal data is:

- processed lawfully, fairly and in a transparent manner in relation to the data subject
- collected for specified, explicit and legitimate purposes and not further processed in a manner that is incompatible with those purposes
- adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed
- accurate and, where necessary, kept up to date

- kept in a form which permits identification of data subjects for no longer than is necessary for the purposes for which the data is processed and
- processed in a manner that ensures appropriate security of the personal data.

Participants must inform the coordinator without delay of any changes to the systems, rules and procedures that were part of the pillar assessment. The coordinator must immediately inform the granting authority.

Pillar-assessed participants that have also concluded a framework agreement with the EU, may moreover — under the same conditions as those above (i.e. not call into question the decision awarding the grant or breach the principle of equal treatment of applicants or beneficiaries) — rely on the provisions set out in that framework agreement.

SECTION 2 RULES FOR CARRYING OUT THE ACTION

ARTICLE 11 — PROPER IMPLEMENTATION OF THE ACTION

11.1 Obligation to properly implement the action

The beneficiaries must implement the action as described in Annex 1 and in compliance with the provisions of the Agreement, the call conditions and all legal obligations under applicable EU, international and national law.

11.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 12 — CONFLICT OF INTERESTS

12.1 Conflict of interests

The beneficiaries must take all measures to prevent any situation where the impartial and objective implementation of the Agreement could be compromised for reasons involving family, emotional life, political or national affinity, economic interest or any other direct or indirect interest ('conflict of interests').

They must formally notify the granting authority without delay of any situation constituting or likely to lead to a conflict of interests and immediately take all the necessary steps to rectify this situation.

The granting authority may verify that the measures taken are appropriate and may require additional measures to be taken by a specified deadline.

12.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28) and the grant or the beneficiary may be terminated (see Article 32).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 13 — CONFIDENTIALITY AND SECURITY

13.1 Sensitive information

The parties must keep confidential any data, documents or other material (in any form) that is identified as sensitive in writing ('sensitive information') — during the implementation of the action and for at least until the time-limit set out in the Data Sheet (see Point 6).

If a beneficiary requests, the granting authority may agree to keep such information confidential for a longer period.

Unless otherwise agreed between the parties, they may use sensitive information only to implement the Agreement.

The beneficiaries may disclose sensitive information to their personnel or other participants involved in the action only if they:

- (a) need to know it in order to implement the Agreement and
- (b) are bound by an obligation of confidentiality.

The granting authority may disclose sensitive information to its staff and to other EU institutions and bodies.

It may moreover disclose sensitive information to third parties, if:

- (a) this is necessary to implement the Agreement or safeguard the EU financial interests and
- (b) the recipients of the information are bound by an obligation of confidentiality.

The confidentiality obligations no longer apply if:

- (a) the disclosing party agrees to release the other party
- (b) the information becomes publicly available, without breaching any confidentiality obligation
- (c) the disclosure of the sensitive information is required by EU, international or national law.

Specific confidentiality rules (if any) are set out in Annex 5.

13.2 Classified information

The parties must handle classified information in accordance with the applicable EU, international or national law on classified information (in particular, Decision 2015/444¹⁵ and its implementing rules).

Deliverables which contain classified information must be submitted according to special procedures agreed with the granting authority.

¹⁵ Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p. 53).

Action tasks involving classified information may be subcontracted only after explicit approval (in writing) from the granting authority.

Classified information may not be disclosed to any third party (including participants involved in the action implementation) without prior explicit written approval from the granting authority.

Specific security rules (if any) are set out in Annex 5.

13.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 14 — ETHICS AND VALUES

14.1 Ethics

The action must be carried out in line with the highest ethical standards and the applicable EU, international and national law on ethical principles.

Specific ethics rules (if any) are set out in Annex 5.

14.2 Values

The beneficiaries must commit to and ensure the respect of basic EU values (such as respect for human dignity, freedom, democracy, equality, the rule of law and human rights, including the rights of minorities).

Specific rules on values (if any) are set out in Annex 5.

14.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 15 — DATA PROTECTION

15.1 Data processing by the granting authority

Any personal data under the Agreement will be processed under the responsibility of the data controller of the granting authority in accordance with and for the purposes set out in the Portal Privacy Statement.

For grants where the granting authority is the European Commission, an EU regulatory or executive agency, joint undertaking or other EU body, the processing will be subject to Regulation 2018/1725¹⁶.

¹⁶ Regulation (EU) 2018/1725 of the European Parliament and of the Council of 23 October 2018 on the protection of natural persons with regard to the processing of personal data by the Union institutions, bodies, offices and agencies

15.2 Data processing by the beneficiaries

The beneficiaries must process personal data under the Agreement in compliance with the applicable EU, international and national law on data protection (in particular, Regulation 2016/679¹⁷).

They must ensure that personal data is:

- processed lawfully, fairly and in a transparent manner in relation to the data subjects
- collected for specified, explicit and legitimate purposes and not further processed in a manner that is incompatible with those purposes
- adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed
- accurate and, where necessary, kept up to date
- kept in a form which permits identification of data subjects for no longer than is necessary for the purposes for which the data is processed and
- processed in a manner that ensures appropriate security of the data.

The beneficiaries may grant their personnel access to personal data only if it is strictly necessary for implementing, managing and monitoring the Agreement. The beneficiaries must ensure that the personnel is under a confidentiality obligation.

The beneficiaries must inform the persons whose data are transferred to the granting authority and provide them with the Portal Privacy Statement.

15.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 16 — INTELLECTUAL PROPERTY RIGHTS (IPR) — BACKGROUND AND RESULTS — ACCESS RIGHTS AND RIGHTS OF USE

16.1 Background and access rights to background

The beneficiaries must give each other and the other participants access to the background identified as needed for implementing the action, subject to any specific rules in Annex 5.

‘Background’ means any data, know-how or information — whatever its form or nature (tangible or intangible), including any rights such as intellectual property rights — that is:

and on the free movement of such data, and repealing Regulation (EC) No 45/2001 and Decision No 1247/2002/EC (OJ L 295, 21.11.2018, p. 39).

¹⁷ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (‘GDPR’) (OJ L 119, 4.5.2016, p. 1).

- (a) held by the beneficiaries before they acceded to the Agreement and
- (b) needed to implement the action or exploit the results.

If background is subject to rights of a third party, the beneficiary concerned must ensure that it is able to comply with its obligations under the Agreement.

16.2 Ownership of results

The granting authority does not obtain ownership of the results produced under the action.

‘Results’ means any tangible or intangible effect of the action, such as data, know-how or information, whatever its form or nature, whether or not it can be protected, as well as any rights attached to it, including intellectual property rights.

16.3 Rights of use of the granting authority on materials, documents and information received for policy, information, communication, dissemination and publicity purposes

The granting authority has the right to use non-sensitive information relating to the action and materials and documents received from the beneficiaries (notably summaries for publication, deliverables, as well as any other material, such as pictures or audio-visual material, in paper or electronic form) for policy, information, communication, dissemination and publicity purposes — during the action or afterwards.

The right to use the beneficiaries’ materials, documents and information is granted in the form of a royalty-free, non-exclusive and irrevocable licence, which includes the following rights:

- (a) **use for its own purposes** (in particular, making them available to persons working for the granting authority or any other EU service (including institutions, bodies, offices, agencies, etc.) or EU Member State institution or body; copying or reproducing them in whole or in part, in unlimited numbers; and communication through press information services)
- (b) **distribution to the public** (in particular, publication as hard copies and in electronic or digital format, publication on the internet, as a downloadable or non-downloadable file, broadcasting by any channel, public display or presentation, communicating through press information services, or inclusion in widely accessible databases or indexes)
- (c) **editing or redrafting** (including shortening, summarising, inserting other elements (e.g. meta-data, legends, other graphic, visual, audio or text elements), extracting parts (e.g. audio or video files), dividing into parts, use in a compilation)
- (d) **translation**
- (e) **storage** in paper, electronic or other form
- (f) **archiving**, in line with applicable document-management rules
- (g) the right to authorise **third parties** to act on its behalf or sub-license to third parties the modes of use set out in Points (b), (c), (d) and (f), if needed for the information, communication and publicity activity of the granting authority

(h) **processing**, analysing, aggregating the materials, documents and information received and **producing derivative works**.

The rights of use are granted for the whole duration of the industrial or intellectual property rights concerned.

If materials or documents are subject to moral rights or third party rights (including intellectual property rights or rights of natural persons on their image and voice), the beneficiaries must ensure that they comply with their obligations under this Agreement (in particular, by obtaining the necessary licences and authorisations from the rights holders concerned).

Where applicable, the granting authority will insert the following information:

“© – [year] – [name of the copyright owner]. All rights reserved. Licensed to the [name of granting authority] under conditions.”

16.4 Specific rules on IPR, results and background

Specific rules regarding intellectual property rights, results and background (if any) are set out in Annex 5.

16.5 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such a breach may also lead to other measures described in Chapter 5.

ARTICLE 17 — COMMUNICATION, DISSEMINATION AND VISIBILITY

17.1 Communication — Dissemination — Promoting the action

Unless otherwise agreed with the granting authority, the beneficiaries must promote the action and its results by providing targeted information to multiple audiences (including the media and the public), in accordance with Annex 1 and in a strategic, coherent and effective manner.

Before engaging in a communication or dissemination activity expected to have a major media impact, the beneficiaries must inform the granting authority.

17.2 Visibility — European flag and funding statement

Unless otherwise agreed with the granting authority, communication activities of the beneficiaries related to the action (including media relations, conferences, seminars, information material, such as brochures, leaflets, posters, presentations, etc., in electronic form, via traditional or social media, etc.), dissemination activities and any infrastructure, equipment, vehicles, supplies or major result funded by the grant must acknowledge EU support and display the European flag (emblem) and funding statement (translated into local languages, where appropriate):



Funded by the
European Union



Co-funded by the
European Union



Funded by the
European Union



Co-funded by the
European Union

The emblem must remain distinct and separate and cannot be modified by adding other visual marks, brands or text.

Apart from the emblem, no other visual identity or logo may be used to highlight the EU support.

When displayed in association with other logos (e.g. of beneficiaries or sponsors), the emblem must be displayed at least as prominently and visibly as the other logos.

For the purposes of their obligations under this Article, the beneficiaries may use the emblem without first obtaining approval from the granting authority. This does not, however, give them the right to exclusive use. Moreover, they may not appropriate the emblem or any similar trademark or logo, either by registration or by any other means.

17.3 Quality of information — Disclaimer

Any communication or dissemination activity related to the action must use factually accurate information.

Moreover, it must indicate the following disclaimer (translated into local languages where appropriate):

“Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or [name of the granting authority]. Neither the European Union nor the granting authority can be held responsible for them.”

17.4 Specific communication, dissemination and visibility rules

Specific communication, dissemination and visibility rules (if any) are set out in Annex 5.

17.5 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 18 — SPECIFIC RULES FOR CARRYING OUT THE ACTION

18.1 Specific rules for carrying out the action

Specific rules for implementing the action (if any) are set out in Annex 5.

18.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such a breach may also lead to other measures described in Chapter 5.

SECTION 3 GRANT ADMINISTRATION

ARTICLE 19 — GENERAL INFORMATION OBLIGATIONS

19.1 Information requests

The beneficiaries must provide — during the action or afterwards and in accordance with Article 7 — any information requested in order to verify eligibility of the costs or contributions declared, proper implementation of the action and compliance with the other obligations under the Agreement.

The information provided must be accurate, precise and complete and in the format requested, including electronic format.

19.2 Participant Register data updates

The beneficiaries must keep — at all times, during the action or afterwards — their information stored in the Portal Participant Register up to date, in particular, their name, address, legal representatives, legal form and organisation type.

19.3 Information about events and circumstances which impact the action

The beneficiaries must immediately inform the granting authority (and the other beneficiaries) of any of the following:

- (a) **events** which are likely to affect or delay the implementation of the action or affect the EU's financial interests, in particular:
 - (i) changes in their legal, financial, technical, organisational or ownership situation (including changes linked to one of the exclusion grounds listed in the declaration of honour signed before grant signature)
 - (ii) linked action information: not applicable

(b) circumstances affecting:

- (i) the decision to award the grant or
- (ii) compliance with requirements under the Agreement.

19.4 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 20 — RECORD-KEEPING**20.1 Keeping records and supporting documents**

The beneficiaries must — at least until the time-limit set out in the Data Sheet (see Point 6) — keep records and other supporting documents to prove the proper implementation of the action in line with the accepted standards in the respective field (if any).

In addition, the beneficiaries must — for the same period — keep the following to justify the amounts declared:

- (a) for actual costs: adequate records and supporting documents to prove the costs declared (such as contracts, subcontracts, invoices and accounting records); in addition, the beneficiaries' usual accounting and internal control procedures must enable direct reconciliation between the amounts declared, the amounts recorded in their accounts and the amounts stated in the supporting documents
- (b) for flat-rate costs and contributions (if any): adequate records and supporting documents to prove the eligibility of the costs or contributions to which the flat-rate is applied
- (c) for the following simplified costs and contributions: the beneficiaries do not need to keep specific records on the actual costs incurred, but must keep:
 - (i) for unit costs and contributions (if any): adequate records and supporting documents to prove the number of units declared
 - (ii) for lump sum costs and contributions (if any): adequate records and supporting documents to prove proper implementation of the work as described in Annex 1
 - (iii) for financing not linked to costs (if any): adequate records and supporting documents to prove the achievement of the results or the fulfilment of the conditions as described in Annex 1
- (d) for unit, flat-rate and lump sum costs and contributions according to usual cost accounting practices (if any): the beneficiaries must keep any adequate records and supporting documents to prove that their cost accounting practices have been applied in a consistent manner, based on objective criteria, regardless of the source of funding, and that they comply with the eligibility conditions set out in Articles 6.1 and 6.2.

Moreover, the following is needed for specific budget categories:

- (e) for personnel costs: time worked for the beneficiary under the action must be supported by declarations signed monthly by the person and their supervisor, unless another reliable time-record system is in place; the granting authority may accept alternative evidence supporting the time worked for the action declared, if it considers that it offers an adequate level of assurance
- (f) additional record-keeping rules: not applicable

The records and supporting documents must be made available upon request (see Article 19) or in the context of checks, reviews, audits or investigations (see Article 25).

If there are on-going checks, reviews, audits, investigations, litigation or other pursuits of claims under the Agreement (including the extension of findings; see Article 25), the beneficiaries must keep these records and other supporting documentation until the end of these procedures.

The beneficiaries must keep the original documents. Digital and digitalised documents are considered originals if they are authorised by the applicable national law. The granting authority may accept non-original documents if they offer a comparable level of assurance.

20.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, costs or contributions insufficiently substantiated will be ineligible (see Article 6) and will be rejected (see Article 27), and the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 21 — REPORTING

21.1 Continuous reporting

The beneficiaries must continuously report on the progress of the action (e.g. **deliverables, milestones, outputs/outcomes, critical risks, indicators**, etc; if any), in the Portal Continuous Reporting tool and in accordance with the timing and conditions it sets out (as agreed with the granting authority).

Standardised deliverables (e.g. progress reports not linked to payments, reports on cumulative expenditure, special reports, etc; if any) must be submitted using the templates published on the Portal.

21.2 Periodic reporting: Technical reports and financial statements

In addition, the beneficiaries must provide reports to request payments, in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2):

- for additional prefinancings (if any): an **additional prefinancing report**
- for interim payments (if any) and the final payment: a **periodic report**.

The prefinancing and periodic reports include a technical and financial part.

The technical part includes an overview of the action implementation. It must be prepared using the template available in the Portal Periodic Reporting tool.

The financial part of the additional prefinancing report includes a statement on the use of the previous prefinancing payment.

The financial part of the periodic report includes:

- the financial statements (individual and consolidated; for all beneficiaries/affiliated entities)
- the explanation on the use of resources (or detailed cost reporting table, if required)
- the certificates on the financial statements (CFS) (if required; see Article 24.2 and Data Sheet, Point 4.3).

The **financial statements** must detail the eligible costs and contributions for each budget category and, for the final payment, also the revenues for the action (see Articles 6 and 22).

All eligible costs and contributions incurred should be declared, even if they exceed the amounts indicated in the estimated budget (see Annex 2). Amounts that are not declared in the individual financial statements will not be taken into account by the granting authority.

By signing the financial statements (directly in the Portal Periodic Reporting tool), the beneficiaries confirm that:

- the information provided is complete, reliable and true
- the costs and contributions declared are eligible (see Article 6)
- the costs and contributions can be substantiated by adequate records and supporting documents (see Article 20) that will be produced upon request (see Article 19) or in the context of checks, reviews, audits and investigations (see Article 25)
- for the final periodic report: all the revenues have been declared (if required; see Article 22).

Beneficiaries will have to submit also the financial statements of their affiliated entities (if any). In case of recoveries (see Article 22), beneficiaries will be held responsible also for the financial statements of their affiliated entities.

21.3 Currency for financial statements and conversion into euros

The financial statements must be drafted in euro.

Beneficiaries with general accounts established in a currency other than the euro must convert the costs recorded in their accounts into euro, at the average of the daily exchange rates published in the C series of the *Official Journal of the European Union* (ECB website), calculated over the corresponding reporting period.

If no daily euro exchange rate is published in the *Official Journal* for the currency in question, they must be converted at the average of the monthly accounting exchange rates published on the European Commission website (InforEuro), calculated over the corresponding reporting period.

Beneficiaries with general accounts in euro must convert costs incurred in another currency into euro according to their usual accounting practices.

21.4 Reporting language

The reporting must be in the language of the Agreement, unless otherwise agreed with the granting authority (see Data Sheet, Point 4.2).

21.5 Consequences of non-compliance

If a report submitted does not comply with this Article, the granting authority may suspend the payment deadline (see Article 29) and apply other measures described in Chapter 5.

If the coordinator breaches its reporting obligations, the granting authority may terminate the grant or the coordinator's participation (see Article 32) or apply other measures described in Chapter 5.

ARTICLE 22 — PAYMENTS AND RECOVERIES — CALCULATION OF AMOUNTS DUE

22.1 Payments and payment arrangements

Payments will be made in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2).

They will be made in euro to the bank account indicated by the coordinator (see Data Sheet, Point 4.2) and must be distributed without unjustified delay (restrictions may apply to distribution of the initial prefinancing payment; see Data Sheet, Point 4.2).

Payments to this bank account will discharge the granting authority from its payment obligation.

The cost of payment transfers will be borne as follows:

- the granting authority bears the cost of transfers charged by its bank
- the beneficiary bears the cost of transfers charged by its bank
- the party causing a repetition of a transfer bears all costs of the repeated transfer.

Payments by the granting authority will be considered to have been carried out on the date when they are debited to its account.

22.2 Recoveries

Recoveries will be made, if — at beneficiary termination, final payment or afterwards — it turns out that the granting authority has paid too much and needs to recover the amounts undue.

Each beneficiary's financial responsibility in case of recovery is in principle limited to their own debt and undue amounts of their affiliated entities.

In case of enforced recoveries (see Article 22.4), affiliated entities will be held liable for repaying debts of their beneficiaries, if required by the granting authority (see Data Sheet, Point 4.4).

22.3 Amounts due

22.3.1 Prefinancing payments

The aim of the prefinancing is to provide the beneficiaries with a float.

It remains the property of the EU until the final payment.

For **initial prefinancings** (if any), the amount due, schedule and modalities are set out in the Data Sheet (see Point 4.2).

For **additional prefinancings** (if any), the amount due, schedule and modalities are also set out in the Data Sheet (see Point 4.2). However, if the statement on the use of the previous prefinancing payment shows that less than 70% was used, the amount set out in the Data Sheet will be reduced by the difference between the 70% threshold and the amount used.

The contribution to the Mutual Insurance Mechanism will be retained from the prefinancing payments (at the rate and in accordance with the modalities set out in the Data Sheet, see Point 4.2) and transferred to the Mechanism.

Prefinancing payments (or parts of them) may be offset (without the beneficiaries' consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

22.3.2 Amount due at beneficiary termination — Recovery

In case of beneficiary termination, the granting authority will determine the provisional amount due for the beneficiary concerned. Payments (if any) will be made with the next interim or final payment.

The **amount due** will be calculated in the following step:

Step 1 — Calculation of the total accepted EU contribution

Step 1 — Calculation of the total accepted EU contribution

The granting authority will first calculate the 'accepted EU contribution' for the beneficiary for all reporting periods, by calculating the 'maximum EU contribution to costs' (applying the funding rate to the accepted costs of the beneficiary), taking into account requests for a lower contribution to costs and CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions (if any). The resulting amount is the 'total accepted EU contribution' for the beneficiary.

The **balance** is then calculated by deducting the payments received (if any; see report on the distribution of payments in Article 32), from the total accepted EU contribution:

{total accepted EU contribution for the beneficiary
 minus
 {prefinancing and interim payments received (if any)}}.

If the balance is **positive**, the amount will be included in the next interim or final payment to the consortium.

If the balance is **negative**, it will be **recovered** in accordance with the following procedure:

The granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to recover, the amount due, the amount to be recovered and the reasons why and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received), it will confirm the amount to be recovered and ask this amount to be paid to the coordinator (**confirmation letter**).

If payment is not made to the coordinator by the date specified in the confirmation letter, the granting authority may call on the Mutual Insurance Mechanism to intervene, if continuation of the action is guaranteed and the conditions set out in the rules governing the Mechanism are met.

In this case, it will send a **beneficiary recovery letter**, together with a **debit note** with the terms and date for payment.

The debit note for the beneficiary will include the amount calculated for the affiliated entities which also had to end their participation (if any).

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

The amounts will later on also be taken into account for the next interim or final payment.

22.3.3 Interim payments

Interim payments reimburse the eligible costs and contributions claimed for the implementation of the action during the reporting periods (if any).

Interim payments (if any) will be made in accordance with the schedule and modalities set out the Data Sheet (see Point 4.2).

Payment is subject to the approval of the periodic report. Its approval does not imply recognition of compliance, authenticity, completeness or correctness of its content.

The **interim payment** will be calculated by the granting authority in the following steps:

Step 1 — Calculation of the total accepted EU contribution

Step 2 — Limit to the interim payment ceiling

Step 1 — Calculation of the total accepted EU contribution

The granting authority will calculate the ‘accepted EU contribution’ for the action for the reporting period, by first calculating the ‘maximum EU contribution to costs’ (applying the funding rate to the accepted costs of each beneficiary), taking into account requests for a lower contribution to costs, and CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions from beneficiary termination (if any). The resulting amount is the ‘total accepted EU contribution’.

Step 2 — Limit to the interim payment ceiling

The resulting amount is then capped to ensure that the total amount of prefinancing and interim payments (if any) does not exceed the interim payment ceiling set out in the Data Sheet (see Point 4.2).

Interim payments (or parts of them) may be offset (without the beneficiaries’ consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

22.3.4 Final payment — Final grant amount — Revenues and Profit — Recovery

The final payment (payment of the balance) reimburses the remaining part of the eligible costs and contributions claimed for the implementation of the action (if any).

The final payment will be made in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2).

Payment is subject to the approval of the final periodic report. Its approval does not imply recognition of compliance, authenticity, completeness or correctness of its content.

The **final grant amount for the action** will be calculated in the following steps:

Step 1 — Calculation of the total accepted EU contribution

Step 2 — Limit to the maximum grant amount

Step 3 — Reduction due to the no-profit rule

Step 1 — Calculation of the total accepted EU contribution

The granting authority will first calculate the ‘accepted EU contribution’ for the action for all reporting periods, by calculating the ‘maximum EU contribution to costs’ (applying the funding rate to the total accepted costs of each beneficiary), taking into account requests for a lower contribution to costs, CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions (if any). The resulting amount is the ‘total accepted EU contribution’.

Step 2 — Limit to the maximum grant amount

If the resulting amount is higher than the maximum grant amount set out in Article 5.2, it will be limited to the latter.

Step 3 — Reduction due to the no-profit rule

If the no-profit rule is provided for in the Data Sheet (see Point 4.2), the grant must not produce a profit (i.e. surplus of the amount obtained following Step 2 plus the action’s revenues, over the eligible costs and contributions approved by the granting authority).

‘Revenue’ is all income generated by the action, during its duration (see Article 4), for beneficiaries that are profit legal entities (— with the exception of income generated by the exploitation of results, which are not considered as revenues).

If there is a profit, it will be deducted in proportion to the final rate of reimbursement of the eligible costs approved by the granting authority (as compared to the amount calculated following Steps 1 and 2 minus the contributions).

The **balance** (final payment) is then calculated by deducting the total amount of prefinancing and interim payments already made (if any), from the final grant amount:

$$\begin{aligned} &\{\text{final grant amount} \\ &\text{minus} \\ &\{\text{prefinancing and interim payments made (if any)}\}\}. \end{aligned}$$

If the balance is **positive**, it will be **paid** to the coordinator.

The amount retained for the Mutual Insurance Mechanism (see above) will be released and **paid** to the coordinator (in accordance with the rules governing the Mechanism).

The final payment (or part of it) may be offset (without the beneficiaries’ consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

If — despite the release of the Mutual Insurance Mechanism contribution — the balance is **negative**, it will be **recovered** in accordance with the following procedure:

The granting authority will send a **pre-information letter** to the coordinator:

- formally notifying the intention to recover, the final grant amount, the amount to be recovered and the reasons why

- requesting a report on the distribution of payments to the beneficiaries within 30 days of receiving notification and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received) and the coordinator has submitted the report on the distribution of payments, it will calculate the **share of the debt per beneficiary**, by:

- (a) identifying the beneficiaries for which the amount calculated as follows is negative:

$$\left\{ \left\{ \begin{array}{l} \text{total accepted EU contribution for the beneficiary} \\ \text{divided by} \\ \text{total accepted EU contribution for the action} \end{array} \right\} \right.$$

$$\left. \begin{array}{l} \text{multiplied by} \\ \text{final grant amount for the action} \end{array} \right\},$$

$$\text{minus}$$

$$\left\{ \text{prefinancing and interim payments received by the beneficiary (if any)} \right\}$$

and

- (b) dividing the debt:

$$\left\{ \begin{array}{l} \text{amount calculated according to point (a) for the beneficiary concerned} \\ \text{divided by} \\ \text{the sum of the amounts calculated according to point (a) for all the beneficiaries identified according to} \\ \text{point (a)} \end{array} \right\}$$

$$\text{multiplied by}$$

$$\text{the amount to be recovered}.$$

and confirm the amount to be recovered from each beneficiary concerned (**confirmation letter**), together with **debit notes** with the terms and date for payment.

The debit notes for beneficiaries will include the amounts calculated for their affiliated entities (if any).

If the coordinator has not submitted the report on the distribution of payments, the granting authority will **recover** the full amount from the coordinator (**confirmation letter** and **debit note** with the terms and date for payment).

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

22.3.5 Audit implementation after final payment — Revised final grant amount — Recovery

If — after the final payment (in particular, after checks, reviews, audits or investigations; see Article 25) — the granting authority rejects costs or contributions (see Article 27) or reduces the grant (see Article 28), it will calculate the **revised final grant amount** for the beneficiary concerned.

The **beneficiary revised final grant amount** will be calculated in the following step:

Step 1 — Calculation of the revised total accepted EU contribution

Step 1 — Calculation of the revised total accepted EU contribution

The granting authority will first calculate the ‘revised accepted EU contribution’ for the beneficiary, by calculating the ‘revised accepted costs’ and ‘revised accepted contributions’.

After that, it will take into account grant reductions (if any). The resulting ‘revised total accepted EU contribution’ is the beneficiary revised final grant amount.

If the revised final grant amount is lower than the beneficiary’s final grant amount (i.e. its share in the final grant amount for the action), it will be **recovered** in accordance with the following procedure:

The **beneficiary final grant amount** (i.e. share in the final grant amount for the action) is calculated as follows:

$$\left\{ \begin{array}{l} \text{total accepted EU contribution for the beneficiary} \\ \text{divided by} \\ \text{total accepted EU contribution for the action} \end{array} \right\} \times \left\{ \begin{array}{l} \text{final grant amount for the action} \end{array} \right\}.$$

The granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to recover, the amount to be recovered and the reasons why and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received), it will confirm the amount to be recovered (**confirmation letter**), together with a **debit note** with the terms and the date for payment.

Recoveries against affiliated entities (if any) will be handled through their beneficiaries.

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

22.4 Enforced recovery

If payment is not made by the date specified in the debit note, the amount due will be recovered:

- (a) by offsetting the amount — without the coordinator or beneficiary’s consent — against any amounts owed to the coordinator or beneficiary by the granting authority.

In exceptional circumstances, to safeguard the EU financial interests, the amount may be offset before the payment date specified in the debit note.

For grants where the granting authority is the European Commission or an EU executive

agency, debts may also be offset against amounts owed by other Commission services or executive agencies.

(b) financial guarantee(s): not applicable

(c) joint and several liability of beneficiaries: not applicable

(d) by holding affiliated entities jointly and severally liable (if any, see Data Sheet, Point 4.4)

(e) by taking legal action (see Article 43) or, provided that the granting authority is the European Commission or an EU executive agency, by adopting an enforceable decision under Article 299 of the Treaty on the Functioning of the EU (TFEU) and Article 100(2) of EU Financial Regulation 2018/1046.

If the Mutual Insurance Mechanism was called on by the granting authority to intervene, recovery will be continued in the name of the Mutual Insurance Mechanism. If two debit notes were sent, the second one (in the name of the Mutual Insurance Mechanism) will be considered to replace the first one (in the name of the granting authority). Where the MIM intervened, offsetting, enforceable decisions or any other of the above-mentioned forms of enforced recovery may be used mutatis mutandis.

The amount to be recovered will be increased by **late-payment interest** at the rate set out in Article 22.5, from the day following the payment date in the debit note, up to and including the date the full payment is received.

Partial payments will be first credited against expenses, charges and late-payment interest and then against the principal.

Bank charges incurred in the recovery process will be borne by the beneficiary, unless Directive 2015/2366¹⁸ applies.

For grants where the granting authority is an EU executive agency, enforced recovery by offsetting or enforceable decision will be done by the services of the European Commission (see also Article 43).

22.5 Consequences of non-compliance

22.5.1 If the granting authority does not pay within the payment deadlines (see above), the beneficiaries are entitled to **late-payment interest** at the rate applied by the European Central Bank (ECB) for its main refinancing operations in euros ('reference rate'), plus the rate specified in the Data Sheet (Point 4.2). The reference rate is the rate in force on the first day of the month in which the payment deadline expires, as published in the C series of the *Official Journal of the European Union*.

If the late-payment interest is lower than or equal to EUR 200, it will be paid to the coordinator only on request submitted within two months of receiving the late payment.

Late-payment interest is not due if all beneficiaries are EU Member States (including regional and local government authorities or other public bodies acting on behalf of a Member State for the purpose of this Agreement).

¹⁸ Directive (EU) 2015/2366 of the European Parliament and of the Council of 25 November 2015 on payment services in the internal market, amending Directives 2002/65/EC, 2009/110/EC and 2013/36/EU and Regulation (EU) No 1093/2010, and repealing Directive 2007/64/EC (OJ L 337, 23.12.2015, p. 35).

If payments or the payment deadline are suspended (see Articles 29 and 30), payment will not be considered as late.

Late-payment interest covers the period running from the day following the due date for payment (see above), up to and including the date of payment.

Late-payment interest is not considered for the purposes of calculating the final grant amount.

22.5.2 If the coordinator breaches any of its obligations under this Article, the grant may be reduced (see Article 28) and the grant or the coordinator may be terminated (see Article 32).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 23 — GUARANTEES

Not applicable

ARTICLE 24 — CERTIFICATES

24.1 Operational verification report (OVR)

Not applicable

24.2 Certificate on the financial statements (CFS)

If required by the granting authority (see Data Sheet, Point 4.3), the beneficiaries must provide certificates on their financial statements (CFS), in accordance with the schedule, threshold and conditions set out in the Data Sheet.

The coordinator must submit them as part of the periodic report (see Article 21).

The certificates must be drawn up using the template published on the Portal, cover the costs declared on the basis of actual costs and costs according to usual cost accounting practices (if any), and fulfil the following conditions:

- (a) be provided by a qualified approved external auditor which is independent and complies with Directive 2006/43/EC¹⁹ (or for public bodies: by a competent independent public officer)
- (b) the verification must be carried out according to the highest professional standards to ensure that the financial statements comply with the provisions under the Agreement and that the costs declared are eligible.

The certificates will not affect the granting authority's right to carry out its own checks, reviews or audits, nor preclude the European Court of Auditors (ECA), the European Public Prosecutor's Office (EPPO) or the European Anti-Fraud Office (OLAF) from using their prerogatives for audits and investigations under the Agreement (see Article 25).

If the costs (or a part of them) were already audited by the granting authority, these costs do not need to be covered by the certificate and will not be counted for calculating the threshold (if any).

¹⁹ Directive 2006/43/EC of the European Parliament and of the Council of 17 May 2006 on statutory audits of annual accounts and consolidated accounts or similar national regulations (OJ L 157, 9.6.2006, p. 87).

24.3 Certificate on the compliance of usual cost accounting practices (CoMUC)

Not applicable

24.4 Systems and process audit (SPA)

Beneficiaries which:

- use unit, flat rate or lump sum costs or contributions according to documented (i.e. formally approved and in writing) usual costs accounting practices (if any) or
- have formalised documentation on the systems and processes for calculating their costs and contributions (i.e. formally approved and in writing), have participated in at least 150 actions under Horizon 2020 or the Euratom Research and Training Programme (2014-2018 or 2019-2020) and participate in at least 3 ongoing actions under Horizon Europe or the Euratom Research and Training Programme (2021-2025 or 2026-2027)

may apply to the granting authority for a systems and process audit (SPA).

This audit will be carried out as follows:

Step 1 – Application by the beneficiary.

Step 2 – If the application is accepted, the granting authority will carry out the systems and process audit, complemented by an audit of transactions (on a sample of the beneficiary's Horizon Europe or the Euratom Research and Training Programme financial statements).

Step 3 – The audit result will take the form of a risk assessment classification for the beneficiary: low, medium or high.

Low-risk beneficiaries will benefit from less (or less in-depth) ex-post audits (see Article 25) and a higher threshold for submitting certificates on the financial statements (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3).

24.5 Consequences of non-compliance

If a beneficiary does not submit a certificate on the financial statements (CFS) or the certificate is rejected, the accepted EU contribution to costs will be capped to reflect the CFS threshold.

If a beneficiary breaches any of its other obligations under this Article, the granting authority may apply the measures described in Chapter 5.

ARTICLE 25 — CHECKS, REVIEWS, AUDITS AND INVESTIGATIONS — EXTENSION OF FINDINGS

25.1 Granting authority checks, reviews and audits

25.1.1 Internal checks

The granting authority may — during the action or afterwards — check the proper implementation of the action and compliance with the obligations under the Agreement, including assessing costs and contributions, deliverables and reports.

25.1.2 Project reviews

The granting authority may carry out reviews on the proper implementation of the action and compliance with the obligations under the Agreement (general project reviews or specific issues reviews).

Such project reviews may be started during the implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the coordinator or beneficiary concerned and will be considered to start on the date of the notification.

If needed, the granting authority may be assisted by independent, outside experts. If it uses outside experts, the coordinator or beneficiary concerned will be informed and have the right to object on grounds of commercial confidentiality or conflict of interest.

The coordinator or beneficiary concerned must cooperate diligently and provide — within the deadline requested — any information and data in addition to deliverables and reports already submitted (including information on the use of resources). The granting authority may request beneficiaries to provide such information to it directly. Sensitive information and documents will be treated in accordance with Article 13.

The coordinator or beneficiary concerned may be requested to participate in meetings, including with the outside experts.

For **on-the-spot visits**, the beneficiary concerned must allow access to sites and premises (including to the outside experts) and must ensure that information requested is readily available.

Information provided must be accurate, precise and complete and in the format requested, including electronic format.

On the basis of the review findings, a **project review report** will be drawn up.

The granting authority will formally notify the project review report to the coordinator or beneficiary concerned, which has 30 days from receiving notification to make observations.

Project reviews (including project review reports) will be in the language of the Agreement.

25.1.3 Audits

The granting authority may carry out audits on the proper implementation of the action and compliance with the obligations under the Agreement.

Such audits may be started during the implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the beneficiary concerned and will be considered to start on the date of the notification.

The granting authority may use its own audit service, delegate audits to a centralised service or use external audit firms. If it uses an external firm, the beneficiary concerned will be informed and have the right to object on grounds of commercial confidentiality or conflict of interest.

The beneficiary concerned must cooperate diligently and provide — within the deadline requested — any information (including complete accounts, individual salary statements or other personal data)

to verify compliance with the Agreement. Sensitive information and documents will be treated in accordance with Article 13.

For **on-the-spot** visits, the beneficiary concerned must allow access to sites and premises (including for the external audit firm) and must ensure that information requested is readily available.

Information provided must be accurate, precise and complete and in the format requested, including electronic format.

On the basis of the audit findings, a **draft audit report** will be drawn up.

The auditors will formally notify the draft audit report to the beneficiary concerned, which has 30 days from receiving notification to make observations (contradictory audit procedure).

The **final audit report** will take into account observations by the beneficiary concerned and will be formally notified to them.

Audits (including audit reports) will be in the language of the Agreement.

25.2 European Commission checks, reviews and audits in grants of other granting authorities

Where the granting authority is not the European Commission, the latter has the same rights of checks, reviews and audits as the granting authority.

25.3 Access to records for assessing simplified forms of funding

The beneficiaries must give the European Commission access to their statutory records for the periodic assessment of simplified forms of funding which are used in EU programmes.

25.4 OLAF, EPPO and ECA audits and investigations

The following bodies may also carry out checks, reviews, audits and investigations — during the action or afterwards:

- the European Anti-Fraud Office (OLAF) under Regulations No 883/2013²⁰ and No 2185/96²¹
- the European Public Prosecutor's Office (EPPO) under Regulation 2017/1939
- the European Court of Auditors (ECA) under Article 287 of the Treaty on the Functioning of the EU (TFEU) and Article 257 of EU Financial Regulation 2018/1046.

If requested by these bodies, the beneficiary concerned must provide full, accurate and complete information in the format requested (including complete accounts, individual salary statements or

²⁰ Regulation (EU, Euratom) No 883/2013 of the European Parliament and of the Council of 11 September 2013 concerning investigations conducted by the European Anti-Fraud Office (OLAF) and repealing Regulation (EC) No 1073/1999 of the European Parliament and of the Council and Council Regulation (Euratom) No 1074/1999 (OJ L 248, 18/09/2013, p. 1).

²¹ Council Regulation (Euratom, EC) No 2185/96 of 11 November 1996 concerning on-the-spot checks and inspections carried out by the Commission in order to protect the European Communities' financial interests against fraud and other irregularities (OJ L 292, 15/11/1996, p. 2).

other personal data, including in electronic format) and allow access to sites and premises for on-the-spot visits or inspections — as provided for under these Regulations.

To this end, the beneficiary concerned must keep all relevant information relating to the action, at least until the time-limit set out in the Data Sheet (Point 6) and, in any case, until any ongoing checks, reviews, audits, investigations, litigation or other pursuits of claims have been concluded.

25.5 Consequences of checks, reviews, audits and investigations — Extension of results of reviews, audits or investigations

25.5.1 Consequences of checks, reviews, audits and investigations in this grant

Findings in checks, reviews, audits or investigations carried out in the context of this grant may lead to rejections (see Article 27), grant reduction (see Article 28) or other measures described in Chapter 5.

Rejections or grant reductions after the final payment will lead to a revised final grant amount (see Article 22).

Findings in checks, reviews, audits or investigations during the action implementation may lead to a request for amendment (see Article 39), to change the description of the action set out in Annex 1.

Checks, reviews, audits or investigations that find systemic or recurrent errors, irregularities, fraud or breach of obligations in any EU grant may also lead to consequences in other EU grants awarded under similar conditions ('extension to other grants').

Moreover, findings arising from an OLAF or EPPO investigation may lead to criminal prosecution under national law.

25.5.2 Extension from other grants

Results of checks, reviews, audits or investigations in other grants may be extended to this grant, if:

- (a) the beneficiary concerned is found, in other EU grants awarded under similar conditions, to have committed systemic or recurrent errors, irregularities, fraud or breach of obligations that have a material impact on this grant and
- (b) those findings are formally notified to the beneficiary concerned — together with the list of grants affected by the findings — within the time-limit for audits set out in the Data Sheet (see Point 6).

The granting authority will formally notify the beneficiary concerned of the intention to extend the findings and the list of grants affected.

If the extension concerns **rejections of costs or contributions**: the notification will include:

- (a) an invitation to submit observations on the list of grants affected by the findings
- (b) the request to submit revised financial statements for all grants affected
- (c) the correction rate for extrapolation, established on the basis of the systemic or recurrent errors, to calculate the amounts to be rejected, if the beneficiary concerned:

- (i) considers that the submission of revised financial statements is not possible or practicable or
- (ii) does not submit revised financial statements.

If the extension concerns **grant reductions**: the notification will include:

- (a) an invitation to submit observations on the list of grants affected by the findings and
- (b) the **correction rate for extrapolation**, established on the basis of the systemic or recurrent errors and the principle of proportionality.

The beneficiary concerned has **60 days** from receiving notification to submit observations, revised financial statements or to propose a duly substantiated **alternative correction method/rate**.

On the basis of this, the granting authority will analyse the impact and decide on the implementation (i.e. start rejection or grant reduction procedures, either on the basis of the revised financial statements or the announced/alternative method/rate or a mix of those; see Articles 27 and 28).

25.6 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, costs or contributions insufficiently substantiated will be ineligible (see Article 6) and will be rejected (see Article 27), and the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 26 — IMPACT EVALUATIONS

26.1 Impact evaluation

The granting authority may carry out impact evaluations of the action, measured against the objectives and indicators of the EU programme funding the grant.

Such evaluations may be started during implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the coordinator or beneficiaries and will be considered to start on the date of the notification.

If needed, the granting authority may be assisted by independent outside experts.

The coordinator or beneficiaries must provide any information relevant to evaluate the impact of the action, including information in electronic format.

26.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the granting authority may apply the measures described in Chapter 5.

CHAPTER 5 CONSEQUENCES OF NON-COMPLIANCE

SECTION 1 REJECTIONS AND GRANT REDUCTION

ARTICLE 27 — REJECTION OF COSTS AND CONTRIBUTIONS

27.1 Conditions

The granting authority will — at beneficiary termination, interim payment, final payment or afterwards — reject any costs or contributions which are ineligible (see Article 6), in particular following checks, reviews, audits or investigations (see Article 25).

The rejection may also be based on the extension of findings from other grants to this grant (see Article 25).

Ineligible costs or contributions will be rejected.

27.2 Procedure

If the rejection does not lead to a recovery, the granting authority will formally notify the coordinator or beneficiary concerned of the rejection, the amounts and the reasons why. The coordinator or beneficiary concerned may — within 30 days of receiving notification — submit observations if it disagrees with the rejection (payment review procedure).

If the rejection leads to a recovery, the granting authority will follow the contradictory procedure with pre-information letter set out in Article 22.

27.3 Effects

If the granting authority rejects costs or contributions, it will deduct them from the costs or contributions declared and then calculate the amount due (and, if needed, make a recovery; see Article 22).

ARTICLE 28 — GRANT REDUCTION

28.1 Conditions

The granting authority may — at beneficiary termination, final payment or afterwards — reduce the grant for a beneficiary, if:

- (a) the beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed:
 - (i) substantial errors, irregularities or fraud or
 - (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or
- (b) the beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants

awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant (see Article 25).

The amount of the reduction will be calculated for each beneficiary concerned and proportionate to the seriousness and the duration of the errors, irregularities or fraud or breach of obligations, by applying an individual reduction rate to their accepted EU contribution.

28.2 Procedure

If the grant reduction does not lead to a recovery, the granting authority will formally notify the coordinator or beneficiary concerned of the reduction, the amount to be reduced and the reasons why. The coordinator or beneficiary concerned may — within 30 days of receiving notification — submit observations if it disagrees with the reduction (payment review procedure).

If the grant reduction leads to a recovery, the granting authority will follow the contradictory procedure with pre-information letter set out in Article 22.

28.3 Effects

If the granting authority reduces the grant, it will deduct the reduction and then calculate the amount due (and, if needed, make a recovery; see Article 22).

SECTION 2 SUSPENSION AND TERMINATION

ARTICLE 29 — PAYMENT DEADLINE SUSPENSION

29.1 Conditions

The granting authority may — at any moment — suspend the payment deadline if a payment cannot be processed because:

- (a) the required report (see Article 21) has not been submitted or is not complete or additional information is needed
- (b) there are doubts about the amount to be paid (e.g. ongoing audit extension procedure, queries about eligibility, need for a grant reduction, etc.) and additional checks, reviews, audits or investigations are necessary, or
- (c) there are other issues affecting the EU financial interests.

29.2 Procedure

The granting authority will formally notify the coordinator of the suspension and the reasons why.

The suspension will **take effect** the day the notification is sent.

If the conditions for suspending the payment deadline are no longer met, the suspension will be **lifted** — and the remaining time to pay (see Data Sheet, Point 4.2) will resume.

If the suspension exceeds two months, the coordinator may request the granting authority to confirm if the suspension will continue.

If the payment deadline has been suspended due to the non-compliance of the report and the revised report is not submitted (or was submitted but is also rejected), the granting authority may also terminate the grant or the participation of the coordinator (see Article 32).

ARTICLE 30 — PAYMENT SUSPENSION

30.1 Conditions

The granting authority may — at any moment — suspend payments, in whole or in part for one or more beneficiaries, if:

- (a) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed or is suspected of having committed:
 - (i) substantial errors, irregularities or fraud or
 - (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or
- (b) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant.

If payments are suspended for one or more beneficiaries, the granting authority will make partial payment(s) for the part(s) not suspended. If suspension concerns the final payment, the payment (or recovery) of the remaining amount after suspension is lifted will be considered to be the payment that closes the action.

30.2 Procedure

Before suspending payments, the granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to suspend payments and the reasons why and
- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the suspension (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

At the end of the suspension procedure, the granting authority will also inform the coordinator.

The suspension will **take effect** the day after the confirmation notification is sent.

If the conditions for resuming payments are met, the suspension will be **lifted**. The granting authority will formally notify the beneficiary concerned (and the coordinator) and set the suspension end date.

During the suspension, no prefinancing will be paid to the beneficiaries concerned. For interim payments, the periodic reports for all reporting periods except the last one (see Article 21) must not contain any financial statements from the beneficiary concerned (or its affiliated entities). The coordinator must include them in the next periodic report after the suspension is lifted or — if suspension is not lifted before the end of the action — in the last periodic report.

ARTICLE 31 — GRANT AGREEMENT SUSPENSION

31.1 Consortium-requested GA suspension

31.1.1 Conditions and procedure

The beneficiaries may request the suspension of the grant or any part of it, if exceptional circumstances — in particular *force majeure* (see Article 35) — make implementation impossible or excessively difficult.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the date the suspension takes effect; this date may be before the date of the submission of the amendment request and
- the expected date of resumption.

The suspension will **take effect** on the day specified in the amendment.

Once circumstances allow for implementation to resume, the coordinator must immediately request another **amendment** of the Agreement to set the suspension end date, the resumption date (one day after suspension end date), extend the duration and make other changes necessary to adapt the action to the new situation (see Article 39) — unless the grant has been terminated (see Article 32). The suspension will be **lifted** with effect from the suspension end date set out in the amendment. This date may be before the date of the submission of the amendment request.

During the suspension, no prefinancing will be paid. Costs incurred or contributions for activities implemented during grant suspension are not eligible (see Article 6.3).

31.2 EU-initiated GA suspension

31.2.1 Conditions

The granting authority may suspend the grant or any part of it, if:

- (a) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed or is suspected of having committed:
 - (i) substantial errors, irregularities or fraud or
 - (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions,

submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or

(b) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant

(c) other:

(i) linked action issues: not applicable

(ii) the action has lost its scientific or technological relevance, for EIC Accelerator actions: the action has lost its economic relevance, for challenge-based EIC Pathfinder actions and Horizon Europe Missions: the action has lost its relevance as part of the Portfolio for which it has been initially selected

31.2.2 Procedure

Before suspending the grant, the granting authority will send a **pre-information letter** to the coordinator:

- formally notifying the intention to suspend the grant and the reasons why and
- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the suspension (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

The suspension will **take effect** the day after the confirmation notification is sent (or on a later date specified in the notification).

Once the conditions for resuming implementation of the action are met, the granting authority will formally notify the coordinator a **lifting of suspension letter**, in which it will set the suspension end date and invite the coordinator to request an amendment of the Agreement to set the resumption date (one day after suspension end date), extend the duration and make other changes necessary to adapt the action to the new situation (see Article 39) — unless the grant has been terminated (see Article 32). The suspension will be **lifted** with effect from the suspension end date set out in the lifting of suspension letter. This date may be before the date on which the letter is sent.

During the suspension, no prefinancing will be paid. Costs incurred or contributions for activities implemented during suspension are not eligible (see Article 6.3).

The beneficiaries may not claim damages due to suspension by the granting authority (see Article 33).

Grant suspension does not affect the granting authority's right to terminate the grant or a beneficiary (see Article 32) or reduce the grant (see Article 28).

ARTICLE 32 — GRANT AGREEMENT OR BENEFICIARY TERMINATION

32.1 Consortium-requested GA termination

32.1.1 Conditions and procedure

The beneficiaries may request the termination of the grant.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the date the consortium ends work on the action ('end of work date') and
- the date the termination takes effect ('termination date'); this date must be after the date of the submission of the amendment request.

The termination will **take effect** on the termination date specified in the amendment.

If no reasons are given or if the granting authority considers the reasons do not justify termination, it may consider the grant terminated improperly.

32.1.2 Effects

The coordinator must — within 60 days from when termination takes effect — submit a **periodic report** (for the open reporting period until termination).

The granting authority will calculate the final grant amount and final payment on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before the end of work date (see Article 22). Costs relating to contracts due for execution only after the end of work are not eligible.

If the granting authority does not receive the report within the deadline, only costs and contributions which are included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

Improper termination may lead to a grant reduction (see Article 28).

After termination, the beneficiaries' obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

32.2 Consortium-requested beneficiary termination

32.2.1 Conditions and procedure

The coordinator may request the termination of the participation of one or more beneficiaries, on request of the beneficiary concerned or on behalf of the other beneficiaries.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the opinion of the beneficiary concerned (or proof that this opinion has been requested in writing)

- the date the beneficiary ends work on the action ('end of work date')
- the date the termination takes effect ('termination date'); this date must be after the date of the submission of the amendment request.

If the termination concerns the coordinator and is done without its agreement, the amendment request must be submitted by another beneficiary (acting on behalf of the consortium).

The termination will **take effect** on the termination date specified in the amendment.

If no information is given or if the granting authority considers that the reasons do not justify termination, it may consider the beneficiary to have been terminated improperly.

32.2.2 Effects

The coordinator must — within 60 days from when termination takes effect — submit:

- (i) a **report on the distribution of payments** to the beneficiary concerned
- (ii) a **termination report** from the beneficiary concerned, for the open reporting period until termination, containing an overview of the progress of the work, the financial statement, the explanation on the use of resources, and, if applicable, the certificate on the financial statement (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3)
- (iii) a second **request for amendment** (see Article 39) with other amendments needed (e.g. reallocation of the tasks and the estimated budget of the terminated beneficiary; addition of a new beneficiary to replace the terminated beneficiary; change of coordinator, etc.).

The granting authority will calculate the amount due to the beneficiary on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before the end of work date (see Article 22). Costs relating to contracts due for execution only after the end of work are not eligible.

The information in the termination report must also be included in the periodic report for the next reporting period (see Article 21).

If the granting authority does not receive the termination report within the deadline, only costs and contributions which are included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

If the granting authority does not receive the report on the distribution of payments within the deadline, it will consider that:

- the coordinator did not distribute any payment to the beneficiary concerned and that
- the beneficiary concerned must not repay any amount to the coordinator.

If the second request for amendment is accepted by the granting authority, the Agreement is **amended** to introduce the necessary changes (see Article 39).

If the second request for amendment is rejected by the granting authority (because it calls into question the decision awarding the grant or breaches the principle of equal treatment of applicants), the grant may be terminated (see Article 32).

Improper termination may lead to a reduction of the grant (see Article 31) or grant termination (see Article 32).

After termination, the concerned beneficiary's obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

32.3 EU-initiated GA or beneficiary termination

32.3.1 Conditions

The granting authority may terminate the grant or the participation of one or more beneficiaries, if:

- (a) one or more beneficiaries do not accede to the Agreement (see Article 40)
- (b) a change to the action or the legal, financial, technical, organisational or ownership situation of a beneficiary is likely to substantially affect the implementation of the action or calls into question the decision to award the grant (including changes linked to one of the exclusion grounds listed in the declaration of honour)
- (c) following termination of one or more beneficiaries, the necessary changes to the Agreement (and their impact on the action) would call into question the decision awarding the grant or breach the principle of equal treatment of applicants
- (d) implementation of the action has become impossible or the changes necessary for its continuation would call into question the decision awarding the grant or breach the principle of equal treatment of applicants
- (e) a beneficiary (or person with unlimited liability for its debts) is subject to bankruptcy proceedings or similar (including insolvency, winding-up, administration by a liquidator or court, arrangement with creditors, suspension of business activities, etc.)
- (f) a beneficiary (or person with unlimited liability for its debts) is in breach of social security or tax obligations
- (g) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has been found guilty of grave professional misconduct
- (h) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed fraud, corruption, or is involved in a criminal organisation, money laundering, terrorism-related crimes (including terrorism financing), child labour or human trafficking
- (i) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) was created under a different jurisdiction with the intent to circumvent fiscal, social or other legal obligations in the country of origin (or created another entity with this purpose)
- (j) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed:

- (i) substantial errors, irregularities or fraud or
- (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.)
- (k) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant (extension of findings from other grants to this grant; see Article 25)
- (l) despite a specific request by the granting authority, a beneficiary does not request — through the coordinator — an amendment to the Agreement to end the participation of one of its affiliated entities or associated partners that is in one of the situations under points (d), (f), (e), (g), (h), (i) or (j) and to reallocate its tasks, or
- (m) other:
 - (i) linked action issues: not applicable
 - (ii) the action has lost its scientific or technological relevance, for EIC Accelerator actions: the action has lost its economic relevance, for challenge-based EIC Pathfinder actions and Horizon Europe Missions: the action has lost its relevance as part of the Portfolio for which it has been initially selected

32.3.2 Procedure

Before terminating the grant or participation of one or more beneficiaries, the granting authority will send a **pre-information letter** to the coordinator or beneficiary concerned:

- formally notifying the intention to terminate and the reasons why and
- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the termination and the date it will take effect (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

For beneficiary terminations, the granting authority will — at the end of the procedure — also inform the coordinator.

The termination will **take effect** the day after the confirmation notification is sent (or on a later date specified in the notification; ‘termination date’).

32.3.3 Effects

- (a) for **GA termination**:

The coordinator must — within 60 days from when termination takes effect — submit a **periodic report** (for the last open reporting period until termination).

The granting authority will calculate the final grant amount and final payment on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before termination takes effect (see Article 22). Costs relating to contracts due for execution only after termination are not eligible.

If the grant is terminated for breach of the obligation to submit reports, the coordinator may not submit any report after termination.

If the granting authority does not receive the report within the deadline, only costs and contributions which are included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

Termination does not affect the granting authority's right to reduce the grant (see Article 28) or to impose administrative sanctions (see Article 34).

The beneficiaries may not claim damages due to termination by the granting authority (see Article 33).

After termination, the beneficiaries' obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

(b) for beneficiary termination:

The coordinator must — within 60 days from when termination takes effect — submit:

- (i) a **report on the distribution of payments** to the beneficiary concerned
- (ii) a **termination report** from the beneficiary concerned, for the open reporting period until termination, containing an overview of the progress of the work, the financial statement, the explanation on the use of resources, and, if applicable, the certificate on the financial statement (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3)
- (iii) a **request for amendment** (see Article 39) with any amendments needed (e.g. reallocation of the tasks and the estimated budget of the terminated beneficiary; addition of a new beneficiary to replace the terminated beneficiary; change of coordinator, etc.).

The granting authority will calculate the amount due to the beneficiary on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before termination takes effect (see Article 22). Costs relating to contracts due for execution only after termination are not eligible.

The information in the termination report must also be included in the periodic report for the next reporting period (see Article 21).

If the granting authority does not receive the termination report within the deadline, only costs and contributions included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

If the granting authority does not receive the report on the distribution of payments within the deadline, it will consider that:

- the coordinator did not distribute any payment to the beneficiary concerned and that
- the beneficiary concerned must not repay any amount to the coordinator.

If the request for amendment is accepted by the granting authority, the Agreement is **amended** to introduce the necessary changes (see Article 39).

If the request for amendment is rejected by the granting authority (because it calls into question the decision awarding the grant or breaches the principle of equal treatment of applicants), the grant may be terminated (see Article 32).

After termination, the concerned beneficiary's obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

SECTION 3 OTHER CONSEQUENCES: DAMAGES AND ADMINISTRATIVE SANCTIONS

ARTICLE 33 — DAMAGES

33.1 Liability of the granting authority

The granting authority cannot be held liable for any damage caused to the beneficiaries or to third parties as a consequence of the implementation of the Agreement, including for gross negligence.

The granting authority cannot be held liable for any damage caused by any of the beneficiaries or other participants involved in the action, as a consequence of the implementation of the Agreement.

33.2 Liability of the beneficiaries

The beneficiaries must compensate the granting authority for any damage it sustains as a result of the implementation of the action or because the action was not implemented in full compliance with the Agreement, provided that it was caused by gross negligence or wilful act.

The liability does not extend to indirect or consequential losses or similar damage (such as loss of profit, loss of revenue or loss of contracts), provided such damage was not caused by wilful act or by a breach of confidentiality.

ARTICLE 34 — ADMINISTRATIVE SANCTIONS AND OTHER MEASURES

Nothing in this Agreement may be construed as preventing the adoption of administrative sanctions (i.e. exclusion from EU award procedures and/or financial penalties) or other public law measures, in addition or as an alternative to the contractual measures provided under this Agreement (see,

for instance, Articles 135 to 145 EU Financial Regulation 2018/1046 and Articles 4 and 7 of Regulation 2988/95²²).

SECTION 4 FORCE MAJEURE

ARTICLE 35 — FORCE MAJEURE

A party prevented by force majeure from fulfilling its obligations under the Agreement cannot be considered in breach of them.

‘Force majeure’ means any situation or event that:

- prevents either party from fulfilling their obligations under the Agreement,
- was unforeseeable, exceptional situation and beyond the parties’ control,
- was not due to error or negligence on their part (or on the part of other participants involved in the action), and
- proves to be inevitable in spite of exercising all due diligence.

Any situation constituting force majeure must be formally notified to the other party without delay, stating the nature, likely duration and foreseeable effects.

The parties must immediately take all the necessary steps to limit any damage due to force majeure and do their best to resume implementation of the action as soon as possible.

CHAPTER 6 FINAL PROVISIONS

ARTICLE 36 — COMMUNICATION BETWEEN THE PARTIES

36.1 Forms and means of communication — Electronic management

EU grants are managed fully electronically through the EU Funding & Tenders Portal (‘Portal’).

All communications must be made electronically through the Portal, in accordance with the Portal Terms and Conditions and using the forms and templates provided there (except if explicitly instructed otherwise by the granting authority).

Communications must be made in writing and clearly identify the grant agreement (project number and acronym).

Communications must be made by persons authorised according to the Portal Terms and Conditions. For naming the authorised persons, each beneficiary must have designated — before the signature of this Agreement — a ‘legal entity appointed representative (LEAR)’. The role and tasks of the LEAR are stipulated in their appointment letter (see Portal Terms and Conditions).

²² Council Regulation (EC, Euratom) No 2988/95 of 18 December 1995 on the protection of the European Communities financial interests (OJ L 312, 23.12.1995, p. 1).

If the electronic exchange system is temporarily unavailable, instructions will be given on the Portal.

36.2 Date of communication

The sending date for communications made through the Portal will be the date and time of sending, as indicated by the time logs.

The receiving date for communications made through the Portal will be the date and time the communication is accessed, as indicated by the time logs. Formal notifications that have not been accessed within 10 days after sending, will be considered to have been accessed (see Portal Terms and Conditions).

If a communication is exceptionally made on paper (by e-mail or postal service), general principles apply (i.e. date of sending/receipt). Formal notifications by registered post with proof of delivery will be considered to have been received either on the delivery date registered by the postal service or the deadline for collection at the post office.

If the electronic exchange system is temporarily unavailable, the sending party cannot be considered in breach of its obligation to send a communication within a specified deadline.

36.3 Addresses for communication

The Portal can be accessed via the Europa website.

The address for paper communications to the granting authority (if exceptionally allowed) is the official mailing address indicated on its website.

For beneficiaries, it is the legal address specified in the Portal Participant Register.

ARTICLE 37 — INTERPRETATION OF THE AGREEMENT

The provisions in the Data Sheet take precedence over the rest of the Terms and Conditions of the Agreement.

Annex 5 takes precedence over the Terms and Conditions; the Terms and Conditions take precedence over the Annexes other than Annex 5.

Annex 2 takes precedence over Annex 1.

ARTICLE 38 — CALCULATION OF PERIODS AND DEADLINES

In accordance with Regulation No 1182/71²³, periods expressed in days, months or years are calculated from the moment the triggering event occurs.

The day during which that event occurs is not considered as falling within the period.

‘Days’ means calendar days, not working days.

ARTICLE 39 — AMENDMENTS

²³ Regulation (EEC, Euratom) No 1182/71 of the Council of 3 June 1971 determining the rules applicable to periods, dates and time-limits (OJ L 124, 8/6/1971, p. 1).

39.1 Conditions

The Agreement may be amended, unless the amendment entails changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants.

Amendments may be requested by any of the parties.

39.2 Procedure

The party requesting an amendment must submit a request for amendment signed directly in the Portal Amendment tool.

The coordinator submits and receives requests for amendment on behalf of the beneficiaries (see Annex 3). If a change of coordinator is requested without its agreement, the submission must be done by another beneficiary (acting on behalf of the other beneficiaries).

The request for amendment must include:

- the reasons why
- the appropriate supporting documents and
- for a change of coordinator without its agreement: the opinion of the coordinator (or proof that this opinion has been requested in writing).

The granting authority may request additional information.

If the party receiving the request agrees, it must sign the amendment in the tool within 45 days of receiving notification (or any additional information the granting authority has requested). If it does not agree, it must formally notify its disagreement within the same deadline. The deadline may be extended, if necessary for the assessment of the request. If no notification is received within the deadline, the request is considered to have been rejected.

An amendment **enters into force** on the day of the signature of the receiving party.

An amendment **takes effect** on the date of entry into force or other date specified in the amendment.

ARTICLE 40 — ACCESSION AND ADDITION OF NEW BENEFICIARIES

40.1 Accession of the beneficiaries mentioned in the Preamble

The beneficiaries which are not coordinator must accede to the grant by signing the accession form (see Annex 3) directly in the Portal Grant Preparation tool, within 30 days after the entry into force of the Agreement (see Article 44).

They will assume the rights and obligations under the Agreement with effect from the date of its entry into force (see Article 44).

If a beneficiary does not accede to the grant within the above deadline, the coordinator must — within 30 days — request an amendment (see Article 39) to terminate the beneficiary and make any changes

necessary to ensure proper implementation of the action. This does not affect the granting authority's right to terminate the grant (see Article 32).

40.2 Addition of new beneficiaries

In justified cases, the beneficiaries may request the addition of a new beneficiary.

For this purpose, the coordinator must submit a request for amendment in accordance with Article 39. It must include an accession form (see Annex 3) signed by the new beneficiary directly in the Portal Amendment tool.

New beneficiaries will assume the rights and obligations under the Agreement with effect from the date of their accession specified in the accession form (see Annex 3).

Additions are also possible in mono-beneficiary grants.

ARTICLE 41 — TRANSFER OF THE AGREEMENT

In justified cases, the beneficiary of a mono-beneficiary grant may request the transfer of the grant to a new beneficiary, provided that this would not call into question the decision awarding the grant or breach the principle of equal treatment of applicants.

The beneficiary must submit a request for **amendment** (see Article 39), with

- the reasons why
- the accession form (see Annex 3) signed by the new beneficiary directly in the Portal Amendment tool and
- additional supporting documents (if required by the granting authority).

The new beneficiary will assume the rights and obligations under the Agreement with effect from the date of accession specified in the accession form (see Annex 3).

ARTICLE 42 — ASSIGNMENTS OF CLAIMS FOR PAYMENT AGAINST THE GRANTING AUTHORITY

The beneficiaries may not assign any of their claims for payment against the granting authority to any third party, except if expressly approved in writing by the granting authority on the basis of a reasoned, written request by the coordinator (on behalf of the beneficiary concerned).

If the granting authority has not accepted the assignment or if the terms of it are not observed, the assignment will have no effect on it.

In no circumstances will an assignment release the beneficiaries from their obligations towards the granting authority.

ARTICLE 43 — APPLICABLE LAW AND SETTLEMENT OF DISPUTES

43.1 Applicable law

The Agreement is governed by the applicable EU law, supplemented if necessary by the law of Belgium.

Special rules may apply for beneficiaries which are international organisations (if any; see Data Sheet, Point 5).

43.2 Dispute settlement

If a dispute concerns the interpretation, application or validity of the Agreement, the parties must bring action before the EU General Court — or, on appeal, the EU Court of Justice — under Article 272 of the Treaty on the Functioning of the EU (TFEU).

For non-EU beneficiaries (if any), such disputes must be brought before the courts of Brussels, Belgium — unless an international agreement provides for the enforceability of EU court judgements.

For beneficiaries with arbitration as special dispute settlement forum (if any; see Data Sheet, Point 5), the dispute will — in the absence of an amicable settlement — be settled in accordance with the Rules for Arbitration published on the Portal.

If a dispute concerns administrative sanctions, offsetting or an enforceable decision under Article 299 TFEU (see Articles 22 and 34), the beneficiaries must bring action before the General Court — or, on appeal, the Court of Justice — under Article 263 TFEU.

For grants where the granting authority is an EU executive agency (see Preamble), actions against offsetting and enforceable decisions must be brought against the European Commission (not against the granting authority; see also Article 22).

ARTICLE 44 — ENTRY INTO FORCE

The Agreement will enter into force on the day of signature by the granting authority or the coordinator, depending on which is later.

SIGNATURES

For the coordinator

Orazio Viele with ECAS id n00224b3 signed in the Participant Portal on 26/04/2023 at 10:36:00 (transaction id SigId-132118-gDrVzpadLAdZrip0knd5zXgl33KfdMqgpOLMRLJoJj8VF61bfvDj7oNjhZ3cVbW3wVbR53vzbIzirVfjLiCzXTj-rS0vSrmBGYCdxRy5uIgXle-JbyBOVZ9tD0p3VMGxshhjrXQazr1KWDwLGg2BGvj26oFBI3394tKmRqf9IOdaPltN2PDLnQcYw8ujzk5a0Gbw). Timestamp by third party at 2023.04.26 10:36:08 CEST

For the granting authority

Signed by Peter PETROV with ECAS id petropt as an authorised representative on 26-04-2023 19:47:22 (transaction id SigId-143196-sDLJD9zKFI5uGpYwVPQpVnzOkkzYnO8mUcj6lznYAsGPZQv2gRfxh7wISqFBBa1uNUqvRYtcdQsaGW1dKqQNmo-rS0vSrmBGYCdxRy5uIgXle-t7JkhmzwbccDm2T09znsVAkNdJFEJv0Aux7ecgIhgW0D4Yc9cKhozVzJA9IWp4iuH1erzzhjVm2LzpVxMXRvIPZu) 2023.04.26 19:47:26 CEST



ANNEX 1



Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT	
<i>Grant Preparation (General Information screen) — Enter the info.</i>	
Project number:	101103998
Project name:	Data-driven Residential Energy Carrier-agnostic Demand Response Tools and Multi-value Services
Project acronym:	DEDALUS
Call:	HORIZON-CL5-2022-D4-01
Topic:	HORIZON-CL5-2022-D4-01-01
Type of action:	HORIZON-IA
Service:	CINEA/C/02
Project starting date:	fixed date: 1 May 2023
Project duration:	36 months

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Staff effort	21
List of deliverables	23
List of milestones (outputs/outcomes)	34
List of critical risks	36
Project reviews	39

PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

DEDALUS will design, develop and demonstrate SSH-driven multi-value energy carrier-agnostic micro (home/apartment)-to macro(building & district-scale) participatory Demand Response (DR) ecosystem, aimed to: (a) facilitate and scale up residential energy consumers massive participation to DR; (b) adapt to a variety of different mono-carrier (electricity, heat) or multi-carrier synergetic scenarios (electricity vs heat and natural gas) at building/district scale, while strengthening social interactions within respective communities. To this aim, we will deploy: 1) a Social Science Framework for multi-dimensional incentives and nudging interventions tools; 2) AI-based individual/building/districts consumers clustering and segmentation algorithms; 3) Open APIs for DR-ready smart appliances; 4) Energy DataSpace adaptation for extended DR interoperability and privacy-preserving DLT/Blockchain Data Governance and flexibility coordination; 5) Digital Twins for consumers-aware DR flexibility planning; 6) Comfort-based flexibility models and tools for serviced apartments for elderly people; 7) flexibility management tools for building and physical/virtual districts energy communities, based on pre-aggregation and shared DR assets; 8) Optimal aggregation of flexible energy assets for district heating and decentralized power2heat 9) Business Sandbox with novel sharing economy social innovation-based business models. DEDALUS solutions will be validated by 5 Front Runners full-scale pilots and replicated by 2 Multipliers in 7 countries, deployed at the interplay of energy, and non-energy (comfort/healthcare, ageing, social innovation) sectors, and operated by different stakeholders (social housing operators, building operators, aggregators, energy communities, retailers/suppliers, DSOs/District Heating Network operators). Capacity building and blueprints will support policy makers and regulators to enable market take-up and EU-wide replication of residential DR.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	ENG	ENGINEERING - INGEGNERIA INFORMATICA SPA	IT	999960488
2	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
3	BEN	CARTIF	FUNDACION CARTIF	ES	999929836
4	BEN	UNIVPM	UNIVERSITA POLITECNICA DELLE MARCHE	IT	999866689
5	BEN	ARC	ARCELIK A.S.	TR	998821708
6	BEN	COMS	COMSENSUS, KOMUNIKACIJE IN SENZORIKA, DOO	SI	937154831
7	BEN	OURPOWER	OURPOWER ENERGIEGENOSSENSCHAFT SCE MIT BESCHRANKTER HAFTUNG	AT	898281208
8	BEN	GREENPOINT	GREEN-POINT 62 GMBH	AT	885248773
9	BEN	BLUEPRINT	BLUEPRINT ENERGY SOLUTIONS GMBH	AT	904096067
10	BEN	FAELLESBO	FAELLESBO	DK	892008024
11	BEN	NEOGRID	NEOGRID TECHNOLOGIES APS	DK	953024419

PARTICIPANTS*Grant Preparation (Beneficiaries screen) — Enter the info.*

Number	Role	Short name	Legal name	Country	PIC
12	BEN	GREEN CITIES	EUROPEAN GREEN CITIES APS	DK	950585936
13	BEN	HERON	HERON SINGLE MEMBER S.A. ENERGY SERVICES	EL	954579911
14	BEN	DOM-X	DOMX IDIOTIKI KEFALAIOUCHIKI ETAIREIA	EL	898645637
15	BEN	ISRAA	ISTITUTO PER SERVIZI DI RICOVERO E ASSISTENZA AGLI ANZIANI	IT	930262690
16	BEN	DYNAME	DYNAME SRL	IT	885390102
17	BEN	CIMNE	CENTRE INTERNACIONAL DE METODES NUMERICS EN ENGINYERIA	ES	999658721
18	BEN	PEUSA	PRODUCTORA ELECTRICA URGELLENCA SA	ES	899979775
19	BEN	ROMUR	ROMUR RENOVALBES SL	ES	885732512
20	BEN	CENTRICA	CENTRICA BUSINESS SOLUTIONS BELGIUM	BE	927493825
21	BEN	TUC	UNIVERSITATEA TEHNICA CLUJ-NAPOCA	RO	999897244
22	BEN	SIN	SMART INNOVATION NORWAY AS	NO	950648016
23	BEN	ICONS	FONDAZIONE ICONS	IT	917939810

LIST OF WORK PACKAGES

Work packages						
Grant Preparation (Work Packages screen) — Enter the info.						
Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
WP1	Coordination and management	1 - ENG	61.00	1	36	D1.1 – Project management plan D1.2 – DEDALUS data management plan D1.3 – DEDALUS first activity report D1.4 – DEDALUS 2nd activity report
WP2	Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR	9 - BLUEPRINT	125.00	1	16	D2.1 – DEDALUS gap analysis and use case capitalisation D2.2 – Active societal participation, emergence, socio-economic enablers and consolidation factors of multi-value and energy carrier-agnostic DR for residential buildings D2.3 – High-level requirements for the DEDALUS social, technological and business framework D2.4 – Enabling conditions for DR in energy-efficient residential buildings
WP3	Technology enablers for residential energy consumer data-driven ‘activation’	3 - CARTIF	163.00	4	28	D3.1 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release) D3.2 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (2nd technology release) D3.3 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (final technology release)
WP4	Value added energy and non-energy data-driven	1 - ENG	174.00	5	29	D4.1 – ‘First wave’ of DEDALUS

Work packages						
<i>Grant Preparation (Work Packages screen) — Enter the info.</i>						
Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
	services for DR in energy-efficient residential buildings					Digital Twins and data-driven services for residential DR D4.2 – "Second wave" of DEDALUS Digital Twins and data-driven services for residential DR D4.3 – "Final wave" of DEDALUS Digital Twins and data-driven services for residential DR
WP5	Demonstration of multi-value and energy carrier-agnostic residential DR, and social acceptance	2 - NTUA	256.00	1	34	D5.1 – DEDALUS evaluation methodology, baseline assessment and pilot implementation plan D5.2 – DEDALUS Pilots' execution documentation and consumers' engagement – pre-pilot phase D5.3 – DEDALUS Pilots' execution documentation and consumers' engagement – full pilot operation phase D5.4 – DEDALUS impact and social acceptance analysis
WP6	People-powered Go2Market and exploitation strategies towards EU replication and market scalability	13 - HERON	91.00	1	36	D6.1 – Preliminary Exploitation plan D6.2 – Intermediate Exploitation Plan D6.3 – Final Exploitation Plan D6.4 – DEDALUS comprehensive replication methodology and roadmap
WP7	Share, connect and attract	23 - ICONS	89.00	1	36	D7.1 – Plan for dissemination and communication D7.2 – DEDALUS Communication Kit D7.3 – Report on Dissemination and

Work packages						
Grant Preparation (Work Packages screen) — Enter the info.						
Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
						Communication activities, including impact assessment

Work package WP1 – Coordination and management

Work Package Number	WP1	Lead Beneficiary	1. ENG
Work Package Name	Coordination and management		
Start Month	1	End Month	36

Objectives

WP1 responsibility is to ensure that all project activities are performed efficiently, in a coordinated and integrated manner, within schedule and budget and in a way that the high research, scientific, technological and stakeholders' expectations are fully met

Description

Description of work

Task 1.1. Project administration and coordination (ENG) [M1-M36]

ENG(17), NTUA(1), CARTIF(1), UNIVPM(1), ARC(1), COMS(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), DYNAM(1), CIMNE(2), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1), ICONS(1)

This task aims at the actual coordination of the project in terms of day-to-day management and interfacing with the EC, including among others a strong project management scheme, Consortium Agreement, management of funds and effort, a common scientific and technical direction within the project and promotion of team development, continuous monitoring of the work progress, consideration of dissemination, standardisation, gender or ethical issues (if applicable), collaboration and reporting channels within the Consortium and with the EC. ENG will set up an interactive, user-friendly internal communication system as the primary channel for knowledge sharing, allowing to store documents, enabling version control and backup capacity, assuring efficient documentation management and direct communication of the implementation and dissemination of information for partners. Exceptions may include pilots' data and stakeholder information (to be stored on a GDPR/ethics-compliant business intelligence system). A detailed Project Management Plan (D1.1), with a GANTT chart and a Work Breakdown Structure, will be completed at a very early stage of the project (M2). It will include a schedule per task, responsible partner related subtasks, related deliverables and dependencies on other tasks. An updated version of this plan will be included in D1.3 and D1.4. The kick-off meeting (M1) will take place either physically in Italy or online, depending on travel conditions. After that, 3 project meetings will take place during the project's lifetime (Austria, Spain, Greece). For sustainability reasons, these meetings will be back-to-back with other physical events organised by the project or held online.

Task 1.2. Technical and scientific management (ENG) [M1-M36]

ENG(2), NTUA(5), CARTIF(1), UNIVPM(1)

The technical co-ordination of the project will be performed by the Technical Committee that is led by the Technical Manager (provided by ENG) in collaboration with the technical WP leaders. In the context of this task, through appropriate communication and meetings, the Technical Committee will co-ordinate and assist WP leaders on technical issues, will undertake initiatives to propose technical solutions and fine-tune technical orientations whenever necessary, and generally will try to ensure consistency, harmonisation and convergence of the scientific and technical activities and results of the project. In collaboration with the Pilot Coordinator (NTUA), clear and coherent integration of the DEDALUS services through demonstrators will be ensured, with the final goal of evaluating their impact from economic, social, and environmental perspective. The Scientific Manager (UNIVPM) shall assist the Project Coordinator and WP Leaders, and report to the former. This includes the management of dependencies between various tasks, review and approval of reports, and resolution of problems of a scientific rather than a technical/administrative nature. During this task, control of the technical work carried out in the related tasks and suggestions of technical modifications and reallocation of resources where necessary will take place.

Task 1.3. Quality control, risk management and Transdisciplinary Advisory Board (ENG) [M1-M36]

ENG(7), NTUA(1)

This task will guide the targeted research and innovation aspects of the work carried out during the project lifespan and will guarantee that they have been produced based on high quality standards. Management and quality assurance procedures will ensure that the project's tasks are aligned with its objectives. The quality management plan (as part of D1.1) will outline quality processes for deliverables: 2-4 internal reviewers (depending on the deliverable nature) will evaluate scientific content, language, and readability: every 6 months, partners will declare availability to review upcoming deliverables; 5 weeks to deadline reviewers will confirm; 4 weeks to deadline a draft will be shared with

reviewers; 3 weeks to deadline reviewers will return their reports; 10 days to deadline deliverable leaders will revise; and 5 days to deadline the deliverable will be copyedited, proofed and submitted to the EC. Quality processes will be agreed in the Consortium Agreement. Potential risks will also be identified, so as to timely initiate corrective actions, if necessary. The relevant activities will be mainly performed by ENG, whereas all the other partners will have to enrol in a peer review rotation mechanism of the project's deliverables, which will be defined in the quality management plan. In order to successfully deal with any issues, difficulties or problems that may arise, a Transdisciplinary Advisory Board will be established. Its collective expertise will draw on a broad range of perspectives from the key scientific disciplines underlying SSH, ICT and data science, energy sector, policymaking, including scientists from all various disciplines relevant to the project scope. Candidates will be nominated by the beneficiaries comprising the Consortium and will be established by the end of the first semester.

Task 1.4. Open science, FAIR data and data management (ENG) [M1-M36]
ENG(4)

This task will continually assess and improve the requirements of knowledge exchange, ensuring research data and outputs are managed in line with the FAIR principles (Section 1.2.8). This entails development of a robust Data Management Plan (D1.2) from the onset of the project, which will be updated annually. It will clearly define what data will be published, as open data might generate a wider interest towards the project. Two additional aspects are also tackled: privacy (to ensure that proprietary or personal data is not externally accessible) and ethics (to avoid the identification of individual persons from data streams, further than performing a correct use of data).

Work package WP2 – Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR

Work Package Number	WP2	Lead Beneficiary	9. BLUEPRINT
Work Package Name	Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR		
Start Month	1	End Month	16

Objectives

WP2 will provide the foundation for DR in energy-efficient residential buildings, considering socio-economic enablers in the design of the DEDALUS solutions.

Description

Description of work

Task 2.1. Gap analysis and use case capitalisation for DR in the residential sector (BLUEPRINT) [M1-M6]
ENG(1), CARTIF(1), OurPower(1), GreenPoint(1), BLUEPRINT(6), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), DYNAM(1), CENTRICA(1), TUC(1)

This initial task will conduct a gap analysis for use case capitalisation in order to develop easy-to-adopt innovation-based models, engaging different actors of the residential sector with different characteristics. The DEDALUS pilots will be analysed and documented according to a standard methodology, to ensure a streamlined workflow of software development across the pilots throughout the project. All use cases will be described according to the standard IEC-62559 (Use Case Methodology). We will define and describe the use cases and requirements associated for reaching the DEDALUS project objectives. To this context, formal analysis of the pilot sites, use case validation, available infrastructure, asset ownership, including governance model for dwellers and community management will serve as foundation for further tasks in WP2 as well as technical WPs 3-4.

Task 2.2. DEDALUS Community building and scenario co-creation for residential DR (SIN) [M1-M16]

ENG(1), OurPower(1), GreenPoint(1), BLUEPRINT(2), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), DYNAM(1), CIMNE(2), CENTRICA(1), TUC(1), ICONS(1)

This task will define the Social Science Framework, consisting of: (i) A combination of deliberative citizen engagement methodology, intersectionality, co-creation and experimental scenario-based research, as well as Design Thinking methodologies, to bring residential energy consumers center stage in the energy markets. The participants will be systematically chosen to include representatives of each profile defined in T2.3. In cases where end user specific feedback is limited, the core identified user segments will be used to define the profiles of end users; (ii) Interface for consumer research in T2.3 for qualitative consumer research. Output from this task will provide insights about customer needs and

will suggest improvements to our services; (iii) Standardised methodology to assess social acceptance, perceived and actual inclusion into community, actual usage of services and perceived constraints (link with T5.8); (iv) Strengthen the knowledge sharing to support business models in T6.1; (v) Training opportunities to ensure that common SSH tools and methodologies are applied across pilots.

Task 2.3. Consumers segmentation and clustering of buildings for optimal DR planning (NTUA) [M1-M16]

ENG(1), NTUA(10), ARC(1), OurPower(1), GreenPoint(1), BLUEPRINT(2), FællesBo(1), GREEN CITIES(1), HERON(4), DOM-X(1), ISRAA(1), CENTRICA(1), TUC(6)

A comprehensive empirical analysis of consumers' decision preferences and underlying cognitive, affective and social factors in the context of residential DR will be conducted. Benefits for individual residents, trust and acceptability, privacy, comfort and ownership would be considered in the profiling part. Mapping of stakeholders and their needs and objectives (per pilot site) will take place to capture the link among building devices vs consumer groups and related preferences, with consumer-oriented perspective. Mapping of technologies/devices present and to be developed in each pilot site during project timeline will be developed, as the same building device may be able to mobilise different flexibility and may be offered along different DR services (e.g. ancillary services vs peak shaving). Mapping of the existing tenders and contracts in each pilot site will also take place to address any constraints or opportunities to deploy additional DEDALUS solutions. As part of DEDALUS concept, on a macro level, this task would also model building as consumer, either: (1) at the same level of individual consumers (e.g. lifts, building-level lighting), or (2) consider the building as aggregated consumer (dwellers consumption). This activity would explore 'Building as Consumer' segmentation to develop different DR flexibility services. Data assessment will follow based on experimental online studies, obtaining representative samples. In this respect, AI-based individual and business-level consumers clustering, and market segmentation algorithms (TE-1) will be used. The characterisation of user groups will be also addressed in terms of socio-demographic and psychological variables.

Task 2.4. Co-designing with consumers innovative social-driven incentive mechanisms and data-driven DR services (DOM-X) [M1-M16]

ENG(1), CARTIF(1), UNIVPM(4), ARC(1), OurPower(1), GreenPoint(1), BLUEPRINT(3), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(7), ISRAA(4), DYNNAME(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1)

Subtask 2.4.1: Co-designing data-driven DR services and nudging interventions (DOM-X): This subtask aims at defining the minimum requirements of the different pilot user interfaces, towards eliciting user preferences in an easy and systematic way and promoting user engagement. Building on the output of T2.1 on the requirements of each supported use case and the core identified consumer groups (T2.3), along with the output of T2.2 about the co-creation sessions with the target pilot users in a workshop format, a minimum set of user preferences will be extracted. In cases where end user specific feedback is limited, the core identified user segments will be used to define the profiles of end users. This will support initial high-level decision-making regarding the design of the data-driven services and nudging interventions for each pilot. Based on a systematic analysis of effectiveness studies on nudging techniques, best practices (output of H2020 NUDGE), design characteristics of nudging techniques and impact of socio-demographic, psychological and contextual variables on the effectiveness of nudging interventions will be mapped. To further adapt the nudging interventions to the needs of end-users, a user-centred approach will be implemented involving end-users. The derived output will be used to drive the adaptation of the existing user interfaces, so as to enable the extraction of key user preferences directly from end users. In addition, various means for increasing user participation (e.g. on time notifications, personalised feedback, contribution to social targets, environmental impact) will also be considered and evaluated for the different considered DR scenario. This process will be carried out separately for each pilot and the respective use cases, with the derived design recommendations and mock-ups being detailed in D2.4 and used within WP4, for upgrading the user interfaces employed in the different pilots (especially in T4.3 and T4.4).

Subtask 2.4.2. Consumer engagement for comfort-based flexibility model definition (UNIVPM): This subtask is based on engaging residential consumers and stakeholders in identifying needs and requirements in relation to comfort-based flexibility model definition. Indoor Environmental Quality (IEQ) will be assessed by the following aspects: thermal, acoustic and visual comfort and indoor air quality, as these aspects influence the health, well-being and performance in people living in indoor environment. Needs for engagement and requirements will be defined through Deliberative Futures Workshops (T2.2) then will be translated in technical requirements (supporting T2.5 activities). The results of this task will be considered in T4.2 when the DR models based on comfort flexibility will be implemented. A specific focus on older people will be considered since there is a particular need of this target group in maintaining comfort conditions in living environments.

Subtask 2.4.3: Financial & non-financial socially driven incentives (ISRAA): This subtask will perform an analysis of the monetary and non-monetary (e.g. social) incentives mechanisms to increase consumer engagement in residential DR. It will be based on outputs from T2.2, T2.3 and previous subtasks. Online studies with experimental variation of different monetary and non-monetary incentives will be conducted in order to test the impact of diverse classic and

behavioural interventions on behaviour change. We will also examine how different interventions impact behaviour of key user clusters (T2.3), in order to develop tailored group-specific interventions.

Task 2.5. Residential consumers business/privacy/cyber-security requirements and functional specifications (ENG) [M3-M16]

ENG(8), NTUA(2), CARTIF(1), UNIVPM(2), ARC(2), COMS(2)

Following the requirements analysis and co-creation sessions, the final task will define the functional specifications for DEDALUS business, privacy and cyber-security framework (e.g. related to the standardised data access and exchange, the security and privacy of the data, taking into account possible ethical issues). The following methodology is proposed: (i) Analysis of each component of the DEDALUS technology enablers and data-driven services; (ii) Analysis of previous projects where relevant matters have been investigated (Section 1.2.4); (iii) Analysis of the pilots' and consumers' needs. This results on a set of in-detail specifications for embracing the incorporation of novel technology tools, services and business models (based on the outcome of the work carried out within this WP) aimed to mobilise citizens, and optimally manage available flexibility, as made available by individual consumers and building clusters.

Work package WP3 – Technology enablers for residential energy consumer data-driven 'activation'

Work Package Number	WP3	Lead Beneficiary	3. CARTIF
Work Package Name	Technology enablers for residential energy consumer data-driven 'activation'		
Start Month	4	End Month	28

Objectives

WP3 will deliver: i) DR-ready smart appliance IoT platform Energy Data Space compliant interoperability layer for residential flexibility mapping among stakeholders; ii) Data/metadata representations for flexibility and DR services characterisation applying a semantic interoperability framework; iii) A suite of ML models for multi-energy flexibility assessment and forecasting in residential buildings with installed smart devices; iv) blockchain-based data exchange and flexibility tokenisation with DR settlement using smart contracts; and v) Privacy, security and data protection framework for sovereign data sharing and smart devices integration in the DR services.

Description

Description of work

Task 3.1. DR-ready smart device & flexible resource data collection (ARC) [M4-M28]

ENG(1), CARTIF(1), ARC(15), OurPower(1), GreenPoint(1), BLUEPRINT(1), NEOGRID(1), HERON(2), DOM-X(4), ISRAA(1), DYNAM(1), CIMNE(3), PEUSA(1), ROMUR(4), CENTRICA(2)

This task focuses on hardware artefacts to ensure technical interoperability at the field level between smart devices. A DR-ready smart appliance IoT platform (TE-2) will be provided with open APIs, service enablers and respective tools aimed for integration in different use cases (WP2). The integration with legacy device types will be achieved through smart controllers employing widely applicable protocols and standards (e.g. BACnet, Modbus, LonWorks, OpenTherm, OCPP). Secure edge gateways (T3.6) will be locally employed to fuse data collected from various devices over the dominant wireless protocols (Wi-Fi, ZigBee, LoRa) and to expose vendor specific APIs in a common format through home automation frameworks (e.g. OpenHAB, HomeAssistant). In this context, the collected data from multiple data sources, such as smart meters, thermostats, sensors, smart appliances, EV chargers, etc. will be harmonised and integrated with open and standardisable connectors over data brokers (e.g. MQTT), REST APIs, and local/external data repositories (e.g. Microsoft SQL Server and openly available European Data Spaces).

Task 3.2. Energy Data Space compliant interoperability layer (ENG) [M4-M28]

ENG(10), CARTIF(7), ARC(2), HERON(2), DOM-X(4)

This task will follow the BRIDGE reference framework and methodology to define a data-driven Energy Data Space compliant interoperability layer (TE-3). We will map the full DR chain, from the DR service consumer request to the control of the flexibility resources (mainly smart devices) on the field, to manage the exchange of information between the business roles (e.g. prosumers or residential buildings - the source of flexibility, and energy market stakeholders, such as aggregators). Such exchange of information will be based on functions (e.g. flexibility offers of prosumers and aggregated flexibility) and interfaces (i.e. the connector to the roles) which will represent the generic business process covered by the proposed set of DR services. In order to demonstrate the level of technical and syntactic

interoperability towards the smart appliances infrastructure (T3.1), as well as and semantic interoperability in services and systems (T3.3), this task will comply with an evaluation and quantification approach based on the Basic Application Interoperability Profiles (BAIOPs) created in accordance with the SGAM.

Task 3.3. DEDALUS flexibility and DR data models and standards (CARTIF) [M4-M28]

CARTIF(10), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), HERON(1), DOM-X(1), ISRAA(1), DYNAMES(1), CIMNE(4), PEUSA(1), ROMUR(4)

This task will adapt and extend H2020 InterConnect, BD4NRG and BRIGHT to set data representations for flexibility modelling and modular DR services according to the available flexibility assets (devices and systems). This will be specified as building blocks considering definitions entailed in SAREF, or other relevant ontologies, and where necessary extending the existing representations with newer ones following the same rationale, thus ensuring an interoperability chain. We will first define metadata representations for flexibility modelling and each DR service, according to the requirements identified in WP2 and the information flows (functions and interfaces) identified in T3.2. Then, a semantic common data model compliant with existing standards and ontologies (e.g. USEF, CIM, EEBUS, OpenADR, SAREF) will be generated to represent such required metadata and facilitate the integration between data from utilities/aggregators and consumers. Finally, the concept of data-driven architecture will be detailed together with the involved protocols and data models. To instantiate/populate the building blocks based on the designed common data model with heterogeneous data sources, mapping languages and translating tools will be explored together with model checking mechanisms that will ensure data completeness (e.g. SHACL shapes). Multi-level data models will be used for: (1) Flexibility modelling and DR services representation via standardised service-based ontologies, data models and languages (T3.4); (2) Supporting the digital representation of consumers preferences embedded in smart contracts within tokenised marketplace (T3.5).

Task 3.4. Multi-energy flexibility modelling for data-driven residential DR (NTUA) [M4-M28]

NTUA(14), CARTIF(4), COMS(3), GreenPoint(1), FællesBo(1), NEOGRID(1), HERON(2), DOM-X(1), ISRAA(1), DYNAMES(1), CIMNE(1), CENTRICA(4)

A methodology will be developed for estimating the potential flexibility that may be offered through heterogeneous multi-energy flexible loads (e.g. electricity, district heating, gas), in the context of a building or district. This methodology will rely on the estimated, predicted or real (if known) abundance of different flexible loads (such as HVAC, heat pumps, EV charging, etc.). We will adapt and extend forecasting ML models and heuristic optimisation algorithms from the H2020 BD4NRG, MATRYCS and I-ENERGY projects (TE-4) to predict and estimate the aggregated flexibility that may be offered by the aggregation of a number of smart devices (T3.1) under different scenarios taking into consideration different seasonal and weather conditions. The building's subsystems will be modelled and simulated in order to mitigate uncertainty and to incorporate the flexible loads that stem from different use patterns of the building. Moreover, the task will focus on investigating how the output of this methodology will be used in the DR services (WP4) to guarantee the interoperability of the tool in different scenarios. The final output of the task will cover possible applications of the flexibility that can be achieved on the demand side in building level, such as allowing the building's operator to control subsystems and providing price signals to various sources of electricity demand (e.g. industrial DR, power-to-heat, EV charging) and heat demand (e.g. DR for district heating network, natural gas).

Task 3.5. Blockchain-based interoperability for flexibility tokenisation & DR settlement (TUC) [M4-M28]

ENG(8), COMS(6), TUC(10)

We will upscale the distributed ledger technology, developed in the H2020 eDREAM, BRIGHT and PHOENIX projects to deploy an overarching blockchain-based solution for privacy-preserving flexibility assessment, P2P flexibility tokenisation and DR settlement using smart contracts (TE-5). The solution will be based on a scalable blockchain-based energy ledger that enables secured, trusted, and flexible data/models sharing among buildings and energy stakeholders (T3.1) while supporting also different interoperability alternatives across Energy Data Spaces and data hubs (T3.2), eventually turning energy data into business assets. We will innovatively consider the integration of different cryptographic techniques, such as homomorphic encryption or zero knowledge proofs to assure the privacy of energy and non-energy data. This will help in assuring the data consistency and transparency needed for data sovereignty in the context of Energy Data Spaces and data hubs, and the secured data exchange in DR settlement. The blockchain infrastructure will be utilized for decentralised parametrisation of prediction models (T3.4) in a way of processing the data within the stakeholder premise (e.g. prosumer, building, or community level) via the secure gateway (T3.6) and sharing in a tamper-proof manner the model via the blockchain network.

Task 3.6. Privacy, security and data protection framework for trusted data sharing and AI (COMS) [M4-M28]

ENG(2), ARC(1), COMS(10)

This task is responsible for developing the tools that will ensure compliance with privacy, security and legal requirements framework as defined in T2.5, taking into account also the newly proposed regulation regarding a legal framework for AI (Sections 1.2.9). That means enabling the pilots to meet the minimum requirements associated with the provision of

DR services and enabling trusted data sharing among DEDALUS services and potential external Data Spaces (i.e. secure REST APIs, encrypted data transfer). Fine-grained identity and access management will be based on KEYCLOAK as developed in H2020 I-ENERGY, while Universal Secure Gateway (TE-6), developed within H2020 PHOENIX and BRIGHT will be leveraged and evolved for the integration and virtualisation of field assets and systems. The features offered by blockchain and smart contracts (T3.5) will be exploited in order to assure that data from different stakeholders and marketplaces taking part in data analytics and the provision of DR services (WP4) are managed according to specific access and usage policies that ensure privacy/confidentiality compliance. Exploiting the Privacy Protection Enforcement (TE-7) component from H2020 PHOENIX, mechanisms for data registration, consent exchange, policy/permission-based data exchange, transactions auditing, and actors' reputation assessment will be supported. Finally, the solution will be expanded from managing data transaction towards enabling a platform for broker-dealer third-party (data processor) services using reusable/decentralized/badge profile identifiers.

Work package WP4 – Value added energy and non-energy data-driven services for DR in energy-efficient residential buildings

Work Package Number	WP4	Lead Beneficiary	1. ENG
Work Package Name	Value added energy and non-energy data-driven services for DR in energy-efficient residential buildings		
Start Month	5	End Month	29

Objectives

WP4 will deliver value-added, data-driven, energy and non-energy services for DR in energy-efficient residential buildings, addressing: Comfort-based flexibility models & DR optimal management tools; Flexibility assessment and pre-aggregation for energy communities/districts; Optimal aggregation of flexible energy resources for district heating management.

Description

Description of work

Task 4.1. Digital Twins for optimal assets vs consumers flexibility planning (ENG) [M5-M29]
ENG(18), HERON(6), CENTRICA(5)

This task aims at designing a cross-commodity DER/DR flexibility-centric DTs (TE-8). It will integrate (1) clustering algorithms (T2.3) of residential user loads (i.e. specific locations, energy communities or even whole countries), combining cross-sectoral fields of expertise to structure a solid (and semantically interpretable) clustering methodology based on a complete set of social-economic, gender, geographic, socio-cultural, and socio-political factors; (2) thermal models based on fully data-driven (e.g. neural networks) or physics-data hybrid models, leveraging real-world assets enabled by building energy management systems; and (3) aggregated models for TRL6 HITL ('Human-In-The-Loop') flexibility assessment and modelling of communities (as opposed to individual residential units/devices), considering heating/mobility needs, and also behavioural response modelling (T3.4) to rank the customers/community's elasticity to dynamic economic and non-economic incentives. The modelling techniques will focus, not just on the average dynamics of flexible assets, but also on modelling uncertainty. The DT will provide information for consumers flexibility planning and will support a cost-effective and tailored-made design of flexibility management and aggregation mechanisms in T4.2-T4.5.

Task 4.2. Optimal trade-off among comfort, energy and building performance (UNIVPM) [M5-M29]
ENG(5), UNIVPM(14), ISRAA(7)

The task will develop a comfort-based flexibility model according to the requirements defined in T2.4, in order to maintain health and well-being during application of DR service. To implement the model, sensors and systems to measure comfort in indoor environments (thermal, acoustic, visual and air quality) and weather station will be integrated within the platform and data stored in cloud. Learning techniques will be applied, considering user behaviour and profile, to implement a comfort-based personalised model to achieve an optimised balance between comfort, energy consumption and building performance. As a result of the task activities, the service will provide a model for flexibility (for DR applications) with comfort-based actions to enable a dynamic reshaping of buildings energy demand. The comfort DR service (TE-9) is based on: Data collection thanks to DEDALUS IoT platform (T3.1), including sensors to measure indoor environmental conditions (air temp, RH, CO₂), energy consumptions (metering and sub-metering, also with smart

plugs) and on-site renewables production, web services to get weather data and grid data; Users interaction: web-based tool to get users profiles/preferences and to allow the communication of flexibility actions from DSO/Aggregator to users; A process that digests all the incoming data and provides the optimal dispatch of flexibility actions for DR.

Task 4.3 Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers (DOM-X) [M5-M29]

ENG(5), NTUA(6), OurPower(6), BLUEPRINT(8), HERON(9), DOM-X(12), CENTRICA(4)

In this context, nudging services for residential consumers (TE-10) and management tools for energy savings and multi-value DR (TE-11) are foreseen. The services/tools will use the historical activity of the household and provide recommendations to shift their demand and offer flexibility to the grid. The aim is to identify the optimal incentives (both in kWh for the utility and the economic/social for the households) so as to enable the consumers to proceed with the DR action. Several time-horizon models will be exploited (T3.4), by taking into consideration building energy profile patterns (electricity and gas), weather conditions and price signals among others. For the purposes of the services/tools, various forecasting techniques will be examined. Algorithms developed for network optimisation (H2020 MATRYCS, BD4RNG, i-ENERGY) will also be exploited to be fitted in the specific sector. Energy disaggregation will be used to validate the DR actions by the consumers and the DR incentivised services. We will make use of open available datasets and the data gathered from the IoT devices installed (T3.1), employing ML (Decision Trees, RNN, LSTM) and probabilistic (FHMM) algorithms and provide accurate decomposition of the energy signal into appliance signals. With transfer learning techniques, it will be possible to reuse developed models in similar conditions when the hardware infrastructure is not available. In this context, it is possible to scale the models from site-specific training data sets to other locations through software solutions, and provide high accuracy results with minimal costs of devices installation in comparison. This service will take the results of the DR incentivised services and transfer the knowledge to other households by deploying and fitting the models in different conditions in different locations.

Task 4.4. Local flexibility management for energy communities and self-consumption/cost optimisation (OurPower) [M5-M29]

ENG(5), CARTIF(8), COMS(8), OurPower(10), BLUEPRINT(8), CIMNE(4), ROMUR(4), TUC(6)

This task focuses on the investigation of possibilities and implementation of means for the optimisation and utilisation of flexibility options at community level (TE-12) or district level (TE-13) in accordance with economic, social and environmental benefits. Leveraging on the P2P energy marketplace platform operated by OurPower as well as blockchain-based interoperability infrastructure for decentralised management of DR developed in T3.5, this task involves the development of pre-aggregation framework for interfacing to local (sub-community) assets and systems, exposing flexibility opportunities to the nudging app (TE-10), and integrating with P2P flexibility markets for the valorisation of residential customers' flexibility via tokenisation. Advanced aggregation approaches leveraging on games theory will be integrated on top of the market considering the cooperation of the prosumers to enable the participation of residential buildings in DR. Fully customizable self-enforcing contracts will be developed and integrated to allow the definition and direct injection of DR signals and track and check the compliance of participants enrolled in DR programs to the desired demand energy profiles and detect potential energy unbalances. Besides utilising sub-metering data in the support of settlement and billing within the community, also new types of beyond financial benefits will be considered as social enablers inferred in WP2. Utilising the Universal Secure Gateway (T3.6) for resource virtualisation and real-time management such framework has good potential of automatising the DR management and settlement process and thereby increasing the participation and adoption of DR (i.e. local or building level sustainability goals, use of tokens for monetisation and rewards, discounts to local cross energy sector businesses, etc.).

Task 4.5. Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorisation (NEOGRID) [M5-M29]

ENG(6), NEOGRID(10)

This task focuses on optimal aggregation of thermal flexibility at individual building and at district level, while trading off with electrical flexibility within a cross-commodity optimisation scenario. Synergies and interaction among thermal and electrical flexibility will be explored and optimised with a view to assess and achieve the optimal aggregation of flexible resources at building and/or at district level to be assessed against end consumers profiles (overall financial remuneration from DR participation, comfort, energy efficiency). The above combined optimisation will be explored along a centralised district heating scenario and along a decentralised Power2Heat heat pump-based scenario. Such aggregation of thermal flexibility will take place from each building with respect to balance the internal indoor environmental comfort/discomfort and cost of heating, and at the same time making the flexibility in heating (of space and domestic hot water) and electricity consumption (ventilation where applicable) available for the district and electricity system. Via role-based access to the management system, various constraints can be applied at different levels of the value chain (tenant, building administrator and district heating operator). At the aggregated level, it will be possible to control, forecast, prepare and monitor the pool of buildings securing that the flexibility can be planned and delivered according to various strategies. As a result of the activities of this task, the following topics will be addressed and

lead to services (TE-13) towards district heating and electricity companies: (i) Regarding temperature/efficiency of district heating: Supporting reduction of the supply temperature in an area, by identifying the limiting installations and correcting performance gaps where possible; Supporting return temperature reduction by improving effectiveness of space and water heating installations in buildings through optimised control and online fault detection; (ii) Regarding network/production bottlenecks: Reducing the peak load of an aggregate of buildings (in a local area); Reducing the contribution of an aggregate of buildings to the district heating system's peak load, via demand reduction within specific time windows; Due to lack of energy, a service for prioritising specific buildings/customer types prior to others will be implemented. Similar topics will be addressed and deployed along such service for the decentralised heating scenario, where heat pumps at individual apartments level will be considered.

Work package WP5 – Demonstration of multi-value and energy carrier-agnostic residential DR, and social acceptance

Work Package Number	WP5	Lead Beneficiary	2. NTUA
Work Package Name	Demonstration of multi-value and energy carrier-agnostic residential DR, and social acceptance		
Start Month	1	End Month	34

Objectives

WP5 is responsible for the overall planning, management, and evaluation of the DEDALUS pilots. We will replicate and validate DEDALUS solutions in 7 pilots (5 Front Runners and 2 Multipliers), having different system needs, consumer needs, economic conditions or different climates. We will assess the impact as well as the potential to scale and replicate the results achieved in the pilots. Focus of this task is also given on assessing the acceptance of the DEDALUS concepts and tools by the residential consumers (and energy stakeholders in general).

Description

Description of work

Task 5.1. Evaluation framework, baseline analysis and pilots' preparation (NTUA) [M1-M13]

ENG(2), NTUA(8), CARTIF(1), UNIVPM(1), ARC(1), COMS(1), OurPower(2), GreenPoint(1), BLUEPRINT(1), FællesBo(2), NEOGRID(1), HERON(2), DOM-X(1), ISRAA(2), DYNAMO(1), CIMNE(3), PEUSA(1), ROMUR(4), CENTRICA(1), TUC(1)

The task will start with the actual engagement of end users and the installation of required digital systems/platforms/applications per pilot to assure interoperability and integration of the energy facilities and assets and to start the pilot monitoring period. We will also analyse the detailed test scenarios (according to WP2) that will comprise the pilots and the methodology to be followed for their evaluation. The evaluation methodology (developed by M10) will be centred on the elaboration of the appropriate performance indicators for the DEDALUS social, technological and business framework (resulting to the DEDALUS baseline analysis as part of D5.1), as well as the user perspective, in terms of compliance with the respective requirements specified in WP2.

Task 5.2. Flexibility pre-aggregation for building-level energy community (Austria) (OurPower) [M14-M34]

ENG(5), NTUA(1), COMS(4), OurPower(10), GreenPoint(8), BLUEPRINT(7)

This task focuses on the demonstration activities related to Pilot 1 in Austria (targeting residential consumers from Wiener Neustadt, Lower Austria), including the involvement of OurPower, GreenPoint, BLUEPRINT and COMS. We will examine the pre-aggregation of individual/small-scale (e.g. EV, heat pump) and community-scale (e.g. PV) demand and generation via local flexibility management platform. More details are provided in Section 1.2.3.1.

Task 5.3. Energy efficiency and DR optimal interactions between district heating and electricity networks in social housing (Denmark) (FællesBo) [M14-M34]

NTUA(1), FællesBo(9), NEOGRID(8), GREEN CITIES(3)

This task focuses on the demonstration activities related to Pilot 2 in Denmark (targeting residential consumers from Herning), including the involvement of FællesBo, NEOGRID, ECG. We will test the link among electricity and heat via the smart ventilation control, providing flexibility at building level to the electricity grid. More details are provided in Section 1.2.3.1.

Task 5.4. Multi-value DR for multi-segments of residential energy (electricity and gas) consumers (Greece) (HERON) [M14-M34]

NTUA(4), ARC(2), HERON(12), DOM-X(9)

This task focuses on the demonstration activities related to Pilot 3 in Greece (targeting buildings from different locations), including the involvement of HERON, DOM-X, NTUA, ARC. We will demonstrate multi-value DR services for both heating and electricity loads in residential buildings. We focus on different segments forming clusters based on demographics (i.e. address, household size etc) and consumption behaviours (e.g. heavy night consumption, earlier in the day heavy load in weekends). More details are provided in Section 1.2.3.1.

Task 5.5. Trading off comfort with energy consumption in DR programs for elderly people living in serviced apartments (Italy) (ICCRA) [M14-M34]

ENG(5), NTUA(1), UNIVPM(5), ISRAA(12), DYNAMES(7)

This task focuses on the demonstration activities related to Pilot 4 in Italy (targeting residential consumers from Treviso), including the involvement of ISRAA, DYNAMES, UNIVPM, ENG. We will be able to combine the centralised management of supply with the personalised individual levels of demand, by employing an interoperability structure guaranteed by the degrees of control and flexibility provided by the devices themselves and the preparedness acquired by the residents. More details are provided in Section 1.2.3.1.

Task 5.6. District-level DR for increasing community-level self-consumption while providing flexibility to local DSO (Spain) (CIMNE) [M14-M34]

NTUA(1), CARTIF(3), ARC(1), CIMNE(18), PEUSA(10), ROMUR(12)

This task focuses on the demonstration activities related to Pilot 5 in Spain (targeting residential consumers from La Seu d'Urgell, Pyrenees), including the involvement of CIMNE, PEUSA, ROMUR and ARCELIC. We will validate DR and electricity flexibility at district level, combined with a district-based electricity storage system and distributed PV rooftop generation plants. More details are provided in Section 1.2.3.1.

Task 5.7. DEDALUS replication across Front Runners and Multipliers [M14-M34]

ENG(4), NTUA(2), CARTIF(1), UNIVPM(1), ARC(1), COMS(1), OurPower(1), BLUEPRINT(1), NEOGRID(1), HERON(1), DOM-X(1), CIMNE(1), CENTRICA(7), TUC(7)

DEDALUS will act as a cutting-edge initiative that will replicate innovative solutions, with the main focus on the value-added energy and non-energy data-driven services for residential DR. More specifically, replication of the services deployed in WP4 is also targeted in DEDALUS, across pilots (through the Front Runners involved in each pilot) and beyond (through CENTRICA and TUC acting as a Replication Multiplier). More specifically, DEDALUS integrates a two-ways replication process: (1) Replicating solutions and lessons learnt from one pilot site to other pilot sites; (2) Replicating pilots' solutions to a Replication Multiplier through CENTRICA and TUC who will 'mirror' some pilots in its market (CENTRICA) or available DR infrastructure (TUC). This way, the project will deal with already known replication weaknesses for increasing residential DR in Europe whereas will carefully examine the parameters that may affect the seamless integration of pilot results in other real/business environments to assure that the activity is realised effectively. The replication activities will be reported in D5.2-D5.3.

Task 5.8. Consumers engagement, social acceptance assessment and impact analysis (SIN) [M14-M34]

ENG(2), NTUA(6), CARTIF(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), DYNAMES(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1)

This task will deal with the engagement of the energy consumers and stakeholders from the energy value chain on the usage of the DEDALUS services (WPs 3-4) considering the particularities of various market roles, as defined in T2.1. Specific awareness and training programmes will be carried out with emphasis on motivating the users on the real benefits of DEDALUS services and explaining the operation both from social, technical and organisational/business perspective. A set of training materials (presentations, brochures, notes, etc.) will be created. Questionnaires, semi-structured interviews and focus groups will be conducted to collect feedback from users with respect to usability and social acceptance aspects. Benchmarking and impact assessment will be provided based on the requirements (WP2), while the strengths and weaknesses of the DEDALUS services will also be seen from this perspective. Three evaluation reports will be produced (D5.2-D5.4), by compiling the conclusions of the different Front Runners while running an iterative demonstration process. The evaluation activities (according to the methodology of T5.1) will be focused both on the correct and useful application of DEDALUS social, technological and business framework and on the impact on the use case implementations. The environmental impact will be also assessed and quantified.

Work package WP6 – People-powered Go2Market and exploitation strategies towards EU replication and market scalability

Work Package Number	WP6	Lead Beneficiary	13. HERON
Work Package Name	People-powered Go2Market and exploitation strategies towards EU replication and market scalability		
Start Month	1	End Month	36

Objectives

WP6 will identify the market impact of DEDALUS results, develop the instruments that will be used to define specific Go2Market, create exploitation and replication strategies based on an exploitation framework which consists of market analysis, assess sharing economy and social innovation business models, provide contributions to standards and map those to feasibility studies.

Description

Description of work

Task 6.1. DEDALUS sharing economy and social innovation-based business models (HERON) [M1-M36]
ENG(2), ARC(1), COMS(1), OurPower(1), BLUEPRINT(1), NEOGRID(2), HERON(4), DOM-X(1), DYNAM(1), PEUSA(1), CENTRICA(1)

This task will develop easy-to-adopt innovation-based models for residential DR, engaging different actors of the energy value chain with different characteristics, and apply/evaluate them through the practical application of DEDALUS pilots. Different business models will capture, propagate, illustrate and share the flows of energy, payments, flexibility services, and system balancing and non-financial segments (e.g. environmental, social cohesion, job creation, improved efficiency of business activities), through the development of single-component diagrams, showing the way that each pilot project works and facilitating the optimisation of procedures and results. The innovation-differentiator factor of business models will be the interactive engagement of the energy consumers during the decision-making process of each business model through semi-structured questionnaires/interviews to explore their emerging effect with the involving parties.

Task 6.2. Contribution to standardisation (CENTRICA) [M1-M36]

ENG(2), CARTIF(1), ARC(1), COMS(1), BLUEPRINT(1), CENTRICA(4)

This task aims at contributing to relevant Standardisation Bodies (SDOs) at the interplay among ICT, smart energy grids, DR and Energy Data Spaces, ranging from SGAM/CIM/IEC and SAREF, to CEN-CENELEC Data Space Architectures and Privacy Standardisation Groups, and IDSA/GAIA-X on standardisable RAS and Connectors through the active involvement of partners. For instance, we will assess existing standards, such as OpenADR, and propose adjustments to improve their practicability, interoperability, privacy and cyber-security. We will build upon already established connections and experience (Table 2.4), both on national and European level, aiming at reinforcing them from lessons learnt and best practices during deployment and validation in pilots. DEDALUS will provide input to the standard adaptation/extension and will create synergies with energy regulatory bodies and network operators' associations, in order to boost data sharing friendly regulatory/business frameworks.

Task 6.3. Go2Market strategy: Market impact (HERON) [M1-M36]

ENG(1), NTUA(1), UNIVPM(1), ARC(1), COMS(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), HERON(4), DOM-X(1), ISRAA(1), DYNAM(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1), ICONS(1)

In parallel with T6.1 and T6.2, this task will evaluate the market impact of DEDALUS outcomes. First of all, we will analyse all partners' contributions map to the identified KERs (Section 2.2.2), through an introductory webinar with all partners followed by guided interviews focusing on partners' exploitation options (e.g. commercial, scientific, other). Based on the Use Case assessment in T2.1, partners will be characterised as 'technical providers' depending on whether they are expected to develop or assist in developing DEDALUS technology, or 'service users', if they are expected to offer a product to end-users. For each of the identified commercially exploitable solutions, a thorough market analysis will be conducted, consisting of a stakeholder analysis, target market assessment and competition and regulation overview (if regulatory barriers exist) with the results captured through a Business Model Canvas. Based on the Market Assessment and Business Model Canvas, a detailed Value Proposition for each of the commercially exploitable DEDALUS outcomes will be drawn. Identified values are then processed in feasibility studies which will evaluate the likelihood of a successful market uptake and identify the pathways for commercial exploitation. The feasibility studies will consist of a techno-

economic model which will calculate financial indicators associated with DEDALUS KERs such as Internal Rate of Return, Net Present Value, among others, and a sensitivity analysis that will assess the impact of the different cost drivers and market growth rates. Data collected through the pilots and through the co-creation events in T2.3 will be complemented with project partner input through the online/hybrid partner workshops described in T6.4.

Task 6.4. Go2Market strategy: Innovation management and exploitation (ICONS) [M1-M32]

ENG(1), NTUA(1), CARTIF(1), UNIVPM(1), ARC(1), COMS(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), DYNAME(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1), ICONS(8)

This task will implement the exploitation activities and consolidate the final exploitation plan. The identification of DEDALUS KERs and the analysis of their market impact in T6.3 will mark the inception of this task. A thorough IPR clinic will assess partners' IP background and foreground, support reaching consensus among IP owners, and define the most effective IPR measures. An exploitation workshop will fine-tune the exploitation strategy at the individual and project levels. This will involve: i) defining the governance and role of partners in driving the project's replication (in line with T6.5) and upscale; ii) evaluating business potential for the project's results, based on the analysis of market opportunities and potential entry barriers (in cooperation with T6.3); iii) monitoring innovation and market trends which can be leveraged in the execution of the project (e.g. desk research and gathering external stakeholders' inputs through small dedicated surveys, discussions, etc.); iv) performing a SWOT analysis, including an assessment of exploitation risks, and related actions; v) agreeing on an overall exploitation strategy for each KER. A continuous feedback cycle with KER owners to ensure consensus will be established. To this aim, IP and IPR issues will be also continuously monitored in case changes that may hamper exploitation occur. The final validation and update of all preliminary individual plans (through questionnaires and direct interviews), and the organisation of a final exploitation workshop (focused on partner roles and governance in driving replication and/or capitalise on potential commercial opportunities) will lead to the consolidation of the final exploitation plan.

Task 6.5. Roadmaps for replication strategies beyond DEDALUS (GREEN CITIES) [M13-M36]

ENG(1), NTUA(2), GREEN CITIES(7), ICONS(1)

Based on the lessons learned from implementing the activities of DEDALUS, blueprints and policy recommendations will be formulated. Focus will be given on adapting and customising the DEDALUS concept and approach of multi-value and energy carrier-agnostic DR for residential buildings towards the replicability and scalability to other regions in the pilot countries and within Europe, considering where possible national, regional and local needs, as well as legislative, regulatory and institutional conditions and the energy market (based on the initial work of T2.1). The roadmap's elaboration is a living, synthesis procedure, based on the work carried out within all WPs. The main steps to be realised towards the roadmap's production include: (1) Formulation of an engagement strategy; (2) Stakeholder dialogues; (3) Co-creating roadmap drafts; (4) Refine roadmaps through a public consultation process. The DEDALUS replication roadmap will also include recommendations on how national and EU regulatory frameworks should be adapted to enable DR in energy-efficient residential buildings. The roadmap will have exemplifying chapters which outline a national roll-out within the pilot countries. The work will start after the first year of the project (M13) and the roadmap will be completed by M25, so that it can be widely disseminated and presented until the end of the project, as well as after its completion. In this context, a first policy event will take place by M25. A final policy event will be envisaged for M36 (minutes to be included in D6.2).

Work package WP7 – Share, connect and attract

Work Package Number	WP7	Lead Beneficiary	23. ICONS
Work Package Name	Share, connect and attract		
Start Month	1	End Month	36

Objectives

WP7 aims to design and implement a highly effective dissemination and communication strategy (according to the general principles outlined in Section 2.2.2) to maximise the impact of DEDALUS. There is an increasing need to fill the gap between technical/scientific 'complex' dissemination to experts (e.g. technical, science and policy experts) and 'simplified' communication to general public target audiences. Towards this direction, this WP will organise a constant dialogue between the project team and target stakeholder groups, turning the project key findings and lessons-learned into shorter and easy-to-access material using common language rather than specialist terms that speak to citizens. More specifically, the strategy will be built around a set of targeted tools and activities, conceived as a way to: (1) Raise

awareness among the broader public and stakeholders on the potential benefits and challenges behind the residential DR solutions developed by the project; (2) Facilitate knowledge transfer around the innovative pool of DR assets provided by the project and tested in 5 pilot sites and 2 multipliers; (3) Engage with key stakeholders, market players and end-users in the EU residential sector to build a multilateral community; (4) Implement public outreach and engagement activities to raise understanding and social acceptance. DEDALUS will contribute, upon invitation by the CINEA, to common information and dissemination activities to increase the visibility and synergies between Horizon Europe supported actions.

Description

Description of work

Task 7.1. Dissemination and communication strategy (ICONS) [M1-M36]

ENG(1), ICONS(4)

An integrated plan for dissemination and communication will outline the strategy for the project to engage its target audience and address key stakeholders to reach the expected impact goals, supporting the uptake and replication of the DEDALUS results, in line with the exploitation pathways defined in WP6 (T6.4). The Plan will be first delivered at M6 (D7.1), including key messages for each identified target group, communication and dissemination measures to be implemented, channels, timelines and monitoring methodologies to assess the effectiveness of the strategy through quantitative KPIs and dedicated outreach and engagement indexes developed by ICONS. Dedicated D&C Working groups will be established in each pilot site and multiplier, with the objective to manage and implement local dissemination and communication activities and thus ensure a closer reach with local stakeholders, under the guidance of a D&C Secretariat (chaired by ICONS as WP7 leader) that will provide coordination at central level.

Task 7.2. Enable public outreach and wide communication (ICONS) [M1-M36]

ENG(1), NTUA(1), CARTIF(1), UNIVPM(1), ARC(1), COMS(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), GREEN CITIES(1), HERON(1), DOM-X(1), ISRAA(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1), ICONS(7)

Public communication will be launched with the design of the project's visual identity at M4, including logo, graphic elements and templates for reporting and presentations. Graphic materials and guidelines for their use will be included in a brandbook, as part of a main communication kit (D7.2) that will also include a project leaflet, poster and presentation video (M8). Online communication channels will be designed and launched following the visual identity: i) project website (M6, anticipated by a landing page at M4); ii) social media channels (LinkedIn and Twitter), supported by a dedicated social media strategy (outlined in D7.1). Public communication tools will be developed to address the general public and mainstream media, including: i) press releases; ii) blog articles and news items on the project website; iii) journalistic articles, written by professional journalists; iii) a 4-episodes audio podcast series, highlighting the main challenges and benefits behind DR for residential buildings; iv) social media regular animation and campaigns, leveraging EU/international celebrations and the media agenda.

Task 7.3. Knowledge transfer and stakeholders dialogue (ICONS) [M6-M36]

ENG(3), NTUA(2), CARTIF(2), UNIVPM(2), ARC(1), COMS(1), OurPower(1), GreenPoint(1), BLUEPRINT(1), FællesBo(1), NEOGRID(1), GREEN CITIES(2), HERON(1), DOM-X(1), ISRAA(1), DYNAMO(1), CIMNE(1), PEUSA(1), ROMUR(1), CENTRICA(1), TUC(1), ICONS(9)

This task is aimed at enabling a smooth transfer of the results and knowledge produced by the project, keeping the key stakeholders regularly informed throughout the different phases of project implementation while creating effective engagement patterns and dissemination pipelines. Actionable knowledge formats and content packaging will be developed through a plurality of tools and channels, including: i) info-packs and factsheets on specific results, oriented to their exploitation; ii) scientific publications and conference proceedings published on peer-reviewed journals and bulletins; iii) a policy brief including recommendations and guidelines for the implementation of existing policy agendas and the definition of future legal framework and policy strategies for DR and energy efficiency; iv) e-newsletters (issued every six months) distributed to web subscribers and accessible via the project website; v) short video-interviews with experts from pilot sites. To foster direct engagement with relevant stakeholders, project partners will participate in different key conferences and thematic fairs to disseminate the results. Workshops, webinars and a final event in the framework of a larger EU conference or initiative will be organised to present the final results.

Task 7.4. Networking and collaboration with fellow projects and initiatives and cooperation with BRIDGE (ENG) [M6-M36]

ENG(3), NTUA(1), CARTIF(1), UNIVPM(1), ICONS(1)

We will create close synergies and cooperation patterns with fellow projects and initiatives, leveraging on existing partnership and communities at EU level working on DR and energy efficiency of residential buildings. Liaison and collaboration with the European Partnership on 'People-centric sustainable built environment' will be established,

contributing to implement its Strategic Research and Innovation Agenda to achieve the expected goals on the built environment. At the same time, contributions to the BRIDGE activities will be secured, in particular with respect to data exchange and interoperability. Clustering with other EU-funded projects will be fostered to find synergies in joint dissemination activities and campaigns, knowledge exchange, active participation in their events, cross-linking and -referencing.

Task 7.5. DEDALUS capacity building programme (UNIVPM) [M12-M36]

NTUA(2), CARTIF(1), UNIVPM(5), GREEN CITIES(3), ICONS(1)

The knowledge, expertise and 'know-how' developed in the framework of DEDALUS will be spread through a series of training activities, using a blended online and face-to-face approach. Contents will be developed by partners with scientific expertise (UNIVPM, CARTIF, NTUA), while ICONS will be in charge for the overall layout and packaging of the training materials, which will be hosted and made accessible through the DEDALUS website. As a result, a Digital Educational kit gathering findings and results from DEDALUS, will be produced, and uploaded on the project website to be used for higher education (university). Expert workshops and webinars will be essential to develop and upgrade skills and competencies for technicians and policy- and decision-makers. Special focus will be on the promotion of start-ups. The workshops will be held in presence/online, or in a mixed hybrid format (with local teams in-presence and the rest of the consortium supporting remotely). A learning-by-doing approach (via hands-on training) will be followed. Specifically: 2 workshops will be addressed to technicians and will focus on DEDALUS solutions for DR in the residential sector to inform about the potential of the project's innovations and to get feedback from an expert audience (including the advisory board from T1.5); 2 workshops (½ day, M25 and M34) will target local authorities, regulators, civil associations and the EU level decision makers to broadcast the potential of DEDALUS DR solutions for residential consumers.

STAFF EFFORT

Staff effort per participant <i>Grant Preparation (Work packages - Effort screen) — Enter the info.</i>								
Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total Person-Months
1 - ENG	30.00	12.00	21.00	39.00	18.00	7.00	8.00	135.00
2 - NTUA	7.00	12.00	14.00	6.00	24.00	4.00	6.00	73.00
3 - CARTIF	2.00	3.00	22.00	8.00	6.00	2.00	5.00	48.00
4 - UNIVPM	2.00	6.00		14.00	7.00	2.00	9.00	40.00
5 - ARC	1.00	4.00	18.00		5.00	4.00	2.00	34.00
6 - COMS	1.00	2.00	19.00	8.00	6.00	4.00	2.00	42.00
7 - OURPOWER	1.00	4.00	2.00	16.00	14.00	3.00	2.00	42.00
8 - GREENPOINT	1.00	4.00	3.00		10.00	2.00	2.00	22.00
9 - BLUEPRINT	1.00	13.00	2.00	16.00	10.00	4.00	2.00	48.00
10 - FAELLESBO	1.00	3.00	2.00		12.00	2.00	2.00	22.00
11 - NEOGRID	1.00	2.00	3.00	10.00	11.00	4.00	2.00	33.00
12 - GREEN CITIES	1.00	4.00			4.00	8.00	6.00	23.00
13 - HERON	1.00	7.00	7.00	15.00	16.00	9.00	2.00	57.00
14 - DOM-X	1.00	10.00	10.00	12.00	12.00	3.00	2.00	50.00
15 - ISRAA	1.00	7.00	3.00	7.00	15.00	2.00	2.00	37.00
16 - DYNAME	1.00	3.00	3.00		9.00	3.00	1.00	20.00
17 - CIMNE	2.00	3.00	8.00	4.00	23.00	2.00	2.00	44.00
18 - PEUSA	1.00	1.00	2.00		12.00	3.00	2.00	21.00
19 - ROMUR	1.00	1.00	8.00	4.00	17.00	2.00	2.00	35.00

Staff effort per participant <i>Grant Preparation (Work packages - Effort screen) — Enter the info.</i>								
Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total Person-Months
20 - CENTRICA	1.00	4.00	6.00	9.00	9.00	7.00	2.00	38.00
21 - TUC	1.00	9.00	10.00	6.00	9.00	2.00	2.00	39.00
22 - SIN	1.00	10.00			7.00	2.00	2.00	22.00
23 - ICONS	1.00	1.00				10.00	22.00	34.00
Total Person-Months	61.00	125.00	163.00	174.00	256.00	91.00	89.00	959.00

LIST OF DELIVERABLES

Deliverables <i>Grant Preparation (Deliverables screen) — Enter the info.</i> <i>The labels used mean:</i> <i>Public — fully open (🚩 automatically posted online)</i> <i>Sensitive — limited under the conditions of the Grant Agreement</i> <i>EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444</i>						
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Project management plan	WP1	1 - ENG	R — Document, report	SEN - Sensitive	2
D1.2	DEDALUS data management plan	WP1	1 - ENG	R — Document, report	SEN - Sensitive	6
D1.3	DEDALUS first activity report	WP1	1 - ENG	R — Document, report	SEN - Sensitive	18
D1.4	DEDALUS 2nd activity report	WP1	1 - ENG	R — Document, report	SEN - Sensitive	36
D2.1	DADALUS gap analysis and use case capitalisation	WP2	9 - BLUEPRINT	R — Document, report	PU - Public	6
D2.2	Active societal participation, emergence, socio-economic enablers and consolidation factors of multi-value and energy carrier-agnostic DR for residential buildings	WP2	22 - SIN	R — Document, report	PU - Public	8
D2.3	High-level requirements for the DEDALUS social, technological and business framework	WP2	1 - ENG	R — Document, report	PU - Public	9
D2.4	Enabling conditions for DR in energy-efficient residential buildings	WP2	9 - BLUEPRINT	R — Document, report	PU - Public	16
D3.1	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release)	WP3	3 - CARTIF	OTHER	PU - Public	12

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (🚩 automatically posted online)

Sensitive — limited under the conditions of the Grant Agreement

EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision [2015/444](#)

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.2	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (2nd technology release)	WP3	3 - CARTIF	OTHER	PU - Public	19
D3.3	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (final technology release)	WP3	3 - CARTIF	OTHER	PU - Public	28
D4.1	'First wave' of DEDALUS Digital Twins and data-driven services for residential DR	WP4	1 - ENG	OTHER	PU - Public	13
D4.2	"Second wave" of DEDALUS Digital Twins and data-driven services for residential DR	WP4	1 - ENG	OTHER	PU - Public	20
D4.3	"Final wave" of DEDALUS Digital Twins and data-driven services for residential DR	WP4	1 - ENG	OTHER	PU - Public	29
D5.1	DEDALUS evaluation methodology, baseline assessment and pilot implementation plan	WP5	2 - NTUA	R — Document, report	SEN - Sensitive	13
D5.2	DEDALUS Pilots' execution documentation and consumers' engagement – pre-pilot phase	WP5	17 - CIMNE	R — Document, report	PU - Public	17
D5.3	DEDALUS Pilots' execution documentation and consumers' engagement – full pilot operation phase	WP5	7 - OURPOWER	R — Document, report	PU - Public	26

Deliverables*Grant Preparation (Deliverables screen) — Enter the info.**The labels used mean:**Public — fully open (🚩 automatically posted online)**Sensitive — limited under the conditions of the Grant Agreement**EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision [2015/444](#)*

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.4	DEDALUS impact and social acceptance analysis	WP5	22 - SIN	R — Document, report	PU - Public	34
D6.1	Preliminary Exploitation plan	WP6	23 - ICONS	R — Document, report	SEN - Sensitive	6
D6.2	Intermediate Exploitation Plan	WP6	13 - HERON	R — Document, report	SEN - Sensitive	24
D6.3	Final Exploitation Plan	WP6	13 - HERON	R — Document, report	SEN - Sensitive	36
D6.4	DEDALUS comprehensive replication methodology and roadmap	WP6	12 - GREEN CITIES	R — Document, report	PU - Public	36
D7.1	Plan for dissemination and communication	WP7	23 - ICONS	R — Document, report	SEN - Sensitive	6
D7.2	DEDALUS Communication Kit	WP7	23 - ICONS	OTHER	PU - Public	8
D7.3	Report on Dissemination and Communication activities, including impact assessment	WP7	23 - ICONS	R — Document, report	PU - Public	36

Deliverable D1.1 – Project management plan

Deliverable Number	D1.1	Lead Beneficiary	1. ENG
Deliverable Name	Project management plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	2	Work Package No	WP1

Description
This deliverable will provide a description of project procedures and adopted tools. Success Criteria: The availability of the report. (T1.1, T1.3)

Deliverable D1.2 – DEDALUS data management plan

Deliverable Number	D1.2	Lead Beneficiary	1. ENG
Deliverable Name	DEDALUS data management plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	6	Work Package No	WP1

Description
This deliverable will provide a description of the data management procedures. D1.2 will ensure open access to research data via a trusted repository under the principle ‘as open as possible, as closed as necessary’. Success Criteria: The availability of the report. (T1.4)

Deliverable D1.3 – DEDALUS first activity report

Deliverable Number	D1.3	Lead Beneficiary	1. ENG
Deliverable Name	DEDALUS first activity report		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	18	Work Package No	WP1

Description
This deliverable will deal with reporting of the first period of project execution, M1-M18. Success Criteria: The availability of the report (T1.1, T1.4)

Deliverable D1.4 – DEDALUS 2nd activity report

Deliverable Number	D1.4	Lead Beneficiary	1. ENG
Deliverable Name	DEDALUS 2nd activity report		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	36	Work Package No	WP1

Description
This deliverable will deal with reporting of the second period of project execution, M19-M36.

Success Criteria: The availability of the report (T1.1, T1.4)

Deliverable D2.1 – DADALUS gap analysis and use case capitalisation

Deliverable Number	D2.1	Lead Beneficiary	9. BLUEPRINT
Deliverable Name	DADALUS gap analysis and use case capitalisation		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	6	Work Package No	WP2

Description

This report will define and describe the use cases, requirements and needs to be considered in the DEDALUS platform.
Success Criteria: The use cases of DEDALUS project are clearly defined according to the standard IEC-62559. (T2.1)

Deliverable D2.2 – Active societal participation, emergence, socio-economic enablers and consolidation factors of multi-value and energy carrier-agnostic DR for residential buildings

Deliverable Number	D2.2	Lead Beneficiary	22. SIN
Deliverable Name	Active societal participation, emergence, socio-economic enablers and consolidation factors of multi-value and energy carrier-agnostic DR for residential buildings		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	8	Work Package No	WP2

Description

The deliverable will report the SSH-driven approach adopted and the relevant co-creation activities.
Success Criteria: The availability of the report providing detailed presentation of the Social Science Framework segmentation and the relevant co-creation activities. (T2.2, T2.3, T2.4)

Deliverable D2.3 – High-level requirements for the DEDALUS social, technological and business framework

Deliverable Number	D2.3	Lead Beneficiary	1. ENG
Deliverable Name	High-level requirements for the DEDALUS social, technological and business framework		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	9	Work Package No	WP2

Description

The deliverable will contain the high-level functional specifications related to the DEDALUS business, privacy and cyber-security framework.
Success Criteria: Availability of detailed specifications for the DEDALUS framework. (T2.5)

Deliverable D2.4 – Enabling conditions for DR in energy-efficient residential buildings

Deliverable Number	D2.4	Lead Beneficiary	9. BLUEPRINT
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Deliverable Name	Enabling conditions for DR in energy-efficient residential buildings		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	16	Work Package No	WP2

Description
The deliverable will present the final versions of the DEDALUS co-creation process, segmentation, socio-economic enablers and high-level requirements and specifications. Success criteria: Availability of final version of the specifications. (T2.2, T2.3, T2.4, T2.5)

Deliverable D3.1 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release)

Deliverable Number	D3.1	Lead Beneficiary	3. CARTIF
Deliverable Name	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release)		
Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	12	Work Package No	WP3

Description
The deliverable will report about the first release of the DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR, including standardised architectural concepts, protocols and data models, flexibility modelling and privacy-preserving DR prediction to support DR aggregation for residential buildings and corresponding to the first technical release MS4. Success criteria: Availability of first version of the technological enablers. (T3.1, T3.2, T3.3, T3.4, T3.5, T3.6)

Deliverable D3.2 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (2nd technology release)

Deliverable Number	D3.2	Lead Beneficiary	3. CARTIF
Deliverable Name	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (2nd technology release)		
Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	19	Work Package No	WP3

Description
The deliverable will report about the second release of the DEDALUS Digital Twins and data-driven services, including standardised architectural concepts, protocols and data models, flexibility modelling and privacy-preserving DR prediction to support DR aggregation for residential buildings and corresponding to the second technical release MS7. Success criteria: Availability of second version of the technological enablers. (T3.1, T3.2, T3.3, T3.4, T3.5, T3.6)

Deliverable D3.3 – DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (final technology release)

Deliverable Number	D3.3	Lead Beneficiary	3. CARTIF
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Deliverable Name	DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (final technology release)		
Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	28	Work Package No	WP3

Description
The deliverable will report about the final release of the DEDALUS Digital Twins and data-driven services, including standardised architectural concepts, protocols and data models, flexibility modelling and privacy-preserving DR prediction to support DR aggregation for residential buildings and corresponding to the final technical release MS10. Success criteria: Availability of final version of the technological enablers. (T3.1, T3.2, T3.3, T3.4, T3.5, T3.6)

Deliverable D4.1 – ‘First wave’ of DEDALUS Digital Twins and data-driven services for residential DR

Deliverable Number	D4.1	Lead Beneficiary	1. ENG
Deliverable Name	‘First wave’ of DEDALUS Digital Twins and data-driven services for residential DR		
Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	13	Work Package No	WP4

Description
The deliverable will report about the first release of the DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the first technical release MS4. Success criteria: Availability of first version of the Digital Twins and the data-driven services. (T4.1, T4.2, T4.3, T4.4, T4.5)

Deliverable D4.2 – "Second wave" of DEDALUS Digital Twins and data-driven services for residential DR

Deliverable Number	D4.2	Lead Beneficiary	1. ENG
Deliverable Name	"Second wave" of DEDALUS Digital Twins and data-driven services for residential DR		
Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	20	Work Package No	WP4

Description
The deliverable will report about the second release of the DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the second technical release MS7. Success criteria: Availability of second version of the Digital Twins and the data-driven services. (T4.1, T4.2, T4.3, T4.4, T4.5)

Deliverable D4.3 – "Final wave" of DEDALUS Digital Twins and data-driven services for residential DR

Deliverable Number	D4.3	Lead Beneficiary	1. ENG
Deliverable Name	"Final wave" of DEDALUS Digital Twins and data-driven services for residential DR		

Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	29	Work Package No	WP4

Description
The deliverable will report about the final release of the DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the final technical release MS10. Success criteria: Availability of final version of the Digital Twins and the data-driven services. (T4.1, T4.2, T4.3, T4.4, T4.5)

Deliverable D5.1 – DEDALUS evaluation methodology, baseline assessment and pilot implementation plan

Deliverable Number	D5.1	Lead Beneficiary	2. NTUA
Deliverable Name	DEDALUS evaluation methodology, baseline assessment and pilot implementation plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	13	Work Package No	WP5

Description
This deliverable will focus on pilot planning by providing a description of the pilots' measurement and verification plan and baseline analysis and by including the steps and processes for pilots' deployment and validation of the DEDALUS solutions, tools, datasets, etc. Success Criteria: The availability of the plan. (T5.1)

Deliverable D5.2 – DEDALUS Pilots' execution documentation and consumers' engagement – pre-pilot phase

Deliverable Number	D5.2	Lead Beneficiary	17. CIMNE
Deliverable Name	DEDALUS Pilots' execution documentation and consumers' engagement – pre-pilot phase		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	17	Work Package No	WP5

Description
This deliverable will provide a description of the pilots' use cases, engagement and social acceptance actions, overall activity monitoring and impact analysis in relation to the pre-pilot cycle running during M14-16 period. Success Criteria: The availability of the report in relation to pre-pilot scope at MS6 pilot pre-phase completion. (T5.2, T5.3, T5.4, T5.5, T5.6, T5.7, T5.8)

Deliverable D5.3 – DEDALUS Pilots' execution documentation and consumers' engagement – full pilot operation phase

Deliverable Number	D5.3	Lead Beneficiary	7. OURPOWER
Deliverable Name	DEDALUS Pilots' execution documentation and consumers' engagement – full pilot operation phase		
Type	R — Document, report	Dissemination Level	PU - Public

Due Date (month)	26	Work Package No	WP5
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Description			
This deliverable will provide pilots' final evaluation, a collection of the best practices collected and guidelines for the DEDALUS adoption in other cases as result of the Large-scale pilot cycle and replication activities running during M30-33 period. Success Criteria: The availability of the report containing the best practices and guidelines in relation to MS11 Large Scale Operation Phase completion. (T5.2, T5.3, T5.4, T5.5, T5.6, T5.7, T5.8)			

Deliverable D5.4 – DEDALUS impact and social acceptance analysis

Deliverable Number	D5.4	Lead Beneficiary	22. SIN
Deliverable Name	DEDALUS impact and social acceptance analysis		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	34	Work Package No	WP5

Description			
This deliverable will provide pilots' final evaluation, a collection of the best practices collected and guidelines for the DEDALUS adoption in other cases as result of the Large-scale pilot cycle and replication activities running during M30-33 period. Success Criteria: The availability of the report containing the best practices and guidelines in relation to MS11 Large Scale Operation Phase completion. (T5.2, T5.3, T5.4, T5.5, T5.6, T5.7, T5.8)			

Deliverable D6.1 – Preliminary Exploitation plan

Deliverable Number	D6.1	Lead Beneficiary	23. ICONS
Deliverable Name	Preliminary Exploitation plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	6	Work Package No	WP6

Description			
This document will include a preliminary business and exploitation plan aimed to carry out a focused market analysis. Success Criteria: The availability of the plan. (T6.1, T6.2, T6.3, T6.4)			

Deliverable D6.2 – Intermediate Exploitation Plan

Deliverable Number	D6.2	Lead Beneficiary	13. HERON
Deliverable Name	Intermediate Exploitation Plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	24	Work Package No	WP6

Description			
This document will include the intermediate version of the exploitation plan aimed to promote and valorise the project outcomes. It will report standardisation activities, DEDALUS data collaboratives and sharing economy business models design.			

Success Criteria: The availability of the report. (T6.1, T6.2, T6.3, T6.4)
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Deliverable D6.3 – Final Exploitation Plan

Deliverable Number	D6.3	Lead Beneficiary	13. HERON
Deliverable Name	Final Exploitation Plan		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	36	Work Package No	WP6

Description

This deliverable will deal with the final version of the Business and Exploitation Plan, focusing on business models validation and market outreach.

Success Criteria: The availability of the report. (T6.1, T6.2, T6.3, T6.4)

Deliverable D6.4 – DEDALUS comprehensive replication methodology and roadmap

Deliverable Number	D6.4	Lead Beneficiary	12. GREEN CITIES
Deliverable Name	DEDALUS comprehensive replication methodology and roadmap		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	36	Work Package No	WP6

Description

This document will present the policy recommendations as well as the visual roadmap for EU-wide replication of the lessons learned from DEDALUS.

Success Criteria: The availability of the report. (T6.5)

Deliverable D7.1 – Plan for dissemination and communication

Deliverable Number	D7.1	Lead Beneficiary	23. ICONS
Deliverable Name	Plan for dissemination and communication		
Type	R — Document, report	Dissemination Level	SEN - Sensitive
Due Date (month)	6	Work Package No	WP7

Description

This document will include the communication and dissemination plan aimed to achieve a high level of visibility to the project outcomes.

Success Criteria: The document should deliver the plan including target groups and stakeholders, strategies to be applied, dissemination meetings and key indicators. (T7.1)

Deliverable D7.2 – DEDALUS Communication Kit

Deliverable Number	D7.2	Lead Beneficiary	23. ICONS
Deliverable Name	DEDALUS Communication Kit		

Type	OTHER	Dissemination Level	PU - Public
Due Date (month)	8	Work Package No	WP7

Description
This document will summarise the Communication material including social media channels, project leaflet, poster and presentation video, as well as the project website to support the project dissemination. Success Criteria: The availability of the DEDALUS website, as well as the accompanying document describing the website structure and summarising communication material. (T7.2)

Deliverable D7.3 – Report on Dissemination and Communication activities, including impact assessment

Deliverable Number	D7.3	Lead Beneficiary	23. ICONS
Deliverable Name	Report on Dissemination and Communication activities, including impact assessment		
Type	R — Document, report	Dissemination Level	PU - Public
Due Date (month)	36	Work Package No	WP7

Description
This document will report the dissemination, communication and capacity building activities performed and the results achieved. Success Criteria: The document should report the achieved results in the context of dissemination, communication and capacity building activities. (T7.1, T7.2, T7.3, T7.4, T7.5)

LIST OF MILESTONES

Milestones					
Grant Preparation (Milestones screen) — Enter the info.					
Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Internal data management system; Quality and management plan	WP1	1-ENG	Online platform ready to be used; Project management plan	2
2	Visual identity; Web and social media release; Cross-promotion; Formulation of the Advisory Board	WP1, WP6, WP7	23-ICONS	Dissemination and communication plan; Project website; social media; Inventory of projects for cross-project collaboration; Confirm members of the Transdisciplinary Advisory Board	6
3	Active societal participation; Novel business models' design; Requirements	WP6, WP2	1-ENG	Use case capitalisation; Consumers analysis and segmentation; Incentive mechanisms; Business Sandbox with novel sharing economy and social innovation-based business models; High-level requirements for the DEDALUS framework	9
4	1st technology development cycle completion	WP4, WP3	1-ENG	'First wave' of DEDALUS solutions (technological enablers and data-driven DR services)	13
5	Socio-economic enablers and consolidation factors of multi-value and energy carrier-agnostic DR for residential buildings	WP2	9-BLUEPRINT	Final versions of the DEDALUS co-creation process, segmentation, incentive mechanisms, Business Sandbox, requirements	16
6	Pre-pilot phase completion	WP5	2-NTUA	Small-scale pilot-site infrastructure deployed along a subset of pilot sites	17
7	2nd technology development cycle completion	WP4, WP3	3-CARTIF	'Second wave' of DEDALUS solutions (full functional implementation)	20
8	Selection of trainees	WP7	23-ICONS	DEDALUS capacity building programme established	24

Milestones <i>Grant Preparation (Milestones screen) — Enter the info.</i>					
Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
9	Full pilot operation phase completion	WP5	2-NTUA	DEDALUS implementation from 5 Front Runners, social acceptance	26
10	Final technology development cycle completion	WP4, WP3	1-ENG	Final release of DEDALUS solutions (final version/iteration of the relevant deliverables in WPs 3-4)	29
11	Large scale operation phase completion	WP5	13-HERON	Final pilot validation from economic, social, and environmental perspective	34
12	Replication of solutions to different geographies	WP5	12-GREEN CITIES	Replication activities from the 7 pilots (including 2 Multipliers)	34
13	Capacity building completion	WP7	23-ICONS	Members of the DEDALUS programme	36
14	Communication report and collaboration with other projects and initiatives; Final exploitation and project completion	WP1, WP6, WP7	23-ICONS	Synergies (e.g. BRIDGE); Dissemination & communication reports; Business impact assessment; Market readiness for DEDALUS exploitation, commercialisation and wider use; Scenarios and application scalability and replicability	36

LIST OF CRITICAL RISKS

Critical risks & risk management strategy			
Grant Preparation (Critical Risks screen) — Enter the info.			
Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	R1: Low engagement of energy consumers and stakeholders from the energy value chain during the co-creation process. Level of likelihood: Medium; Level of severity: Medium	WP5, WP2	Partners have large networks, experience in engaging key stakeholders. DEDALUS beneficiaries from the pilots are already members of the Consortium. Robust and rigorous community building methodology (T2.2) and communication and dissemination plan (T7.1) will be elaborated, promoting project with all possible means. To maximise participation, we will use existing knowhow in establishing effective communication platform. The Advisory Board members will be asked to support if needed (T1.3).
2	R2: DEDALUS solutions fail to conform to user's needs and requirements and/or are not appropriate. Level of likelihood: Low; Level of severity: Medium	WP4, WP2, WP3	The Consortium includes technology providers with strong knowledge on ICT domain. The development of the DEDALUS solutions will follow an iterative approach to integrate end users' feedback. The work plan contains extensive analysis and specification of the different components (T2.5). Preliminary versions of the components (WPs 3-4) will be released during the project. These results will be verified by the Consortium. Additionally, the development of the components is fed with the results of WP5, thus these risks are reduced to manageable levels. Most tools are provided to DEDALUS with a certain maturity level, as indicated in Section 1.3.
3	R3: Insufficient or ineffective pilots' support and/or poor feedback gathered for pilot evaluation Level of likelihood: Medium; Level of severity: Medium	WP5	The early planning of the pilot preparation and the evaluation methods and means for the collection of network data and user feedback will leave little room for this risk occurring (WP5). If the risk occurs, SSH/scientific partners will undertake more intensive user support, update accordingly the supporting material and even improve the DEDALUS solutions, e.g. in terms of user friendliness. The partners will devise new/more persistent evaluation means (i.e. phone interviews, F2F meetings) in order to get the needed feedback and will also exploit their large partner networks to this purpose.
4	R4: Replication of solutions to different geographies is not as expected. Level of likelihood: Medium; Level of severity: High	WP6	There are dedicated tasks in the workplan assuring that all the parameters and particularities of the pilot sites are being considered in the design of the scenarios. Partners responsible for the implementation of the digital tools and the incorporation of the market structures and business models have a proven vast experience on interoperability and standardisation matters that will safeguard this technology and innovation transfer from one pilot site to other.
5	R5: Reluctance and lack of motivation of energy	WP5	A dissemination and communication strategy will be developed early in the project in order

Critical risks & risk management strategy <i>Grant Preparation (Critical Risks screen) — Enter the info.</i>			
Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
	consumers and stakeholders to data sharing and reuse. Level of likelihood: Medium; Level of severity: Medium		to identify the correct audiences and venues and devise strategies to reach them. Professional workshops, networking events, or other motivating initiatives would be organised. The project's pilots are carefully selected in order to represent the widest possible spectrum of stakeholders.
6	R6: DEDALUS results are not on the path to be considered useful after the project finishes Level of likelihood: Low; Level of severity: Low	WP6, WP7	The project partners' participation in other projects provide a potential community of users: both the scientific and the industrial partners, have high motivation to keep using the DEDALUS technology due to its benefits for their day-to-day activities and/or their business agendas, and are well-connected with user communities that promise a broader uptake of the results
7	R7: The running H2020 projects that are expected to contribute in DEDALUS are not delivering as expected. Level of likelihood: Low; Level of severity: Medium	WP7	DEDALUS is planning to build on and exploit other current projects' results to assure compliance with existing work and established concepts. The fact that many members of the consortium are actively contributing to those projects reassures their commitment to deliver the expected results, and safeguards the knowledge transfer to a satisfactory level of detail.
8	R8: Lack of coordination with other actors, wrong expectations. Level of likelihood: Low; Level of severity: Medium	WP2, WP7	DEDALUS is oriented to co-creation (WP2). A specific task has been designed for cross-fertilisation activities with BRIDGE and relevant projects/initiatives (T7.4).
9	R9: New technology is launched during the project, changing the market. Level of likelihood: Low ; Level of severity: Medium	WP6, WP4, WP3	The Innovation Manager (T6.4) detects this market shift and, together with the Technical Manager, contacts each technology provider on the given area to conduct and analysis of the new technology and assess its impact on DEDALUS activities. If detected as relevant, the topic will be included in the Management Board and then voted for adoption. In affirmative case, the relevant technical WPs (WPs 3-4) will change the scope to cover it.
10	R10: Business adoption of DEDALUS proposals is not as expected. Level of likelihood: Low; Level of severity: Medium	WP6	WP6 focuses on monitoring business adoption and providing a roadmap for tools and pilots to exploit their results. They will work closely with pilots to guarantee the adoption, at demonstrator side, is full and the way to marketise the result is also clear for the future beyond the project.
11	Partner(s) unable to Contribute Level of likelihood: Low; Level of severity: High	WP5, WP1, WP6, WP4, WP2, WP3, WP7	Rigorous project management. Failure of individual participants will lead to immediate assessment of capabilities and task reassignment (T1.1). Partners have adequate range of technical skills and can take over tasks.

Critical risks & risk management strategy <i>Grant Preparation (Critical Risks screen) — Enter the info.</i>			
Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
12	R12: Partners' contribution to outputs not delivered on time Level of likelihood: Medium; Level of severity: Low	WP5, WP1, WP6, WP4, WP2, WP3, WP7	Quality management, early directions for timely outputs. Monthly calls on WPs (reviews/ appropriate revisions) for corrective actions. 10 of 26 deliverables in first year indicating issues; 14 milestones to track progress.
13	R13: Low scientific and technical quality of deliverables Level of likelihood: Low; Level of severity: Hight	WP5, WP1, WP6, WP4, WP2, WP3, WP7	Rigorous quality management. First deliverable drafts one month prior to deadline, for review, detailed feedback by Coordinator, 2-4 reviewers. Partners have excellent track record for scientific/technical quality (T1.3).
14	R14: End user privacy and security insufficiently addressed. Level of likelihood: Low; Level of severity: Low	WP5, WP1, WP6, WP4, WP2, WP3, WP7	We have detailed the ethical requirements prior to submission based on different EU projects engaging end-users. Specific tasks have been included in the work structure to address these issues (T1.4 and T4.1).
15	R15: Project implementation affected by extreme developments. Level of likelihood: Low; Level of severity: Medium	WP5, WP1, WP6, WP4, WP2, WP3, WP7	The Consortium shall identify appropriate methodological alternatives that ensure that the project is not brought to a complete standstill (e.g. due to COVID-19) and can continue producing results.

PROJECT REVIEWS

Project Reviews			
Grant Preparation (Reviews screen) — Enter the info.			
Review No	Timing (month)	Location	Comments
RV1	21	TBC	First review meeting
RV2	36	TBC	Final review meeting

HISTORY OF CHANGES			
VERSION	PUBLICATION DATE	CHANGE	SECTION
0.9	02.01.2023	First draft of the DoA based on the proposal text	
0.9	02.01.2023	Included in the WP7 description the following sentence: DEDALUS will contribute, upon invitation by the CINEA, to common information and dissemination activities to increase the visibility and synergies between Horizon Europe supported actions.	Sec 3.1, table 3.2b, WP7 description
0.9	02.01.2023	Deliverable names and due dates for D1.3 and D1.4 has been changed.	Sec 3.1, table 3.2b, WP1 description
	02.01.2023	Included the following statement: <i>All depreciation costs for equipment, infrastructure or other assets in the project are in compliance with Article 6 and will be recorded in the appropriate beneficiary's accounts, purchased in accordance with Article 6.2.C of the grant agreement and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.</i>	Sec 3.1
0.9	02.01.2023	To address GAP comment 2.1: (i) the DEDALUS methodological approach for consumer-aware tools, DTs and services has been further elaborated and described in sec. 1.2.1, (ii) the feedback loop among the 'end-user data-driven model' and the 'assets' phases has been defined more in detail in sections 1.2.2 and 1.3.1 (iii) the methodological steps of the proposed co-creation process have been more elaborated and described in sec.1.2.2 for the design of a participatory residential DR framework	Sec 1.1.1. Sec. 1.2.1, Sec. 1.2.2, Sec. 3.1
0.9	02.01.2023	To address GAP comment 2.2, Tables 2.3 and 2.5 have been updated accordingly, providing a mapping between communication/dissemination activities and target audiences.	Sec 2.2.1 (Table 2.3), Sec 2.2.1.1 (Table 2.5)
0.9	02.01.2023	To address GAP comment 2.3, the timing of the three cycles has also been added to the structure of the work plan.	Sec 3.1
0.9	02.01.2023	D1.3 and D1.4 due dates have been changed/corrected to M18 and M36 respectively to align with Reporting periods end dates (M18, M36).	Sec. 3.2 tab. 3.2.b
0.95	13.01.2023	Deliverables description has been extended as for GAP request 4.12	Part A
0.97	06.02.2023	Tab 3.1h modified in accordance with GAP comment 4.1	Sec. 3.1

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Term	Definition	Term	Definition	Term	Definition
AI	Artificial Intelligence	EV	Electric Vehicle	MS	Milestone
BEMS	Building Energy Manag. Systems	EO	Expected Outcome	P2P	Peer-to-Peer
DER	Distributed Energy Resource	HVAC	Heating, Ventilation and Air Conditioning	PV	Photovoltaic
DH	District Heating	ICT	Information & Communication Technology	REC	Renewable Energy Community
DHN	District Heating Network	IoT	Internet of Things	RES	Renewable Energy Sources
DT	Digital Twin	IEQ	Indoor Environmental Quality	SAREF	Smart Applications REference
DLT	Distributed Ledger Technology	KER	Key Exploitable Result	SGAM	Smart Grid Architecture Model
DR	Demand Response	KPI	Key Performance Indicator	SSH	Social Sciences & Humanities
DSO	Distribution System Operator	LEC	Local Energy Community	TE	Technology Enabler
ESCO	Energy Service Companies	LEI	Long-term Expected Impact	VPP	Virtual Power Plant
EU	European Union	ML	Machine Learning	WP	Work Package

In the European Union (EU), households (namely the residential sector) represented 27,4% of the final energy consumption [1]. The residential loads often contribute significantly to seasonal and daily peak demand. To support this variation, utilities companies need to increase their generation, in order to avoid interruptions in power supply. Generally, 20% of the power generation capacity is latently available to meet the peak demand that occurs for approximately 5% of the time [2].

Historically, matching electricity supply and demand was relatively straightforward, with large and controllable power plants on the one hand and relatively easy to predict demand on the other. But in recent years, this scenario is starting to change, due to a number of reasons: (i) The **use of smaller, variable and less predictable decentralised renewable generation** is increasing, affecting directly the energy matrix; (ii) Europe attempts to disentangle its demand from **Russian gas and fuels** (with a looming supply gap for the coming winter), while keeping up with its decarbonisation targets [3]; (iii) **Climate change**, and the resulting temperature changes (like the recent European heat waves), affect electricity markets both on the demand and the supply sides.

Matching electricity supply and demand at all times is becoming more challenging and the electricity system needs more flexibility. A **flexible consumption** (i.e. demand following supply) could - next to energy efficiency as 'first fuel' - play an important role in the residential sector. **Residential Demand Response (DR)** can reduce the need for fossil fuel power plants and help integrate renewable energy onto the electric grid by providing increased stability and management.

The residential sector in Europe offers huge energy demand flexibility, mainly due to the huge thermal energy storage and power management capabilities of the buildings. Smart meters, connected appliances or in-home displays open new possibilities for innovation on the demand side. For instance, DR can be about shifting electric heating and air conditioning away from electricity price peaks, charging or discharging an electric car at times of interesting prices or delaying the use of a washing machine while being rewarded for reducing consumption at peak times. Complementary to the huge potential for residential flexibility, **leading-edge Information and Communication Technologies (ICTs)**, such as Internet of things (IoT), Artificial Intelligence (AI), Distributed Ledger Technology (DLT)/blockchain/smart contracts, edge computing and big data analytics [4], enable a great opportunity to develop the residential DR market.

However, **this important flexibility potential currently remains unexploited**. DR at residential level is often overlooked with the focus being the industrial and commercial sectors. Expanding DR to cover also the residential sector gives rise to a wide range of socio-economic and technological challenges:



'Socio-economic' Challenges

Participation	Empowered consumers can increasingly participate in DR by adapting their consumption habits. However, the residential consumers, firstly, need to become energy literate and understand the current plans and technologies (i.e. how the energy market works, how they can be part of it and how they can benefit from engaging in DR), though customised tools that meet their needs and bring consumers, technologies, regulation, and markets together. It is becoming clear that effective and interactive SSH-driven tools and methodologies can strengthen the human connection and interaction, awareness and preparedness of the residential consumers on the opportunity to value their flexibility and to ultimately reduce their energy bills or be rewarded to modify their consumption accordingly.
Multi-Value DR	At the same time, the lack of and immaturity of incentive and appropriate financial and/or non-financial compensation mechanisms hinders the unleashing of the residential flexibility potential. Due to the still predominant presence of incumbent stakeholders which operate the electricity network and/or play the role of energy suppliers and/or flexibility aggregators, the flexibility markets are rather illiquid, with a very low level of competitiveness. The value from participation in DR is not perceived as sufficiently clear by residential consumers and is not able to be propagated in a most appropriate way to them. This brings inevitable market distortions with extra profit for non-independent aggregators, and with a very low level of financial benefits for participation from the residential consumers. Moreover, actual flexibility markets are not able to allow smaller pool of aggregated residential flexibility to participate due to a high minimal threshold, which created a high entry barrier to overcome to deploy effective residential DR programs and significant market failures to deal with. Main challenge here is to design a multi-value DR concept which may bring and trade-off beyond financial (e.g. social and/or wellness) against financial compensation to the participants.

Multi-Carrier DR	So far, DR has focused almost exclusively on the electricity . Electrical and non-electrical (e.g. heat and/or gas) loads flexibilisation may have an interesting potential to support optimised and more energy-efficient operation of non-electricity network, like district heating networks, while looking at potential trade-off with residential consumers thermal (and more general) comfort. It becomes of utmost importance to explore a multi-carrier (or energy carrier-agnostic) DR approach , and consider a number of synergies and interaction scenarios from heat and/or gas energy carriers towards electricity , while considering residential buildings as the appropriate level where such synergies may take place. This means that same flexibility action may simultaneously bring revenues from both electricity and district heating value chain stakeholders.
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‘Technological’ Challenges

Extended Interoperability	The advent of cutting-edge ICTs is undoubtedly representing a fundamental enabler to deploy successful DR at residential level, while bringing a significant reduction of the ICT DR infrastructure. In this respect, residential-level appliances are becoming smarter and smarter, and hence capable to advanced sensing and local control and to support operational flexibility. However, beyond the lack of technical interoperability , the main problem consists of having smart appliances still not DR-ready . They are not able to interact with DR/aggregator/utilities infrastructure to allow an optimal yet dynamic and closer to real-time coordination to really support smart energy grids applications. Moreover, DR requires consumers, suppliers, other DR service providers and distribution network operators to exchange increasing amounts of detailed data. Most consumers will be comfortable sharing their data if they are confident that it is stored securely in a way that safeguards privacy, while at the same time exchanging consumption energy data sharing against additional non-financial benefits (e.g. flexible loads device maintenance) [5]. In that respect, the ongoing EU-level investment on deploying Energy Data Space can be seen as an emerging opportunity to streamline personal energy consumption data sharing in more interoperable, secure and trusted way.
DR Services & Business Models	Despite a variety of apps, tools and application for activating consumers have been largely deployed in the last years, the lack of a multi-value multi-commodity perspective for residential DR when combined with a data-driven angle, as compliant to the Energy Data Space approach, is bringing center stage the challenge of designing energy carrier-agnostic data-driven services . These services may be deployed over different individual energy networks (namely electricity network, district heating network), and eventually take into due consideration interaction and synergies of non-electricity carriers (heat, gas) towards electricity. Building-level flexibility pre-aggregation may also become a promising way for achieving the minimum threshold imposed by many flexibility marketplaces for accepting flexibility bids and may be considered as a service offered to the energy community represented by tenants and/or dwellers, where buildings as physical assets overlap with Local Energy Communities (LEC). However, flexibility categorisation and clustering of buildings has not sufficiently explored, and building-level pre-aggregation suffer from insufficient and rather immature residential building energy (electricity and heat) flexibility profiling and planning.

1.1. OBJECTIVES & AMBITION

The overall vision of DEDALUS is to design, develop and demonstrate SSH-driven **multi-value energy carrier-agnostic DR ecosystem**, tailored to **optimise and manage automated DR in residential buildings** (Fig 1.1). We combine leading-edge ICTs technologies [IoT, AI, DLT/blockchain, edge computing big data] with social/behavioural dimension, and with sharing economy and **value stacking governance and business models**.

The overarching goal is to deploy a **social, technological and business framework**, aimed to: (i) facilitate and scale up residential energy consumers massive participation to DR; (ii) adapt to a variety of different **mono-carrier** (electricity, heat) or **multi-carrier** synergetic scenarios (electricity vs heat, electricity vs natural gas), with a view to take into full consideration synergies and interaction among the participating energy commodities.

- **Social layer:** Designate socio-economic enablers, which leverage on a combination of qualitative and quantitative SSH-driven research and inspiring principles from Design Thinking methodologies, for engaging residential energy consumers in DR at building-level through appropriate design of beyond-financial DR incentives, such as the personal comfort.
- **Technological layer**, consisting of:
 - A data-driven stack of technology enablers **to enable smart loads residential appliances into DR-ready flexible ones**, which will leverage on and contribute to ongoing deployment of the **Energy Data Space** along Horizon ENERSHARE (ENG, NTUA, COMS, SIN). It includes: (i) **technology adaptation, harmonisation and alignment of most prominent DR interoperability efforts and protocols** (such as EFI, Open ADR2.0, USEF, EC BRIDGE Interoperability of flexibility assets) **to enable DR-ready relevant assets**. At the same

time, respective APIs will be made available through [Open APIs via the resulting \(planned\) Reference Implementation of the Energy Data Space](#); (ii) **privacy preserving blockchain-based solutions** technology

- Novel **multi-value data-driven DR flexibility management services and Building/District Digital Twins**, exploiting available potential at different levels of flexibility aggregation (apartment-level, building-level, district-level and multiple buildings within a virtual building cluster, i.e. not geographically bounded buildings). These services will consider flexibility as depending on one hand both technical characteristics of the flexibilised assets and the habits of the targeted end consumers (individual residential users, buildings or groups of buildings). On the other hand, they will consider service bundling (e.g. peak shaving bundled with ancillary services), and/or cross-commodity synergies to widen the exploitable flexibility while leveraging on the same assets' portfolio and on unchanged DR infrastructure costs. They will be implemented over a set of complementary use cases, addressing a variety of interactions with electricity grids and other energy (District heating, decentralised Power2Heat, natural gas) and non-energy vectors (buildings, comfort/wellness/health, elderly people/ageing), and respective DR infrastructures.

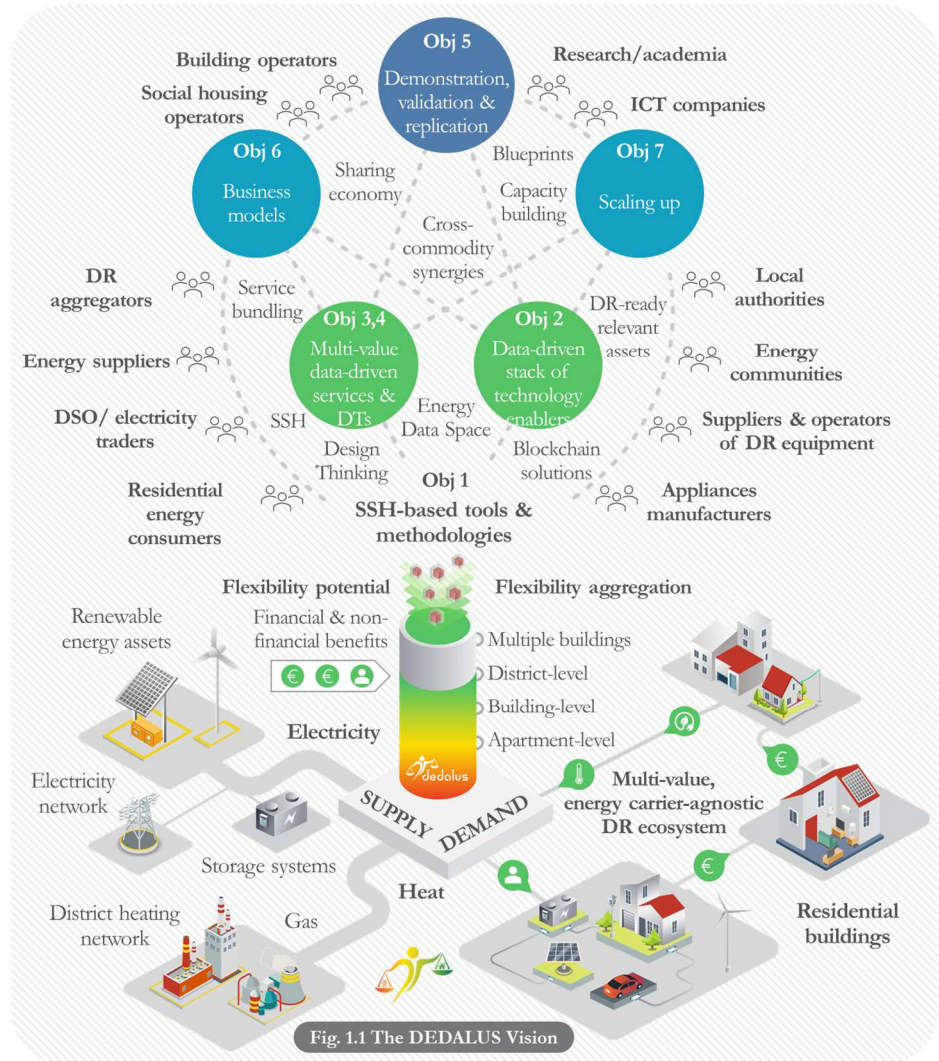


Fig. 1.1 The DEDALUS Vision

- **Business Layer**, where new building-centered DR business models will be designed and validated, while combining: (i) **Building-level DR service offering cost reduction** through larger socialisation of the DR infrastructure (and DR-ready device costs deployment) than possible at individual apartment level; (ii) **revenue increases from service bundling and/or cross-commodity services (electricity vs heat) and/or cross-value chain services (energy vs healthcare/comfort, ageing)** may represent significant enablers.

1.1.1. PROJECT OBJECTIVES

DEDALUS vision will be accomplished by the realisation and fulfilment of a set of high-level **Scientific, Technical and Business Objectives (Obj.)**, leading to **9 Key Exploitable Results (KERs)** and relevant **Key Performance Indicators (KPIs)**, as follows:

Obj. 1

WP2

To make use of SSH-based tools and methodologies to facilitate the engagement of energy consumers in the residential DR

We will make use of SSH-based tools and methodologies, such as deliberative citizen engagement, intersectionality, co-creation and experimental scenario-based research, as well as Design Thinking methodologies, in order to support the design of an innovative concept **for participatory, multi-value and energy carrier-agnostic residential DR**. DEDALUS will pave the way for designing multi-factor financial and non-financial socially driven incentives, aiming at: (i) additional financial remuneration by offering the same flexibility to different markets (flexibility bundling), such as electricity and heating marketplace; (ii) trading-off financial vs comfort and other non-financial

preferences of homogeneous clustered groups of consumers; (enabling social facilitation services and benefits to the local community of people living in the same buildings or groups of buildings (e.g. reducing energy poverty, via energy sharing of locally generated RES with more vulnerable people). Nudging techniques and multiple concepts, such as socio-economic privilege, gender, age and behavioural analyses, will be mobilised to promote an inclusive engaging dynamic with residential energy consumers, in order to identify a number of social, environmental, geographical, financial, and personal preferences, and related KPIs against which buildings and consumer characteristics will be segmented and clustered thanks to AI-based algorithms.

KERs	KPIs
Social Science Framework, innovative multi-dimensional incentives and nudging interventions for residential DR (KER1); AI-based individual and building-level-level consumers clustering and market segmentation algorithms (KER2).	▪ > 1.500 stakeholders (65% consumers/35% business roles) in at least 10 co-creation sessions ▪ Clustering based on an understanding of the following factors: social-economic, gender, sociocultural, and socio-political factors

Obj. 2

WP3

To adapt and extend Horizon ENERSHARE Energy Data Space Interoperability and Trust layer, combining harmonisation and further extension of ongoing interoperability efforts, with privacy-preserving blockchain-based enablers for trusted, sovereign secure and participatory flexibility data sharing, in order to enable DR-ready user-consumption based flexibilised smart appliances

We will leverage on existing interoperability efforts from on-going or recently completed initiatives (SGAM, EFI, USEF, CIM, EEBUS, SAREF, OpenADR 2.0), projects (H2020 MATRYCS, BRIGHT, BD4NRG and Horizon ENERSHARE and DigiBUILD), as well as H2020 OpenDEI and EU-level BRIDGE cluster cross-sector reference architectures, to deploy an adaptation of the Horizon ENERSHARE Energy Data Space Layer. Such flexibility interoperability layer will be able to effectively support and enable large scale participatory automated and seamless multi-value and energy carrier-agnostic DR for residential buildings. It will be deployed through a set of extended data models and Open APIs, which will facilitate turning over smart appliance/devices/assets into flexible and DR-ready ones, which will be the basis for the flexibility exploitation at building level together with the prediction algorithms for that flexibility. Moreover, we will adapt and evolve TRL 5-6 technology developed in H2020 eDREAM, PHOENIX, BRIGHT and BD4NRG projects to deploy a TRL7 blockchain-based multi-purpose technology enabler for cross-cutting multi-stakeholders' privacy-preserving data sharing, privacy-preserving flexibility assessment and forecasting, P2P flexibility tokenisation and DR settlement using multi-criteria smart contracts. Cryptographic techniques will be integrated with blockchain to achieve the data consistency and transparency needed for data sovereignty in the context of Energy Data Spaces and data hubs, as well the secured data exchange and interoperability in DR. A blockchain-driven marketplace will be set up and evolved for allowing the P2P trading of the flexibility against other energy and non-energy assets, enabling the control, aggregation (i.e. prosumers, buildings, community, etc.), and valorisation of small-scale residential flexibility via tokenisation. Finally, fully customizable smart contracts will be developed and integrated for near real-time beyond financial settlement of DR to increase the trust, acceptability, and participation in DR of residential consumers (e.g. tokens for compensation, discounts to local business, building sustainability, etc.).

KERs	KPIs
Open APIs for DR-ready smart appliance IoT platform (KER3); Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange (KER4).	▪ Identification of and interoperability with at least 4 Open Standards, standardisable interfaces and formal languages/ontologies, based on the Basic Application Interoperability Profiles (BAIOPs) created in accordance with the SGAM. ▪ Open APIs and extended data Models: ≥ 10 ▪ 15-20% increased share of aggregated flexibility mobilised by piloted assets

Obj. 3

WP4

To combine H2020 eDREAM, BRIGHT, MATRYCS and HORIZON DigiBUILD building-level data-driven models, algorithms and flexible assets DTs with end-users' consumption and beyond consumption profiling models (e.g. comfort, healthcare, ageing, poverty/vulnerability) and deploy TRL6 HITL ('Human-In-The-Loop') flexibility assessment and valorisation DTs

We will adapt and evolve data-driven models, algorithms and flexible assets DTs developed in H2020 eDREAM, BRIGHT, MATRYCS and Horizon DigiBUILD projects, end-users multi-value energy consumption models and beyond consumption user profiling, such as comfort, healthcare, ageing, poverty/vulnerability, in order to deploy innovative end-user-driven DTs for flexible assets. Final outcomes will consist of DTs for optimal assets vs consumers flexibility planning, and data-driven flexibility assessment and valorisation models and algorithms which trade-off financial against non-financial (e.g. comfort) criteria for DR participation. These human-oriented models, algorithms and DTs will increase the predictability of energy consumers' participation in DR, as well as the

KERs

Digital Twins for optimal assets vs consumers flexibility planning and decentralised DR management (KER5); Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments (KER6).

KPIs

- Increased predictability by 15% of DR behaviour and flexibility potential to mobilise
- ≥ 2 Digital Twins

Obj. 4

WP4

To deliver data-driven, value stacking cross-stakeholder/cross-commodity/cross-value chain DR algorithms, models, and ICT-based services, for DR in energy-efficient residential buildings

Suitable multi-value data-driven services will be adapted, deployed and validated to support the optimal flexibility aggregation for DR aggregators, building managers, suppliers and/or DR service providers. In particular, we will deploy: (i) mono-carrier electrical flexibility assessment, valorisation and optimal management at building-level, while leveraging on building-level pre-aggregation; (ii) mono-carrier optimal electrical energy and flexibility management at district-scale or virtual cluster of buildings, based on shared DR assets (e.g. district-scale shared battery storage) and/or self-consumption; (iii) cross-domain heat-centered and electricity-centered decentralised flexibility aggregation and assessment (e.g. District Heating Network, heat pumps, natural gas boilers) to optimise the overall electrical and heat flexibility, and achieve optimal data-driven control of flexible multi-energy assets. We will investigate closed-loop tools enabling near real-time feedback sharing on the performance monitoring of the implemented activation strategy, with a view to eventually take informed adaptive future decisions on alternative options. Such services and the respective products that derive from the consumer energy consumption (and production for prosumers) profiles, takes also into consideration the profile of the consumer in other closer domains (like healthcare/comfort/wellness, ageing, as well as aspects related to 'energy poverty') and through a digital (data-driven) supported process will be concretised and published in local or broader markets.

KERs

Electrical flexibility management tools for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets (KER7); Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralised cross-commodity electricity vs heat management (KER8).

KPIs

- Number of validated data-driven aggregation models and services for residential DR: ≥ 7

Obj. 5

WP5

To demonstrate, validate and replicate DEDALUS solutions through a number of different pilots, located in different geographical and climatic areas and operated by different stakeholders

We will demonstrate, validate and replicate DEDALUS solutions in 7 countries across Europe, representing very different social and economic situations, addressing a variety of interactions with energy and non-energy vectors (e.g. buildings, storage, comfort/wellness/health, elderly people) and operated by different stakeholders (social housing operators, building operators, energy communities, retailers/suppliers, DSOs/District Heating Network operators): (i) 5 Front Runners demonstrators (Austria, Denmark, Greece, Italy and Spain), in which the solutions will be tested initially, to be later replicated in other locations; (ii) 2 Multipliers (Ireland, Romania), where the approach followed in the Front Runners will be transferred and adopted. These pilots will be evaluated through definition and validation of operational, regulation conformity, legal and technical performance indicators.

KERs

Best practices from demonstration of innovative social, technical (at TRL 6-7) and business solutions for market parties (All KERs).

KPIs

- Number of consumers actively engaged in residential DR: ≥ 500
- Different types of actors engaged by DEDALUS services: ≥ 10
- Number of technologies that reached TRL6-7 by the end of the project ≥ 14
- At least 95% satisfaction accruing from pilot users

Obj. 6

WP6

To explore and validate new business models which combine inspiring principles from sharing economy, with multi-carrier system-level perspective and with social innovation and non-financial benefits capture for DR in energy-efficient residential buildings

Novel sharing economy and social innovation-based business models will be designed and validated, which will leverage on combining both DR infrastructure cost socialisation and unit DR cost reduction, among a larger number of B2B and B2C stakeholders (ranging from PPP shared flexibility assets ownership, to buildings as community full ownership or multiple energy utilities shared ownerships) stakeholders, with cross-stakeholders and/or cross-commodity and/or cross-value chain (e.g. healthcare, comfort) stacking of value-added services. We will identify through the field demonstrations and the applied business models - where the current regulatory framework (either EU-wide or as locally applied) might not be sufficient to allow - the service offerings that are tested. This might be either in the form of allowing consumers to market their potential flexibility in a market or even on the market design

itself (e.g. allowing P2P services between consumers). We will identify a set of actions to be followed in order to address such regulatory/administrative barriers and/or gaps.

KERs	KPIs
Business Sandbox with novel sharing economy and social innovation-based DR business models (KER9)	▪ Deliver at least 4 new business models ▪ Recommendations in each of the pilot countries and at EU level developed

Obj. 7

WPs 6-7

To allow effective consumer participation in DR and further market opportunities by providing input to regulatory bodies to stabilise and harmonise the regulatory framework for residential DR, ensuring an efficient communication and dissemination, as well as alignment and synchronisation with European relevant initiatives and projects

DEDALUS will build on a strong legacy of SSH-led energy research and experimentation to deepen, operationalise and exchange knowledge with communities, stakeholders and other EU funded projects and initiatives. The aforementioned research and tools will set in place multiple dissemination methods (see Section 2.2) raising the awareness to increase the impact and to ensure the efficient advancement of novel market designs and business models, focusing on the effective consumer participation in residential DR. DEDALUS aims also at the activation from the very beginning of the project and continuous enlivening of links with relevant standardisation bodies, through the active involvement of consortium partners (e.g. CENTRICA, ENG, BLUEPRINT, CARTIF, ARC), which address ICT, smart energy grids, DR and Energy Data Spaces, with a view to provide input for the standard adaptation/extension, and with regulatory bodies to provide inputs to stabilise and harmonise the regulatory and business frameworks for residential DR.

KERs	KPIs
Promote the project results, bringing them one step closer to the market, ensuring dialogue with potential funders and/or customers (All KERs).	▪ 1 project exploitation plan, individual exploitation plans for each partner, 1 SWOT analysis and 1 sustainable plan for after project operation ▪ At least 7 different dissemination/communication channels set-up & maintained ▪ At least 200 organisations and 1,000 individuals following the project's channels ▪ Contribution to 2 standards, synergies with more than 30 relevant projects/initiatives and at least 3 cross-fertilisation activities with BRIDGE ▪ At least 4 virtual and physical workshops with 150 trainees

1.1.2. AMBITION

Multi-vector DR for explicit and implicit flexibility management

The majority of technological solutions and business models for DR are tailored so far to exploit electrical flexibility against electricity consumptions models from flexible electrical loads and smart appliances. However, electricity is only one of the energy carriers which residential consumers make use, together with heating and natural gas carriers. The optimal operation of non-electrical devices and energy infrastructures from one hand opens up to make available some shares of flexibility towards non-electrical commodity marketplaces. While natural gas accounted for 31.7% of the EU final energy consumption in households [6], the pioneering approach of demand aggregation for natural gas suppliers has only been considered by a few recent initiatives, mainly by U.S. utilities National Grid pilot, Southern California Gas pilot [7-8]. Especially in Northern European countries, space heating is mainly supported through district heating networks, whose optimal operation can make available certain level of flexibility to be offered to future heat local flexibility marketplaces or to be used to improve energy efficiency of the building. Moreover, certain devices as part of non-electrical infrastructures, such as ventilation units for district heating, or gas boilers and heat pumps are electrical loads, which may make available additional flexibility. Hence also the amount of electrical flexibility is strictly linked to the optimal operation of non-electrical energy networks and this aspect becomes of utmost important if we consider the building level. As from our knowledge DR frameworks and solutions mostly consider only the electrical flexibility, while heat-centered DR has not reached an appropriate technology maturity level. DR at building level does not usually consider synergies and interaction with other energy carriers.

Beyond the state-of-the-art

Addressed in WP4

Electrical flexibility management tools will be provided for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets (KER7). DEDALUS will also span over cross-commodity (or commodity-agnostic) DR throughout its activities, ranging from scenarios and use cases, technology enablers upscaling, adaptation, deployment and field pilot validation. The energy-carrier agnostic or cross-commodity characteristics of residential DR will be designed based on a variety of use cases and piloted scenarios. The overarching concept of a cross-commodity and carrier agnostic DR will allow to increase the optimal operation of the energy system as a whole, while at the same time will enable heat flexibility prediction and management to be upscaled and brought closer to the full market deployment, hence widening the DR concept to

SSH-based tools and methodologies for facilitating energy consumers participation in residential DR

By nature, the citizen participation in the energy market is a type of social innovation that can bring tremendous benefits for citizens (energy consumers) by placing them closer to the energy transition and fulfilling Europe's decarbonisation goals [9]. DR provides an opportunity for residential consumers to play an active role in this transition, especially in the operation of the electric grid. However, consumers overestimate technical complexities and potential disruptions in their comfort or daily routines, as they are not able to fully understand the value and benefits of flexibility. Although some of the household consumers may be aware of very basic forms of implicit demand side flexibility when faced to time-discriminating tariffs (day/night tariffs, etc), they are probably not aware of their full individual and aggregated flexibility potential and benefits, not only economic, but also social and environmental, and the significant impact aggregation can have. Whereas normative rational actor models assume that consumers are primarily driven by the goal to increase financial benefits, research from the SSH has shown that other factors, such as environmental concerns, social norms, community trust, awareness and energy independence play an important role in the willingness to participate in emerging decentralized energy systems and business models [10-11]. While interdisciplinary research [12] emphasises the need to (i) take the diverse consumer preferences into account and (ii) to develop monetary and non-monetary incentive structures that address the diverse consumer preferences, research developing and testing such tailored and consumer (group) specific incentives is scarce. As a result, a large potential of residential DR remains unexploited [13-14].

Beyond the state-of-the-art

DEDALUS aims to provide a Social Science Framework, innovative multi-dimensional incentives and nudging interventions (KER1) for energy consumers' activation and participation in residential DR. This framework will be an effective tool to unlock flexibility potential from the residential buildings, enabling consumers to take full control of their energy usage. This framework outlines techno-economic and socio-institutional requirements for enabling DR in energy-efficient residential buildings. It ensures that the residential consumers take the front seat and co-create innovations that enable them to play an active role in DR mechanisms and fair allocation of incurred costs and benefits. We will develop the means for collecting the feedback (elaboration of questionnaires, organising focus group sessions and workshops) and for processing/translating the feedback into the functional requirements for the technology development.

AI-based models for buildings clustering considering consumers socio-behavioural features

The behaviour of residential customers and socio-economic factors impact the building energy consumption, variation and flexibility that can be used in DR [15]. Understanding the consumers behaviour in correlation with the energy use in everyday life is an important step towards unlocking and aggregating building level flexibility, enabling participation of residential buildings in DR. Existing efforts have focused on achieving segmentation of consumers using their sociodemographic characteristics and little attention has been given to behaviour patterns and schedules which could be different for each building due to occupants' lifestyles or preferences. In this context the building type, location size and consumers energy profiles are not enough for clustering the buildings as they do not consider the heterogeneous social enablers behind residential customers' energy use and engagement in DR. The identification of homogeneous clusters of buildings or residential customers is addressed using regression models and ML algorithms (mainly k-means clustering algorithm) which find the least variations within each group. However, such algorithms take as input the energy consumption profiles of several households of residential building and they detect consumption similarities within these households with limited consideration of behaviour and preferences limiting the acceptability of DR solutions as well as residential buildings participation in DR.

Beyond the state-of-the-art

Addressed in WP2

DEDALUS will progress the existing state of the art by developing ML-based clustering models that are innovatively considering behavioural and social-related features on top-of energy-related ones (KER2). Deep learning models will be used to analyse time-correlated cross-carrier energy data together with the social/non-temporal data related to residential customer behaviour to enable the efficient clustering of buildings. This will enable the innovative consumers behavioural segmentation as well as the buildings segmentation on thermal and electrical flexibility bands to be used simultaneously in DR by enabling their aggregation as well as portfolio optimisation. The clustering based on social behavioural/preferences features will allow the offsetting and compensation of social enablers towards residential building participation in cross-carrier and potentially cross-energy sector DR. Several clustering algorithms with heterogeneous characteristics will be investigated to find which ones better fit: connectivity-based (hierarchical clustering, single-linkage models, etc.), centroid-based (k-means, k-means++, fuzzy k-means models, etc.), distribution-based (Gaussian mixture models) and density-based models (DBSCAN, OPTICS, etc.). DEDALUS envisions creating a solid, feature-based, and Data Space compliance methodology for the selection of the most appropriate parameters that models should consider.

Building and grid operations have so far been designed isolated, with non-collaborative bodies, rarely considering interactions between their data models and standards [16]. As a result, their interoperability (at both technical and semantic levels), or lack thereof, leads to technically complex, labour-intensive, and cost-prohibitive integration efforts, exhibiting challenges for promoting the widespread deployment of DR services [17]. Although numerous efforts toward standard communication protocols (e.g. BACnet, KNX, and Modbus) have been proposed to address technical interoperability issues, the adoption of smart home devices and enabling DR technologies is still vendor-dependent and based on proprietary interfaces [18]. Despite initiatives from industry, academia, and standards entities to establish DR-related information models (e.g. SAREF, openADR) at the semantic interoperability level for both data exchange and domain specification purposes, they have not been widely implemented or explored in terms of semantic sufficiency [19]. The latter refers to methodologies that can determine whether a pilot site is capable of providing a given service through data compliance verification mechanisms, which is a critical driver in enabling replicability, adaptability, and scalability of the DR implementation. Accurately forecasting energy assets is also critical for balancing supply, demand and for scheduling distribution networks. The problem becomes more demanding when there is insufficient amount of data to train models [20].

Beyond the state-of-the-art

Addressed in WP3

DEDALUS proposes adaptable and modular interoperability tools and methodologies based on standardised architectural concepts, protocols and data models, following the Bridge reference documentation, to integrate smart devices and enable DR services (KER3). At the technical interoperability level, hardware artefacts will be developed to connect heterogeneous data from existing standardised and proprietary communication protocols, data brokers and local/external data repositories, depending on each pilot's infrastructure. At the semantic interoperability level, a methodology that leverages relevant ontologies and standards (e.g. SAREF, openADR) will be proposed to provide building controller and operator stakeholders with a reference implementation of adaptable and modular semantic models. Such a methodology will be based on workflows for developing the semantic models that can capture data specifications of the various DR services defined in the project, as well as assisting ETL tools to streamline cross-domain integration to populate the models based heterogeneous data sources. For generating such adaptable, modular data models that fulfil the DR service-specific data requirements, DEDALUS will use graph model constraints languages (e.g. SHACL) for creating model view definitions for required semantic model data. The use of SHACL will also allow for completeness checking mechanisms that can ensure semantic sufficiency while plugging available smart device's data to deliver given DR service per pilot, promoting the widespread adoption of the proposed tools and methodologies. Finally, we will use physics informed data modelling techniques for the assessment and forecasting of the energy flexibility of different energy assets to be then used in DR programs. We will not start from scratch, but we aim to scale up and re-use some of the models developed in H2020 BD4NRG, MATRYCS, I-ENERGY projects. Such models go beyond current state of the art by combining data driven deep neural network models with technical characteristics and mathematical functions to improve the flexibility prediction. Finally, we will comply with an evaluation and quantification approach based on the Basic Application Interoperability Profiles (BAIOPs) created in accordance with the SGAM.

Blockchain-based privacy-preserving electrical and thermal DR

The energy systems are transitioning to cooperative decentralised scenarios, in which peer-to-peer (P2P) coordination among prosumers, building operators and energy stakeholders is required to adjust the demand according to the available generation while exploiting the local non-grid-owned flexible loads [21]. In this context, the adoption of blockchain technology for the implementation of DR programs may bring significant advantages in terms of transparency, openness, improving audit capabilities through peers' consensus, and decentralisation [22-23]. Nevertheless most of state-of-the-art approaches are focused on P2P energy trading, and little attention is provided to other smart grid use case, such as DR [24]. The blockchain-based digitisation via tokenisation could unlock more small-scale energy flexibility while considering the buildings operational constraints, prosumers' preferences and social enablers enacting the compensation and offsetting of flexibility shifting and behaviour change increasing their individual and community level of engagement [25-26]. The transparency of the blockchain-based system is most of the time a desired feature that brings openness and audit capabilities. However, in the energy domain, the prosumer monitored energy data are sensitive and rather private information thus, a private solution is required. State of the art solutions are addressing these issues by means of Zero Knowledge Proofs and cryptography enabling the validation without the disclosure of actual monitored values [27]. Finally, blockchain and smart contracts can provide a decentralised control mechanism for DR without the need for an intermediary and the near real-time settlement of DR programs. Blockchains and smart contracts provide a better approach to energy exchange and the settlement of related transactions [28]. With these technologies, the tracking of energy production, imbalances, and flexibility

efforts can be automated and quickened. Furthermore, settlements are not only managed autonomously by smart contracts, but financial transactions can be provided after the verification of the delivery (or not) of flexible service.

Beyond the state-of-the-art

Addressed in WP3

In DEDALUS, we will go beyond the existing state of the art by providing blockchain-based tools for the decentralisation of energy or heat DR programs management, while assuring the privacy of sensitive data (KER4). In more detail, we will provide a scalable blockchain-based energy ledger that enables the secured, trusted, and privacy-preserving sharing of flexible data/models by innovatively integrating NoSQL decentralised DBs and homomorphic encryption algorithms. The energy ledger will handle beyond electrical energy data, providing the core infrastructure for the management of DR programs on other energy carriers, such as heat. On top the following tools will be provided: (i) privacy-preserving DR forecasting tool allowing the run of ML models at the edge nodes an exchange of learned parameters (decentralised parametrisation) on the blockchain network, (ii) decentralised management and control of DR tool based on fully customizable self-enforcing contracts and (iii) energy and financial settlement of DR tool which will innovatively consider beyond financial DR settlement solutions and different the social enablers facilitated by the tokenisation and digitisation features brought by the blockchain (e.g. incentives, building sustainability, discounts to local cross energy sector businesses, etc.).

Digital Twins for consumer-aware flexibility planning

Digital Twins (DTs) can be used to successfully model all components of a smart grid and simulate behaviour through complex analytics with the general objective of applying optimisation actions through IoT actuators [29]. DTs have been proposed for managing customer profiles and raise their awareness [30]. Gathering Customer Relationship Management (CRM) data regarding customer preferences, satisfaction and behaviour DTs are constructed and fed to train and run big data analytics to generate reports for customer-awareness. Using historical customer data, a DT profile can be constructed and used as base for creating customer-centred targeted offers. Currently, the flexibility profile of assets and devices, such as heat pumps, EVs, hot water boilers, is not coupled with information relating the user's wishes in terms of comfort, convenience and wellbeing [31]. Moreover, DT are facing two main challenges at this stage: (i) although DT are praxis in large tertiary and industrial buildings, where large effort and budget can be spent for model training and validation, their utilisation in contexts where a fast replication is needed and the budget is short (as in the residential sector), requires easy models that can be trained based on few input known parameters during decision making phase and that are resilient enough as to learn based on monitoring data acquired, during operation; (ii) DT need a robust IT infrastructure that enables communication with field devices on the on hand, data analysis and communication with service providers on the other hand, relying on secure privacy practices. This guarantees that monitoring data is translated into solid information on assets operation including user behaviour, useful to devise facility management and flexibility services.

Beyond the state-of-the-art

Addressed in WP4

DEDALUS aspires to contribute beyond current state of the art (KER5) with (i) effective clustering algorithms of residential user loads, supported by social-science driven insights, (ii) thermal models based on advanced neural networks, leveraging real-world assets as monitored in DEDALUS, and enabled by BEMS, (iii) aggregated models for HITL ('Human-In-The-Loop') flexibility assessment and modelling of communities (as opposed to individual residential units/devices). The modelling techniques focus is not just the average dynamics but also modelling uncertainty, i.e. a statistical description of stochastic behaviour as it has a severe impact on any decision-making logic. Real-world data collection will rely on edge devices and behind-the-meter devices, building further on state-of-the-art connectivity technology. We combine insights from 3 different dimensions (business, technological and social), to introduce a holistic DT where a user's profile in terms of behaviour and their mapped set of flexibility assets, reflecting various commodities (electricity, heat), are coupled. DEDALUS will be able to explore how best to answer questions related to which types of DR services fits the user best, while on another, whether there are flexibility assets that are a better match to a certain user profile and how this affects then the DR services that would be attractive to this user.

Comfort-based flexibility for DR and aggregators

In [32] is explained the value of considering the indoor comfort in DR programs to improve the sustainability of the building while maintaining the well-being of the residents. As an example, a real-time optimisation mechanism has been proposed to guarantee the thermal comfort maintenance by using users' feedback, minimise the electricity cost (reduction of 10.7% of annual costs) and peak load demand (reduction of 26%) [33]. The working principle is based on the concept that, in function of the electricity cost in the balancing/ancillary markets, the DSO can ask for flexibility actions by the users, who subscribed the service. The actions concern the change of heating/cooling setpoints, the shift of appliances start/stop and others (e.g. charge and discharge electric storage if available), that will be developed within the project, to optimise the energy use obtaining flexible energy consumption when required by the DSO/Aggregator.

DEDALUS will provide a platform for flexibility (for DR applications) with comfort-based actions to enable a dynamic reshaping of buildings energy demand (KER6). This will allow to enable the mechanism between energy managers and Aggregators/DSO to reduce the peak loads when less convenient, balancing the deviation between the demand and response curves and taking into account the health and well-being of residents. Performing the suggested actions, the users' comfort will be shifted of a small amount that will have to be considered acceptable, without jeopardising well-being and health. Each action done by the user according to the required flexibility will be rewarded by the service provider with access to other services, virtual coins or discount on the energy bill. The deployment of the service will start with an investigation of the contextual factors (type of users, energy profiles, RES, grid characteristics etc.) that will allow the optimal design of the service. After the feasibility study, the installation of the system and required sensors will be performed. Given the potential application to groups of dwellings or buildings in the same district, the sharing of devices will be possible, for instance building integrated generation and storage plants, as well as climate and metering stations for an entire district. After the preliminary tests, continuous refining of the system will be done to improve its effectiveness. In this service, the concept of comfort will be as wider as possible to generate a large spectrum of actions that can be required. This will be done by investigating several comfort aspects: in addition to thermal, visual, acoustic, IAQ comfort, the comfortable sensation in scheduling the use of appliances and other equipment will be considered. The fine tuning of the actions (e.g. change of heating/cooling setpoint that can be applied with respect to the one usually adopted by the user) will be potentially done by adopting learning techniques on the data acquired during the service operation, that can take into account the different behaviours of users which depend on their lifestyle (e.g.: home as working place, aged occupants, etc.). The information profile of consumption combined with generation forecast and storage performance, associated to market energy prices, will be used as the input for AI based services, which will be devoted to make an optimal management and distribution of renewable generation and energy between users to foster DR impact. The results of the investigation will be preparatory for the definition of new business models for DSO/Aggregators.

Value-stacking sharing economy and data-driven organisational and business models

Sharing economy models, as promoted by the EU, are leading paths for sustainability [34]. Back in 2012, when Seoul self-declared the world's first sharing city [35], the win-win effects of a collaborative economy started becoming clear. In recent years and in addition to the classical ride and accommodation sharing marketplaces (e.g. Uber, Airbnb), the energy sector has also been in the centre of the discussion. A Dutch start-up created an energy marketplace to enable that prosumers and pure consumers could exchange locally sourced power [36]. This power sharing idea was also studied in the literature in a scaled-up perspective [37]. Moreover, a number of initiatives, mostly related to Smart Cities, and projects (H2020 BD4NRG, BRIGHT, Horizon ENERSHARE) are actually exploring innovative ways for -making participation of residential consumers to DR programs more attractive, through looking at beyond financial incentives and compensation.

Beyond the state-of-the-art

Addressed in WP6

We will combine cost reduction effects due to socialisation of DR infrastructure aiming B2B and/or B2C or hybrid forms with new parallel value stacking revenue streams, along the following dimensions: (i) bundled mono-carrier electrical and/or thermal DR services to different operators in different flexibility markets (e.g. offering at the same time ancillary services to power network operators and peak shaving services to aggregators/suppliers/ESCOs); (ii) cross-commodity DR services through bidding both thermal and electrical flexibility from the same asset into different flexibility markets (electrical flexibility and heat marketplace), while being compensated twice for the same action; (iii) cross-value chain services, which trade off monetary remuneration for DR participation against comfort and wellness/healthcare, which becomes of utmost importance for elderly or other vulnerable groups, or pure energy services deployed to local-community B2C social services, such as services tailored to alleviate energy poverty (e.g. providing the surplus of electricity to poor people or elderly people) or to support elderly people (e.g. through exchanging energy payment against 'home-based' caring services for elderly isolated people). The above dimension will be integrated with the ownership one into multi-layered business models (KER9). We will hence explore and validate different ownership dimension, including fully private, public-private and in particular community-level hybrid mixed shared ownerships.

1.1.3. R&I MATURITY

In this section, we provide an overview of the **existing technological solutions, tools and services (TE-x)** at TRL 5-6 that partners have developed and DEDALUS can build upon them, extend them, and adapt them (TRL 6-7) to 7 real-life pilots.

Task 2.2

TE-1	Consumers clustering and segmentation tools	TRL 6 → TRL 7	KER 2	HERON, NTUA,
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HERON and NTUA have developed specific clustering models for the segmentation of consumers in the context of H2020 MATRYCS, BD4NRG and I-ENERGY. In DEDALUS, the aim is to further extend the features of the developed models and to introduce new AI-based individual and building-level consumers clustering and market segmentation algorithms. This will enhance the decryption and deeper understanding of buildings, consumption patterns, attitudes and preferences, towards the goal of limiting consumption and managing DR in a prosumer's society. The enrichment of the models will be conducted by HERON and NTUA, in close collaboration with TUC and SIN, combining cross-sectoral fields of expertise to structure a solid clustering methodology based on a complete set of socio-economic, gender, geographic, socio-cultural, and socio-political factors.

Task 3.1

TE-2	DR-ready connected smart appliances control and interoperability	TRL 5 → TRL 7	KER 3	ARC, DOM-X
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ARC produces connected home appliances (e.g. washing machines, dryers, dishwashers, refrigerators, ovens, air conditioners, smart TVs, etc.). Currently, these appliances can be controllable remotely with the help of HomeWhiz IoT Platform. The platform and ARC's connected devices can communicate with Wi-fi & Bluetooth technologies to provide data transmission in the existing system. In addition, DOM-X provides an interoperable home-IoT gateway that enables smart control of appliances using the dominant wireless protocols (Wi-Fi, ZigBee, LoRa) and vendor specific APIs exposed in common formats through home automation frameworks (e.g. OpenHAB). In DEDALUS, the aim is to enhance these connected products by adding DR capabilities. In this way, home appliances will have a grid connection and they can respond to the signal, while maintaining a level of service to the consumer. The resulting DR-ready smart appliance IoT platform will be able to send datasets of connected home appliances (e.g. energy usage, appliance usage frequency and times, user behaviour, and preference) to feed prediction models and DR services.

Tasks 3.2-3.3

TE-3	Residential DR/flexibility interoperability building blocks	TRL 6 → TRL 7	KER 4	CARTIF, ENG
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We will provide, as part of Tasks 3.2 and 3.3, the Horizon ENERSHARE Energy Data Space Layer and a set of building blocks for data enhancement and interoperability. These are already part of H2020 MATRYCS, BRIGHT, BD4NRG and Horizon ENERSHARE and DigiBUILD projects. In DEDALUS, the interoperability building blocks will be improved from different point of views: (i) customised and tailored for the project needs and pilot requirements (residential DR/flexibility interoperability); (ii) released to be easily adopted by pilots and integrated in their data collection and processing pipelines; (iii) evolved to provide their functionalities in a full as-a-service mode to be used as data interoperability services in Energy Data Space in multiple provisioning approaches (e.g. centralised/decentralised-federated).

Task 3.4

TE-4	Flexibility modelling tools for residential DR	TRL 5 → TRL 7	KERs 5-8	NTUA, CARTIF, CENTRICA
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A library of state-of-the-art AI data-driven tools and ML/DL models has been developed within the context of MATRYCS, BD4NRG and I-ENERGY, in order to estimate the potential flexibility of individual or aggregated consumers. In DEDALUS, a flexibility modelling tool will be delivered, integrating and expanding these models, and including means to compare and evaluate the aggregation levels and effectiveness of the aggregated flexibility amongst different pilots under the DEDALUS project.

Task 3.5

TE-5	DLT/Blockchain/Smart contracts solution for decentralized machine learning, P2P flexibility tokenisation, and DR settlement	TRL 5 → TRL 6	KER 4	TUC, ENG, COMS
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In H2020 eDREAM, PHOENIX, BRIGHT and BD4NRG projects, we have developed a scalable second tier energy ledger allowing the storing and digitally fingerprinting of energy data. Integration with metering devices is achieved using self-enforcing smart contracts allowing the generation of associated energy transactions which are then registered and shared in an immutable and tamper-proof manner on the blockchain. Advanced DR control mechanism is implemented using smart contracts allowing the injection of DR signals defined directly or via P2P flexibility trading to programmatically define the expected energy flexibility to be delivered as well as the associated remuneration. At the same time, a near real-time DR statement process is developed considering the monitored data to activate the appropriate financial remuneration (or other forms of settlement) for the flexibility providers. The smart contracts will check and compare the monitoring data with the corresponding commitments, and based on

these values, the actual settlement will be generated. In **DEDALUS**, the solution will be advanced to support the implementation of privacy preserving process, by integrating (i) ML models and (i) consent- or policy-driven data sharing framework with blockchain technology, inherently enabling traceability over any state change. The local building-level data will be used at the edge (on premise) to train the models and only shared outside when in compliance with data protection rules of data owner. Upon such data exchange, the blockchain will only store transaction hashes, while the actual data will reside in an off-chain data store. Next, decentralised DR control mechanism will be enhanced to support explicit or implicit residential DR and integration of smart contracts will be supported for different monitoring and control devices of DR automation. Different tokenisation and settlement strategies will be investigated and adopted, aiming at the enablement of cross-carrier flexibility integration and different rewarding schemes that go beyond financial remuneration.

Task 3.6

TE-6 Universal Secure Gateway		TRL 6 → TRL 7	KERs 4, 7	COMS
The Universal Secure Gateway (USG) is a hardware and software solution in the context of H2020 PHOENIX and BRIGHT aimed at enabling arbitrary integration of field assets and legacy systems with cloud backend and data analytics services, enhancing security over legacy protocols, and adopting standardised APIs to allow for the interoperability of traditionally disparate systems. Flexible combination of north and south bound connectivity interfaces together with edge computing capability allow for instantiating and migrating virtualised functions (microservices) according to the instantaneous needs of active services. In DEDALUS , the existing building blocks will be reused and enhanced to provide a plug & play solution for the pre-aggregation, orchestration and activation of flexibility via DR. The overall ambition is to enable the energy community manager to gather in real-time data from various sources, such as assets/systems, grid infrastructure, market and weather forecast, and make informed decision based on analytics services. Integrating the results of TE-3, the solution will support local, on-premises private data processing as specified by IDSA. In the sense of software implementation, the LF Energy open-source projects FledgePOWER and Super Advanced Meter will be considered for gateway, and FlexMeasures will be considered as framework for local flexibility orchestration. Additionally, open-source home automation frameworks, such as openEMS, openHAB and HomeAssistant will be considered for semantic interoperability with flexible DER.				
TE-7 Secure and Privacy/Confidentiality Preserving Data Transactions		TRL 6 → TRL 7	KER 4	COMS

COMS has developed within H2020 PHOENIX a blockchain-based Privacy Protection Enforcement (PPE) component that enables consent-driven sharing and exchange of personal (GDPR-compliant) and confidential data among stakeholders and with data marketplaces. It provides data registration, consent exchange, data access and usage policy definition, provision of permission-based mechanisms and protocols, comprehensive auditing functionalities, and actor reputation assessment. Coupled with a service mesh like implementation approach, granular end-to-end data transaction policies may be imposed. This allows for private, secure, sovereign, and transparent data management over its entire data lifecycle. In **DEDALUS**, the data management framework based on PPE will be (i) advancement towards self-enforcing data usage policies definition through automated smart contract agreement mechanisms among multiple stakeholders, and (ii) expanded from managing data transaction towards enabling a platform for broker-dealer third-party (data processor) services using reusable/decentralized/badge profile identifiers.

Task 4.1

TE-8 Digital Twins for consumer-aware optimal flexibility planning		TRL 5 → TRL 6	KER 5	ENG, CENTRICA, HERON
Flexibility-centric DTs assets have been delivered, like thermoelectric water heaters, heat pumps, behind-the-meter and community battery storage and EV, based on data-driven and physics-informed flexibility models developed and demonstrated in past and on-going EU projects (H2020 MATRYCS, BD4NRG, InterConnect, I-ENERGY). Integrating results from TE-4, in DEDALUS , ENG and CENTRICA are expected to enhance the DTs, by: (i) Integrating business and socio-economic dimensions (on the top of the pre-existing technical dimension from past projects), as well as heating and mobility sectors, to create an accurate virtual representation of flexibility, providing planning functionalities for DR services at the individual and aggregated level; (ii) Applying seamlessly integrated/interoperable in the Energy Data Space. Moreover, a virtual net-metering service ('RES everywhere every time for residential consumers') will be delivered by HERON to boost residential flexibility potential in order to meet market or environmental objectives. Through digital twinning, a virtual storage asset will be added to complement HERON's ENergy Autonomy (EN.A) program ¹ under two possible configurations: i) behind the meter in support of EN.A's 500 kW PV, and ii) a stand-alone asset providing shared storage using the residual part of a				

¹ Participation in the HERON's ENergy Autonomy (EN.A) program gives access to the revenues of HERON's and TERNA ENERGY's PV assets through a flat fee which can be re-adjusted to offset all bill related costs and turn HERON customers into zero billers.

primarily merchant storage asset. While both configurations provide flexibility options to end-users seeking to minimise their carbon footprint on hourly or quarterly basis, they do so through different operational strategies leading to distinct business models. DTs will be provided to the Danish, Greek, Italian, Irish and Romanian pilots.

Task 4.2

TE-9 Comfort-based flexibility models	TRL 5 → TRL 6	KER 6	UNIVPM, ISRAA
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The development of comfort-based methodologies and devices for energy management in new and renovation projects have been implemented in previous H2020 projects. In H2020 NewTREND and RenoZEB, the tools for comfort evaluation and monitoring (both data management platform and dedicated sensor networks, including the patented solutions Comfort Eye and Comfort Air) have been developed. These tools, providing accurate real time estimation, can help in managing DR comfort-based approaches, but up-to-date no specific solutions have been developed in this field. It is worth considering that the comfort dimension is very relevant to include also the social dimension. Quite often this component is not properly considered in energy management strategies, but it has been already demonstrated that e.g. in buildings comfort-based HVAC control strategies are able to provide energy savings up to 20%, with a great impact on user satisfaction and on productivity in tertiary buildings. Considering that, **in DEDALUS**, comfort-based flexibility models for the specific needs of DR will be developed. Sensors to monitor comfort (thermal, acoustic, visual and air quality) and learning techniques will be developed at TRL6. A specific focus on requirements and needs related to comfort for aging people will be considered in the Italian pilot, while the replication of the comfort-based flexibility model to other pilots (e.g. Denmark, Spain, Ireland) will be explored.

Task 4.3

TE-10 Nudging Apps for near real-time closed-loop residential consumers activation and multi-vector DR	TRL 5-6 → TRL 7	KERs 1, 8	DOM-X, OurPower& BLUEPRINT
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DOM-X offers a commercial solution for managing the space heating demand of natural gas. The cost-effective and universal heating controller of DOM-X is attached with legacy natural gas boilers to upgrade them into smart flexible assets able to participate in energy markets, by building on a set of open standards (Opentherm, Modbus) and widely applicable control modes (modulation, ON/OFF) for enabling minimum interoperability. The DOM-X cloud-based energy management platform analyses the data collected from edge controllers, and delivers optimal control instructions for improved energy efficiency through load reduction and energy system flexibility through explicit DR [38]. The overall system was extended and demonstrated in the context of H2020 BIGG and BRIGHT projects, while its smartphone app was extended with nudging features in the context of the H2020 NUDGE project. Moreover, in serve-U project (funded by Austrian Research Promotion Agency FFG), OurPower started development of a nudging app to serve as information channel for its end-users and energy communities. **In DEDALUS**, DOM-X will extend the energy management platform, nudging app and algorithms (demand forecasting, flexibility evaluation) to support electrical residential HVAC appliances (ACs, heat pumps) as well, delivering a unique solution for third parties (building managers, energy suppliers) interested in multi-vector DR and decentralised power2heat management. OurPower and BLUEPRINT will upgrade the nudging app to incorporate new functionalities, such as: (i) provide recent and forecast values of generation and consumption to users; (ii) offer advice to change use of energy; (iii) provide answering channel for participants (Button ‘Sure, I can react and help’/‘No, sorry, no option for me’). Communication channel will also include community dialogs/online event notifications for feedback and will continually improve app UX. While nudging apps will be directly deployed in the Greek and Austrian pilots, the design and UX will also be provided to other pilots (Spain and Romania) to accelerate SSH engagement and DR programme.

TE-11 Multi-value DR for virtual cluster of buildings	TRL 6 → TRL 7	KER 7	HERON, NTUA
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HERON has deployed a smart metering platform based on InfluxDB time-series database and MQTT, and collects/maintains market and system data from the Greek Market Operator and TSO respectively, integrating it in a proactive DR solution. HERON is also developing a remote device monitoring app that can provide smart-meter analytics and can be expanded to asset specific monitoring and scheduling. So far, HERON’s developed or under development solutions (in the context of H2020 InterConnect, iFlex and BIGG projects) either rely on specific (project issued) appliances or provide generic advice based on the aforementioned market data e.g. ‘Please reduce your consumption due to high electricity market prices’. **In DEDALUS**, HERON will expand its platform and app to integrate data from Non-Intrusive Load Monitoring (NILM) devices, in order to monitor disaggregated loads from its customer portfolio. This will enable the virtual clustering of consumers and buildings, and provide advanced features beyond ON/OFF categorisation in the Greek pilot. First, we can provide electricity customers information that allows them to become more aware of their energy consumption, and thereby, more energy efficient. Second, we will allow detecting flexible loads from the virtual cluster of buildings and appraise their flexibility potential,

critical when issuing advice at appliance level, given that one may expect that consumers are less likely to change their habits (if for example they use the washing machine the same time every day, as opposed to consumers with no specific usage patterns). The replication of the models to the Italian, Spanish and Irish pilots will be studied.

Task 4.4

TE-12	P2P digital platform for trading pre-aggregated building flexibility	TRL 6 → TRL 7	KER 7	COMS, OurPower, BLUEPRINT, TUC, ENG
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This tool is a nation-wide market driven citizen engagement platform for RES power operating in Austria, with the objective to provide citizens full market access and make it as simple and attractive to use, and as economic as possible. The technological tool and processes of the platform are designed for optimisation of community generation and consumption in order to provide the widest possible market access at lowest investment and cost for users, optimise balancing/marketplace cost, i.e. processes, community self-sufficiency, balancing, independence of external markets, trading of surplus or residual needs. **In DEDALUS**, it will be enhanced in order to demonstrate and replicate the ability of energy community manager to model, pre-aggregate and activate local assets' flexibilities for buildings and residential customers. In this sense, it will also integrate with TE-3 providing secure interoperability and virtualisation interfaces among legacy assets/systems and the orchestrator, and TE-5 blockchain solution allowing P2P data/data services and aggregation of small-scale flexibility from residential sector to be offered to different stakeholders overcoming potential DR barriers. It is expected that the tool will be tested in a real environment with a high number of users beyond GreenPoint demo (until the end all tenants in 80 apartments and GreenPoint as a building operator will be engaged) across Austria - and include other green and smart buildings/quarter/residential areas and their new smart devices that can be used in a DR strategy (PV, heating optimisation, storage and EV-chargers) to additional 4 building sites in Austria that are in the process of becoming members of OurPower platform, as well as the Romanian pilot.

TE-13	District-level DR for self-consumption and cost optimisation	TRL 6 → TRL 7	KER 7	CARTIF, CIMNE
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The tool has been developed in the context of H2020 MUSEGRIDS, EN-TRACK and SUNHORIZON projects. It is focused on the control of different storage devices (thermal and electric) that can be installed at user or district level and that provide the required flexibility for the energy system optimisation. The optimisation objectives will usually be to maximize the self-consumption of the overall system, minimise the peaks of energy exchanged with the grid or the cost minimisation. The energy system is optimised at global level while the demand requirements and device control are considered at user level. It can provide control signals or recommendations in a day-ahead time horizon and apart of the predicted demand it can accept restrictions from the user (or the smart devices) about the preferred operation scheduling. **In DEDALUS**, it is expected that the tool can be upgraded and tested in a real environment (in the Spanish pilot, replicated by the Austrian pilot) with a high number of users, including new smart devices that can be used in a DR strategy (not only storage), such as smart chargers and heat pumps, etc. In this sense, it will be necessary to develop new models for the users' demand (especially thermal demand) and include in the optimisation algorithm the restrictions and models associated to the new smart devices (integrating results from TE-2 and TE-3). In addition, the management of optimal sharing coefficients among users of the renewable power generated and stored to handle DR will be also an upgrade. In a second stage, the research will be focus on the convergence of the optimisation algorithm for the high number of users.

Task 4.5

TE-14	Aggregation of flexible energy resources for district heating management	TRL 5 → TRL 7	KER 8	NEOGRID
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With its product PreHEAT, NEOGRID makes energy management possible for house and buildings owners and large-scale monitoring allowing optimised individual and aggregated controlling. This consists of a data acquisition platform and online control interface. It enables the user to monitor, plan and even shift energy consumption based on knowledge of price, average consumption patterns, and intelligent weather forecast management. **In DEDALUS**, NEOGRID will extend its solution with new services, where flexibility in how an energy-efficient building is heated can be used towards the district heating system, so either a building or a pool of buildings can move their energy consumption in time to minimise the collective load on the energy system. Furthermore, the solution will be extended to include ventilations systems connected to the electrical grid and include the flexibility from these systems toward the electrical grid, i.e. having one solution which can handle flexibility towards district heating and electricity systems and at the same time maintain the required indoor climate. This solution will be tested in the Danish pilots, while the possible replication by the district network of the Austrian pilot case will be analysed.

1.2. METHODOLOGY

1.2.1. OVERALL CONCEPT

DEDALUS will focus on: (i) **bringing individual buildings or clusters of buildings, either geographically bounded (e.g. districts) or virtual cluster of buildings**, to the aggregation level suitable for mobilising higher latent flexibility and hence making it convenient for utilities and building operators to confront attractive flexibility services and residential DR engagements schemes; (ii) deploying an **adaptable end-to-end data-centered framework** which will span and adapt along two different aggregation dimensions: from **micro** (i.e. individual apartments) to **macro** (individual buildings, districts, virtual aggregation of buildings), as well as from **mono-carrier** (electricity, heat) to **multi-carrier** (electricity vs heat, electricity vs natural gas); (iii) designing and validating **novel sharing economy cross-value chain services** and underlying **business models**, where existing stakeholders, such as traditional utilities, or new stakeholders, including building operators, energy communities, social housing operators, care/wellness service providers, may eventually take over the role of **DR service and/or solution providers**, eventually in a combined way with utilities and independent aggregators.

The DEDALUS concept articulates around the reciprocal interaction among social, technological and business layers to enable an effective consumer participation in residential DR:

Social Layer

The Social Layer will provide the **Social Science Framework** adopting a variety of established **SSH methodologies** to examine consumer preferences and behaviour, and to facilitate consumer activation in the context of residential DR (**KER1**). Experimental online studies will be conducted to investigate the key drivers of and barriers to the active participation in DR. Based on this analysis key consumer groups will be identified and characterised (i.e. individual and building-level consumers clustering and market segmentation, using AI-based algorithms (**TE-2**), along a variety of different criteria and KPIs, including social, financial, and environmental concerns and goals, technology friendliness, living location, and demographic characteristics) (**KER2**). Building on these insights, **tailored monetary and non-monetary multidimensional interventions** that specifically address the concerns and preferences of the identified consumer clusters will be developed and tested. Deliberative citizen engagement and inspiring principles from Design Thinking methodologies will be exploited, and we will foster the dialogue among the different actors (energy consumers, suppliers, aggregators, and other involved market stakeholders), facilitate experience exchanges and ultimately achieve a **co-creation and co-implementation** process for the design of data-driven DR services and nudging interventions. More specifically, **closed-loop near real-time nudging services** will be delivered, which provide near real-time information to the end consumer (**TE-10**). We will also address information asymmetry within the civil society, familiarising the wider public with the energy market, and highlighting the role they can play in it, with **easy-to-adopt efficient business models** (**KER9**).

Technological Layer

The Technological Layer will include (**Fig 1.2**):

- A cross-cutting adaptation of **Interoperability** and **Trust** layers of Horizon ENERSHARE **Energy Data Space** to support cross-commodity and carrier-agnostic value stacking DR concept (**KER4**). In particular, we will adapt the **Energy Data Space Interoperability Layer** (**TE-3**) through integrating **IoT sensing and communication platforms** for smart home/buildings appliances (**TE-2** from ARC or local business platforms from the pilot partners) with DR protocols and standardised interfaces (EFI, USEF, OpenADR 2.0, IFC, SGAM), aiming to turn smart buildings' electricity and heat consuming appliances into DR-ready devices and enabling seamless multi-commodity or cross-commodity interoperability (**KER3**). The resulting **Interoperable DR Manager** will combine adapted integration of USG (**TE-6**, from COMS) and home automation framework (**TE-2** from ARC/DOM-X), to deploy a privacy-preserving **edge-level stack** capable of supporting: (i) effective management of DR in a semi-automated and/or automated way; (ii) integration with smart metering infrastructure; (iii) extended interoperability with OpenADR2.0, EFI, SAREF. With a view to support data analytics services and cost-effective interoperable smart home automation, the edge-level stack will support near real-time gathering of consumption and other flexibility-related contextual data from behind-the-meter IoT-enabled appliances and devices, or dedicated gateways.

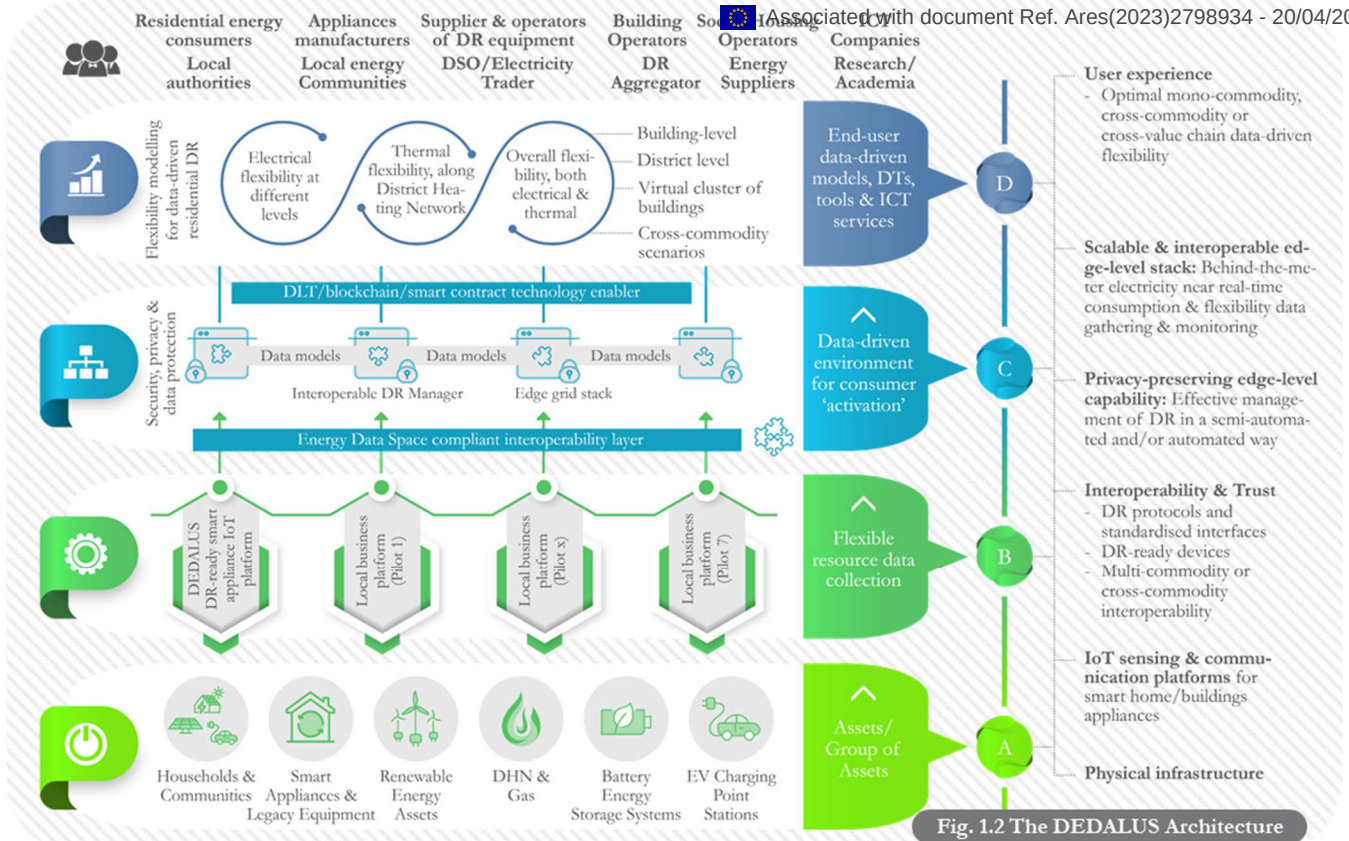


Fig. 1.2 The DEDALUS Architecture

- We will upscale H2020 BD4NRG, BRIGHT, PHOENIX and Horizon ENERSHARE multi-purpose **DLT/blockchain/smart contract** for consent-based data sharing (TE-7) and decentralized DR management (TE-5) to deliver: (i) privacy and sovereignty preserving data sharing and secure decentralised data governance capabilities; (ii) participatory tokenised marketplace for exchange and compensation of DR participation with financial and non-financial incentives (KER4).
- On top of the above-described Energy Data Space DR adaptation, we will deploy advanced AI algorithms for flexibility modelling (TE-4) to support the development of a variety of end user data-driven models, DTs, tools and ICT services (KERs 2, 5-6).

The **DEDALUS methodology** is based on the acknowledgement that for optimal control of flexible devices (e.g., HVAC, whitegoods, EV charging), it is necessary to know the energy/flexibility characteristics of the device, as well as the typical usage of the device by the end user. For example, in order to use the building as a thermal buffer to optimize electricity consumption, it is necessary to know in which conditions how long the heating can be turned off before the temperature in the building becomes unacceptable to the end user. In addition, it necessary to make forecasts of when the user turns on the thermostat.

Such tools build on the integration and linkage of data-driven models for individual end user behaviour and for community behaviour together with the respective preferences (e.g., thermal comfort for temperature, reasonable price for flexibility provisioning) with data-driven models for asset flexibility behaviour and incentives, designed for the type of users as identified in the DEDALUS Social Layer. Data-driven modelling of individual and community-level behaviour will be carried out, based on AI-based ML for determining what are the climatic, economic and social conditions under which consumers find it suitable to participate in DR programs (e.g., what is the minimal temperature which one consumer would consider unacceptable, etc.). Such modelling will be done in a multi-commodity environment, i.e., we will consider power and heat as well. Data-driven modelling of thermal and electrical flexibility assets will be done as well in parallel.

DTs of the end user will be hence combined with incentive and end dimensions specifically designed for that category of user within a given community and will be mapped to a set of flexibility assets. The associated flexibility profiles will be then combined via multi-dimensional aggregation mechanism with the user profile to realize the user's "DT"; in the same set of flexibility assets both commodities electricity & heat may be reflected

- Based on the above, a number of tools, DTs and ICT services will be deployed, such as:
 - **Consumers (individual and building-level) clustering and segmentation models**, and AI-based data-driven tools (TE-1), where a variety of individual and aggregated individual and building-scale consumer preferences will be taken into consideration (e.g. comfort, age, socio-economic situation, sensitivity to increased green energy procurement and healthier quality of life, location of the buildings).

- **Cross-sector consumers aware DTs for flexibility modelling and planning (TE-8)**, which includes mono-carrier (electricity and heat), and cross-carrier (electricity vs heat, electricity vs natural gas) flexibility modelling, based on data available respectively from pilots. Planned consumption/energy demand from consumers utilising given flexibility assets will be taken into consideration, as well as other consumers' preferences.
- **Personal comfort-based flexibility assessment models and algorithms (TE-9)**, where exploitable flexibility will be traded-off with personal comfort and wellness.
- **Optimal mono-commodity, cross-commodity or cross-value chain data-driven flexibility management services** for optimising (KERs 6-8):
 - o **Electrical flexibility** at: (i) **Building-level** by leveraging on building-scale flexibility pre-aggregation (Building-as-a-Community Flexibility Management) (TE-12) and/or trading-off with personal comfort to offer **personalised cross-value chain care services to elderly people (TE-9)**; (ii) **District level (TE-13)**, while operating self-consumption models linked with community-level shared resources, such as district-level battery storage (District-as-a-Community); (iii) **Virtual cluster of buildings (TE-11)**, which can be clustered based on similar characteristics (size, age, share of RES local generation); (iv) Within **cross-commodity scenarios (TE-10 and TE-14)**, such as electricity vs heat and electricity vs natural gas, to take into due consideration electricity consumption of non-electrical infrastructures devices (e.g. ventilation units for DHN, heat pumps, natural gas boilers for domestic hot water and space heating).
 - o **Thermal flexibility**, along District Heating Network (TE-14) for individual or geographically bounded groups of buildings, deployed by social housing operator.
 - o **Overall flexibility**, both electrical and thermal (TE-14), where the same flexibility control strategy will result in both thermal and electrical flexibility to be double rewarded in the two respective electricity and heat flexibility marketplaces.

Business Layer

The Business Layer will provide a Business Sandbox (KER9) which will include the design and validation of:

- **New hybrid business models**, which:
 - Hybridise and combine energy communities with buildings to optimally exploit and manage exploitable flexibility at building, district and virtual cluster of buildings level, while considering collective-based aggregation of flexible assets.
 - Combine cost reduction through extended DR infrastructure costs socialisation (i) at local community level and/or (ii) through a joint venture and a shared ownership of different private stakeholders (e.g. joint investment among building operator, social housing operator and an aggregator) and/or a shared PPP Public-Private ownership, with new revenue streams by designing:
 - o New cross-stakeholders mono-commodity and/or cross-commodity services to increase the compensation from DR participation from simultaneous bidding into different flexibility marketplaces (e.g. ancillary services vs intra-day marketplace for electrical flexibility, or intra-day electrical and heat flexibility marketplace).
 - o Innovative cross-domain services, such as optimal comfort/wellness services for the benefits of elderly people or people affected by energy poverty.
- **New market roles** for new and/or existing stakeholders, such as: (i) Building Operators/Managers **community developer** and/or **facilitator**, in charge for mobilising individual consumers' flexibility at individual and collective building or district level. Such role may be undertaken by energy players engaging groups of consumers, such as ESCOs or independent aggregators or smart home appliance manufacturers which will invest in smart finer-grained energy consumption and flexibility monitoring devices to offer DR services, or DSO for local network congestion management; (ii) **local energy communities managers/orchestrators/service providers**, or **community-centred VPP operators** which combine in a hybrid way top down aggregation strategies aimed to minimise risks for insufficient flexibility provisioning to their respective markets with bottom up community-driven P2P strategies, where dynamic coalitions of prosumers may be willing to aggregate based on a prior information exchange. This new role could be played by traditional aggregator at the extent to trade off their increased entrepreneurial risk against increased flexibility assets portfolio. A P2P market design for automated smart contract-based energy and non-energy sharing and trading will be investigated where coalitional games from games theory will allow multi-dimensional incentive sharing and splitting among dynamically created energy prosumers coalitions.
- Best practices, blueprints/roadmap drawing from a variety of legal **community-level arrangements and governance models at building/district scale (All KERs)**, ranging from energy cooperatives to not-for-profit trusts/foundations, consumers collectives, or public-private partnerships, or public-owned sustainable

developments hubs, delivering multiple services with a  to provide input to regulatory bodies for the adherence to Clean Energy Package provisions.

Thanks to the above-described concept, DEDALUS expected outcomes consist of DR services for both heating and electricity loads in residential buildings. By combining hardware and software solutions, it will be offered the ability to provide implicit and explicit DR services, aiming to test and validate flexibility solutions in different energy vectors in the building level. Considerations regarding energy efficiency, comfort levels and monetary incentives will also be made to unlock demonstrable flexibility solutions, prompting both consumers and energy players. The multi-attribute segmentation of the pilot participants will lead to DR solutions that are likely to have more impact, given that they will be more relevant to the users. In turn, this increases scalability and market potential, as it could lead to specific products aimed at specific types of consumers.

1.2.2. PROJECT METHODOLOGY

DEDALUS considers the adoption of an **incremental Minimum Viable Product** approach. The proposed validation methodology will go through different yet incremental stages, starting from an initial smaller-scale validation taking place in a controlled environment where a subset of functionalities and tools will be deployed and validated over a reduced amount of data/assets from all pilot sites aimed to provide early feedback and allow refinements and adaptations. Secondly, we will deploy a technical implementation of the entire DEDALUS framework along with all the data/assets made available by the pilots. Final validation will incorporate scalability analysis, where the DEDALUS framework will be validated against the larger datasets/assets, which in the meantime will have been incrementally made available by all pilots.

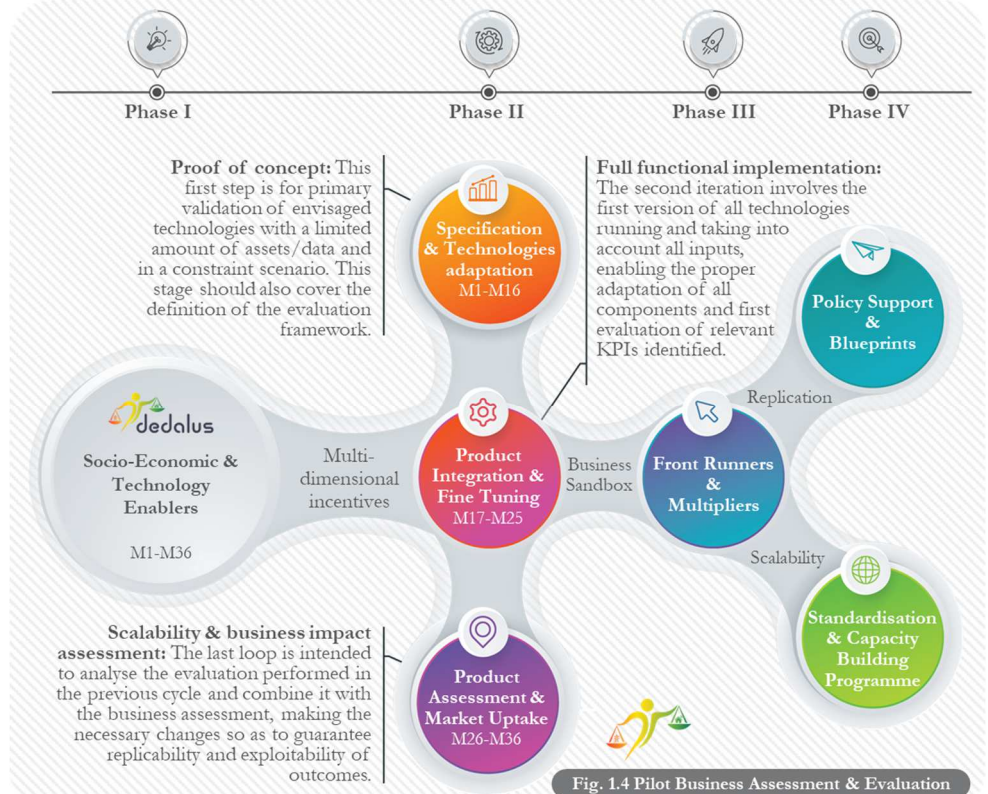


Fig. 1.4 Pilot Business Assessment & Evaluation

Phase I

Residential consumers groups will be mobilised and engaged (WP2) through a SSH-based co-creation process by leveraging on personalised cluster-level multi-dimensional incentives (WP2) to actively participate as flexible consumers towards the optimal management and operation of renewable energy systems. This co-creation process will consist of five individual yet complementary methodological steps so as to ensure the successful design of the participatory residential DR concept: (i) progressive mobilization, engagement and enlivening of all the targeted stakeholders, (ii) identification of current barriers and recognition of energy consumers' needs, (iii) co-exploration of realistic solutions for residential DR, (iv) refinement of proposed ideas and finalization of the co-designed framework, (v) cross-stakeholder validation of the finalized framework. Hence, consumers can access the electricity markets with clean energy flows, based on a user-operator 'alliance' that offsets the variability of renewable energy by harvesting distributed flexibility from DR, active control and distributed storage. This will be explored by promoting new business models (WP6) for flexibility at levels, commencing from P2P local markets to ancillary, flexibility, retail and wholesale marketplaces.

Phase II

A full development cycle is implemented in each phase (Fig. 1.4):

- **Step 1 || Specification & Technologies adaptation (M1-M16):** It will include the market analysis and the design of business models, as well as the definition of scenarios, use cases, the elicitation of user requirements and system requirements, dynamic assessment, modelling and definition of system architecture (WP2). The DEDALUS development processes will start by collecting insights through residential consumers' and energy stakeholders' activities (WP2), as well as based on the analysis of the project's pilots. This phase will be complemented by

information from linked research initiatives from partners (Associated with relevant standards (7993), as well as project-specific, pilots and data/assets requirements, and will be further analysed through requirements engineering techniques (WP2). The outcome of this activity will be a Business Sandbox with novel sharing economy and social innovation-based business models (WP6) and a set of requirements (WP2), each split into a set of prioritised functionalities for the DEDALUS development activities, as needed for enhancing the project's technology capabilities and conducting the pilots. Early technology and asset assessment (WP3 and WP4) will be followed by preliminary validation of key solution ideas in local, existing small-scale pilot-site infrastructure deployed along a subset of pilot sites (**Pre-pilot Phase**). Along this phase early **consumer-aware DTs for flexibility modelling and planning will be developed along WP4**, where consumers behaviours and preferences will be combined and integrated with energy and flexibility characteristics and potentials of physical flexibility assets. Evaluation (M14-M16) will start via prototyping and testing referring to the planned pilots, to identify potential limitations and drawbacks by involving a group of selected users (WP5).

- **Step 2 || Product Integration & Fine Tuning** (M17-M25): Based on feedback from Step 1, the business models, scenarios and requirements will be refined (WP2 and WP6), and a second implementation of advanced functionalities for technological components will be released (M20). This phase will focus on the delivery of all the technology enablers and on their deployment on the pilot sites. A pilot validation campaign (**Full Pilot Operation Phase**) will be performed, based on the release of all the building blocks and technology enablers of the project's concept (WP3 and WP4). DEDALUS complete framework will be deployed at pilot site through controlled environments, decoupled from 'production' environments. DEDALUS components will be evaluated (M21-M25), during the full pilot operation phase, for both usability and performance levels and suitable feedback will be propagated towards the consumer (WP2), the "asset" (WP3) and the overall consumer-aware tools (WP4) aimed to improve performance of data-driven models for flexible consumer behaviours and flexibility assets modelling and planning (WP5).
- **Step 3 || Product Assessment & Market Uptake** (M26-M36): It will take into consideration the evaluation results of Step 2 with a view to enable the refinement of component and system level technology consolidation, through scalability-driven functionality enhancements (M26-M29) using the DEDALUS solutions. Overall, the final version of DEDALUS will have been concluded latest by M29, resulting in the final version/iteration of the relevant deliverables in WP3 and WP4. A final pilot validation (M30-M33) will be carried out (WP5), mostly focused on the scalability (WP6) and will provide the necessary benchmark to perform the business validation (M34-M36) based on a techno-economic assessment of the business benefits gained by each pilot (WP7).

Phase III

In order for the **DEDALUS demonstration and validation activities by Front Runners** (WP5) to be efficiently supported by relevant technology enablers (WP3 and WP4) and also the business assessment, a three-loop cycle structure has been designed (covering each topic respectively) in a way that they are continuously feeding the process while providing incremental improvements along the way (Fig. 1.4): Pilot Cycle 1 (Pre-pilot Phase, M14-M16); Pilot Cycle 2 (Full Pilot Operation Phase, M21-M25); Pilot Cycle 3 (Large Scale Pilot Operation Phase, M30-M33). In parallel with the pilot activities, the **DEDALUS Multipliers** in Ireland and Romania (WP5) is committed to use technologies and outcomes tested in pilots and revisit the development processes and define new business models.

Phase IV

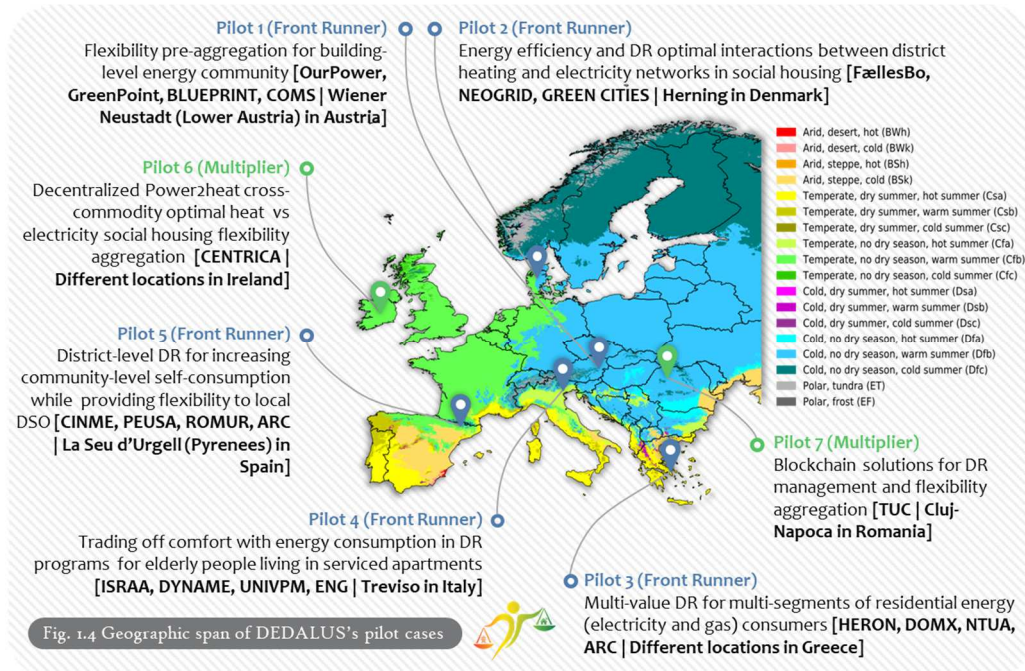
DEDALUS will also deliver best practices and blueprints drawing from a variety of legal community-level arrangements and governance models at building/district scale (WP6), and implement a comprehensive capacity building programme for energy stakeholders, regional/local authorities, energy communities, etc. (WP7).

1.2.3. DEMONSTRATION AND VALIDATION

The DEDALUS framework will be applied, implemented, and validated in **7 real-life pilots (5 Front Runners and 2 Multipliers)**, across **7 countries** (Fig. 1.4). The large geographical coverage of the pilot sites aims to support the large-scale EU-wide replicability and market take-up of solutions in different socio-economical contexts to maximise the impact of DEDALUS services across Europe.

We will cover to the largest possible extent the electricity and energy value chains, while considering also other non-energy sectors, such as buildings, mobility, storage, financial, comfort/wellness/health, older people. The identified technologies (Section 1.1.3) to be upscaled and tested resolute from an exhaustive mapping of pilot capabilities, based on their business models, available infrastructure, covering areas and ability to upscale the solutions. We will follow a two-fold approach in order to implement, test, validate and upscale the multi-value services into the pilots. By leveraging the availability to deploy multiple services to each pilot, a **Front Runner** for each service is adopted, followed by the **Multiplier**.

More specifically, replication of the services is targeted in DEDALUS, across demos (through the Front Runners involved) and beyond (through CENTRICA and TUC acting as Replication Multipliers). This approach will allow utilising the available infrastructure of the pilots and the technical solutions, so as to implement and validate the services in multiple sites, aiming to maximise the replication potential of DEDALUS solutions.



This holistic approach (Table 1.2) will enable to test the non-exhaustive set of DEDALUS social-economic and technological solutions which could be expanded by third parties in the future.

Table 1.1. DEDALUS pilots

		Front Runners					Multipliers	
		Pilot 1	Pilot 2	Pilot 3	Pilot 4	Pilot 5	Pilot 6	Pilot 7
		Austria	Denmark	Greece	Italy	Spain	Ireland	Romania
Region	Northern European							
	Central European							
	Southern European							
Energy Carriers	Electricity							
	District Heating							
	Natural Gas							
Sectors	Energy (consumption)							
	Energy (RES production)							
	Buildings (residential)							
	Mobility							
	Storage							
	Financial							
	Comfort/Wellness/Health							
	Older People							
Stakeholders *	Residential Energy Consumers (Electricity, DHN, Gas)							
	Appliances Manufacturer			ARC		ARC		
	Building Operators	GreenPoint			ISRAA		CENTRIC	
	DR Aggregator							
	Energy Community	OurPower						
	Energy Supplier			HERON				
	DSO/Electricity Trader				DYNAMIE	PEUSA		
	Local authorities							
	ICT Company	COM+Blue			DOM-X	ENG		
	Research/Academia				NTUA	UNIVPM	CINME	TUC
	Social Housing Operators		FællesBo					
	Supplier & Operator of DR equipment		NEOGRID			ROMUR		
Technology Innovations **	Consumers clustering and segmentation tools (TE-1)							
	DR-ready connected smart appliances control and interoperability (TE-2)							
	Residential DR/flexibility interoperability building blocks (TE-3)							
	Flexibility modelling tool for residential DR (TE-4)							
	DLT/Blockchain/Smart contracts solution (TE-5)							
	Universal Secure Gateway (TE-6)							
	Secure and Privacy/Confidentiality Preserving Data Transactions (TE-7)							
	Digital Twins for consumer-aware optimal flexibility planning (TE-8)							
	Comfort-based flexibility model (TE-9)		FR	FR	FR		Mult	Mult
	Nudging Apps for near real-time closed-loop residential consumers activation and multi-vector DR (TE-10)	FR		FR		Mult		
	Multi-value DR for virtual cluster of buildings (TE-11)			FR		Mult		
	P2P digital platform for trading pre-aggregated building flexibility (TE-12)	FR + Mult						Mult
	District-level DR for self-consumption and cost optimisation (TE-13)	Mult				FR	Mult	
	Aggregation of flexible energy resources for district heating management (TE-14)	Mult	FR					

* Other stakeholders will also support the pilot phase and have already expressed their interest, such as: Herning Municipality and Verdo Varme Herning (District Heating Company) in Denmark; Treviso Municipality, Province of Treviso, ENEL (electricity provider), Dolomiti Energia (gas provider) and Nice S.p.A. - Home Automation Systems in Italy; Municipality of La Seu d'Urgell and CEGASA (energy storage manufacturer) in Spain; Social housing bodies, such as Westmeath County Council, Respond Housing and Cluid Housing, and Daikin as technology provider in Ireland; Electrica Furnizare (the local energy company) in Romania.

** FR = Front Runner, Mult = Multiplier



AUSTRIA

Flexibility pre-aggregation
for building-level energy community

Pilot Context: OurPower is a cooperative based in Vienna, Austria, operating a P2P marketplace for matching RES electricity generated by its members with consumers. The platform provides all necessary services – accounting, settlements and contracting according to legislation. Enabling in the present scheme a nationwide P2P electricity trading based on standard market procedures allows economic valuation on top of statistical load profiles, without necessitating real-time access to smart metering data. The challenges associated with the recently established regulation of energy communities are however associated with technical aggregation (i.e. 15-min generation and consumption data) and the ability to provide participants of formal energy communities the means to easily exchange energy, power and flexibilities. OurPower has 520 members and gathers 12 energy communities, among those also GreenPoint, which is a privately owned multi-unit building complex in Wiener Neustadt. The complex has 84 private rentable serviced apartments spread over the area of 5.345 m² and 20 commercial premises distributed at 1.795 m². It is equipped with 226 kW_p rooftop and balcony PV system, 200 kWh battery storage, slow and fast EV charging stations, and baseline BEMS. On an annual average, around 30% surplus electricity is generated and traded on OurPower platform. Nearly 800 customers with annual consumption of 4,6 GWh/a buy electricity from marketplace, thereof 620 households, 113 farmers and 48 SMEs.

Scope: Leveraging on the longstanding partnership and with the ambition to actively involve residents in trading-off comfort, economic, social and environmental impacts, the pilot aims at utilising a building-level DR programme. Technically this means utilising and coordinating energy assets in the building to smoothen the statistical load profile and/or offer excess flexibility in a DR scheme (TE-4). The solution is composed of Customer Cockpit with nudging app for customer engagement (TE-10), Community Manager platform with flexibility management system (TE-12). Specifically, the ambition is to utilize data analytic models and forecasting (TE-4) for triggering the nudging app actions, utilize Universal Secure Gateway (TE-6) as the infrastructure to pre-aggregate individual assets/systems of the building and potentially automate DR procedures, and trade flexibility on energy marketplace. The platform will be further developed to promote principles such as self-generation, efficient use of RES-based energy, increase of energy savings, and collective generation and sharing – including EV charging and new services such as operating leasing for electric vehicles (Tesla). With all these processes involving great amount of data from various sources, specific attention will be devoted to the adoption of by-design privacy preservation (TE-5).

Existing infrastructure and flexible assets: 84 Flats = 5.344,46 m²; 20 Commercial = 1.795,02 m²; 226 kW_p PV-Plant (collective generation); 200 kWh Battery-Storage; 10 AC Chargers; 2 DC 50 kW Chargers; District heating; Mobile app for charging stations; BEMS for systems monitoring; OurPower P2P RES trading platform; Integrated dashboard with accounting and billing (sub-module of OurPower platform).

Legacy equipment upgrade and new equipment: Existing customers' and generators' side metering systems (DSO installed) as well as other own- or third-party equipment in the building (storage, chargers) have to be refitted, enforced, or replaced by adequate communication, control and/or additional metering systems to allow for online data readout. optimized forecasting and balancing, and active assets management in a DR scheme. GreenPoint will also enlarge the battery storage capacity to 1 MWh during the project implementation.



DENMARK

Energy efficiency and DR optimal interactions
between district heating and electricity networks in social housing

Pilot Context: In the Danish pilot, smart heating management and flexibility services will be demonstrated to apartment buildings of totally 2.000 m² as part of totally renovation of 692 flats from 4 renovated housing departments, 55.101 m² in total) in Herning municipality. The pilot departments of 2,000 m² will be provided by FællesBo, namely a social housing company, acting as administrator between the district heating company (Verdo Varme Herning) and the individual household.

Scope: The aim is to explore the link among electricity and heat via the smart ventilation control, providing flexibility at building level to the electricity grid (TE-8). In addition to the electricity flexibility, one very actual scenario would be to investigate in what extent the heat residential consumers will be able to reduce and shift their planned heating consumption, while scarifying a share of their comfort (TE-9) with a Russian gas supply reduction scenario. This will be achieved via data driven solutions where thermal inertia in the building will be used as storage at building level (TE-14). Another important aspect will be to design a compensation framework where energy (heat) consumption reduction could be compensated.

Existing infrastructure and flexible assets: In each building, we will have a renovated district heating installation with a district heating unit and weather compensation controller. Furthermore, the following assets will be available (building renovation measures to be carried out for all 4 pilot housing departments, funded by the social housing

company's own funds): Insulation of outer walls/cellars/roofs; Insulation of hot domestic water and heating pipes within the buildings; Balanced mechanical ventilation system; New installation for domestic water and electric installations; LED lighting in all common areas; Electronic thermo-states on all radiators controlled by innovative smart home unit; Building integrated PV for power production; PV-systems, totally 2,000 m² for all 4 departments; Energy, water and electricity meters which can be logged via 3rd party system (Neogrid); Temperature and humidity sensors in each apartment – if not available, they will be installed within DEDALUS.

Legacy equipment upgrade and new equipment: Installation of local controller connecting to building components making secure communication between the building and NEOGRID backend possible. In the building, a controller will be installed enabling a secure two-way communication between heating installation components and NEOGRID backend will be established. On each radiator in the apartments, Electronic thermo-states on all radiators controlled by innovative smart home unit will be installed and communication to NEOGRID backend will be established.



Multi-value DR for multi-segments of residential energy (electricity and gas) consumers

Pilot Context: In Greece, there is no DSO smart metering infrastructure, nor policies or market mechanisms to actively support DR solutions at mass scale. Therefore, for energy retailers, it is critical to develop new business models that will reward load shifting behaviour and create additional incentives for energy efficiency and demand management, while relying on privately owned IoT equipment (smart thermostats/appliances/meters) deployed at the consumers' side. In this context, the Greek pilot will be deployed at 100 residential setups (in 5 different cities in Greece), all of which are supplied with electricity and natural gas from HERON and are equipped with cost-effective and universal IoT devices of DOM-X.

Scope: The aim is to test multi-vector DR solutions, considering implicit DR for electrical home appliances and explicit DR for flexible HVAC loads, evaluating the replicability and scalability of residential DR, while minimising the need for complex grid infrastructure and high-end building equipment. Advanced AI-based data analytics and load disaggregation algorithms will be employed to monitor disaggregated loads and provide advanced categorisation beyond ON/OFF level. The above activities will result in multi-vector DR based on HVAC appliances and nudging services (TE-10) using appropriate comfort-based flexibility models (TE-9), as well as multi-value DR for electrical loads (TE-11), with a key result being their validation and replicability at the supplier's client base level. Furthermore, by matching consumption with RES-based generation (TE-8) through HERON's virtual net-metering product, the reduction of total energy carbon footprint will be targeted by optimally managing the demand across energy carriers. The DR solutions will be evaluated across different clusters of pilot participants, allowing HERON to test the viability of implicit and explicit DR focused on different consumer segments and ultimately of its client base. In turn, this will calibrate incentives and rewards specific to customer segments, contributing towards the design of data driven business models.

Existing infrastructure and flexible assets: The 100 pilot households are equipped with the DOM-X heating controllers, upgrading a plethora of legacy HVAC appliance types, such as natural gas boilers, split unit ACs and heat pumps, into smart flexible assets able to participate in explicit DR, building on open standards (Opentherm, ModBus) and widely applicable control modes (ON/OFF, power modulation, etc.), across energy vectors (natural gas, electricity). HERON's smart-metering platform is used to monitor the electricity consumption at household and plug level for advanced load disaggregation. Data captured across both vectors are made available for real-time analysis and dashboard visualisation to HERON. The DOM-X smartphone app is used for remote management of space heating/cooling, the collection of user preferences (comfort settings, schedules, etc.) and the participation of end consumers in explicit multi-vector DR for HVAC appliances. The smartphone app of HERON is used to test implicit DR, by considering the real-time consumption of electrical appliances (white goods) and data collected from the commercial virtual net-metering product (EN. A.).

Legacy equipment upgrade and new equipment: Installation of new heating controllers for electrical HVAC units, smart plugs, home gateways and controllers extending the currently deployed infrastructure; Replacement of white goods by ARC smart appliances (up to 20 air conditioners).

Organisations involved: HERON (the largest independent electricity and natural gas supplier in Greece), DOM-X (Technology provider of energy management solutions for retrofitting and interoperability of legacy building systems), NTUA (data models, AI services provider and added value services), ARC (provider of DR-ready smart appliances).



Trading off comfort with energy consumption in DR programs for elderly people living in serviced apartments

Pilot Context: Borgo Mazzini Smart Cohousing (BMSC) is located in the historical town centre of Treviso, with an urban grid dating back to medieval times. BMSC is composed by various separate buildings that communicate between them via shared gardens and communal spaces. BMSC hosts nearly 80 residents (elderly people) living in 1-bedroom and 2-bedroom flats, each with a surface of 40-50 m². Each flat is provided with all necessary facilities for independent living: kitchen appliances and full-furnished bathroom in accordance with the Design4All principles, in order to accommodate people that might develop some form of disability in the future. BMSC flats present a high degree of comfort for their residents but the management of heating and cooling systems, as well as the supply of energy and water is conducted exclusively by ISRAA as the only owner and property manager. This centralised management of the supplies and regulation of indoor temperature often risks of creating differences in terms of perception and comfort between different residents who may have diverse needs.

Scope: Through active residents' engagement, we will initiate a participatory co-design process of utility planning that takes into account both energy and non-energy benefits by placing more importance on the personal comfort and on the value that is attributed to the resulting efficiency by the users (TE-9). ISRAA will work in collaboration with DYNAMÉ (as energy broker) for the optimisation of the technical and economical options that will empower the DR approach adoption considering all the potential features that it allows (TE-8). This bottom-up approach aims to cultivate genuine interest, accountability and care on the real energetic demand based on the individual habits of each resident. A small minority is connected to the gas supply for heating and cooking appliances, while the rest of all BMSC is entirely dependent solely on electricity supply, with heating supported by heat pumps. The bill for each building therefore varies according to the type of infrastructure, both including gas and electricity or just electricity, the age of the systems installed and, of course, on the consumption behaviour of each resident (TE-11).

Existing infrastructure and flexible assets: Wall boilers, thermal power stations, fan-coils (fan coil, convective heater, induction cabinet), air conditioning and combined production systems (chiller, heat pump unit, mono-multisplit, VRF-VRV), mono-multi split system indoor units, and VRF-VRV, evaporative tower/cooling.

Legacy equipment upgrade and new equipment: The pilot foresees the appliance of new devices for the monitoring and optimisation of consumption in all the BMSC flats with the aim of establishing a personalised tracking of users' needs and habits in terms of energy demand.



District-level DR for increasing community-level self-consumption while providing flexibility to local DSO

Pilot Context: The Spanish pilot, at La Seu d'Urgell in the Pyrenees, involves all the agents in the DR value chain, from end-users, to energy retailers involving also energy management service providers and the local City Hall. This is remarkable as DR does not only benefit the energy consumers affected by the current energy crisis in Europe, but it is also helpful for small size local electricity retailers, like PEUSA in Spain, which are in risk of disappearing due to the economic stress of participating in the current high volatile electricity markets, which have been shook up by the effect of Ukraine's war in the gas prices. Is the combination of these needs that permits to settle a fruitful pathway for the development of DR at residential level. In this pilot the management of the energy resources and the deployment of the DR services will be controlled and operated following a district approach, taking advantage not only from the assets installed at each building individually but an aggregation of consumers, generation and storage of several blocks. To be more specific in this study case the pilot counts on 13 dwellings, being overall 1400m² new social housing promoted by the local city hall, two collective PV self-consumption systems (involving around 50 kW) and a district-level battery (120 kWh). All assets are placed within a distance smaller than 500 m, to respect the Spanish regulation on collective self-generation and storage to be managed at district level.

Scope: The aim is to promote DR and electricity flexibility at district level, combined with a district-based electricity storage system and PV generation plants (TE-13). The management of the demand will be focused on the operation of flexible assets, such as electrical heat pumps and mobility electrical chargers. We will test how residential consumers can get benefit from new time of use tariffs and collective self-consumption and dynamic storage to implement explicit and implicit DR, without affecting their comfort and energy supply needs (TE-9). This community level DR scheme will become a good trade-off between user's needs, efficient energy use and active participation of end users in the electricity market (TE-10 and TE-11).

Existing technologies/infrastructure: The data of generation and consumption of electricity is already collected by PEUSA and one department of CIMNE, which applies data analytics and AI algorithms for energy management. Moreover, the social housing dwellings to be integrated in this energy district count already on electrical heat pumps that will be linked to the control platform. There is a cloud-based application for managing the EMS of building energy assets, hosted by ROMUR and an IoT platform provided by ARCELIK to connect/interact with active loads

Legacy equipment upgrade and new equipment: The devices to be installed are mainly a centralized energy storage in the range of 120 kWh, integrated nearby a building by ROMUR Renovables, using a new development of ESS specially devoted for DR applications, manageable HVAC systems, able to be controlled remotely, 15 smart

1.2.3.2. DEDALUS MULTIPLIERS



IRELAND

Decentralised Power2heat cross-commodity optimal heat vs electricity social housing flexibility aggregation

Pilot Context: The main motivation of this pilot is to develop a framework to deploy DR in a social housing context to fight energy poverty. The pilot will focus on DR at the level of three different social housing bodies, each with 10 houses equipped with air-to-water heat pumps.

Scope: Optimal control policies will be tested based on algorithms that simultaneously learn a Digital Twin (TE-8) of the dynamics of the building and the optimal control policy to be adopted. The objective of the control policies is to maximally valorise the available flexibility to fight energy poverty, while still preserving the end-users personal comfort requirements (TE-9). The pilot will also allow to test the project's innovative solution to make smart assets ready for DR solutions. The installed air-to-water heat pumps have the technical capabilities to be used in DSR but are not yet ready for it at this stage. Besides the technological advancements, the investigation will also provide insight in the optimal way for social housing bodies to offer flexibility to aggregators in order to maximise the valorisation (TE-13).

Existing infrastructure and flexible assets: The social housing units are equipped with an air-to-water heat pump (AtW HP), which is smart but not yet ready to be deployed in a DR solution. No interaction with an aggregator is in place at this stage. The existing sensor is a coarse electrical power sensor.

Legacy equipment upgrade and new equipment: To make the DSR solutions ready, a more accurate heat meter and local steering box should be installed.



ROMANIA

Blockchain solutions for DR management and flexibility aggregation

Pilot Context: TUC operates the students' hostel on the university campus where students can rent apartments on yearly basis. It hosts around 2.700 students covering a net floor area of 17.376 m².

Scope: The scope of this pilot is to validate on top of the available pilot infrastructure a blockchain framework for DR program management (TE-8 and TE-12). We will explore: (i) The use of blockchain energy ledger to immutability store and share the energy (and non-energy) data provided by sensors from hostel flats and buildings, reinforcing citizen trust, privacy and offering audit features; (ii) The automated definition of DR programs using smart contracts and signals agreed via P2P flexibility trading making citizens aware of the value of their flexibility; (iii) Aggregation of flexibility in decentralised fashion via blockchain driven coordination to meet energy services constraints; (iv) The DR program energy and financial settlement via lockable and non-fungible tokens enabling the integration of flexible heterogeneous sources (electricity, heat, cross energy sector services).

Existing infrastructure and flexible assets: To monitor the electricity use, BEMS have been installed. For the moment, BEMS is not provided with a control function, but the system could be extended. Some of the main energy assets (chiller, water pumps, ventilation system, etc.) could be switched off from the electric panel by the building administrator or technical staff. Later this year, within SunHorizon project, the roof of the restaurant (1.150 m²) will be used for installing solar thermal, covering a total area of 220 m² with aggregated power 132 kW_{th}, which corresponds to 174 MWh/yr. Furthermore, a heat pump of 50 kW, a thermal storage of hot water equal to 10,000 lt.

Legacy equipment upgrade and new equipment: -

1.2.4. RELATION TO NATIONAL/INTERNATIONAL RESEARCH/INNOVATION ACTIVITIES

The consortium brings vast amount of experience by leveraging on the achievements of a variety of completed or quasi-completed projects, funded by Horizon, H2020, FP7, Intelligent Energy Europe, and related EU initiatives.

Relevant EU/national funded projects

- We will leverage on the socio-technological data-driven enabling frameworks deployed in the H2020 **BRIGHT** (ENG, TUC, DOM-X, COMS, CENTRICA) and **H2020 eDREAM** (ENG, TUC) specifically tailored to consumer engagement and energy communities. Also we will capitalise, upscale and validate blockchain based tools for privacy/confidentiality preserving data processing developed in H2020 **PHOENIX** (COMS) to support decentralised DR response management which empowers prosumers to value and trade their flexibility in a P2P manner, reinforcing their trust and increasing their engagement.
- We will leverage on and adapt the Common European Energy Data Space deployed within the **Horizon ENERSHARE** (ENG, NTUA, COMS, SIN common partners) and **DigiBUILD** (ENG, NTUA, CARTIF, UNIVPM) projects to provide a Building-level DR adaptation of the Energy Data Space, through expanding and

validating additional, use cases and adapting technological building blocks of the Energy Data Space to enable automated DR

- We will capitalise knowledge and IT developments (e.g. GIS-based tool capabilities) from H2020 **REWARDHeat** and national Spanish project **HySGRID+** (CARTIF), interaction capabilities with guided questionnaires from H2020 **ATELIER** (CARTIF) and community-based decision-support systems from H2020 **LocalRES** (CARTIF, CENTRICA). Results from H2020 **SocialRES** (CARTIF) will be used for a better understanding of the socio-economic, -cultural, -political and gender factors that influence the consumers' behaviour in the energy system.
- H2020 **BD4NRG** (ENG, NTUA, CARTIF, COMS, CENTRICA), **PLATOON** (ENG, COMS) and **MATRYCS** (NTUA, ENG, COMS, CARTIF, ICLEI) will be leveraged for adapting electricity-centred cross-stakeholder cross-domain AI/ML models, big data analytics and other solutions for flexibility platforms. DTs of different physical assets developed in past and on-going projects, such as **InterConnect** (HERON) will be extended to integrate new business and social dimensions to better represent flexibility and planning functions for DR services. Results regarding advanced management platforms to unlock grid flexibility will be also leveraged from H2020 **X-FLEX** (BLUPERINT). We will also expand the Austrian Climate Energy Fund project **serve-U** (OurPower), for flexible optimisation models and automated engagement and nudging of energy community participants. The Austrian national funded project **SEED** (BLUEPRINT) that focused on promotion of sector-coupled energy communities provides catalogue of technical elements supporting technical WPs and policy recommendations.
- The experience from **I-ENERGY** (NTUA, ENG, CARTIF, HERON, COM), exploring applications on AI-as-a-Service, Energy Analytics and digital services will contribute to the optimal management of electric power and value chain of energy systems. This knowledge will be further enriched by the H2020 **frESCO** innovative business models expertise (CARTIF), combining and rewarding local flexibility for buildings through energy efficiency and demand side management practices.
- To expanded and maximise the pool of assets relevant for the provision of DR services and enable secure integration of disparate legacy and new systems, we will exploit the H2020 **PHOENIX** (COMS) implementation of Universal Security Gateway aimed to provide interoperability and resource virtualisation framework.
- Being part of the **Auto-DAN** (ARC, CARTIF) project will provide an enhanced baseline to improve the data-driven environment of the project by bringing its activities on automated, dynamic, and continuous energy performance assessment derived from the monitoring of the energy consumption at the building level.
- The participation of ENG and ED to H2020 **OpenDEI** and **OneNet** projects will enable strong alignment with OpenDEI Reference Architectures and with the inspiring principle for cross-sector Data Space design. We will extend the former H2020 **InterConnect** (HERON), **BD4NRG**, **BRIGHT** and **PlatONE** (ENG), data-driven energy services frameworks with a view to provide and deploy an effective open, scalable and interoperable implementation of Data Space compliant tools.
- We will leverage outcomes of H2020 **POWERPOOR** (NTUA) and **DECIDE** (OURPOWER, HERON, DOM-X) projects focused on energy communities. H2020 **SHAREs** (OurPower) will be leveraged for best practice in replication to Multipliers.
- FFG Austrian national funded project **DigitalEnergy4All** (OURPOWER) will provide knowhow on managing interdisciplinary group of experts. We will also exploit state-of-the-art research from psychology and behavioural economics from **NUDGE** (DOM-X) project.
- UNIVPM is partner of **BIMSPEED** focused on developing a combination of methodologies and tools with one central information source at its core: the Building Information Model (BIM), a digital representation of a building. This model will be the catalyst for a smarter, more efficient, method of deep renovation for the residential building sector.
- Comfort-based flexibility models will be based on the results of H2020 **NewTREND** and **RenoZEB** (UNIVPM).
- **DigiBUILD** (ENG, NTUA, UNIVPM, CARTIF) is adapting recent comfort and well-being measurement technologies to increase monitoring performance and ability to predict human preferences. DigiBUILD will also contribute to understand how data can contribute to the development of 'Comfort Performance Contracts'.

EU Initiatives

We will liaise and synchronise with work in **BRIDGE** (the European cluster of Smart Energy grids funded innovation projects on H2020), providing significant contribution to the Business Model and Consumer Engagement working groups on DR solutions for residential consumers. Moreover, BLUEPRINT provides advisory and IT services to European agencies, such as **ACER** (the EU Agency for the Cooperation of Energy Regulators), the **Energy Community Secretariat** and participates in **ETIP SNET** (the European Technology Platform for Smart Grids), with active role in leadership of WG4 (Digitalisation and Customer Engagement). CENTRICA is an Executive Member of **smartEn**, the European business association integrating the consumer-driven solutions of the clean energy transition. ENG will also establish a common collaboration ground with Flexible Generation WG3 of

the ETIP SNET. ENG is also founding member of **BDVA/DEFACTO** (Big Data Value Association) and **ETWARE Foundation**, as well as one of the companies in the board of **GAIA-X** for data infrastructure and **IDSA** (International Data Space Association). An alignment and coordination with these initiatives will be constantly ensured. An Energy on Vertical Task Force (TF7) has been launched within BDVA (Dr. Bertoncini from ENG acting as Deputy Chair of TF7), with a view to harmonise the actual on-going work on big data management and AI along the different H2020 and Horizon projects.

Global Initiatives

NTUA is actively involved in the **Clima-Med** project ('Acting for climate in South Mediterranean' which has as objective to establish the regional Covenant of Mayors Mediterranean or CoM-Med), as well as in the '**Support to the Secretariat of the Global Covenant of Mayors for Climate and Energy**' project. NTUA also contributes to the coordination of the **EU-GCC Clean Energy Technology Network**.

1.2.5. INTERDISCIPLINARITY & TRANSDISCIPLINARITY

DEDALUS envisages the co-creation of knowledge among people from several fields with different backgrounds, crossing traditional boundaries between different disciplines. DEDALUS is by default interdisciplinary, as it combines insights from different scientific fields to deliver a full interoperable, interconnected, and trusted residential DR. We integrate the following disciplines: social sciences such as sociology, psychology and political sciences for engaging with energy consumers, local communities and other key stakeholders horizontally and vertically, e.g. through co-creation and Design Thinking methodologies (WP2); Physics, mathematics, data and computer science, along with specific areas of science related to energy, financial, engineering, healthcare, water and mobility sectors for the design and implementation of the data-driven residential energy carrier-agnostic DR tools and multi-value services (WP3 - WP4) and demonstration (WP5); Economics for the development of viable business models and exploitation plans (WP6); Education science for organising and implementing workshops/capacity building activities (WP7); Political sciences for better understanding of regulatory measures (WP2, WP6) and facilitating the decision making process and pursuing collaborative multi-level governance (WP1 and WP7).

The above disciplines are supported by a consortium of partners that covers the entire bandwidth of knowledge and expertise necessary for the successful implementation of all planned activities (Section 3.2). Transdisciplinarity will be promoted in the project's co-creation activities with energy consumers and stakeholders from the energy value chain, who will significantly co-define the aspired use cases (WP2) and related services (WP4). Transdisciplinarity also requires the additional challenge of differentiating and integrating various bodies of knowledge and inputs from a wide range of stakeholders [39]. We will pilot a collaborative process of knowledge production, based on profound transdisciplinary experiences of the partners which thus draws from scientific and societal bodies of knowledge alike, to contribute to a multi-dimensional understanding of the role of the residential DR in trusted data exchange among participants, and to enable scientists to combine disciplinary theories, methods, perspectives with an overarching aim of arriving to common ground.

1.2.6. INTEGRATION OF SOCIAL SCIENCES AND HUMANITIES

DEDALUS adopts a socio-technological innovation-inspired systemic approach to the design, development, uptake, scaling up and replication of multi-value and energy carrier-agnostic residential DR, which considers the technological, market and social dimensions of upscaling processes jointly and in an explicit manner.

The proposed approach brings social sciences centre stage and makes use of a co-creation and Design Thinking methodologies for participatory co-design of value-added data-driven services, increasing the 'energy democracy' and contributing towards eradicating energy poverty in the EU. In addition to this, a multi-dimensional incentive framework will be utilised for a comprehensive incentive design aimed to understand the creation of value from the energy value chain for residential consumers, while trading off with either privacy infringement risk management and/or regulatory constraints. Hence, multidimensional and beyond financial incentive design may stimulate and boost individuals/communities and/or energy business stakeholders to align activities, which consider both economic and beyond economic (focus sustainability) dimensions. By adopting such a multidimensional framework, we will be able to take into due consideration not only financial incentives for the monetisation of flexibility services but also environmental, comfort, social incentives, as well as the possibility for exchanging and/or compensating heterogeneous assets and services (security, home automation, etc.). The proposed methodology will allow end user groups' engagement at the demo sites, who will be continuously involved along the project phases, with a view to actively participate to the incremental service designs and validation campaigns, as well as for taking part to the social assessment activities.

The proposed approach represents a departure from traditional upscaling and market structuration practices, and DEDALUS embraces it as one of its fronts of innovations ambition (see Section 1.1.3). In practical terms, the outcome of this approach will allow to deploy data-driven residential energy carrier-agnostic DR tools and multi-

value services fully meeting user expectations in terms of functionality and design. Social sciences play also a relevant role in DEDALUS in applying C2B2B and C2B2C business models appropriate to energy flexibility provision, with a view to design what could be shared (assets, data-driven services), reward mechanisms design (pay per consumption, flat subscription-based fee), and especially how distribute and socialise revenues and benefits along all the energy participating stakeholders.

1.2.7. GENDER DIMENSION

Gender will be a focal point in DEDALUS as it is central in the formation, response and responsibility bearing of energy transitions [40]. More specifically, participation in energy communities empowers consumers in their everyday lives. Nevertheless, numerous factors, such as gender, may affect individuals' willingness and acceptability to participate. The 2016 report 'Gender and energy' [41] highlights several aspects that have been considered in designing DEDALUS. The report also refers to energy availability and affordability trends affecting women, men, and other genders differently. The gender dimension is even more glaring in the field of ICT where, according to a recent World Economic Forum report, women hold a paltry 22% of jobs with an even lower number occupying senior positions in the field. This gap is evident in ML research as well with only 12% female representation. With a global estimate of 15% of women choosing careers in Science, Technology, Engineering and Mathematics (STEM) compared to 18% in Europe [42], it is obvious that more female participation in these fields is urgently needed. Moreover, energy industry is gender-agnostic, while social factors, such as cultural backgrounds, age, disabilities and specific needs, income, educational background and geographic location are generally considered to affect individuals' energy consumption awareness and behaviour. Gender parity would additionally consolidate women's and other genders' right to equality and represent a significant social achievement. The energy sector is influenced by a set of persistent gender inequalities: gaps in energy access; gaps in the energy labour market; gaps in decision-making and to a greater extent, the attainment of SDG goals particularly Goal 5, 7, 8 and 13.

Therefore, the inclusion of women in the field of energy decisions making in the household and policy making level has the potential to promote more equitable, sustainable economic growth, climate change mitigation and a scale down health effect if conventional energy usage.

DEDALUS will include the cross-cutting priority of gender on three core axes: (i) the composition and configuration of the consortium; (ii) the theoretical, thematic and methodological structure of the project; (iii) the research focus across the whole project duration. We will take the following actions: 50% of consortium members identify as women and other genders, including leadership roles (WP leaders); The execution of dissertation research related to DEDALUS will be promoted to the women and other genders PhD students of the R&D partners involved in the project. During the stakeholder engagement process, we will take into account the opportunities of participation based on gender identity; the composition of the pilot participants will be based on the gender distribution of the population (at maximum degree, targeting the 50% of women and other genders share of participation in the project activities). We will also create synergies with relevant initiatives, such as the Global Women's Network for the Energy Transition [43], an international non-profit organisation to empower women in the energy sector through interdisciplinary networking, advocacy, training, coaching, and mentorship (BLUEPRINT, a female-founded and managed small-size SME, has been a permanent member). As part of communication and dissemination measures, in order to both support research and innovation for energy communities, as well as open career opportunities, we will co-host annual event 'Energy Is Female' (OurPower with support from BLUEPRINT has been organising it in previous years). The diversity of the necessary competencies (social sciences, energy economics, technology and law), the elimination of the tension between technology and social sciences in the implementation of research projects and the demand for flexibility in interdisciplinary project work will be emphasised.

However, we acknowledge that gender representation does not automatically lead to gender equality; so, we will make dedicated efforts to consult gender experts inside and outside our consortium to ensure that a gender lens is included in a meaningful way throughout the project. Evaluation and feedback forms will include questions on gender as well as other issues of intersectionality where relevant, including those addressing whether the end-consumers/prosumers or stakeholders experienced being subject to gender differentiation or felt somehow discriminated based on gender or any other criterium (e.g. socio-economic situation, culture, age, etc.).

1.2.8. OPEN SCIENCE & RESEARCH DATA MANAGEMENT

We will follow open science practices in line with the Horizon Europe principles. DEDALUS will offer open access to its scientific publications either submitting them to the Open Research Europe or ensuring open access, selecting either self-archiving/'green' open access or open access publishing/'gold' open access, to facilitate dissemination and reuse of the project's results. In order to ensure that data management is done according to FAIR principles and in adherence with the GDPR guidelines, there will be a thorough Data Management Plan (Month 6, Task 1.4), which will be continuously revised.

- **Types of data/research outputs:** DEDALUS as project  observes data services for energy end-users and operators, is expected to generate, and process a variety of data. All generated data will be stored in an easily accessible way by both humans and software, as appropriate. Except from the existing data which will mainly stem from the demos, DEDALUS will produce a large volume of data as research output. Generally, the generated data falls into two categories: Organisational data, which are relevant to the implementation of the innovation actions and technical and scientific data, including raw and processed experimental data, publications, and software.
- **Findability of data/research outputs:** The generated data within DEDALUS will be traceable and locatable by means of unique identification mechanisms. All files will be uniquely identifiable by using standardised name conventions and file versioning. Additionally, all generated research data will be annotated with metadata following specific standards (e.g. the Dublin Core generic metadata standard) in order to describe the wide range of networked resources. All datasets will be shared in publicly accessible disciplinary repositories, such as Zenodo, using descriptive metadata in order to ensure public traceability.
- **Accessibility of data/research outputs:** DEDALUS will use different tools to make data accessible by different users, either participating or not to the consortium. DEDALUS reports will have different dissemination levels. Public reports will be shared via public facilities and confidential reports will be shared among the consortium. Scientific publications will be open access and they will be uploaded to public repositories. Data will be accessible between the partners through a common space for the necessary data exchange and communication (MS Teams). Certain research data considering the demo users are sensitive due to privacy and data protection issues and therefore will be kept confidential or anonymised before being accessible.
- **Interoperability of data/research outputs:** Suitable standards for data and metadata creation along with appropriate vocabularies will be developed. Data interoperability is a very significant aspect for DEDALUS, as it is intended to develop semantic Data Models and ontologies to ensure interoperability. The analysis of standards and interoperable Data Models will be performed in the **WP3**, facilitating data sharing from different sources.
- **Reusability of data/research outputs:** Data will be made available as soon as possible and published on the project's web site and/or other repositories (e.g. Zenodo). Access will be permitted to anyone, in order to use a data copy which may be shared or adapted (remixed, transformed, or build upon existing material).
- **Curation and storage/preservation costs:** The Project Coordinator is responsible to maintain the project document repository, the Scientific Manager is responsible for the quality of any scientific data outcomes and each partner is responsible for the recoverability of own generated data. The costs of making data FAIR cannot be determined yet, as it depends on the amount of data which will be generated.

Finally, at least two of the project's main outputs will be published under open-source licence (e.g. CC-BY, ASL), see Table 2.7, and will be made available as a GitHub repository via the European Open Science Cloud.

1.2.9. ARTIFICIAL INTELLIGENCE SAFETY AND LIABILITY

The newly proposed regulation regarding a legal framework for AI (EC COM/2021/206) takes a risk-based approach to solve AI present challenges in relation to safety and liability. AI-based solutions have a different lifecycle than traditional cyber-physical systems and it is important to ensure that security, traceability and explainability are handled throughout the lifecycle: development, training, deployment, and operation. DEDALUS will: (i) Implement systematic software development practices focusing on quality management and establishing confidence in the maturity of the data-collection (**WP3**), training processes and deployment of AI (**WP4**); (ii) Prioritise explainable AI (and if necessary, at the expense of accuracy) in different use cases (**WP5**); (iii) When applicable, use robust inference solutions as countermeasures to deal with unmodelled phenomena or adopt uncertainty representations (**WP3**); (iv) Conduct risk assessment during the design phase, mapping the AI-based systems internal decision-making process to the system control - both in terms of decisions made by the AI system, but also on humans in proximity to the system - using Failure-Attack-Countermeasure graph [44] and mitigation strategies from on-going initiatives as ETSI Industry Specification Group (ISG) Secure AI (SAI) [45] (**WPs 3-4**); (v) Before deployment, assess reliability and robustness with the Digital Twin, conducting test demarcation, which is critical for AI systems, as it is difficult to predict system behaviour for scenarios vastly different from what it has been trained to and tested against (**WP4**); (vi) Ensure that humans are active part of the design and safety assessment of AI in the pilots validation (**WP5**), in particular undergo functional testing to validate safety mechanisms, and make sure that safety of installations and humans is always the highest priority.

1.2.10. COMPLIANCE TO THE 'DO NO SIGNIFICANT HARM' (DNSH) PRINCIPLE

DEDALUS is fully compliant with DNSH Principle since its activities as well as the expected life cycle impact of the innovations at a commercialisation stage, substantially contribute in a positive way and/or do not significantly harm any of the 6 environmental objectives set out in Art.9 and 17 of the EU Sustainable Finance Taxonomy Regulation.

- **Climate change mitigation:** DEDALUS ‘contributes substantially’ pursuant to Taxonomy Regulation, to Articles 10.1/a, 10.1/b, 10.1/c, 10.1/d and 10.1/h by reducing dependences on fossil fuels through the promotion of renewable energy use.
- **Climate change adaptation:** DEDALUS ‘contributes substantially’, pursuant to Taxonomy Regulation, to Articles 11.1/a by promoting RES without increasing the risk of an adverse impact on people and nature.
- **Sustainable use and protection of water and marine resources:** DEDALUS has no or an insignificant foreseeable impact on the sustainable user and protection of water and marine resources principle.
- **Transition to a circular economy:** DEDALUS has no or an insignificant foreseeable impact on the transition to a circular economy principle.
- **Pollution prevention and control:** DEDALUS ‘contributes substantially’, pursuant to Taxonomy Regulation, to Articles 14.1a by supporting the transition of building stock towards zero pollution.
- **Protection and restoration of biodiversity and ecosystems:** DEDALUS has no or an insignificant foreseeable impact on the protection and restoration of biodiversity and ecosystem principle.

SECTION 2: IMPACT

2.1. PROJECT'S PATHWAYS TOWARDS IMPACT

In this section, we provide a detailed overview of all contributions of the project to the **Expected Outcomes (EO)** of HORIZON-CL5-2022-D4-01-01, as well as the **relevant Long-term Expected Impacts (LEI)** specified in the **Destinations** in the work programme. The expected **Key Exploitable Results (KER)** of the project implementation are presented in Fig. 2.1, detailing the potential end-users as well as the validation to ensure the TRL desired. In order to achieve this challenging maturity level, different results will exploit and adapt existing solutions developed in previous European projects and other initiatives where consortium partners are involved (Section 1.2.3).

Key Exploitable Results (KERs)	Target groups & end users	WP	Obj	TE	TRL
 KER1- Social Science Framework, innovative multi-dimensional incentives and nudging interventions for residential DR	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Building/facility managers; Energy communities; Local authorities; Researchers	WP2	O1/5/7	All, especially TE-9	6-7
 KER2- AI-based individual and building-level consumers clustering and market segmentation algorithms	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Researchers	WP2	O1/5/7	TE-1	7
 KER3- Open APIs for DR-ready smart appliance IoT platform	Data Marketplaces Operators; Data Space Operators; Data Value Chain Stakeholders; Smart appliances manufacturers; Suppliers & operators of DR equipment; Providers of local business platforms and assets/group of assets; Researchers	WP3	O2/5/7	TE-2	7
 KER4- Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange		WP3	O2/5/7	TE-3, TE-5, TE-6, TE-7	6-7
 KER5- Digital Twins for optimal assets vs consumers flexibility planning and decentralised DR management		WP4	O3/5/7	TE-4, TE-8	6
 KER6- Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders), and other Energy Value Chain & Data Value Chain Stakeholder; Energy communities; Local authorities; Researchers	WP4	O3/5/7	TE-4, TE-9	6-7
 KER7- Electrical flexibility management tools for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets		WP4	O4/5/7	TE-4, TE-11, TE-12, TE-13	7
 KER8- Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralised cross-commodity electricity vs heat management		WP4	O4/5/7	TE-4, TE-10, TE-14	7
 KER9- Business Sandbox with novel sharing economy and social innovation -based DR business models	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Local authorities; Energy communities; SMEs; Policy makers; Researchers; Regulators	WP6	O5/6/7	All	6-7

 Fig. 2.1 DEDALUS Key Exploitable Results

In WP1 (management), WP5 (pilot demonstration) and WP7 (dissemination/exploitation) all KERs are addressed.

2.1.1. EXPECTED OUTCOMES (EO)

EO1

Increased potential benefits, trust and acceptability of DR solutions for residential consumers

Baseline: DR programs can be valuable to residential customers. When deployed strategically, DR programs can help consumers by readapting energy demand to their needs, and put consumers in full control of their energy consumption. Additionally, DR programs can provide incentives to help offset the costs of adopting smart home technology or building-management system upgrades that may not have been considered otherwise. Finally, DR programs can act as a catalyst for consumer energy awareness and engagement, as well as, thanks to the capability for managing latent flexibility, for increased energy efficiency at home/building level. Despite its huge potential, residential DR is still lagging behind due to a number of barriers (too much large flexibility threshold to participate in flexibility markets, lack of transparency in value for money for residential consumers, low financial compensation in the DR remuneration, excessive hardware/software infrastructure costs and high costs for DR verification, making unattractive DR participation from residential consumers, high privacy breaches consumers concerns from energy data and consumption profiles sharing). All of them contribute to undermine the trust and the acceptability of DR programs in front of residential consumers.

Scientific outcomes: DEDALUS plans to leverage on latest ICT advancements on big data, IoT, edge computing, blockchain, to deploy more granular sensing capabilities for assessing and modelling exploitable flexibility from smaller scale home/building-scale assets to support residential DR. We will deploy more effective data-driven consumer profiling models, both at individual and at building level (**KER2**), which allow to design, develop and deploy innovative ICT-based DR flexible controllable assets aggregation services. These services will allow

residential consumers to participate to flexibility markets, hence increasing the efficiency of the electricity grids, while at the same time increase self-consumption of locally generated renewable energy. In addition to this, the cross-domain residential building-level DR concept will widen the DR scope to non-electricity carriers. Innovative heat flexibility models and/or multi-energy flexibility models and heat or energy-as-a-whole flexibility valorisation algorithms (KER6, KER7, KER8), as well as consumer-aware Digital Twins for flexibility planning (KER5) will be developed to increase the overall amount of flexibility. The above planned results from DEDALUS will significantly increase the potential benefits of DR towards increased efficiency of building operation, of energy distribution and usage, as well as increase share of self-consumed renewable energy. Moreover, the DR adaptation of the trust layer of the H2020 ENERSHARE Energy Data Space (KER4) will enable to put energy consumers in full control of their data, while guaranteeing them on appropriately managing the access, sharing and exchange of their energy consumption data, hence contributing to increase trust and acceptability of DR programs.

Economic/technological outcomes: DEDALUS will adapt the Horizon ENERSHARE Energy Data Space to provide extended interoperability for smart residential smaller scale appliance and evolve them towards DR ones (KER4). Thanks to the DEADALUS extended interoperability, data models and Open APIs, a large number of appliances can become DR-ready (KER3) and can participate via pre-aggregation at building or a district-scale to flexibility marketplace via aggregators (KER7). At the same time, such interoperability layer will allow DR-ready appliance to become controllable and actuate DR programs strategies. DEDALUS planned services are tailored to provide DR services to improve operation of smart electricity grids and District Heating Network, while at the same time increasing the efficiency of the energy distribution and the energy system as a whole (KER8). This will result also in a reduced carbon footprint of the energy distribution, which is also due to consumer flexible behaviour.

From the consumer side, main expected outcome will be reducing the operation cost of the building without the tenants noticing it. Services (KER6, KER7, KER8) will be based on data from the building behaviour and deduct the flexibility potential, which then is used to deliver DR services to the surrounding environment. Tenants are kept in the loop by inputting their energy (electrical, heating) demand requirements which then is used to deliver the best service, eventually through required participation in DR programs.

DEDALUS will also contribute to smooth the EU-Russia crisis, through bringing more and more center stage the importance of untapping the potential for DR, where, undoubtedly residential DR has significant room to upscale. If European governments subsidise energy use this winter and supply stays inelastic, EU consumers will compete with each other for limited supply, push up prices and make suppliers richer. So, it is urgent to think about how to leverage DR for residential consumers. This would not only allow Europe to better navigate what promises to be a difficult winter but also seize on the crisis to introduce possibly lasting effects in terms of DR. Those are significant, considering the important role of such measures in lowering the investments required to reach the objectives of the European Green Deal (KER9).

Societal outcomes: DEDALUS will increase the actual involvement of building residents in the planning and management of their energy usage according to their real needs and to the desired expected consumption of suppliers. The SSH-based tools and methodologies (KER1) through which they will be engaged intends to create a truly user-centred approach that aims at triggering positive behaviour and responsibility in each individual; thus sparking an initial social change, supported by the assistive technology, which can be disseminated to the surrounding community. We expect this participative approach to foster a higher degree of trust in the effectiveness of the technological devices, in the convenience of sustainable behaviour and environmental solutions and, lastly, in the participants' own capabilities of consumption management. DEDALUS envisages a cross-commodity DR concept and underlying ecosystem which considers multi-dimensional incentives and nudging interventions for DR participation (KER1). The P2P blockchain marketplace (KER4) for energy, flexibility and other tokenised barter-like asset/service exchange, can contribute to reinforce social relationships within the building and/or districts as local communities, and will result in an increased community identity and social cohesion, while at the same time offering a P2P based social service provisioning (KER8) (for example to alleviate energy poverty among community members and/or building residents).

Scale/significance:

	KPI	Target	Verification
KPI _{1-SC1}	Increased buildings energy efficiency due to DR programs measured in the pilots	0 Y ₁ , 5-10% Y ₂ , 10%-15% Y ₃ , 16-18% of reduced energy bills for residential consumers Y ₅₊	WP2 & WP5 Deliverables
KPI _{1-SC2}	Increased energy data sharing by active consumers	10% Y ₁ , 20-22% Y ₂ , 30-35% Y ₃ , 45% Y ₅₊	WP3 Deliverables
KPI _{1-ET1}	Energy system cost reduction, by providing flexibility services to electrical, gas and district heating grids	0 Y ₁ , 5-10% Y ₂ , 10%-15% Y ₃ , 16-18% Y ₅₊	WP4 & WP5 Deliverables
KPI _{1-ET2}	N° of consumers engaged in residential DR	>80 Y ₂ , >500 Y ₃ , >800 Y ₅₊	
KPI _{1-SO1}	Reduction of number of complaints from tenants regarding temperature comfort and indoor climate	0 Y ₁ , 15-25% Y ₂ , 26%-35% Y ₃ , 50%, 60% Y ₅₊	

Advanced asset control and aggregation approaches that enable the participation of residential buildings in commercial DR

Baseline: Recently, flexibility services have been emerging in most forms, aiming to contribute to the balance of power supply and demand and with it explore the possibility of providing new energy services to the grid, as well as, allowing the exploitation of community-based building-scale energy services. Different flexibility sources and aggregation principles have been modelled, however mostly addressing larger assets size, which are characteristic for the scale of Construction & Infrastructure (C&I) sector. The modelling of exploitable aggregated flexibility from smaller scale assets at residential level has been carried out without taking into due consideration the effective usage and consumption profiles of specific assets and by specific groups of residential consumers. Moreover, the interdisciplinary behaviour and synergies among them in the context of cross-sector synergies have been hard to model in an accurate and dynamic way despite the availability of several data sources regarding monitoring the appliances, devices and equipment. Flexibility estimation, management, and activation in an optimal way therefore remains a challenge and calls for the combination of different flexibility services (e.g. DR, mobility, thermal, etc.) and their aggregation at different scales to improve the value of flexibility towards the power system. This multi-level and multi-domain flexibility modelling introduces additional challenges, namely in the exploitation of distributed optimisation schemes. Furthermore, demand-side flexibility that combines multiple energy vectors (electricity, natural gas, etc.) successfully, while rigorously modelled as part of academic research [23], tends to be evaluated in simulated case studies and rarely as part of even small-scale real-world test cases. In this context, a recent survey in DR for multi-energy systems has identified challenges in integrating demand-side resources across multiple energy vectors due to limited capabilities of current power systems [24].

Scientific outcomes: DEDALUS will pursue the improvement of AI-based algorithms for the modelling, management and use of flexibility in an optimal aggregated way (KER2) in an end-to-end perspective from micro (individual apartments/home, buildings) and macro (aggregators, energy communities and grid) scale levels. In fact, it will provide AI-based tools and services for the estimation and use of flexibility covering the end-user, aggregators, energy communities and system operators' perspectives. In particular, multi-energy and cross-sector modelling and aggregation of flexibility will be key to contribute to the empowerment of end-users and the successful deployment of energy communities in the system (KER5, KER7, KER8).

Economic/technological outcomes: DEDALUS aims to increase flexibility options for consumers, who otherwise may have considered that their homes may either not be 'smart' enough to have flexibility potential or that it could be very expensive to set it up. For example, for countries where smart metering has not yet been provided by a DSO, DEDALUS will develop forecasting and optimisation algorithms that are essential for DR, as services to be deployed using commercial products that are easy to install, maintain and operate. DEDALUS aims to increase flexibility pool and therefore flexibility's resonance with the wider public, by upgrading their legacy HVAC appliances into smart flexible assets that can directly or indirectly participate in energy markets (KER3), offering additional benefits such as monitoring and visualisation of energy efficiency improvements, a singular comfort setpoint (KER6) irrespective of the heat source and tangible environmental benefits by increasing heat electrification while keeping overall costs the same. DEDALUS builds upon consortium's existing infrastructure, combining partners' fragmented products into integrated solutions that bring together electricity, heat and gas consumption, RES generation and storage under a granular data-driven ecosystem that can monitor and control home environment integrating sensors, controllers and other equipment belonging to possibly different vendors (KER3). Consumers will be given options on how they offer their flexibility with DEDALUS services respecting their comfort limits (KER5) and engagement capacity through data-driven solutions (KER6, KER7, KER8) that can mitigate possibly disrupting impact on daily routines. Finally, DEDALUS flexibility solutions will be fully replicable, deployed and tested in countries of varying regulatory maturity, built on a modular approach that can include or discard features depending on the regulatory environment or a partner's business plans and objectives. Thanks to DEDALUS activities and resulting technology enablers (KER3, KER4, KER5, KER7), a larger number of smart residential appliances and devices will become DR-interoperable and will contribute to deliver effective aggregation approaches and services suitable to effectively manage and exploit flexibility at residential level via DR-based automated control.

Societal outcomes: DEDALUS will allow to provide consumers (i.e. individual citizens, vulnerable consumers, energy communities) with a great deal of flexibility and give impetus to act and participate in demand side flexibility markets. At the same time, new business roles will be enabled for existing building and energy value chain stakeholders (KER9), collaborating in innovative ways (i.e. energy cooperatives, independent aggregators, technology providers, DSOs, District Heating Network Operators), and deliver better services to their customers, allowing them to benefit from the flexibility potential embedded in their appliances and energy assets. Individual consumers will be directly engaged from the very beginning through multi-dimensional beyond financial comfort-based incentive design (KER1, KER6) with a view to bring their energy consumption fully in their hands, while

increasing their awareness on providing a significant contribution to support increased efficiency of the energy system as-a-whole and become active socially-engaged players to reinforce the social cohesion of the local building-scale community.

Scale/significance:

	KPI	Target	Verification
KPI _{2-SC1}	N° of validated data-driven aggregation models and services for residential DR	5 Y ₂ , 7 Y ₃ , >10 Y ₅₊	WP2 & WP4 Deliverables
KPI _{2-ET1}	N° of scientific publications for automated DR with SSH scientists	1 Y ₁ , 2 Y ₂ , 4 Y ₃	WP7 Deliverables
KPI _{2-SO1}	N° of multi-dimensional incentives co-designed	2 Y ₁ , 3 Y ₂ , 4 Y ₃	WP2 Deliverables
KPI _{2-SO2}	Increased share of aggregated flexibility mobilised by piloted assets	0% Y ₁ , 5-10% Y ₂ , 15%-20% Y ₃ , >25% of consumption Y ₅₊	WP4 & WP5 Deliverables
KPI _{2-SO3}	Increased ratio of exploited vs planned aggregated flexibility validated through pilots	0% Y ₁ , 7-10% Y ₂ , 13%-18% Y ₃ , > 25% of DR transaction cost reduction Y ₅₊	

EO3

Expanded pool of assets relevant for DR in the residential sector

Baseline: Confidence in smart applications, the use of Distributed Energy Resources (DERs), such as RES generation and storage equipment, and, above all, flexible loads open up new ways of interacting with the energy markets for any kind of user. In this sense, new added value DR-based services are emerging for residential consumers, which leverage on their engagement and may contribute to increase flexibility and efficiency of the energy grids, while allowing reduced energy bills for customers and eventually building operation cost reduction. However, the exploitation of such potential is currently prevented because there are still numerous technical barriers, given the lack of interoperability of solutions and the dependence on numerous actors for their integration. Smart home devices and solutions are characterised by their extreme fragmentation, diversity of standards and proprietary interfaces, constraining significantly cross-domain interoperability. This leads to ad-hoc and customised modelling of DR services that are not replicable and hardly adaptable, hindering DR deployment at scale within the residential sector. Open communication interfaces and information exchanges are needed to ensure that heating control systems, hot water production units, storage systems, electric cars, cooling systems, household appliances, etc. can participate in DR.

Scientific outcomes: DEDALUS uses adaptable and modular methods and tools to ensure data interoperability among smart devices and technologies constituting the DR chain (KER3), and promote scalability based on minimum functional requirements per DR service using model view definitions of semantic models.

Economic/technological outcomes: DEDALUS explores the deployment of smart hardware artefacts and SGAM-based architecture specification to integrate acquired data from heterogeneous sources, complying with minimum security and privacy requirements (KER3). At a higher level, DEDALUS supports quality assurance when delivering the proposed DR services, achieved by verifying compliance between semantic models generated based on the DR service-specific data requirements and pilots' available data and technologies. This adaptable and modular approach allows for high data quality and less error-prone replication processes with minimal human intervention based on seamless integrated smart devices and explicit semantic model definitions. New smart devices will be deployed according to the newly defined data and semantic models to improve residential DR usability with replicable and adaptable solutions. Improved choice and features on the devices will enable customers to have more options for managing their energy consumption by using smart home appliances. The natural economic benefits of this solution will provide smart appliance manufacturers to achieve affordability and higher market penetration for DR-ready smart appliances and increase market share. On the other hand, understanding and experiencing interoperability architecture with the help of DEDALUS will provide easy DR integration capability to the device manufacturers. This will lead to reducing the time it takes to extend this technology to other smart products. This is another positive economic impact for the manufacturers to produce DR-ready assets.

Societal outcomes: By fostering co-creation of the solutions through participatory processes (KER1), DEDALUS will promote the adoption and usage of flexibility from DERs and households/buildings equipment (e.g. smart appliances, heat pumps, EVs, batteries energy storage unit), empowering customers to actively participate in the flexibility markets. The ultimate value of DEDALUS interoperability modular (KER4) and adaptable approach is to provide well-defined methodologies and assisting tools to help building controller and operator stakeholders reducing efforts necessary in delivering connected smart devices and enable DR services at scale, expanding pool of available DR-ready relevant assets. Increased number of users will access DR services as more and more smart devices will be available in the market. When a significant number of users along EU could unleash their DR potential this will lead to a deep change in the use of energy and resources availability. For the customer side, reducing their energy consumption during peak hours and easily managing their own energy consumption will provide save money

on their electricity bills as economic benefits but also bring great benefits to the community by taking part in the smart grid load balancing.

Scale/significance:

	KPI	Target	Verification
KPI _{3-SC1}	N° of adapted and harmonized Open APIs and extended Data Models	3 Y ₁ , 7 Y ₂ , 10 Y ₃	WP3 Deliverables
KPI _{3-SC1}	N° of Basic Application Interoperability Profiles (BAIOPs) created in accordance with the SGAM	3 Y ₁ , 7 Y ₂ , 10 Y ₃	
KPI _{3-SC1}	N° of adapted interoperable Open Standards, standardisable interfaces and formal languages/ontologies	1 Y ₁ , 2 Y ₂ , 4 Y ₃	
KPI _{3-ET1}	Increased number of DR-interoperable assets type	5 Y ₁ , 7 Y ₂ , 12 Y ₃ , >20 Y ₅₊	
KPI _{3-ET2}	Increased modularity and adaptability in replicating DR services measured through reduced share of estimated setup time for replication	0% Y ₁ , 5-8% Y ₂ , 10%-13% Y ₃ , >18-20% Y ₅₊	WP3 & WP5 Deliverables
KPI _{3-ET3}	Reduced share of pollution at local level assessed through pilots' validation	0% Y ₁ , 5-8% Y ₂ , 10%-13% Y ₃ , >18-20% of reduced NOx and SOx Y ₅₊	WP5 & WP6 Deliverables

2.1.2. LONG-TERM EXPECTED IMPACTS (LEI)

LEI1

Technological and socio-economic breakthroughs for achieving climate neutrality and the transition to zero pollution of the building stock by 2050, based on inclusive and people-centric R&I

Baseline: It is estimated that roughly 75% of European buildings are not energy efficient while 85-95% of current buildings will still be in use until 2050. Reducing energy consumption of buildings and increasing their energy efficiency is of utmost importance for achieving climate neutrality and zero pollution for the building stock by 2050. While building renovation is undoubtedly one of the most effective longer term strategy to achieve such objective towards the European Green Deal, the adoption of 'softer' strategies and policies, such as buildings and assets smartening, digitisation and DR-oriented flexibilisation, may significantly contribute towards climate neutrality of building stock in a combined way with deep renovation in reduced times. However, DR solutions, despite really promising in that sense, have been so far mostly tailored to Commercial & Industrial sector, while residential DR is lagging behind.

Scientific Outcomes: DEDALUS will fully support the transition of building stock towards zero pollution and climate neutrality, through demonstrating the viability of 'soft' beyond renovation approaches which will enable buildings residents to actively participate to the energy market and accordingly contributing to the cost reduction of the energy system as a whole. DEDALUS will hence leverage on advanced digitalisation of buildings, providing a data-driven stack of technology enablers to enable smart loads residential appliances into DR-ready flexible ones (KER3). DEDALUS introduces new AI-based individual and building-level consumers clustering and market segmentation algorithms (KER2). This will enhance the decryption and deeper understanding of buildings performance, consumption patterns, attitudes and preferences. We will enable the innovative consumers behavioural segmentation as well as the buildings segmentation on thermal and electrical flexibility bands to be used simultaneously in DR by enabling their aggregation as well as portfolio optimisation. The clustering based on social behavioural/preferences features will allow the offsetting and compensation of social enablers towards residential building participation in cross-carrier and potentially cross-energy sector DR. Moreover, DEDALUS will deploy data-driven innovative consumer-aware cross-commodity and commodity-agnostic DR flexibility aggregation models tools and services (KER5, KER6, KER7, KER8) aimed to demonstrate the technical and business viability of residential DR to contribute to the building stock zero pollution.

Economic/technological outcomes: DEDALUS will adapt the Energy Data Space Interoperability Layer through integrating IoT sensing and communication platforms for smart home/buildings appliances (KER4). The DEDALUS approach will be based on co-creation techniques to ensure people-centric technology development and upgrading socio-economic breakthroughs. We will establish valuable synergies between SSH methodologies and digital solutions/infrastructure (including AI and DT) to co-design, with active energy consumers and technology developers, novel sharing economy business models (KER9) for flexible assets ownership and boosting cross-sector flexibility integration. This will pave the way for future scientific work in green multi-commodity markets supported by blockchain tech (e.g. trading RES surplus/DER flexibility by non-energy assets and services) (KER3) that increase residential consumers acceptance of flexibility products due to an augmented perception of social benefits (KER1), beyond economic benefits (e.g. reduction in monthly electricity bill).

Societal outcomes: The DEDALUS approach will be based on people-centric and responsible R&I to move forward towards climate neutrality of the building stock and the energy system as a whole. In particular co-creation techniques will be used to ensure people-centric technology development and upgrading socio-economic breakthroughs (KER1). We will establish valuable synergies between SSH methodologies and digital solutions/infrastructures (including AI and DT) to co-design, with active energy consumers and technology

developers, novel sharing economy business models (KER9) for flexible assets ownership and boosting cross-sector flexibility integration. Blockchain technologies (KER3) will be effectively leveraged and adapted to deploy a digital marketplace for the reciprocal exchange and compensation of tokenized heterogeneous energy and non-energy assets, which will result in a participatory DR-ready building-scale community of building residents. Thanks to this technology enabler, DEDALUS will facilitate the inclusion of non-financial social-oriented levers as potential enablers for residential consumers to participate to DR, and at the same time increasing the social cohesion among building residents. Finally, the DEDALUS blueprints/roadmap and the capacity building programme will be the primary mechanisms for the creation, growth and capacity building of relevant stakeholders, in order to facilitate further uptake of residential DR.

Scale/significance:

	KPI	Target	Verification
KPI _{LEI1-SC1}	N° of scientific papers published in data-driven services & sharing economy business models	1 Y ₁₊ , 6 Y ₂₊ , 12 Y ₃₊	WP7 Deliverables
KPI _{LEI1-ET1}	N° of technologies that reached TRL6-7 by the end of the project	≥ 1 Y ₁ , ≥ 8 Y ₂ , ≥ 14 Y ₃ , ≥ 20 Y ₅₊	WP3 & WP4 Deliverables
KPI _{LEI1-ET2}	Increased predictability of DR behaviour and flexibility potential to mobilise	≥ 5% Y ₁ , ≥ 8% Y ₂ , ≥ 15% Y ₃ , ≥ 20% Y ₅₊	WP3 & WP4 Deliverables
KPI _{LEI1-SO1}	Different types of actors engaged by DEDALUS services	≥ 6 Y ₁ , ≥ 8 Y ₂ , ≥ 10 Y ₃ , ≥ 15 Y ₅₊	WP2 & WP5 Deliverables
KPI _{LEI1-SO2}	Deliver new business models	2 Y ₂ , 4 Y ₃ , 8 Y ₅₊	WP6 Deliverables

LEI2

Building stock that effectively combine energy efficiency, renewable energy sources and digital and smart technologies to support the transformation of the energy system towards climate neutrality

Baseline: Enrolling individual Energy consumers to become active players in the energy transition through facilitating their participation to DR, where suitably combined and integrated with local deployment of renewable energy sources, can significantly contribute to a more sustainable energy system, through enabling more decentralized self-consumption at community level, and to an increasing energy efficiency of the overall energy system, and building too. Increased digitisation of buildings and smartening of home appliance can become a fundamental enabler to support the green energy system transitioning towards climate neutrality.

Scientific Outcomes: A key objective of DEDALUS is to combine DR with energy efficiency as ‘first fuel’ and decentralised renewable generation. We will demonstrate how and to what extent advanced digital technologies can facilitate consumer activation and participation in residential DR, through the deployment of innovative residential consumers-based data-driven consumption, flexibility and aggregation models (KER2), services, Digital Twins and other tools (KERs 5-8). The above technology enablers will steer the transition towards a more decentralised and greener energy system, thanks to the deployment and the validation of a number of technology enablers which boost local integration of renewables and their improved controllability on consumption side to participate to automated DR.

Economic/technological outcomes: DEDALUS technology enablers will be leveraging on the Data Space compliant adaptation (KER3, KER4), which will allow to deliver improved learning models and related algorithms, as well as Digital Twins for combined assets versus consumers flexibility planning (KER5). We will extend a set of open APIs and specs to reconcile, harmonise, and extend existing automation-oriented standards (e.g. IEC standards), business DR standards (OpenADR2.0), ontologies (SAREF) and languages (EFI), aiming to provide extended semantic interoperability over a wide portfolio of legacy and emerging assets, including data (KER4). This augmented interoperability among different sectors, assets, digital platforms, and systems will foster new data-driven energy services (KERs 6-8) and business models (KER9), enabled by standardised and open interfaces and increased data availability

Social Outcomes: DEDALUS portfolio of data-driven services and enablers (KER4, KER5, KER6, KER7, KER8) and digital marketplace (KER5), will contribute to increase social cohesion and acceptance of investments in smartened DR-ready interoperable appliance, renewable technologies, as an effective way to optimize The consumption of renewable locally generated at local level, hence contributing to decarbonize the energy system, improving the local air quality and demonstrating the concept of a future energy-driven citizenship (with lower electricity bills, carbon footprint, and higher resilience). The enhanced participation of a set of actors in DEDALUS (from individual residents, building communities, local renewable energy communities, aggregators, DSOs, etc.) covers a full spectrum of social implication in the energy system, with special emphasis in empowering residential sector and RES self-consumption, focused on green investments in physical and human infrastructure

The seamless integration of RES and demand-side multi-energy flexibility assets (as planned in the Spanish and Austrian pilot, and in the Irish and Romanian multiplier), when combined with building-level pre-aggregation of residential thermal and electrical loads through trading off financial versus social and wellness/comfort benefits (as

in the Italian pilot), will improve significantly health and welfare of European citizens in total, with special attention of chronic bronchitis, and other health benefits. In addition, DEDALUS by improving the air quality and noise reduction, will advance smarter and more sustainable towns and communities with increased wellbeing, attracting citizens to live there, and thus boosting indirectly the local low carbon economy.

Moreover it is worth considering that the comfort dimension is very relevant to include also the **social dimension** in the project. Quite often this component is not properly considered in energy management strategies, but it has been already demonstrated that e.g. in buildings comfort-based HVAC control strategies are able to provide energy savings up to 20 %, with a great impact on user satisfaction and on productivity in tertiary buildings. DEDALUS will provide innovative cross-domain services, such as optimal comfort/wellness services for the benefits of elderly people or people affected by energy poverty (KER6). The inclusion of the lifestyle will allow the positioning of the occupant at the centre of the system with maximum engagement, optimising the comfort balancing/unbalancing with the minimum cost in terms of energy and discomfort in function of the occupants' attitude in accepting less comfort to save money (enthusiast user) or requiring specific comfort condition (e.g. aged occupants) even at a slightly higher cost. Innovative multi-dimensional incentives (KER1) that incorporate and combine financial and non-financial yet social dimensions, such as carbon emission reduction, improved comfort, provisions of non-energy social services to alleviate energy poverty, will be designed and validated through sharing-economy -based fair incentive sharing splitting.

Scale/significance:

	KPI	Target	Verification
KPI _{LEI2-SC1}	Carbon emissions reduction (environmental benefit) for the energy grids measured at piloted local community level	0% Y ₁ , 10-15% Y ₂ , 15%-20% Y ₃	WP5 & WP6 Deliverables
KPI _{LEI2-ET1}	Increased rate of self-consumption of locally generated renewable energy	0% Y ₁ , 20-25% Y ₂ , 30%-35% Y ₃	

2.1.2. BARRIERS/OBSTACLES

DEDALUS aspires to bring significant impacts, which are achievable either within the project lifecycle or after full-scale adoption of the project outcomes at cross-stakeholder level by a wide range of energy players. The Consortium, on its behalf, is fully committed to mobilise all resources required in order to realise the envisaged impact. However, although the project impacts are solid and tangible, there always may exist external factors and barriers over which the project Consortium has limited control upon and that may affect the project impacts. Such external assumptions and barriers can be found in the following table.

Table 2.1: PEST Analysis

External Factor	How DEDALUS will address the factor Barriers (T= technological, B= Business, S= Social, R=Regulatory, E=Economic)
Insufficient engagement of residential energy consumers and reluctance to share personal data (S), (R), (E)	One of the most critical barriers is the limited number of engaged residential consumers, due to a number of reasons: i) citizens do not understand how they can become active consumers, or, in the case of already active consumers, they are not timely informed on how much effective has been their activation; ii) insufficient capacity of energy value chain and market failure in propagating the value of consumer activation and designing an appealing financial incentive which consumer should value as relevant against the upfront incurred hardware/software costs for their activation. From a regulatory point of view, demand-side response is not allowed to participate in the provision of services towards system operators, in many countries and also the minimum capacity for participating in more centralised energy markets is too high for small consumers to reach without organised aggregation. iii) residential consumers are often reluctant to share their energy consumption data due to the potential risk of privacy breaches, which will prevent energy stakeholders to develop and deploy data-driven optimal DR services. All these, in combination with the complex processes to access the market (such as prequalification) limit significantly the opportunities of consumers to easily market their potential and to have access to alternative and lucrative revenue streams. We will leverage on a co-creation process supported by a remarkable critical mass of SSH partners (SIN, ICONS, ISRAA) which involve end user from early stage to design appropriate multi-dimensional financial and non-financial incentives in the form of attractive business models which facilitate durable consumer engagement in DR with other service providers and data sharing. Moreover, we will make use of blockchain technologies adaptation for reducing the flexibility bid sizes through aggregation micro-contracts set up. Last but not least, the implementation of multi-value services with a strong focus on social services for poorer and more vulnerable people (e.g. elderly people) (e.g. energy 'donation' to poor people, by

	exchanging with energy asset third party flexibility asset maintenance, e-mobility, power-to-heat) will also create new drivers for consumer engagement.
Limited adoption of open standards, Open APIs, and interoperability to support data-driven DR services and seamless data exchange and transparency (T)	DEDALUS will implement existing standards and will promote adaptation of them to deliver extended interoperability and achieve compatibility with existing legacy data platforms/hubs within the electricity sector, across-energy vectors and across different domains. We will build on open-source tools, technology building blocks and platforms to facilitate legacy and proprietary energy and beyond data hubs/platforms to be federated with a minimal effort. Widely used standards and protocols such as IEC standards (IEC 61850, CIM market), SAREF ontology for smart energy, EFI and OpenADR for flexibility and DR management, and IDSA and GAIA-X Data Space compliant building blocks and connectors will be deployed leading to an Energy Data Space DR adapted implementation. Active communication with all relevant stakeholders will focus on the common benefits from open interfaces and endorse involvement in DEDALUS results and the engagement of more end-users. Moreover, consortium partners (ENG, CENTRICA, CARTIF, ARC) will make use of their prominent and active position in leading-edge standardisation committees to contribute to turn on OpenADR, EFI, IDSA and GAIA-X as open standards for EU-wise data sharing in the energy and beyond domains for supporting DR in energy-efficient residential buildings.
High upfront investment costs (E)	High investment costs necessary for residential DR represents a significant barrier for full exploitation of the DR potential by decentralised (e.g. local communities, aggregators) stakeholders. The benefits of DR must outweigh consumers' efforts as well as investment costs in e.g. automation equipment or communication tools. Although, there are funds available and favourable conditions in loan packages in many countries, risk aversion of banks concerning loans or investment support for renewables for individual and communities is a major barrier to financing. To deal with such barriers, we will investigate and validate through our planned pilot activities alternative hybrid financing and business models, such as the sharing-economy ones (e.g. shared asset ownership and rewards at community level, as well as blockchain decentralised platforms for shared assets trading)
New business models for data-driven energy services (Business Models immaturity) (E), (S)	We will leverage on SSH approaches and methodologies for co-designing value-added data-driven services and multi-dimensional sharing-economy-based business models for establishing attractive remuneration schemes and routes for marketing the flexibility for individual consumers and communities. These will be facilitated through a DLT based smart-contract concept for simplifying the transactions and making market participation easier and more attractive for small consumers. Peer-to-peer market designs will also be favoured by this approach. Data sharing, as essential prerequisite to deploy optimal consumer activation services will be simplified through the DR-ready smart appliance IoT platform of DEDALUS, thus building consumer trust and willingness to share their data. As an additional mitigation strategy we will continuously keep involved energy consumers in validating both the multi-dimensional incentives and the business models, as well as appropriate awareness campaigns will be put in place to increase individual business consumers feeling as an essential stakeholder to contribute to greening the energy systems
Lack of jobs and growth (underinvestment in AI and data-driven services) in energy sector (S)	DEDALUS will develop a variety of consumer and DR-oriented data-driven services, ranging from basic services to support Energy Data Providers and Energy Service Providers (aggregators, community managers, ESCOs, suppliers) to added-value cross-value chain services across energy sectors and/or across energy and non-energy sectors (like Digital Twins applications). The estimated growth of Data Market Value and, above all, of EU-27 Data Economy is creating a fertile ground for innovative digital skills to cope with the market request, which may create a high potential for local jobs creation as enabled by DEDALUS digital tools. However, to sustain the requested rate of digital skills, educational policy makers, academic institutions need some time for adapting their programs to the AI and data-driven digital skill request. We will mitigate such potential issues through focused and dedicated workshops and awareness campaigns tailored to education-oriented policy-makers, including local stakeholders, to inform them on the need for a prompt and immediate adaptation of the university programs and local community-level specialised training activities, which will include also activities to support local entrepreneurship.
Insufficient speed of technology transfer (E)	The spreading of our outcomes and the business opportunities for consumer participation may be hindered by an insufficient speed of technology transfer. In that respect DEDALUS will develop both individual exploitation and consortium-level exploitation plans which coordinate with and synchronise with ongoing leading-edge initiatives carried out by some

	common partners along on-going projects (ENERSHARE, BD4NRG, ONENET, PLATONE, I-ENERGY)
Regulatory uncertainty and immaturity leading to high risk of exclusion of and small-scale consumer from flexibility service-provisioning to the grid (R)	DEDALUS will showcase how consumers through different levels of aggregation can gather a high enough capacity to support individual and community-level self-consumption optimisation and/or provide flexibility to the grid. This will provide the opportunity of replacing large-scale conventional and often expensive power plants with more flexible, cleaner and cheaper sources of flexibility, creating at the same time additional revenue streams for end-consumers and creating fertile ground for additional business models. DEDALUS will perform a detailed impact assessment and results evaluation and will include scalability and replicability analyses which will generate concrete recommendations towards policy-makers, aiming to contribute to new regulation that will boost the role of the consumers.

2.2 MEASURES TO MAXIMISE IMPACT

The DEDALUS project will rely on an integrated and holistic strategy for Dissemination, Exploitation and Communication, designed to meet the goals set by the expected outcomes and long-term impacts of the project. The strategy, outlined in the Plan released as D7.1, will address key players, enablers and stakeholders in the energy and buildings ecosystems, developing tailored measures to ensure their regular engagement through the different project stages, thus creating feedback loops as a way to facilitate the future uptake, replication and scalability of the DEDALUS results.

2.2.1. DISSEMINATION AND COMMUNICATION OF RESULTS

The dissemination activities are expected to ensure that the project's advancements are widely diffused to the intended targeted audiences with appropriate mechanisms in a timely manner, and that the key stakeholders for the project's exploitation and market uptake are early engaged and actively participating to the various project's implementation phases. Dissemination is instrumental to effectively promote the exploitation activities, while it is closely related to the communication activities (defined in [Section 2.2.1.1](#)) and consistent with the IP/IPR management measures (see [Section 2.2.3](#)). The project consortium aims to implement a comprehensive strategy and conduct effective dissemination, communication, and exploitation activities from the very early stages of the project's implementation. All partners are committed throughout the project to mobilise the appropriate stakeholders in order to multiply the effects of dissemination and exploitation activities. Moreover, dedicated D&C Desks will be set up at pilot sites by local partners (Spain, Greece, Italy, Denmark, Austria, Romania), aiming to boost outreach and engagement at local level under the coordination of a central D&C Secretariat (chaired by ICONS as WP8 leader).

An initial list of these **target groups** for DEDALUS, which at the same time are **key actors** that will directly benefit from the project results and outcomes, is depicted in the following table.

Table 2.2. Target Audiences for the DEDALUS Project

Target Group	Description	Interest in the Project
End-users – Residential consumers	Building owners, tenants, residents and citizens, prosumers	Raise awareness on how DEDALUS results can produce a significant impact on their day-by-day through reduced operating costs, greater comfort, improved lighting and temperature controls and overall increased operational efficiency, thus helping to cut their carbon footprint. Participate to the DEDALUS co-creation activities.
Building Value chain stakeholders & social housing operators and other non-energy value chain stakeholders	Building Operators/Managers, Building Management Systems Solution & Service Providers, ICT/cloud operators, Care Management service providers, Health/wellness service providers, Security/safety service providers,	Leverage on DEDALUS results to promote new business models for cross-sector flexibility and interoperability; Exploitation of project's open-source results; Inspiration for new ideas and applications
Energy value chain stakeholders	Power network operators, District Heating network operators, ESCOs, retailers, aggregators, suppliers	Leveraging DEDALUS results to contribute to redesign their business models with a view to push cooperation with building management stakeholders; Exploitation of project's open-source results; Inspiration for new DR ideas and applications.

IT Industry Players & SMEs	IT companies, web entrepreneurs, software engineers of solutions for big data and energy domains	Exploitation of project's digital assets and open-source results to support the smart readiness of residential buildings for DR solutions; Inspiration for new business ideas and applications; Integration of project results in corporate digital platforms to increase commercial offer.
Industry Associations & Technology Clusters	European initiatives and clusters (like BDVA, IDSA, AIOTI and FIWARE, ECTP, ETIP-SNET), research communities, associations, federations (like IMS, EFFRA, IFIP, IEEE, NEM, ECTP)	Inclusion of project's results to collaborative research activities (roadmap, white papers etc.); Dissemination of project's results to their members; Bilateral participation in events for knowledge exchange
Researcher communities, RTOs and Academia	Individuals engaged in research initiatives and/or working in research/academic institutes	Further advance the project's research; Extension/reuse of the project's innovative technologies to other application domains, inspiration for future research initiatives based on the project's concept and results; Participation in the project's events
Policy Makers & Standardisation Organisation (SDO)s	Policy-makers at any level like EC Directorates and Units, Ministries and Governments, Regulatory Agencies, Standardisation Organisations (CEN, ISO, ETSI, etc.)	Evaluation of the project's Social-Technological-Economic-Environmental-Political (STEEP) aspects; Definition of future research and innovation directions based on project's acquired knowledge; Inputs for standardisation activities
Financial Sector	Banks, investors, development agencies, Real estates	Increasing the financial opportunities arising from district construction/renovation investment; Promotion and replication of innovative financing schemes for districts renewal
Public authorities	Local authorities, city planners, mayors, municipalities and their national associations	Definition of green procurement tender; Utilise the project's results to define new sustainable energy agendas for residential districts ; Understand the benefits offered by the proposal; Take part in the activities of the project

The project plans to address relevant scientific audience (including audience both from the Smart Grids and the energy area) and *strategic decision makers* (such as stakeholders, EU energy ministers and general secretariats of investment, head of research departments, professors etc.) in related research facilities (both in industry and academia). The Consortium will disseminate the DEDALUS research outcomes, concepts and provide guided research directions and stimuli via selected events, magazines, workshops, and conferences (Table 2.3).

Table 2.3. DEDALUS Dissemination Activities

Type	Indicative Examples	KPI	Target Groups
Exhibition stands in the industry innovation events/fairs	Sustainable Places; BILT Europe (Digital Built Week Europe); European Sustainable Energy Week; Enlit Europe (the new brand for European Utility Week); European Big Data Value Forum; Big Data Conference Europe; Conference on Artificial Intelligence (AI) and Big Data in Engineering Applications; IDSA conferences	≥ 5	Energy value chain stakeholders, IT Industry Players & SMEs
Publication in highly ranked international journals and magazines	AI: Foundations and Trends in Machine Learning, Journal of Machine Learning Research, Machine Learning, Artificial Intelligence, Journal of Artificial Intelligence Research, IEEE Transactions on Pattern Analysis and Machine Learning, IEEE Transactions on Neuronal Networks and Learning Systems.; Energy: Applied Energy; Energy; Energy Policy; IEEE Transactions on Smart Grid; IEEE Transactions on Power Systems; IEEE Transactions on Power Delivery, Energy, Energy Policy; Advances in Applied Energy; Energy Conversion and Management; Renewable and Sustainable Energy Transition; Energies; Sustainability; IEEE Transactions on Industrial Informatics; International Journal of Electrical Power and Energy Systems.	≥ 12	Researcher communities, RTOs and Academia
Contributions in international	ICT: International Conference on Artificial Intelligence and Advanced Manufacturing; International Conference on Machine Learning; International Conference on Information, Intelligence,	≥ 15	Researcher communities, RTOs and Academia

peer-reviewed conferences	Systems and Applications; IEEE Consumer Electronics Conference Energy: IEEE Innovative Smart Grid Technologies; IEEE SmartGridComm; Industrial and Commercial Power System Technical Conference; Power Systems Computation Conference; IEEE Power and Energy Society for Innovative Smart Grid Technologies Conference; International Conference on Electricity Distribution; Conference on European Energy Market; CIGRÉ Conference; SDEWES (Sustainable Development of Energy, Water and Environment Systems) Conference; International conference on Applied Energy.	
Special issues/Thematic Workshops Organisation	Dedicated workshops to engage specific target groups with respect to a range of topics Typically arranged within the scope of larger international conferences to promote wider discussion with stakeholders. The objectives of these workshops are: i) Share with different stakeholders, project objectives and results achieved; ii) Define potential supporting measures to extend the use of project results and foster replication actions: ‘Energy Sources, Part B: Economics, Planning and Policy’; ‘International Conference on Information, Intelligence, Systems and Applications’	Researcher communities, RTOs and Academia ≥ 2
Cluster of European projects and other related projects and initiatives	EC DG Energy BRIDGE Data Management WG, Regulation WG, Business Models WG; EC DG Connect H2020 Open DEI Energy Vertical WG on Data Frameworks Architectures; GAIA-X Technical Working Groups and Energy vertical WG, and General Assemblies/Conferences; BDVA/DAIRO TF7 Energy Vertical WG	Industry Associations & Technology Clusters, Researcher communities, RTOs and Academia ≥ 10
Targeted Meetings with policy makers	Meetings with regional/national data and energy policy-makers and regulators to inform them about the benefits/impacts of different deployment and policy scenarios	Policy Makers & Standardisation Organisation (SDO)s ≥ 5

All scientific publications will be available as preliminary versions in the secure area of the webpage with the option of guided forum discussion, in order to engage actively both the consortium and the Transdisciplinary Advisory Board members in scientific discussions and enhance the quality and relevance of the publications. Following the general open strategy of DEDALUS, the open access publishing option will be pursued for scientific journals and conferences to maximise the target audience and the impact of the relevant publications.

Disseminating results to industry and society: DEDALUS will address industrial targets at an international level, exploiting already established contacts within the consortium, especially via industrial partners and technology providers. Some of the currently planned activities include: Participation in **major industrial and trade events** to make presentations, distribute project dissemination material and built potential partnerships with other exhibitors; Publishing **press releases and articles** in specialised trade newspapers & magazines; **Release exploitation-oriented info-packs and factsheets** on the different project’s innovative solutions, collected into a final **Innovation Handbook**; Distribute a six-monthly **e-newsletter**, providing major news and updates from the DEDALUS activity and pilot trials; Develop a set of **short video** interviews with experts from the pilot sites; Organising **focused workshops and public relations events**, and inviting European/international community to discuss issues of interest for the project; **Organise joint events and dissemination campaigns with sister projects and other relevant EU initiatives pursuing similar goals**, also by applying to the Horizon Results Booster to receive support through portfolio dissemination services.

Contributions to Standards Development Organisations (SDOs): DEDALUS will promote new work items in relevant SDOs from the very beginning and throughout all the project lifetime and thereby actively participate in work to standardise the interfaces, APIs, business models and architecture relating to DEDALUS framework. Through the memberships of many DEDALUS partners (ENG, NTUA, CARTIF), the project will contribute to relevant standardisation activities in the respective working groups dealing with big data/buildings (Table 2.4).

Table 2.4. DEDALUS Standardisation Activities

Area	SDO Body Group or Industry Initiative	DEDALUS Contribution	Partners
Smart Grids/DR	Mandate 490: Smart Grid Coordination Group (SG-CG) composed by ESOs (CEN, CENELEC and ETSI) working to	Focus on promoting a Reference Architecture supporting multi-party energy data exchange and	ENG, COMS, CENTRICA

	deliver Smart Grid Architecture Model (SGAM), CENELEC TC59X WG07; IEC TC57 WG21; IEC 61850 communication protocols; IEC 61970/IEC 61968 (Common Information Model) for data exchange among network operators; IEC 62746 for BMS smart grid interaction, which includes OpanADR2.0; USEF (Universal Smart Energy Framework), Open ADR Alliance, SmartEN (CENTRICA as member)	full interoperability of data-driven models with main Smart Grid standards	
	ETSI/oneM2M SAREF, EFI	Contribution to evolve SAREF to support cross-stakeholder data models	ENG
	ETIP SNET Technology Platform WG2 (storage technologies and sector interfaces), WG3 (flexible generation) and WG4 (digitalisation of the electricity system and customer participation)	We will liaise with WGs 2-4 by promoting Open APIs and interoperable open architectures, etc.	ENG, BLUEPRINT
Digitalisation in Buildings	European Construction Technology Platform (ECTP) Materials & Sustainability Committee (Coordinated by UNIVPM)	Digitalisation processes of the built environment: standards and tools to integrate data on future climate conditions and risks (adaptation to climate change) into all aspects of decision-making along the built environment value chain.	CARTIF
		Contribution to LCA (Life-cycle Assessment) working group with the Commission Services	UNIVPM
Interoperability of appliances	Home Connectivity Alliance	Standardised communication protocols and interoperability of different types of connected products to share information in a secure way	ARC
Open API	NGSI-LD, Context Information Management API, defined by ETSI ISG CIM	Contribution to extend Context Information Management to support SAREF and off-grid data	ENG

2.2.1.1. COMMUNICATION STRATEGY AND MEASURES

The DEDALUS communication strategy will determine how messages are communicated to broad stakeholders' group and the civil society, leveraging on mainstream media to multiply the outreach and achieve strategic goals. Key messages on the DEDALUS action and rationale will be elaborated and transferred using a simple laymen language, aiming to raise awareness on the social, environmental and economic impacts of the DR solutions developed by the project, thus enhancing their trust and acceptability. Social acceptance will be achieved by communicating key results with a storytelling approach, transforming the more complex jargon into easy-to-understand contents addressed to less technical audiences. Communication activities will start from the very beginning, targeting all audiences, from the general public to more expert ones and will be executed through the development of dedicated tools. The following table summarises the basic planned means of communication.

Table 2.5. Communication methods of DEDALUS

Method		
Project web site	All activities are accompanied with short reports announcements, photos, news and links to downloads (e.g. for the project public deliverables), etc. Beyond the reactive approach following activities, a proactive approach by an overall editorial plan, including the website as well as mailings, to set special articles or series like interviews and trial site promotion will help to keep up an interesting up-to-	All 1 (>15.000 visits)

	date content during the whole project life.		
Visual and content identity	A coordinated graphic identity in line with DEDALUS values, including the design of a logo, graphic elements and templates.	1 Brandbook; > 1 template for.doc and.ppt	All
Social media	We will start by promoting DEDALUS through partners' social networks, and create social media accounts for the project (e.g. Twitter, LinkedIn) promoting authorised discussions and exchanges with online communities. DEDALUS will establish a single hashtag (e.g.#DEDALUS), which can be complemented by more topic-oriented hashtags (e.g. #energycitizenship, #decarbonisation, #energytransition).	> 500 followers in total	End-users – Residential consumers, Energy value chain stakeholders, Policy Makers & Standardisation Organisation (SDO)s
Leaflet	Electronic and hard copies of the project leaflet comprehending a general overview of the project, its challenges and expected impacts in different languages (beyond English) to reach large audiences in different countries, including those where the DEDALUS demonstrators will be carried out (Italy, Spain, Greece, Denmark, Austria).	1 leaflet, (6 language versions)	All
Posters	A poster will be designed and printed to exhibit at partners' premises and use at events where the project takes presence.	> 1 poster	All
Institutional Presentation	Project standard Presentation will be created at the beginning of the project, containing basic information about the project (activities, objectives, partnerships, events).	>1 ppt presentation	All
Presentation video	Short video to introduce the project rationale and objectives, hosted on the website and YouTube and other social media.	>200 views on Youtube	End-users – Residential consumers
Podcast series	A short podcast series using a storytelling approach to explore and dive into the different aspects and challenges behind DR solutions, adopting the end-users perspective	>4 episodes	End-users – Residential consumers
Press releases	Press releases will be issued to specialised and general media channels at key project milestones (kick-off, major achievements, etc.). A press/media kit will be developed containing detailed press releases, videos (e.g. of project demos), publishable images from the project, few short papers (devoted to some key theme/topic of the project).	≥ 8	All
Talks in workshops	Invited talks in workshops and international events of reference as to communicate the project experimentation platform and solutions (T8.3).	On invite	Researcher communities, RTOs and Academia, IT Industry Players & SMEs

2.2.1.2. INDIVIDUAL DISSEMINATION PLANS

To complement the join dissemination plan (Section 2.2.1.1) and the join exploitation plan (Section 2.2.2.1), this section provides a short description of the individual activities planned by each category of partners (Table 2.6).

Table 2.6. Individual Dissemination Plans

Category	Partners	Dissemination plan
Building & social housing operators	FællesBo, ISRAA, GreenPoint	Building & social housing operators, participating to DEDALUS mostly as pilot partners (Front Runner or Multipliers), will disseminate and communicate the project outcomes in the domain of the organisations' activities and to the respective associations (Housing Europe, EDSO for Smart grids, SmartEN, BEUC, etc.). Through the demonstration activities they will bring together a group of stakeholders to advise building value chain stakeholders which may take over a significant role central for the massive deployment of residential DR.

DR Aggregators/Solution/Service Providers, Energy Community Managers, Other Energy Stakeholders	CENTRICA, HERON, OurPower, DYNAME, PEUSA	Energy stakeholders will participate to DEDALUS activities mostly as pilot partners (Front Runner or Multipliers). They will disseminate and communicate the project outcomes in the domain of the organisations' activities and to the respective associations (EDSO for Smart grids, SmartEN, BEUC, ICLEI, etc.).
Smart Appliance Manufacturers and Smart Home/Flexible Assets s Integrator Providers & integrator of DR-ready relevant assets	ARC, ROMUR	The results obtained from the project will be presented in workshops, conferences, and exhibition fairs. The use of ARC social media (LinkedIn, Instagram, Twitter, etc.) will provide access to a huge list of audiences, stakeholders, and channels to disseminate project results. In addition, innovations can be published in the scientific literature, dedicated journals and magazines.
Large IT service & tech providers, Third Party SMEs Data-driven DR Service Developers/Providers	ENG, BLUEPRINT, COMS, DOM-X, NEOGRID	ICT companies will put effort on project results dissemination according to a twofold strategy. Scientific results will be disseminated via journals and conferences publishing, following a consolidated process for scientific production. On other hand an internal dissemination action will be put in place towards internal business units. The latter has the goal to activate, enforce and consolidate the knowledge transfer between the R&D department and the production lines. SMEs will disseminate major outcomes of DEDALUS internally to their organisations and by publishing papers in international conferences
Policy makers and SSH stakeholders	GREEN CITIES, SIN, ICONS	Such partners will disseminate and communicate the DEDALUS concept, methodology and findings in the domain of the organisations' activities. Publication of interesting reports in related magazines (at least 4 during the project implementation) of interest will also be considered. They will use their connections with other foundations and initiatives to spread the word (joint events, conferences, etc.).
Leading research and technological development institutions	NTUA, CARTIF, UNIVPM, TUC, CIMNE	The research project partners will benefit from the dissemination of the results in specialised scientific publications and at national/international conferences, as well as in lectures and in training researchers in an early career stage (master students, PhDs, Post-Docs). They will also communicate the progress and the results of DEDALUS project beyond the project's own community (including media) and show the benefits of research informing the public and society at large. They will also participate in workshops and/or international conferences in order to come in direct contact with other academic institutions, enterprises, researchers and other projects' stakeholders and exchange current practices and opinions. The website will be a platform for direct communication and dissemination of the results.

2.2.2. EXPLOITATION OF RESULTS

The project exploitation strategy will comprise of a bouquet of exploitation activities: (1) The identification of the innovative exploitable assets, whether these are technological or added value services which the project will deliver through its results to its target users; (2) The conduction of a thorough market analysis (which will comprise of an initial and a final analysis) which will aim at the identification of the market towards which DEDALUS is targeted; (3) The documentation of an analytical IPR management strategy based on the principles outlined in the project Consortium Agreement which will guide joint and individual exploitation capabilities of the project partners; (4) The analytical definition of all possible commercial and non-commercial exploitation models, which have been preliminary identified from the proposal stage; (5) The analytical definition and evaluation of the sustainability and viability of possible business models and alternative solutions that may be followed for the provision of the project solution and services to the identified stakeholders, including licensing schemes, pricing, etc.; (6) The validation of the aforementioned exploitation activities through the relevant, market driven DEDALUS demos.

DEDALUS overall **exploitation strategy** combines three different patterns: (1) **Joint exploitation and wider uptake of project's main technological outcomes**; (2) **Exploitations of the data-driven services/applications**; (3) **Individual Exploitation Plans**. An initial incremental roadmapping exercise for joint exploitation (pattern 1) of the DEDALUS exploitable results may consists of: **Joint identification and commercialisation of different bundles of services and/or enablers or individual enablers tailored to specific building and energy value chain target groups (KERs)**; **Joint exploitation of the technological, business and social layers of the DEDALUS**, which

aligns to DEDALUS consortium playing the role of **Data-Driven DR Residential Consumer Activation Ecosystem Facilitator**.

An initial plan of the various exploitable results, in relation to the potential ways of service provisioning and pricing strategy is depicted in the table below:

Table 2.6. Preliminary business-plan for individual exploitation results

#	KERs	Owner(s)	Pricing	Additional Services
1	Social Science Framework, innovative multi-dimensional incentives and nudging interventions for residential DR	SIN, ICONS, ISRAA, DOM-X, pilot partners	Open Source	Training, Consulting
2	AI-based individual and building-level-level consumers clustering and market segmentation algorithms	HERON, NTUA, TUC, SIN	Open Source/fee/licencing	Training, Consulting, Extension
3	Open APIs for DR-ready smart appliance IoT platform	ARC, DOM-X	Open Source/fee/licencing	Training, Consulting, Extension
4	Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange	ENG, COMS, TUC, CARTIF	Open Source/fee/licencing	Training, Consulting, Extension
5	Digital Twins for optimal assets vs consumers flexibility planning and decentralised DR management	ENG, CENTRICA, HERON, NTUA	Open Source/fee/licencing	Training, Consulting, Extension
6	Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments	UNIVPM, ISRAA	Open Source/fee/licencing	Training, Consulting, Extension
7	Electrical flexibility management tools for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets	OurPower, BLUEPRINT, COMS, TUC, HERON, NTUA, CARTIF, CIMNE	Open Source/fee/licencing	Training, Consulting, Extension
8	Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralised cross-commodity electricity vs heat management	NEOGRID, DOM-X	Open Source/fee/licencing	Training, Consulting, Extension
9	Business Sandbox with novel sharing economy and social innovation-based DR business models	All partners	Open Source	Training, Consulting

For the joint exploitation of the project's results, DEDALUS will use the Communities from BD4NRG, BRIGHT, MATRYCS, ENERSHARE and DigiBUILD projects to bring together the necessary industry players and realise an Energy Data Ecosystem in which organisations and individuals can unlock the full value of their data for increasing energy system flexibility.

The communities will be further upgraded during the DEDALUS lifetime, offering additional services through the DEDALUS integrated consumer-centric DR system: (1) Access to data analytics, Digital Twins, data-driven flexibility/DR services for residential individual and building-scale consumers and (2) Training, consulting, and technical support services. These services will be offered in the form of complementary assets to the KERs listed above.

Exploitation of the applications/services: The services/applications validated and replicated in 7 countries will provide early showcases of the project's functionalities, and they will be gradually advanced against maturity and market readiness towards a viable route to market.

Individual partners exploitation plans: An alternative method of capitalisation on the DEDALUS's outcomes for all its partners is by internal exploitation the expertise and know-how they will gain on the exploitation of new market structures and business model design for a residential DR. The development of detailed exploitation plans by each individual partner, in-line with each partner's business and research strategy. In most cases, these plans concern the improvement of existing products and services of the partners.

Go-to-Market Strategy: The commercialisation of the DEDALUS related products and services will be realised in three phases, including post-project exploitation, and may be aligned with incremental patterns for joint exploitation

- **A pilot and (business) validation phase**, which will be carried out as part of the project's work plan (M31-M36). It will validate the initial services from both a technical and a business perspective, while at the same time

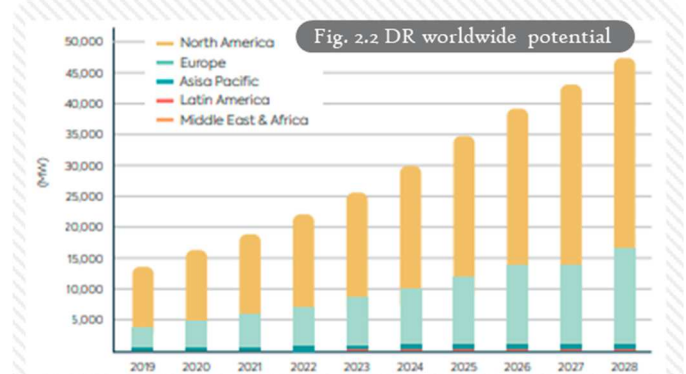
establishing a detailed business plan. 100+ additional participants (incl. both consortium members, participants to the background projects of the partners) will be registered to the project's ecosystem.

- **A market launch phase (M36-M60)**, which will start at the end of the project, implementing the established business plan and attracting the first customers (i.e. service/solution providers outside the consortium). Key to this phase will be the proper exploitation of the combined DEDALUS results within the partners' business networks. At the end of this phase, there will be 250+ additional participants to the project's ecosystem. This phase aligns with DEDALUS playing the role of technology/solution/service providers.
- **A global expansion phase (beyond M60)**, where DEDALUS will become the world's top ecosystem for advanced consumer-centric energy systems. As part of this phase, the emphasis will be in global networking and attraction of 1,500+ participants.

2.2.2.1. MARKET ANALYSIS

Residential DR is an emerging market in Europe, with an estimated 1.3GW of assets already contracted today. France accounts for much of this success, with the leading provider Voltalis boasting 900MW, while the Nordics, the Netherlands, Germany and Switzerland trail behind. The residential sector has huge untapped potential with over 200GW of installed assets of which HVAC and hot water heaters account for 97 per cent. However, accessing these assets for DR is challenging; less than one per cent of all existing residential electric HVAC and hot water assets have inbuilt connectivity. International Energy Agency [46] estimates smart DR could provide 185 GW of system flexibility, roughly equivalent to the currently installed electricity supply capacity of Australia and Italy combined. This could save USD 270 billion of investment in new electricity infrastructure that would have otherwise been needed. In the residential sector alone, 1 billion households and 11 billion smart appliances could actively participate in interconnected smarter electricity systems, allowing these households and devices to alter when they draw electricity from the grid. Indeed, DR has a large market potential on a global scale, with Europe and North America being the two largest markets up to 2030 [47]. The EU's DR potential, for example, is estimated to be up to 57 GW of small-scale flexibilities brought to markets by 2030, compared to a BAU of 30 GW (Fig. 2.2) [48].

According to the same report, there are certain lessons learnt when it comes to the market size and the impact of digitalisation on the consumer engagement and DR: (1) Digitalisation can help integrate variable renewables and DERs by enabling grids to better match energy demand to times when the sun is shining, and the wind is blowing.



In the EU alone, increased storage and digitally-enabled DR could reduce curtailment of solar photovoltaics and wind power from 7% to 1,6% in 2040, avoiding 30 million tonnes of carbon dioxide emissions in 2040; (2) rolling out smart charging technologies for electric vehicles could help shift charging to periods when electricity demand is low and supply is abundant. This would provide further flexibility to the grid while saving between USD 100 billion and USD 280 billion in avoided investment in new electricity infrastructure between 2016 and 2040; (3) digitalisation can facilitate the development of distributed energy resources, such as household solar PV panels and storage, by creating better incentives and making it easier for producers to store and sell surplus electricity to the grid. New tools such as blockchain could help to facilitate peer-to-peer electricity trade within local energy communities. However, the DR and flexibility service provisioning has slowed down in the last years worldwide [54].

Beyond the above-mentioned need for increased security of supply due to the current geopolitical situation with Russia, which is putting a strong pressure on the decentralised optimal management of newly electrified demand via local consumers engagement, a strong enabler for DR revamping is definitely the flourishing of the Data Economy.

The power of data and data sovereignty is clearly on the rise, as reflected in the Data Strategy of the EC for data sharing and support of European Data Spaces [49]. EC DG Energy has made some provisions (€32 million) to support the concrete implementation of the European Common Energy Data Space, through a targeted H2020 Call (deadline in October 2021), where 4 sister projects have been retained for funding. ENERSHARE (ENG, NTUA, COMS, SIN as common partners) is one of these projects and will represent the technological workbench to enable effective DR for residential buildings to be deployed successfully. It is worth mentioning that the EU27 data market value is expected to grow by 10,7% compared to 2019, to reach €83 billion by 2025, whereas the Value of Data Economy shows a similar pattern and will reach a value of €432 billion [50]. There are a number of SMEs that provide platform services that help energy communities visualise how electricity is being produced and consumed within the community, and some also show how they could optimise the renewables mix at the household or community level, and how local flexibility trading would impact their economic and environmental objectives. This also opens up

opportunities for community members to participate and feel more engaged. Such an interface could give community members insight into how their assets and those of the community energy assets are performing. The highly promising market trend for data sharing and data exchange is rapidly creating an enabling ecosystem and framework which facilitates evolving towards a DR community/consumer-based Data Space environment, which, in accordance with the most recent European Data Strategy, will boost a high number of SMEs developing data-driven energy and non-energy services to support active and informed consumers to participate to energy markets, eventually via data-driven DR.

2.2.2.2. INITIAL BUSINESS MODEL AND BUSINESS PLAN

In order to better describe the initial business model approach for DEDALUS, a segmentation based on the ways potential customers will use the planned KERs is preferable.

Table 2.8. Business Model

<u>PARTNERSHIPS</u>	<u>KEY ACTIVITIES</u>	<u>VALUE PROPOSITION</u>	<u>RELATIONSHIPS</u>	<u>CUSTOMER SEGMENTS</u>
Residential Energy Consumers Building Operators/Managers Building Owners Building Tenants Social Housing Operators Cross-sector Building/District-scale Energy Communities Aggregators/Flexibility Service Providers/DR Managers Suppliers/retailers Cross-sector flexibility Providers (district heating network operators, water network operators) Cross-sector Service Providers (Care/Wellness Service Providers for elderly people) System operators (TSO, DSO) CT Service/Technology Providers/Integrators/Data Space Developers Smart Home/Automation Solution/Service Providers	Major exploitable assets (see Tab 2.7) <u>KEY RESOURCES</u> Qualified employees R&D Capabilities Access to use cases & demos and relevant datasets, services, energy assets/services Multiplier effect provided by partners corporate strategies	Value capture for data-driven tools and energy services for cross-domain/domain-agnostic residential DR Deploy Digital Twins for flexibility planning/DR Build profitable flexibility/DR services for cross-stakeholders/cross-commodity/cross-value chain flexibility management	Local Technical Assistance Open, collaborative community ecosystem building Training Sessions & API Documentation/Best Practices <u>CHANNELS</u> Web based services Open APIs Apps repository	Building Owners/Tenants as Active energy consumers and energy communities Building Management Value chain B2B Stakeholders Energy and non-energy value chain B2B Stakeholders/ICT/Smart Home/Automation Service & Solution Providers Other assets providers
<u>COST STRUCTURE</u> Economy of scale Fixed Costs Personnel, marketing, lab Variable Costs Promotion of web & mobile apps; Promotional offers; Wages on customer support		<u>REVENUES</u> Access membership flat Fees and Pay-per-transactions, Revenues share from Third party developers, Infrastructure Providers, Models/Algorithms Developers Selling of software licences for the usage of tools and services, models and algorithms		

Towards this direction, the main identified customer segments are: (i) Individual residential energy consumers, eventually organised through collective forms in energy communities, willing to participate in electrical and/or heat and or cross-domain DR programmes and/or gaining benefits from the self-consumption optimisation and/or flexibility provisioning to the electricity grids and/or district heating network.. They could be interested to share their flexibility and their energy consumption/flexibility data to get beyond financial and beyond financial compensation, not only from DR participation but eventually also from data sharing; (ii) B2B building value chain stakeholders, including building owners, building operators/managers, Building Management Systems Technology/Service Providers, Social Housing Operators, Real Estate Stakeholders, interested to utilise and integrate some of the DEDALUS KERS in their tools/ICT systems and/or service offering, with a view to make available their building- or district-scale flexibility to participate to DR programs through aggregators); (iii) B2B energy and non-energy (es elderly care management) value chain stakeholders/utilities, including aggregators, retailers, DSOs, electricity traders, suppliers, interested to make use and leverage a number of KERs to deploy energy and/or non-energy services for residential DR; (iv) ICT & Smart Home/automation service/technology providers/integrators, and innovative SMEs Apps developers which access energy data from consumers, ML learning models/algorithms to develop third party data-driven analytics digital tools/services on top of virtualised automation hardware/devices; (v) Other asset providers (e.g. cloud providers, ML models/algorithms, SSH methodologies). All the identified customers groups will have a two-way interaction with the DEDALUS ecosystem and hence they can be considered at the same time as ‘demand’ and ‘supply’ customers for the potential venture.

As for the revenue streams, they can derive from combining digital tools service licencing, offered to aggregators, utilities, building stakeholders, smart appliance manufacturers, and revenue sharing from commercial agreements with third party external developers (or a combination of the above). The pricing models could be a combination of subscription, pay-per-transaction (pay-per-usage), or a combination. The business model (see Table 2.8 from CANVAS model), especially the pricing part will be tested and revised in the due course of the project, mainly in order to adapt on up-to-date real-life conditions.

2.2.2.3. INDIVIDUAL EXPLOITATION PLANS

The consortium has already identified the different categories of stakeholders that can take advantage of the DEDALUS results and services.

Table 2.9. Individual Exploitation Plans

Category	Partners	Exploitation plan
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Building & social housing operators	FællesBo, ISRAA, GreenPoint	They will widely use the project's results in order to improve their business performances and open new business opportunities. More specifically, the integration of already operational tools and technologies with the project's data governance, energy analytics tools, as well as the offering of new services and intuitive analytics constitute additional targets.
Energy Value chain stakeholders (DR Aggregators/Solution/Service Providers, Energy Community Managers, Other Energy Stakeholders)	CENTRICA, HERON, OurPower, DYNAM, PEUSA	Independent Aggregators (CENTRICA), and other utilities, including retailers, suppliers, traders, energy community developers/managers, power network operators are mostly interested to validate new building-scale aggregation approaches with a view to widen the pool of controllable residential assets, design new DR programs tailored to attract residential sector to a massive participation to DR.
Smart Appliance Manufacturers and Smart Home/RES Assets Integrator Providers	ARC, ROMUR	They are interested to leverage on DEDALUS principal outcomes to widen their respective commercial offer towards flexible and DR-ready white goods, smart home solutions and towards smarter renewable assets.
Large IT service & tech providers, Third Party SMEs Data-driven DR Service Developers/Providers	ENG, BLUEPRINT, COMS, DOM-X, NEOGRID	The industrial partners of the consortium are mainly interested in the commercial exploitation model, providing the project results to the end users. They are interested in the technological exploitation model, delivering products and/or services built on top of the project results.
Policy makers and SSH stakeholders	GREEN CITIES, SIN, ICONS	They will be interested to promote the demonstration of consumer-centric business models in their regions and engage citizens. They can capitalise on the project's results and best practices identified, to design European, national, or regional policies.
Leading research and technological development institutions	NTUA, CARTIF, UNIVPM, TUC, CIMNE	They are mainly interested in the research exploitation model, aspiring re-utilisation of the research know-how acquired in future research activities. From a commercial exploitation model, they are interested in founding spin-offs and start-ups to commercially exploit the developed research results.

2.2.3. MANAGEMENT OF INTELLECTUAL PROPERTY

Open-Source Licenses Handling: While a plethora of components and modules (mainly coming from the academic institutions and research organisations) are provided open source, a set of the results are based upon proprietary/legacy components and modules that are not provided open source. The consortium will follow a multi-level approach towards managing the knowledge produced: The solutions will be copyright protected using a licensing scheme that will not violate the terms and conditions of the discrete components comprising it; The components that can be provided open source will be delivered under such a license (e.g. CC-BY); The respective knowledge producer holds the rights to the knowledge they have produced, while sharing the software implementation with the rest of the consortium. We will examine the software licenses of all algorithms, components, and modules to be used, and decide under which license (or combination of licenses) the solutions will be released. For the components and modules that can be delivered open source, the Apache Software License (ASL) seems to be the prominent candidate, as it allows the redistribution of the program's source code in any form (compiled binary or plain text) without posing any limitations to the distributor. With regards to the components and modules that cannot be delivered open source, they will be copyright protected but of course will be freely available to consortium members.

IPR Management at Consortium Level: Dissemination and use of knowledge generated in DEDALUS is governed by the terms of the Grant Agreement and the terms of the Consortium Agreement. Regarding ownership of the foreground, it is agreed that foreground will be the property of the beneficiary carrying out the work generating that foreground. Beneficiaries will also define the background needed for the purposes of the project development in a written agreement and, where appropriate, may agree to exclude specific background. In order to make sure that these terms are followed, to avoid disputes and to facilitate business planning, the Management Board will maintain an IPR Directory throughout the lifetime of the project. This document will list all items of knowledge relating to the work of the project (both background know-how and results developed in the project), and make explicit for each item its owner, nature, status and dissemination and protection measures. An initial version of the IPR directory will be created at the start of the project. The basic principle on which we are agreed from the proposal stage is that research and development results must be available to a large audience to facilitate wide adoption of project results, while in the meantime having options in place for rewarding those that invested.

Patents Landscape Analysis: The project needs to consider whether the project's outcomes can be exploited without

infringing on existing patents claims and if its innovations can be patented. A preliminary analysis of the patents indicates that no patents/applications address a consumer-centric energy system which is our main innovation.



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2.3. SUMMARY

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Challenge 1: Need for strengthening the human connection and interaction, awareness and preparedness of the residential consumers on the opportunity to value their flexibility and DR participation Challenge 2: Lack of and immaturity of incentive and appropriate financial and/or non-financial compensation mechanisms hinders the unleashing of the residential flexibility potential. Challenge 3: Need for exploring a multi-carrier (or energy carrier-agnostic) DR approach, and consider a number of synergies and interaction scenarios from heat and/or gas energy carriers towards electricity, while considering residential buildings as the appropriate level where such synergies may take place. Challenge 4: Besides the lack of technical interoperability, the main problem consists of having smart appliances still not DR-ready. Challenge 5: Energy Data Space can be seen as an emerging opportunity to streamline personal energy consumption data sharing in more interoperable, secure and trusted way. Challenge 6: Flexibility categorisation and clustering of buildings has not been sufficiently explored and building-level pre-aggregation suffers from insufficient and rather immature residential building energy (electricity and heat) flexibility profiling and planning.	Result 1: Social Science Framework, innovative multi-dimensional incentives and nudging interventions for residential DR (KER1) Result 2: AI-based individual and building-level-level consumers clustering and market segmentation algorithms (KER2) Result 3: Open APIs for DR-ready smart appliance IoT platform (KER3) Result 4: Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange (KER4) Result 5: Digital Twins for optimal assets vs consumers flexibility planning and decentralised DR management (KER5) Result 6: Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments (KER6) Result 7: Electrical flexibility management tools for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets (KER7) Result 8: Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralised cross-commodity electricity vs heat management (KER8) Result 9: Business Sandbox with novel sharing economy and social innovation-based DR business models (KER9)	Exploitation: Partners will integrate results within their existing products or services and for the creation and offering of new ones (TE-1 - TE-14, Section 1.1.3). Open-source results will be exploited by any interested stakeholder. Dissemination towards the scientific community and energy, mobility, ICT, water and wellness industry: Scientific publications (also to the Open Research Europe platform), Participation in conferences and events, Liaison and cross-fertilisation with stakeholders, projects, the BRIDGE and other initiatives; Set up of Stakeholders Hub, Factsheets; direct contributions to standardisation working groups (ISO/IEC, ETSI, etc.) Communication towards citizens: Digital presence (web site, social media, and inclusion in e-newsletters); Organisation of events and workshops in the pilot sites; Publications for different types of general audience (in different languages where appropriate).

TARGET GROUPS	OUTCOMES	IMPACTS
Building stakeholders 3 <u>early adopters</u>: FællesBo (social housing company); ISRAA (public sector care provider); GreenPoint (building operator) Energy stakeholders 4 <u>early adopters</u>: CENTRICA (DR aggregator); PEUSA (DSO/electricity trader); HERON (producer, energy supplier and electricity trader); OurPower (energy community); DYNAME (energy trader and broker) Providers & operators of DR-ready relevant assets 2 <u>early adopters</u>: ARC (appliance manufactures), ROMUR (BESS integrator) SSH stakeholders 3 <u>early adopters</u>: SIN, ICONS, as well as ISRAA team Policy makers 1 <u>early adopter</u>: GREEN CITIES ICT/Energy/Buildings service providers 5 <u>early adopters</u>: ENG, BLUEPRINT, COMS, DOM-X, NEOGRID Scientific community Researchers in the buildings, ICT and energy sectors (NTUA, CARTIF, UNIVPM, TUC, CIMNE) All residential energy consumers (e.g. electricity, DHN, gas) & communities	Validated use cases available and synchronised with DG BRIDGE Energy Data Management (Interoperability of flexibility assets Task force)with GAIA-X Energy Vertical WGs Uptake of Energy Data Space Technology Enablers as Open-Source components in GitHub by at least 50 Data Service Providers/App Developers Uptake of data-driven algorithms, models, tools and services for residential DR by: (i) 10 aggregators/supplier/retailers/ESCOs (e.g. EDP Commercial, PPC, ACEA Energia, Umbria Energy, ENEL-X, etc.) (ii) 5 Buildings Operators (VEOLIA, ACCIONA, ENGIE, etc.), 10 Energy Communities (eNostra, Energy@Work, etc.) (iii) 5 Smart home automation/BMS/Smart Asset Manufactures& Integrators/(e.g. Siemens, Honeywell, etc.) (iv) 5 DSOs (ASM Terni, ARETI, HEDNO, etc.) Cross-stakeholder services and Digital Twins uptake by at least 7 stakeholders (e.g. smartEn, EDSO, REScoop.eu, BEUC, ICLEI, ECTP, Housing Europe, EU Building Stock Observatory), interested to adopt DEDALUS best practices and blueprints.	Scientific: New scientific discoveries will consist of: (a) Finer-grained learning data models, data-driven energy services and Digital Twins to mobilise DR in energy-efficient residential buildings, by leveraging on a large share of DR-interoperable assets (b) A Social Science framework that leverages on a combination of deliberative citizen engagement methodology, intersectionality, co-creation and experimental scenario-based research, as well as Design Thinking methodologies, to bring residential energy consumers center stage in the energy markets. Economic: (i) 15% increased piloted buildings energy efficiency and reduced energy bills for residential consumers due to DR programs (ii) 15% cost reduction of the overall energy system (iii) >25% reduction of DR transaction costs after 5 years Technological: (i) Harmonise and extend existing automation-oriented standards (e.g. Family if IEC standards), business DR standards (OpenADR2.0), and languages (EFI) (ii) 12 DR-ready smart residential asset types after 3 years (iii) >=7 new aggregation models and data-driven services and >=2 Digital Twins to facilitate consumer activation and market participation Societal: (i) > 500 consumers activated in residential DR programs after 3 years (ii) 45% increased share of energy consumption data by active consumers after 5 years (iii) 15-20% increased share of aggregated flexibility mobilised by piloted assets (iv) >18-20% carbon emission reduction in the areas of the 7 pilots/multipliers due to increased participation of residential consumers to DR programs

SECTION 3: QUALITY & EFFICIENCY OF THE IMPLEMENTATION

3.1. WORK PLAN AND RESOURCES

The DEDALUS narrative is made feasible through a work plan (Fig. 3.1) of **7 Work Packages (WPs)**, as presented in Table 3.1. We integrate effective and interactive SSH-driven frameworks and methodologies, guided by stakeholder knowledge and experience, with novel data-driven methods and tools. Thus, impact-driven, ready-to-use knowledge is made possible by an interdisciplinary ‘needs-based’ approach to research and technological activities.

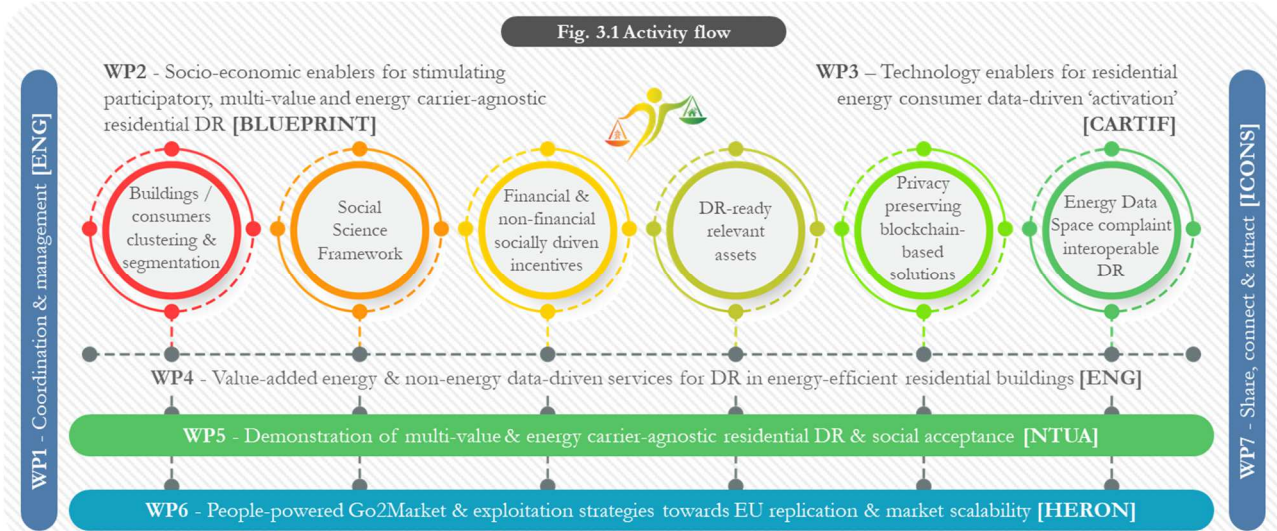


Table 3.1: Structure of the work plan and its relation to the DEDALUS Objectives (Obj.) and Key Exploitable Results (KER)

	Obj.	KER	Short description
WP1	All	All	WP1 will guarantee the adequate development of the project and will tackle both scientific and administrative coordination measures.
WP2	Obj. 1	KERs 1-2	WP2 will establish the “user”-oriented basis for the development of the DEDALUS socio-economic and technological framework, providing groundwork for energy market, consumers’ analysis, flexibility categorisation and clustering of buildings. We ultimately achieve a co-creation and co-implementation process for the design of financial/non-financial incentives mechanisms to promote demand flexibility in energy-efficient residential buildings.
WP3	Obj. 2-3	KERs 3-4	WP3 will deliver “asset”-oriented technology enablers (DR-ready smart appliance IoT platform, data-driven Energy Data Space compliant interoperability layer and DLT/Blockchain solutions), in order to deliver a data-driven, decentralised and privacy preserving DR environment, as well as facilitate the integration of smart devices and other flexibility sources present in residential buildings within the local business platforms.
WP4	Obj. 4	KERs 5-8	Leveraging from the WP3 data-driven environment, WP4 aims to implement, suitable value-added energy and non-energy data-driven consumer-aware tools, applications and services for human-aware DR in energy-efficient residential buildings.
WP5	Obj. 5	All	The resulting DEDALUS solutions (WPs 3-4) will be demonstrated, validated and replicated in 7 real-life pilots (Front Runners and Multipliers). Emphasis will be given on the engagement of residential energy (electricity, DHN, gas) consumers and social acceptance assessment.
WP6	Obj. 6-7	KER 9	In WP6, a Business Sandbox with novel sharing economy and social innovation-based business models will be delivered. On the base of pilot evaluation results and lessons learnt (WPs 2-5), we will develop exploitation plans to create a concrete methodology and define the necessary steps towards sustainability. A roadmap on the replicability and scalability at EU level will be delivered. The DEDALUS demonstration and validation activities take place through a three-loop structure: the Pre-pilot Phase (M14-M16), the Full Pilot Operation Phase (M21-M25) and the Large Scale Pilot Operation Phase (M30-M33).
WP7	7	All	As part of the horizontal activities, WP7 sets the bases for the communication and dissemination of the DEDALUS results/findings. WP7 includes also alignment with relevant European projects and initiatives (e.g. BRIDGE) on consumer and citizen engagement, as well as the DEDALUS capacity building programme.

TIMING OF DIFFERENT WORK PACKAGES AND THEIR COMPONENTS

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
WP1 Coordination and management	ENG																		R																		R
T1.1 Project administration and coordination	ENG																																				
T1.2 Technical and scientific management	ENG		1.1																1.3																		1.4
T1.3 Quality control, risk management and Transdisciplinary Advisory Board	ENG																																				
T1.4 Open science, FAIR data and data management	ENG						1.2																														
WP2 Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR	BLUEPRINT																																				
T2.1 Gap analysis and use case capitalisation for DR in the residential sector	BLUEPRINT						2.1																														
T2.2 DEDALUS Community building and scenario co-creation for residential DR	SIN																																				
T2.3 Consumers segmentation and clustering of buildings for optimal DR planning	NTUA									2.2								2.4																			
T2.4 Co-designing with consumers innovative social-driven incentive mechanisms and data-driven DR services	DOM-X																																				
T2.5 Residential consumers business/privacy/cyber-security requirements and functional specifications	ENG										2.3																										
WP3 Technology enablers for residential energy consumer data-driven ‘activation’	CARTIF																																				
T3.1 DR-ready smart device and flexible resource data collection	ARC																																				
T3.2 Energy Data Space compliant interoperability layer	ENG																																				
T3.3 DEDALUS flexibility and DR data models and standards	CARTIF																																				
T3.4 Multi-energy flexibility modelling for data-driven residential DR	NTUA													3.1						3.2									3.3								
T3.5 Blockchain-based interoperability for flexibility tokenisation & DR settlement	TUC																																				
T3.6 Privacy, security and data protection framework for trusted data sharing and AI	COMS																																				
WP4 Value-added energy and non-energy data-driven services for DR in energy-efficient residential buildings	ENG																																				
T4.1 Digital Twins for optimal assets vs consumers flexibility planning	ENG																																				
T4.2 Optimal trade-off among comfort, energy and building performance	UNIVPM																																				
T4.3 Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers	DOM-X																																				
T4.4 Local flexibility management for energy communities and self-consumption/cost optimisation	OurPower																																				
T4.5 Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorisation	NEOGRID																																				
WP5 Demonstration of multi-value and energy carrier-agnostic residential DR, and social acceptance	NTUA																																				
T5.1 Evaluation framework, baseline analysis and pilots’ preparation	NTUA																																				
T5.2 Flexibility pre-aggregation for building-level energy community (Austria)	OurPower																																				
T5.3 Energy efficiency and DR optimal interactions between district heating and electricity networks in social housing (Denmark)	FællesBo																																				
T5.4 Multi-value DR for multi-segments of residential energy (electricity and gas) consumers (Greece)	HERON																																				
T5.5 Trading off comfort with energy consumption in DR programs for elderly people living in serviced apartments (Italy)	ICCRA																																				
T5.6 District-level DR for increasing community-level self-consumption while providing flexibility to local DSO (Spain)	CIMNE																																				
T5.7 DEDALUS replication across Front Runners and Multipliers	TUC																																				
T5.8 Consumers engagement, social acceptance assessment and impact analysis	SIN																																				
WP6 People-powered Go2Market and exploitation strategies towards EU replication and market scalability	HERON																																				
T6.1 DEDALUS sharing economy and social innovation-based business models	HERON																																				
T6.2 Contribution to standardisation	CENTRICA																																				
T6.3 Go2Market strategy: Market impact	HERON						6.1																														6.3
T6.4 Go2Market strategy: Innovation management and exploitation	ICONS																																				
T6.5 Roadmaps for replication strategies beyond DEDALUS	GREEN CITIES																																				6.4
WP7 Share, connect and attract	ICONS																																				
T7.1 Dissemination and communication strategy	ICONS						7.1																														
T7.2 Enable public outreach and wide communication	ICONS									7.2																											

Distribution of costs: DEDALUS plans a total of **959 PMs** effort over a **36 months' period** and has a calculated total budget of € **7.359.075** for an EC funding request of € **5.999.801,25**. The budget includes the following costs: Personnel Costs (€ **5.444.660** | 74%); Other Direct Costs (€ **442.600** | 6%): € **174.400** for travel and subsistence, € **111.000** for equipment (hardware/infrastructure/software) to support the operation of the DEDALUS services and pilot implementation, € **157.200** for consumables and the organisation of the project meetings, co-creation process, capacity building workshops and other project events, publications, Certificate Financial Statements (CFS); Indirect Costs (€ **1.471.815** | 20% of the total budget).

Distribution of efforts:

- DEDALUS socio-economic enables [WP2 | 13%], with particular emphasis on the Social Science Framework, multi-dimensional incentives and nudging techniques (**KER1**), the individual and building-level consumers clustering (**KER2**), as well as the co-creation process.
- Innovative technology enablers for residential energy consumers 'activation' and DR aggregation, including Open APIs for DR-ready smart appliance IoT platform (**KER3**) and Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange (**KER4**) [WP3 | 17%], as well as data-driven DR services (**KERs 6-8**) and DTs (**KER5**) [WP4 | 18,1%].
- Demonstration, replication and validation activities [WP5 | 26,7%], outlining the importance that has been given by the partners to demonstration and replication.
- Business modelling, standardisation and exploitation activities [WP6 | 9,5%], with a view to deliver a Business Sandbox with novel sharing economy and social innovation-based DR business models.
- Dissemination, communication, networking and capacity building [WP7 | 9,3%], and management activities [WP1 | 6,3%, in line with traditional approach for EU projects].

All depreciation costs for equipment, infrastructure or other assets in the project are in compliance with Article 6 and will be recorded in the appropriate beneficiary's accounts, purchased in accordance with Article 6.2.C of the grant agreement and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.

Table 3.2g. 'Purchase costs' items (€)

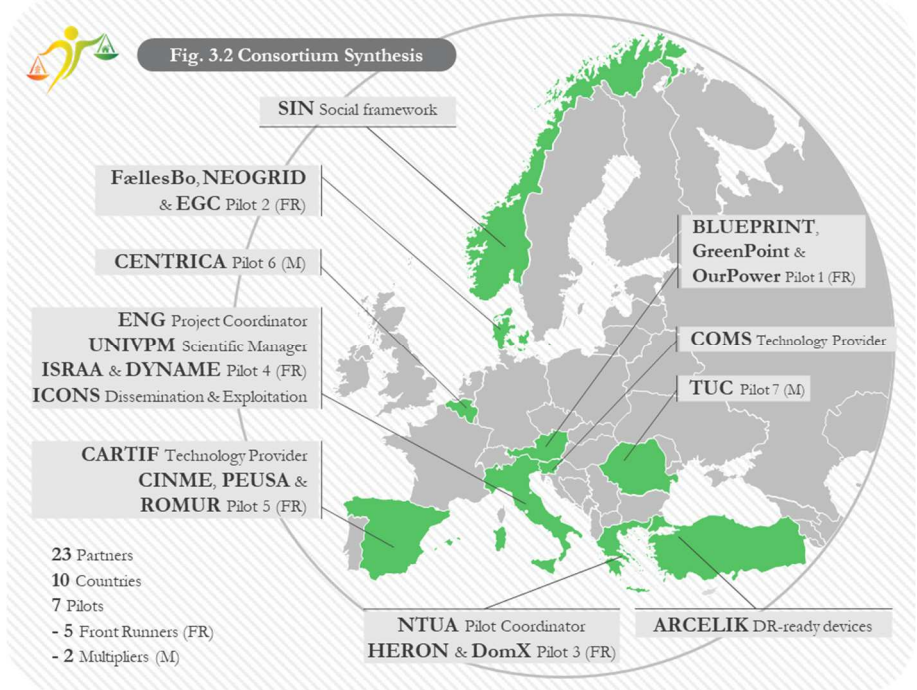
19. ROMUR	26.000	Justification
Other goods, works & services	19.500	Consumables (electronics, etc. to develop the heating components for the Spanish pilot)
Travel & subsistence	6.500	€ 3,000 for participation in project meetings (1 person, at least 3 meetings during the project and average € 1000 / person); € 2,000 for participation in Review meetings (1 person, 2 review meetings during the project and average € 1000 / person); € 1,500 for participation to events related to the project's activities
20. CENTRICA	61.000	Justification
Equipment	45.000	€ 30,000 for more accurate metering equipment (1.000 euro per heat pump x 30 housing units) € 15,000 for local steering box for heat pumps (500 euro per heat pump x 30 housing units)
Travel & subsistence	9.500	€ 4,000 for participation in project meetings (1 person, at least 4 meetings during the project and average € 1000 / person) € 2,000 for participation in Review meetings (1 person, 2 review meetings during the project and average € 1000 / person); € 2,000 for participation to standardization meetings € 1,500 for participation to events related to the project's activities
Other goods, works & services	6.500	€ 3,500 for Open access publications; € 1,000 for Co-creation events; € 1,000 for pilot local events for energy consumers engagement; € 1,000 for participation in external conferences, printing and other promotional activities

3.2 CONSORTIUM AS A WHOLE

DEDALUS consists of **23 partners** from **10 Countries** (8 Member States and 2 associated countries), including leading ICT technology providers, universities and research institutions, SMEs, consumers representatives and a large variety of building and energy and non-energy stakeholders, as well as networks advancing policy innovation and thought leadership within the energy community (Fig. 3.2).

Consortium synthesis draws upon its members' expertise at the interplay among energy, buildings, smart technologies and devices, Artificial Intelligence, Data Spaces and policy research. The consortium will also value evidence stemming from the

inter- and transdisciplinary integration of expertise, knowledge co-production, open science, and insights from SSH, which comprise key dimensions of the project. The consortium has the capability to organise the whole value chain of innovation up to the release of the project centred in average at the required **TRL 6-7**.



Building Stakeholders

FællesBo is a social housing company, including tenants owned cooperatives with around 5.000 apartments (2.000 apartments are planned to be renovated). FællesBo will be responsible for the Danish pilot (**T5.3 leader**), acting as administrator between the district heating company (Verdo Varmer Herning) and the individual household; **ISRAA** is a public sector care provider, providing full support to around 900 citizens in later life stages that live in 4 nursing serviced apartments located in Treviso, and home-care services to roughly 1.000 citizens living in their own houses in and around Treviso. ISRAA will be responsible for the Italian pilot (**T5.5 leader**); **GreenPoint** is building owner and operator of the residential building where the Austrian pilot will be demonstrated (**T5.2**), where OurPower plays the role of energy supplier for all the tenants and energy community developer and manager.

Energy Stakeholders

CENTRICA is a leading European independent DR aggregator specialised in automated DR, and Executive Member of smarten (<http://www.smart.eu>). CENTRICA is responsible for the Irish multiplier pilot (**T5.7**) and will lead the standardisation activities (**T6.2 leader**). CENTRICA will also contribute to flexibility modelling (**T3.4**), DTs (**T4.1**) and decentralised power2heat management (**T4.3**); **PEUSA** manages the LV electricity network of several villages around La Seu d'Urgell. PEUSA has a trader unit and is also promoting collective PV self-consumption installations. These installations are supported by massive indirect incentivised DR actions, automatically delivered by opt-out messaging services. In this context, PEUSA will strongly support the Spanish pilot implementation (**T5.6**); **HERON** is a group of companies engaging in the production, supply and trading of electricity with a clientele of over 260.000 and access to 1,5 GW installed RES capacity through TERN Energy. HERON will be responsible for the Greek pilot (**T5.4 leader**), upscaling its smart metering platform (TE-11, as part of **T4.3**) and the DT 'RES everywhere every time for residential consumers' (TE-8, as part of **T4.1**). HERON will also contribute to the consumers segmentation and clustering of buildings (TE-1, as part of **T2.3**) and is leading the people-powered Go2Market strategy (**WP6** and **T6.3 leader**); **OurPower**, one of the pioneers of energy transition in Austria, is an energy cooperative which operates a P2P marketplace for energy demand-supply matching and manages an association of 12 energy communities. OurPower will be responsible for the Austrian pilot (**T5.2 leader**), upscaling (**T4.4 leader**) its P2P digital platform (TE-12) and Nudging app (TE-10); **DYNAMIE**, as a leading-edge innovative energy trader and broker interested to enter in the flexibility market, participate to the Italian pilot (**T5.5**).

Smart Appliance Manufacturer & Smart Home/RES Assets Integrator Provider

ARC, as smart appliances manufacturer, will be responsible for the DR-ready smart appliance IoT platform (**T3.1 leader**), upscaling its HomeWhiz Platform (TE-2) by adding DR capabilities. ARC will support the Greek and Spanish pilot case (**T5.6** and **T5.6**), providing connected home appliances to show interoperability and flexible

remote-control options; **ROMUR**, acting as BESS integrator and IoT solution provider for BMS, will contribute to the Spanish pilot (T5.6).

SSH partners

SIN, as a renowned distributed research center with a leading-edge focus on SSH, will be responsible for the SSH-based activities in DEDALUS, coordinating scenario co-creation for residential DR (T2.2 leader) and social acceptance assessment (T5.8 leader). **ICONS**, as a leading pioneering organisation combining science communication/dissemination, social and business innovation, will be responsible for the dissemination and communication activities (WP7 leader), as well as innovation management (T6.4 leader).

Policy making

GREEN CITIES, as a European network of cities, social housing and consulting companies, will carry out European level replication (T6.5 leader). **GREEN CITIES** will also support FællesBo and NEOGRID in carrying out the demonstration activities (T5.3) and being responsible for the DK replication and dissemination activities.

Large ICT company

ENG, leading on advanced ICT and data management technologies (among the top 10 IT groups in Europe), will act as Project Coordinator (WP1 leader), bringing its experience from various previous related projects. **ENG** will be in charge for the value-added services and DTs (WP4 and T4.1 leader), as well as Energy Data Space compliant interoperability layer (T3.2 leader) and Blockchain-based solutions (T3.5), including TE-3, TE-5, TE-8 and TE-12 upscaling. **ENG** is also leading the networking and collaboration with fellow projects and initiatives and cooperation with **BRIDGE** (T7.4 leader). **ENG** actively support the pilot activities of the Austrian (T5.2), Danish (T5.3) and Italian (T5.5) pilots, as well as multiplier pilots (T5.7).

SMEs (Energy and non-energy service providers)

BLUEPRINT, an Austrian engineering and ICT services company, leads WP2 (T2.1) and support OurPower in technology adaptation (TE-10 and TE-12, as part of WP4), and pilot implementation (T5.2); **COMS**, as software developer with proven experience in data collection and data analysis, will be responsible for privacy, security and data protection (T3.6 leader), upscaling the Universal Secure Gateway (TE-6) and blockchain-based Privacy Protection Enforcement component (TE-7). **COMS** will contribute to the technological adaptations for Blockchain/Smart contracts solution (TE-5, as part of T3.5) and P2P digital platform for trading pre-aggregated building flexibility (TE-12, as part of T4.4). **COMS** will be actively involved in the Austrian pilot (T5.2); **DOM-X**, as an innovative IoT start-up offering smart and flexible natural gas boilers retrofitting solutions, will be actively involved in the Greek pilot (T5.4) and will be in charge for the multi-vector DR management platform and nudging app (TE-10, T4.3 leader). **DOM-X** will be also responsible for co-designing data-driven DR services and nudging interventions (T2.4 leader) and support the activities for interoperability (T3.2) and DR-ready relevant assets (T3.1); **NEOGRID** is a developer and provider of data-driven energy efficiency and flexibility services for heat commodity and district heating. **NEOGRID** will upscale its product PreHEAT (TE-14) towards the optimal aggregation of flexible energy resources for district heating management (T4.5 leader) and will support FællesBo in the information and follow up activities towards the tenants, who have had installed the DR equipment (T5.3).

Leading research technology & academy institutions

NTUA (participating through the Decision Support Systems Lab) has strong experience on AI-based algorithms for flexibility aggregation, DR and other energy applications. **NTUA** is undertaking effort with a view to support consumers segmentation and clustering of buildings (T2.3 leader, incl. TE-1), multi-energy flexibility modelling (T3.4 leader, incl. TE-4), as well as manage the demonstration activities (WP5 leader) as Pilot Coordinator; **CARTIF** (through the Energy Division), bringing its expertise in the areas of energy system modelling, will be responsible for the DEDALUS flexibility and DR data models and standards (WP3 and T3.3 leader, incl. TE-3) and will also support the pilot implementation in Spain (T5.6) through the technology adaptation for district-level DR (TE-13, as part of T4.4); **UNIVPM**, bringing its experience on human comfort measurements, algorithms and behavioural modelling techniques, will contribute to the Italian pilot (T5.5), and will deliver the comfort-based flexibility model (TE-9, T4.2 leader). **UNIVPM** will contribute to consumer engagement for comfort-based flexibility model definition (T2.4) and will be responsible for DEDALUS capacity building programme (T7.5 leader). **TUC** will bring its experiences from H2020 eDREAM, BRIGHT and CATALYST projects to further evolve the privacy-preservice blockchain technology stack (through TE-5, leading on T3.5), while at the same time will contribute to building-level consumer segmentation (T2.3, incl. TE-1), and data-driven cross-domain DR assessment and optimised management (TE-12, as part of T4.4). **CIMNE**, as one of the most relevant research institutions driving the smart energy transition in Spain, is leading the Spanish pilot activities (T5.6 leader), providing district-level DR for self-consumption and cost optimisation (TE-13, as part of T5.6).

SECTION 4: ETHICS SELF-ASSESSMENT

4.1 ETHICAL DIMENSION OF THE OBJECTIVES, METHODOLOGY AND LIKELY IMPACT

In DEDALUS a set of procedures will be adopted to protect the privacy of the involved human end-users. Information managed by the project consortium will be carefully controlled with restriction policies where appropriate.

Applicants will particularly take into account the following provisions:

- Ethical standards and guidelines compatible with, and equivalent to, those of Horizon Europe will be rigorously applied, regardless of the country in which DEDALUS demonstration is carried out.
- All participants will receive introductory descriptions about the project and the purpose of the studies - the purpose and procedure of the project's activities will be introduced in an understandable way.
- It will be emphasized that it is the potential participants' choice about whether to be in the study.
- All participants will be informed of their right to privacy and the extent to which participation in this project may impact their lives – and the mechanisms the researchers have put in place to protect participant privacy through processes of anonymization and data storage and security.
- Participants will be informed about the duration and effort to participate in any activity.
- Participants will be made aware of their withdrawal rights, which they could enforce at any time and informed that, if they wish, any personal data can be destroyed.
- Contact information to the project's stakeholders will be provided

The compliance with ethical codes will be ensured by the consortium via ethical review processes which will be continuously reported.

When involving human beings in the project, information on privacy and confidentiality, and whether they comply with national and EU legislation will be provided to project participants. Information Sheets and Informed Consent Forms will also be provided and these will be in language and terms understandable to the participants. They will include the right of participants to know that participation is voluntary, the degree of risk and burden involved in participation, who will benefit from participation, the procedures that will be implemented in the case of incidental findings. Participants will also know that they can ask questions and receive understandable answers before making a decision, and withdraw themselves and their data from the project at any time without any negative consequence. Upon signing, the subject or the legally authorized representative will receive a copy of these forms, and the original ones will be held in the subject's research record. Only anonymised or aggregated data (completely disjoined from people identification and profiles) will be processed related to the participation in project workshops and events, as well as the training sections.

As a general rule, during dissemination, only anonymised or aggregated data (completely disjoined from people identification and profiles) will be processed and made available. Whenever for specific reasons personal data will need to be processed, the interested partner (e.g. the pilot leader) will remain responsible for the data provided during its own research and will appoint a Data Protection Officer who is in charge of ensuring the full respect and compliance with GDPR. The partner will also be required to provide evidence of the authorisation to process personal data before access to or use of such data be granted. If defined by the European and national legislative framework, such authorisation will need to be requested by the specific partner to the appropriate competent authority in the partners' countries. In case of use of specific platforms such as for instance Twitter, LinkedIn, Facebook, Google cloud, etc. used by the project to share information containing personal data, the responsible partner for the specific activity will obtain a Data Processing Addendum/Agreement for data processing

Finally, the methodology carried out during DEDALUS does not involve clinical trials, involvement of children, or protection of the data collected; and its expected impacts do not involve environmental damage, stigmatisation of particular social groups, political or financial adverse consequences, misuse, etc.

4.2 COMPLIANCE WITH ETHICAL PRINCIPLES AND RELEVANT LEGISLATIONS

The DEDALUS consortium is fully aware that the project's activities may have the potential to generate ethical, fundamental rights, privacy and data protection implications and is fully committed to adhering to the highest ethical, fundamental rights and legal standards, as recognized at the European Union and International level. In particular, activities will be carried out to ensure that the proposal is compliant with the highest ethical principles and fundamental rights dictated in:

- the Helsinki Declaration
- the European Code of Conduct for Research Integrity (ECCRI, 2011);

- the EU Charter on Fundamental Rights (CFREU, 2010);
- the UNESCO Universal Declaration on Bioethics and Human Rights (2005);
- the European Convention for the Protection of Human Rights and Fundamental Freedoms (ECHR, 1950);
- the Universal Declaration of Human Rights (UDHR, 1948).

With regard to the rights to privacy and to the protection of personal data, DEDALUS will adhere to:

- The General Data Protection (GDPR) Regulation (EU) 2016/679;
- The Data Protection Directive (1995/46/EC) and the Directive on Privacy and Electronic Communications (2002/58/EC).
- The EU Charter on Fundamental Rights (art. 7 and 8);
- The European Convention for the Protection of Human Rights and Fundamental Freedoms (art. 8);
- The CoE Convention No. 108 for the Protection of Individuals with regard to Automatic Processing of Personal Data (1981);
- The International Covenant on Civil and Political Rights (ICCPR, 1966).

The Consortium is also aware that it is likely that the European data protection framework will change within the lifetime of the project and is committed to complying with any new privacy and data protection regulation.

Finally, the Consortium will comply with all relevant national and local regulations applicable to the conduct of the project activity.

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ANNEX 2

ESTIMATED BUDGET FOR THE ACTION

	Estimated eligible ¹ costs (per budget category)									Estimated EU contribution ²				
	Direct costs								Indirect costs	Total costs	EU contribution to eligible costs			Maximum grant amount ⁶
	A. Personnel costs		B. Subcontracting costs	C. Purchase costs			D. Other cost categories	E. Indirect costs ³	Funding rate % ⁴		Maximum EU contribution ⁵	Requested EU contribution		
	A.1 Employees (or equivalent) A.2 Natural persons under direct contract A.3 Seconded persons		A.4 SME owners and natural person beneficiaries	B. Subcontracting	C.1 Travel and subsistence	C.2 Equipment	C.3 Other goods, works and services	D.2 Internally invoiced goods and services	E. Indirect costs					
	Forms of funding	Actual costs	Unit costs (usual accounting practices)	Unit costs ⁷	Actual costs	Actual costs	Actual costs	Actual costs	Unit costs (usual accounting practices)	Flat-rate costs ⁸				
	a1	a2	a3	b	c1	c2	c3	d2	e = 0,25 * (a1 + a2 + a3 + c1 + c2 + c3)	f = a + b + c + d + e	U	g = f * U%	h	m
1 - ENG	607 500.00	0.00	0.00	0.00	14 500.00	0.00	6 700.00	0.00	157 175.00	785 875.00	70	550 112.50	550 112.50	550 112.50
2 - NTUA	379 600.00	0.00	0.00	0.00	6 500.00	0.00	9 500.00	0.00	98 900.00	494 500.00	100	494 500.00	494 500.00	494 500.00
3 - CARTIF	192 000.00	0.00	0.00	0.00	6 500.00	0.00	3 000.00	0.00	50 375.00	251 875.00	100	251 875.00	251 875.00	251 875.00
4 - UNIVPM	200 000.00	0.00	0.00	0.00	7 500.00	0.00	19 000.00	0.00	56 625.00	283 125.00	100	283 125.00	283 125.00	283 125.00
5 - ARC	170 000.00	0.00	0.00	0.00	9 700.00	10 000.00	1 000.00	0.00	47 675.00	238 375.00	70	166 862.50	166 862.50	166 862.50
6 - COMS	214 200.00	0.00	0.00	0.00	7 500.00	0.00	3 000.00	0.00	56 175.00	280 875.00	70	196 612.50	196 612.50	196 612.50
7 - OURPOWER	325 920.00	0.00	0.00	0.00	6 500.00	15 000.00	3 500.00	0.00	87 730.00	438 650.00	70	307 055.00	307 055.00	307 055.00
8 - GREENPOINT	132 000.00	0.00	0.00	0.00	6 500.00	0.00	2 000.00	0.00	35 125.00	175 625.00	70	122 937.50	122 937.50	122 937.50
9 - BLUEPRINT	288 000.00	0.00	0.00	0.00	6 500.00	0.00	2 000.00	0.00	74 125.00	370 625.00	70	259 437.50	259 437.50	259 437.50
10 - FAELLESBO	189 530.00	0.00	0.00	0.00	7 500.00	0.00	3 000.00	0.00	50 007.50	250 037.50	100	250 037.50	250 037.50	250 037.50
11 - NEOGRID	273 900.00	0.00	0.00	0.00	7 500.00	15 000.00	1 000.00	0.00	74 350.00	371 750.00	70	260 225.00	260 225.00	260 225.00
12 - GREEN CITIES	151 800.00	0.00	0.00	0.00	9 500.00	0.00	10 000.00	0.00	42 825.00	214 125.00	100	214 125.00	214 125.00	214 125.00
13 - HERON	313 500.00	0.00	0.00	0.00	6 500.00	0.00	24 200.00	0.00	86 050.00	430 250.00	70	301 175.00	301 175.00	301 175.00
14 - DOM-X	300 000.00	0.00	0.00	0.00	6 500.00	6 000.00	24 600.00	0.00	84 275.00	421 375.00	70	294 962.50	294 962.50	294 962.50
15 - ISRAA	166 500.00	0.00	0.00	0.00	6 500.00	15 000.00	3 000.00	0.00	47 750.00	238 750.00	100	238 750.00	238 750.00	238 750.00
16 - DYNAME	120 000.00	0.00	0.00	0.00	6 400.00	0.00	2 000.00	0.00	32 100.00	160 500.00	70	112 350.00	112 350.00	112 350.00
17 - CIMNE	211 200.00	0.00	0.00	0.00	6 500.00	0.00	5 000.00	0.00	55 675.00	278 375.00	100	278 375.00	278 375.00	278 375.00
18 - PEUSA	105 110.00	0.00	0.00	0.00	6 500.00	5 000.00	0.00	0.00	29 152.50	145 762.50	70	102 033.75	102 033.75	102 033.75
19 - ROMUR	140 000.00	0.00	0.00	0.00	6 500.00	0.00	19 500.00	0.00	41 500.00	207 500.00	70	145 250.00	145 250.00	145 250.00
20 - CENTRICA	342 000.00	0.00	0.00	0.00	9 500.00	45 000.00	6 500.00	0.00	100 750.00	503 750.00	70	352 625.00	352 625.00	352 625.00
21 - TUC	253 500.00	0.00	0.00	0.00	7 500.00	0.00	3 000.00	0.00	66 000.00	330 000.00	100	330 000.00	330 000.00	330 000.00
22 - SIN	147 400.00	0.00	0.00	0.00	7 500.00	0.00	1 000.00	0.00	38 975.00	194 875.00	100	194 875.00	194 875.00	194 875.00
23 - ICONS	221 000.00	0.00	0.00	0.00	8 300.00	0.00	4 700.00	0.00	58 500.00	292 500.00	100	292 500.00	292 500.00	292 500.00
Σ consortium	5 444 660.00	0.00	0.00	0.00	174 400.00	111 000.00	157 200.00	0.00	1 471 815.00	7 359 075.00		5 999 801.25	5 999 801.25	5 999 801.25

¹ See Article 6 for the eligibility conditions. All amounts must be expressed in EUR (see Article 21 for the conversion rules).

² The consortium remains free to decide on a different internal distribution of the EU funding (via the consortium agreement; see Article 7).

³ Indirect costs already covered by an operating grant (received under any EU funding programme) are ineligible (see Article 6.3). Therefore, a beneficiary/affiliated entity that receives an operating grant during the action duration cannot declare indirect costs for the year(s)/reporting period(s) covered by the operating grant, unless they can demonstrate that the operating grant does not cover any costs of the action. This requires specific accounting tools. Please immediately contact us via the EU Funding & Tenders Portal for details.

⁴ See Data Sheet for the funding rate(s).

⁵ This is the theoretical amount of the EU contribution to costs, if the reimbursement rate is applied to all the budgeted costs. This theoretical amount is then capped by the 'maximum grant amount'.

⁶ The 'maximum grant amount' is the maximum grant amount decided by the EU. It normally corresponds to the requested grant, but may be lower.

⁷ See Annex 2a 'Additional information on the estimated budget' for the details (units, cost per unit).

⁸ See Data Sheet for the flat-rate.

ANNEX 2a

ADDITIONAL INFORMATION ON UNIT COSTS AND CONTRIBUTIONS

SME owners/natural person beneficiaries without salary (Decision C(2020) 7115¹)

Type: unit costs

Units: days spent working on the action (rounded up or down to the nearest half-day)

Amount per unit (daily rate): calculated according to the following formula:

{EUR 5 080 / 18 days = **282,22**}
multiplied by
{country-specific correction coefficient of the country where the beneficiary is established}

The country-specific correction coefficients used are those set out in the Horizon Europe Work Programme (section Marie Skłodowska-Curie actions) in force at the time of the call (see [Portal Reference Documents](#)).

HE and Euratom Research Infrastructure actions²

Type: unit costs

Units³: see (for each access provider and installation) the unit cost table in Annex 2b

Amount per unit^{*}: see (for each access provider and installation) the unit cost table in Annex 2b

^{*} Amount calculated as follows:

For trans-national access:

$$\frac{\text{average annual total trans-national access costs to the installation (over past two years)}^4}{\text{average annual total quantity of trans-national access to the installation (over past two years)}^5}$$

For virtual access:

$$\frac{\text{total virtual access costs to the installation (over the last year)}^6}{\text{total quantity of virtual access to the installation (over the last year)}^7}$$

Euratom staff mobility costs⁸

Monthly living allowance

Type: unit costs

¹ Commission [Decision](#) of 20 October 2020 authorising the use of unit costs for the personnel costs of the owners of small and medium-sized enterprises and beneficiaries that are natural persons not receiving a salary for the work carried out by themselves under an action or work programme (C(2020)7715).

² [Decision](#) of 19 April 2021 authorising the use of unit costs for the costs of providing trans-national and virtual access in Research Infrastructure actions under the Horizon Europe Programme (2021-2027) and the Research and Training Programme of the European Atomic Energy Community (2021-2025).

³ Unit of access (e.g. beam hours, weeks of access, sample analysis) fixed by the access provider in proposal.

⁴ In exceptional and duly justified cases, the granting authority may agree to a different reference period.

⁵ In exceptional and duly justified cases, the granting authority may agree to a different reference period.

⁶ In exceptional and duly justified cases, the granting authority may agree to a different reference period.

⁷ In exceptional and duly justified cases, the granting authority may agree to a different reference period.

⁸ [Decision](#) of 15 March 2021 authorising the use of unit costs for mobility in co-fund actions under the Research and Training Programme of the European Atomic Energy Community (2021-2025).

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

* Amount calculated as follows from 1 January 2021:

{**EUR 4 300** multiplied by
country-specific correction coefficient** of the country where the staff member is seconded}⁹

**Country-specific correction coefficients as from 1 January 2021¹⁰

EU-Member States¹¹

Country / Place	Coefficient (%)
Bulgaria	59,1
Czech Rep.	85,2
Denmark	131,3
Germany	101,9
Bonn	95,8
Karlsruhe	98
Munich	113,9
Estonia	82,3
Ireland	129
Greece	81,4
Spain	94,2
France	120,5
Croatia	75,8
Italy	95
Varese	90,7
Cyprus	78,2
Latvia	77,5
Lithuania	76,6
Hungary	71,9
Malta	94,7
Netherlands	113,9
Austria	107,9
Poland	70,9
Portugal	91,1
Romania	66,6
Slovenia	86,1

⁹ Unit costs for living allowances are calculated by using a method of calculation similar to that applied for the secondment to the European Commission of seconded national experts (SNEs).

¹⁰ ⚠ For the financial statements, the amount must be adjusted according to the actual place of secondment.
The revised coefficients were adopted in the Decision authorising the use of unit costs for the Fusion Programme co-fund action under the Research and training Programme of the European Atomic Energy Community 2021-2025. They are based on the 2020 Annual update of the remuneration and pensions of the officials and other servants of the European Union and the correction coefficients applied thereto (OJ C 428, 11.12.2020) to ensure purchasing power parity. The revised coefficient are applied as from 1 January 2021 through an amendment to the grant agreement.

¹¹ No correction coefficient shall be applicable in Belgium and Luxembourg.

Slovakia	80,6
Finland	118,4
Sweden	124,3

Third countries

Country/place	Coefficient (%)
China	82,2
India	72,3
Japan	111,8
Russia	92,7
South Korea	92,3
Switzerland	129,2
Ukraine	82,3
United Kingdom	97,6
United States	101,4 (New-York) 90,5 (Washington)

Mobility allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 600** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Family allowance

Type: unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 660** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Education allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

*Amount calculated as follows from 1 January 2021:
{**EUR 283.82** x number of dependent children¹²}

¹² For the estimated budget (Annex 2): an average should be used. (⚠ For the financial statements, the number of children (and months) must be adjusted according to the actual family status at the moment the secondment starts.)

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ETHNICON METSOVION POLYTECHNION (NTUA), PIC 999978142, established in
HEROON POLYTECHNIOU 9 ZOGRAPHOU CAMPUS, ATHINA 157 80, Greece,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Ioannis CHATJIGEORGIOU with ECAS id nchatjio signed in the
Participant Portal on 03/05/2023 at 07:48:42 (transaction id SigId-
191323-uzL0VMpLuMafdwLZpxIisNKOtPWntC2s9htopN5GvsWhyzzIV
K78RQqy2K1gVpsKozzuXHLIs8mXj4FTFxaj0m8-
rS0vSrmBGYCdXRy5uIgXle-3MKF8Fzr2taQqANgnw50i8WVeKfEoCy19
WLRCT3fLuzJryR1a7y9M5JChHrtcs5GS0eWoKkzr6wGNR7oj7PiYGW).
Timestamp by third party at
2023.05.03 07:48:46 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

FUNDACION CARTIF (CARTIF), PIC 999929836, established in PQ TECNOLÓGICO BOECILLO 205, BOECILLO 47151, Spain,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Julio GONZALEZ with ECAS id ngonzaou signed in the Participant Portal on 04/05/2023 at 09:04:22 (transaction id SigId-207007-exrbk WHHxOHUmiuUtXSdx8LiuE2FAT6ZzfParxF1cSX3iOANAQrXp7IsGDGzH23plra5hUszzU6HOxjzNAJ1sEX-r50vSrmBGYCdxRy5uIgXle-ae4fdn apY5FD2R7tzWHV3byIXmD2dI8zug4ge4ixMqxjP13TUTSTLsu9zLnIA OI3UYBPNs7FMEdyCTcYrV7P42G). Timestamp by third party at 2023.05.04 09:04:31 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

UNIVERSITA POLITECNICA DELLE MARCHE (UNIVPM), PIC 999866689, established in
PIAZZA ROMA 22, ANCONA 60121, Italy,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Gian Luca Gregori with ECAS id n002zvrr signed in the Participant Portal on 28/04/2023 at 14:24:14 (transaction id SigId-168503-yaECxgEsUFdjFFmFaEA9suPAocccqr7D2PqSTzhdxtpv7EX5pF4DdmzT6by5Hmu4N3LWSpzQB2spiXclEi9LTom-rS0vSrmBGYCdXRy5uIgXle-zI59FuMZ7F3FqYY6g2ZzK14oppqWcAmeJwLawimYo4oAo35NzPmUGms9ZZQYOhChmtLzYqKoFbFZY4AGmP6zoCM). Timestamp by third party at 2023.04.28 14:27:33 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ARCELIK A.S. (ARC), PIC 998821708, established in KARAAGAC, SUTLUCE BEYOGLU CAD. NO:2-6, ISTANBUL 34445, Türkiye,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Evrin ÖZGÜL with ECAS id noezguev signed in the Participant Portal on 27/04/2023 at 14:21:05 (transaction id SigId-153053-zrv4X9QpASth74V1v1Zq4Nnp4H6O9fG4UYACQHjRXzXulffTpD3RU9xQdU1396EdGkmze8ZWjipB9HJzSGKCQjW-rS0vSrmBGYCDxRy5uIgXle-zW4pNoiHNqHYIBNRTeglsGsyTykRfyhpQq00403dHAL9ZhFGxj53kzzdwemVszN9D63r6Fy7yjj1ZCOKqpSEXiy). Timestamp by third party at 2023.04.27 14:21:12 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

COMSENSUS, KOMUNIKACIJE IN SENZORIKA, DOO (COMS), PIC 937154831,
established in BREZJE PRI DOBU 8A, DOB 1233, Slovenia,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Miha SMOLNIKAR with ECAS id nsmolnmi signed in the Participant Portal on 28/04/2023 at 10:37:39 (transaction id SigId-163261-zSH0mVpafpXAOJCqspUGUCc0j0OjppKE3zTOeAy8TasfAl6G9L6dVqzTyf0zPe25DoVjMbQrKhNXZtK2f3SBeVy-r50vSrmBGYCdXrY5uIgXle-r38rG1umabpWIoZj30nYbAX95QAThIxynoy011iNAzZ4uZ0zTsZs0OnfGzzPV7Hsr2UQ4i5FiWAzdDyI5DcuWgW). Timestamp by third party at 2023.04.28 10:37:45 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

OURPOWER ENERGIEGENOSSENSCHAFT SCE MIT BESCHRANKTER HAFTUNG (OURPOWER), PIC 898281208, established in MARIAHILFER STRASSE 1D/13, WIEN 1060, Austria,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Ulfert Hoehne with ECAS id n0031xrd signed in the Participant Portal on 27/04/2023 at 09:36:57 (transaction id SigId-146515-D9yNICrHRZAd6zyPzi3mZiBvxRdGgHIuVmXjrPNeT7OBWH3WbOA4Pu09EPpmAwCI3acMamiPzzO46H60Ejjzj2d-rS0vSrmBGYCdxRy5uIgxle-rUtrLoBnx5dnnbqeTedVsklgt9Mn5JlQXgmo2jLF4vbtCbVr5HrIvRtQnG4zS5sIfxddbduv74I4kritPN63s0). Timestamp by third party at 2023.04.27 09:37:02 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

GREEN-POINT 62 GMBH (GREENPOINT), PIC 885248773, established in POTTENDORFER STRASSE 62/EG04, WIEN 2700, Austria,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Vitaliy Kryvoruchko with ECAS id n00ao7lz signed in the Participant Portal on 27/04/2023 at 00:42:06 (transaction id SigId-143767-2Wxy RXYOQI4HCUYZ5zzbAK4ztWLVeFOxfZ44LjZ6pBYCSkRFGQMhNFD4h PJznc3DrNAmHELwk4dSHI543H1cCFO-rS0vSrmBGYCdXRy5uIgXle-0 SvWnSLfAL4h5uDXFKT381Sb6VDXxouuRzPT9MTj3j88zbzNa67FVHfy 0pZUAHKL2HXLiqxETJwgfzuj89IAMC4). Timestamp by third party at 2023.04.27 00:42:10 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

BLUEPRINT ENERGY SOLUTIONS GMBH (BLUEPRINT), PIC 904096067, established in AM GESTADE 3, WIEN 1010, Austria,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Elena BOSKOV-KOVACS with ECAS id nboskoel signed in the Participant Portal on 27/04/2023 at 04:00:17 (transaction id SigId-144111-kgmmqHPH1HYDrFMBDIw6Dtzq0h7r5hVE0HgCcOrfrvuBE2dWeHFIM5DQX5Q1617u653TuSip3LPtqASMwUhB5i-rS0vSrmBGYCdXrY5uIgXle-ZQqS5YS3y2RYU9C9XunLX60PRxUvi8mtxo aEzjly8AzrzZQzo9bglt1BQeSTTrCUEPVztkrK57jb2v2KDuQO6Pb).
Timestamp by third party at
2023.04.27 04:00:22 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

FAELLESBO (FAELLESBO), PIC 892008024, established in NYGADE 20, HERNING 7400, Denmark,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Kaj Louis Mortensen with ECAS id n0096mi7 signed in the Participant Portal on 20/06/2023 at 12:51:09 (transaction id SigId-39147-DO1MIb OM4EWJVeSP5isR3ameq46FAVgSRzkzm1mYDcC1QA70uOFPzUwXU4 Usn446Yua9j21zXEUCPgIzw3OI9xV-jpJZscgsw0K4epxy2XYw0y-XHwFp TGJQRJU0nftD7x5Hix1ebbVfhYkq4neCVfyx5eOWzpzzzoHbEIIOY9y9eX Mia9ru1q2tFsi3i1zyJBjapm). Timestamp by third party at 2023.06.20 12:51:17 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

NEOGRID TECHNOLOGIES APS (NEOGRID), PIC 953024419, established in NIELS JERNES VEJ 10, AALBORG OST 9220, Denmark,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Henrik LUND STAERMOSE with ECAS id nlundshe signed in the Participant Portal on 27/04/2023 at 09:38:37 (transaction id SigId-146563-jzjf5giEjXEjWL1HJkjdXA0zQGdHVBZHny57Wzhbi8Dyybn4q4a-ALuZPBPq4zVVCgTZBU1heRsGELpQm284Va-rS0vSrmBGYCdXy5uIgXle-Kgm3ttstLGGZ4picHkT4La0Ft1j7SK9hUxldjkHTj9D1diEELIKzMXAOVZR3i4X4YUY9pQTlTmIgUXtypbexO).
Timestamp by third party at
2023.04.27 09:38:42 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

EUROPEAN GREEN CITIES APS (GREEN CITIES), PIC 950585936, established in NORSGADE 17, AARHUS C 8000, Denmark,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Elsebeth TERKELSEN with ECAS id nterelse signed in the Participant Portal on 04/05/2023 at 08:39:18 (transaction id SigId-206717-pE9eL OVWesUvdfeZ827zNS9SwMWAtZkx91p32zT6faTZdqluenAv0WsUzfVy m2dRwzOyeZ3I7iaW2Edkc4D3hqy-rS0vSrmBGYCdxRy5uIgXle-3zGxs RBQinoKUuFjTxM4qbxzczwzboiQ0dnNCBxi2SQy2Izdhs0Azq0WioOM bBaewGkE77cjCnVCbXEwKKMzsJam). Timestamp by third party at 2023.05.04 08:39:22 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

HERON SINGLE MEMBER S.A. ENERGY SERVICES (HERON), PIC 954579911, established in LEOFOROS MESOGEION 85, ATHINA 115 26, Greece,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Loukas Dimitriou with ECAS id n009cah1 signed in the Participant Portal on 05/05/2023 at 09:07:44 (transaction id SigId-220502-nZc0n3 GF3dhQwFzLA8LvWKxG7YZBNQhNmmmmKds3NbzcC0oCziVAymaDUS xMNVIO40NDzlOKqhX7xYWszO44K9y4-rS0vSrmBGYCdxRy5uIgXle-m8 A8T69wFzLX5cOnjD8HoxzrDtKUvwS0BZ98izqH4C5wyvXN8DqQx5S1p pLHj1rDMexnPs4PZYxhgUa2srhg8IW). Timestamp by third party at 2023.05.05 09:07:53 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

DOMX IDIOTIKI KEFALAIOUCHIKI ETAIREIA (DOM-X), PIC 898645637, established in STR. SARAFI 48E, THESSALONIKI 551 32, Greece,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Elisavet Apostolia Efstathiou with ECAS id n002unru signed in the Participant Portal on 27/04/2023 at 13:24:15 (transaction id SigId-152016-Nasv2YiZT18QrQpdKEinFeZzub8YKFHdV1xRxjQJ8AfnsCBX4bc-ae2oXXraqWXbei2BxALvZ2k0Lhk9zeMhhua-r50vSrmBGYCdxRy5uIgXle-dvrv6h2PLEHi7ur6exMhzUQNPghxTKUwj39yGIBsaNblZ9LHSzxnNNOIg0G5IWq9ak2GmSkmOUoazmGkM7D8b8G). Timestamp by third party at 2023.04.27 13:24:19 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ISTITUTO PER SERVIZI DI RICOVERO E ASSISTENZA AGLI ANZIANI (ISRAA), PIC 930262690, established in BORGO MAZZINI 48, TREVISO 31100, Italy,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Giorgio PAVAN with ECAS id npavgior signed in the Participant Portal on 27/04/2023 at 14:57:50 (transaction id SigId-153993-U5U40 zLvJ2zQkkhM29xzPg3cvKncCRuqQvjFBwXjVjymKeovd3iTNWdCDmRT X8M1dOJ65zNjHdOEzGtrMM1ktoj-r50vSrmBGYCdxRy5uIgXle-wxqaC 8ithqtFkwrzREMVdfzQZDF5hGXjNIq37NPmCGpXhTsQgqh2hu2LeoRi MY4HRylc3wWLU7y1RvxcHHzQgcm). Timestamp by third party at 2023.04.27 14:57:56 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

DYNAME SRL (DYNAME), PIC 885390102, established in VICOLO CADORE 45, SAN VENDEMIANO 31020, Italy,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ADRIANO FREZZA with ECAS id n00akevh signed in the Participant Portal on 27/04/2023 at 15:49:27 (transaction id SigId-155287-m6K6 iO9dosAfO3S0vehsEtuzXKP5Fi1nX1nB7GraAAzaUBCXOZQzfABVvZB OWJN2AVV7ZIpxrO876zRSRV0HAa4-r50vSrmBGYCdxRy5uIgXle-Bcio vxOYqzZcmzjQOY8zOWDjIYfbcgO7Sq0g8WSXFloQCjYJ8QwBeszmB wVoDWy0Rd9LZC52ajzry6S6aitjyL4). Timestamp by third party at 2023.04.27 15:49:32 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CENTRE INTERNACIONAL DE METODES NUMERICS EN ENGINYERIA (CIMNE),
PIC 999658721, established in C GRAN CAPITAN, EDIFICI C1, CAMPUS NORD UPC SN,
BARCELONA 08034, Spain,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Anna FONT JORNET with ECAS id nfontjan signed in the Participant Portal on 27/04/2023 at 12:22:49 (transaction id SigId-150916-r6V14rFvizqzRTDwLzSzsoNctDxtzQLynS7lrl9FT7oBpEzpB6fS0QiaaCFzIBnIWn8jjsuiBzealltskcBXs10G-rS0vSrmBGYCdxRy5uIgXle-Jy8QeHMxznuZGr0AYSEjvsNYBUwGjeDOcd10zHTI4KP1ccNijk1PxChOev6nEuIGN7zVeF4KmgL4KUWFFbN2sNK). Timestamp by third party at 2023.04.27 13:22:53 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

PRODUCTORA ELECTRICA URGELLENCA SA (PEUSA), PIC 899979775, established in PASEO JOAN BRUDIEU, 17, LA SEU D'URGELL 25700, Spain,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Jordi Duat with ECAS id n002o0bz signed in the Participant Portal on 03/05/2023 at 08:45:22 (transaction id SigId-191966-uEgb8tfnzg wK1tGzzW5wtgjhn2JeJuuzjuznKTV7h3VAu3QcAOWAILcUYJ4ko2GST lmmHteczLeiI47xZdmrI7tG-rS0vSrmBGYCdxRy5uIgXle-1UyNE2B3Q kWHfzcO2Y1umjZrWUvr3XGcqSlzSPTljEEzLTXMtW0t3qb6qeCQuWz V9mEX2IKu64e36R6UEBp59qu). Timestamp by third party at 2023.05.03 08:45:27 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ROMUR RENOVALBES SL (ROMUR), PIC 885732512, established in URB SAN PATRICIO 97, ALGINET 46230, Spain,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Norberto RODRIGUEZ with ECAS id n00aa0kx signed in the Participant Portal on 27/04/2023 at 08:57:39 (transaction id SigId-145792-zkVJ2xueW6RnytX1dpiu9B5blqzuQfCpodBRMsHg4zkHYMzVqHgZIrBbzKQEu3sTGh0oYiO66PkiyeFlwxizfIE0-rS0vSrmBGYCdxRy5uIgXle-ThfPZVmb2HoVtl0FEZ5gZzsYH3zRFzqlRFVzh1TIBQIN9IzGzwJtCqqUzOyjiyQdwYOMSdducvRGYzwuhIaUJDn66).
Timestamp by third party at
2023.04.27 08:57:48 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CENTRICA BUSINESS SOLUTIONS BELGIUM (CENTRICA), PIC 927493825, established in RODERVELDLAAN 2, ANTWERPEN 2600, Belgium,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Stavros Sachinis with ECAS id n00bkker signed in the Participant Portal on 02/05/2023 at 20:06:21 (transaction id SigId-189762-11h WTTe3RjSivF3btkON03fQTrB1zpCE8cwZihgwBLzznFK3HqZhEci95p ha6ZUwEeLjBkM9tCrGmOAW0ECNVG-rS0vSrmBGYCdXRy5uIgXle-u JfYHkostazgM1JhhRbyaa11nGvF6eDAeFNrIXAnwUQy3RVcpito4pXZ FaKSjSsQEYR6RUUKiiUtSS7cxbnqve). Timestamp by third party at 2023.05.02 20:06:29 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

UNIVERSITATEA TEHNICA CLUJ-NAPOCA (TUC), PIC 999897244, established in STR
MEMORANDUMULUI 28, CLUJ NAPOCA 400114, Romania,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Florin Oniga with ECAS id n004fh1x signed in the Participant Portal
on 27/04/2023 at 12:22:45 (transaction id SigId-150912-HiYgtUi96Mh
404puDjrxT6PazgWKvcQrRzYryOUjy5WQ0YnS6zSWnzPzzOJH2zzZAR
Q60zJG6PMzMzQY5H94xM0mv-rS0vSrmBGYCdXRY5uIgXle-hM0gBY4
gWqOPDziGTkwwzbrP8YtLRQ9SzgF9W0mzUheskV11r72c2vABfzUH
w88uhBDzwDNICbo3rNUwQdl7B1e0). Timestamp by third party at
2023.04.27 12:22:50 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

SMART INNOVATION NORWAY AS (SIN), PIC 950648016, established in HAKON MELBERGS VEI 16, HALDEN 1783, Norway,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) and the European Climate, Infrastructure and Environment Executive Agency (CINEA) ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Joseph Negreira with ECAS id nnegrejh signed in the Participant Portal on 26/04/2023 at 21:23:04 (transaction id SigId-143427-SzscEQAxo75Rj4oj1x2FeXalHVkTgr5myhvgzMYUt4wmEu6YxyfDMr3GLH5f8jRFaul63mp3ZxbndzKeYajW93W-rS0vSrmBGYCdxRy5uIgXle-uzVzgVcijg4F8boUarLnCnqMjupmNDFLS8sZ6cQpogyN6ks8ZTePsUvGwX4sD2wiM2zgxB4Kow0okw0AgF7EqfG). Timestamp by third party at 2023.04.26 21:23:09 CEST

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

FONDAZIONE ICONS (ICONS), PIC 917939810, established in PIAZZA DELLA VITTORIA 1, LODI 26900, Italy,

hereby agrees

to become beneficiary

in Agreement No 101103998 — DEDALUS ('the Agreement')

between ENGINEERING - INGEGNERIA INFORMATICA SPA (ENG) **and** the **European Climate, Infrastructure and Environment Executive Agency (CINEA)** ('EU executive agency' or 'granting authority'), under the powers delegated by the European Commission ('European Commission'),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

Mario MARTINOLI with ECAS id nmartfjq signed in the Participant Portal on 26/04/2023 at 20:56:30 (transaction id SigId-143362-PDqsa thEyqHRdEba3ToyA4KWfogh1nuAjQeEeba90CL0Ngq8y4RmZHjtGp5 tuF6DtzWGA5uWGjbztM18PzOboN0-rS0vSrmBGYCdXRy5uIgXle-eMf R8ZKJznvoZjoxKxmwwH0wVXDFRznyYZjzqfVAv11HjachbrDdbTzP3vR CSMaby211v3mj7teQMvaEKCHygs3). Timestamp by third party at 2023.04.26 20:56:35 CEST

ANNEX 4 HORIZON EUROPE MGA — MULTI + MONO

FINANCIAL STATEMENT FOR [PARTICIPANT NAME] FOR REPORTING PERIOD [NUMBER]

	Eligible ¹ costs (per budget category)															EU contribution ²				Revenues				
	Direct costs															Indirect costs	Total costs	EU contribution to eligible costs			Total requested EU contribution	Income generated by the action		
	A. Personnel costs		B. Subcontracting costs	C. Purchase costs			D. Other cost categories							E. Indirect costs ²	Funding rate % ³	Maximum EU contribution ⁴		Requested EU contribution						
	A.1 Employees (or equivalent)		A.4 SME owners and natural person beneficiaries	B. Subcontracting	C.1 Travel and subsistence	C.2 Equipment	C.3 Other goods, works and services	D.1 Financial support to third parties	D.2 Internally invoiced goods and services	D.3 Transnational access to research infrastructure unit costs	D.4 Virtual access to research infrastructure unit costs	[OPTION for HE PCP/PPR: D.5 PCP/PPR procurement costs]	[OPTION for Euratom Programme Cofund Actions: D.6 Euratom Cofund staff mobility costs]	[OPTION for HE ERC Grants: D.7 ERC additional funding]	[OPTION for HE ERC Grants: D.8 ERC additional funding (subcontracting, FSTP and internally invoiced goods and services)]	E. Indirect costs								
A.2 Natural persons under direct contract																								
A.3 Seconded persons																								
Forms of funding		Actual costs	Unit costs (usual accounting practices)	Unit costs ⁵	Actual costs	Actual costs	Actual costs	Actual costs	/ Actual costs	Unit costs (usual accounting practices)	/ Unit costs ⁵	/ Unit costs ⁵	/ Actual costs	/ Unit costs ⁵	/ Actual costs	/ Actual costs	Flat-rate costs ⁶							
		a1	a2	a3	b	c1	c2	c3	/ d1a	d2	/ d3	/ d4	/ d5	/ d6	/ d7	/ d8	e = 0,25 * (a1 + a2 + a3 +b+ c1 +c2 + c3 +d1a+d2+d3 +d4 / +d5)+d6 / +d7 / +d8)	f = a+b+c+d+e	U	g = f*U%	h	m	n	
XX – [short name beneficiary/affiliated entity]																								

The beneficiary/affiliated entity hereby confirms that:

The information provided is complete, reliable and true.

The costs and contributions declared are eligible (see Article 6).

The costs and contributions can be substantiated by adequate records and supporting documentation that will be produced upon request or in the context of checks, reviews, audits and investigations (see Articles 19, 20 and 25).

For the last reporting period: that all the revenues have been declared (see Article 22).

① Please declare all eligible costs and contributions, even if they exceed the amounts indicated in the estimated budget (see Annex 2). Only amounts that were declared in your individual financial statements can be taken into account lateron, in order to replace costs/contributions that are found to be ineligible.

¹ See Article 6 for the eligibility conditions. All amounts must be expressed in EUR (see Article 21 for the conversion rules).

² If you have also received an EU operating grant during this reporting period, you cannot claim indirect costs - unless you can demonstrate that the operating grant does not cover any costs of the action. This requires specific accounting tools. Please contact us immediately via the Funding & Tenders Portal for details.

³ See Data Sheet for the reimbursement rate(s).

⁴ This is the *theoretical* amount of EU contribution to costs that the system calculates automatically (by multiplying the reimbursement rates by the costs declared). The amount you request (in the column 'requested EU contribution') may be less.

⁵ See Annex 2a 'Additional information on the estimated budget' for the details (units, cost per unit).

⁶ See Data Sheet for the flat-rate.

ANNEX 5

SPECIFIC RULES

CONFIDENTIALITY AND SECURITY (— ARTICLE 13)

Sensitive information with security recommendation

Sensitive information with a security recommendation must comply with the additional requirements imposed by the granting authority.

Before starting the action tasks concerned, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task. The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary.

For requirements restricting disclosure or dissemination, the information must be handled in accordance with the recommendation and may be disclosed or disseminated only after written approval from the granting authority.

EU classified information

If EU classified information is used or generated by the action, it must be treated in accordance with the security classification guide (SCG) and security aspect letter (SAL) set out in Annex 1 and Decision 2015/444¹ and its implementing rules — until it is declassified.

Deliverables which contain EU classified information must be submitted according to special procedures agreed with the granting authority.

Action tasks involving EU classified information may be subcontracted only with prior explicit written approval from the granting authority and only to entities established in an EU Member State or in a non-EU country with a security of information agreement with the EU (or an administrative arrangement with the Commission).

EU classified information may not be disclosed to any third party (including participants involved in the action implementation) without prior explicit written approval from the granting authority.

ETHICS (— ARTICLE 14)

Ethics and research integrity

The beneficiaries must carry out the action in compliance with:

- ethical principles (including the highest standards of research integrity)

¹ Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p. 53).

and

- applicable EU, international and national law, including the EU Charter of Fundamental Rights and the European Convention for the Protection of Human Rights and Fundamental Freedoms and its Supplementary Protocols.

No funding can be granted, within or outside the EU, for activities that are prohibited in all Member States. No funding can be granted in a Member State for an activity which is forbidden in that Member State.

The beneficiaries must pay particular attention to the principle of proportionality, the right to privacy, the right to the protection of personal data, the right to the physical and mental integrity of persons, the right to non-discrimination, the need to ensure protection of the environment and high levels of human health protection.

The beneficiaries must ensure that the activities under the action have an exclusive focus on civil applications.

The beneficiaries must ensure that the activities under the action do not:

- aim at human cloning for reproductive purposes
- intend to modify the genetic heritage of human beings which could make such modifications heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed)
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer, or
- lead to the destruction of human embryos (for example, for obtaining stem cells).

Activities involving research on human embryos or human embryonic stem cells may be carried out only if:

- they are set out in Annex 1 or
- the coordinator has obtained explicit approval (in writing) from the granting authority.

In addition, the beneficiaries must respect the fundamental principle of research integrity — as set out in the European Code of Conduct for Research Integrity².

This implies compliance with the following principles:

- reliability in ensuring the quality of research reflected in the design, the methodology, the analysis and the use of resources
- honesty in developing, undertaking, reviewing, reporting and communicating research in a transparent, fair and unbiased way

² European Code of Conduct for Research Integrity of ALLEA (All European Academies).

- respect for colleagues, research participants, society, ecosystems, cultural heritage and the environment
- accountability for the research from idea to publication, for its management and organisation, for training, supervision and mentoring, and for its wider impacts

and means that beneficiaries must ensure that persons carrying out research tasks follow the good research practices including ensuring, where possible, openness, reproducibility and traceability and refrain from the research integrity violations described in the Code.

Activities raising ethical issues must comply with the additional requirements formulated by the ethics panels (including after checks, reviews or audits; see Article 25).

Before starting an action task raising ethical issues, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task, notably from any (national or local) ethics committee or other bodies such as data protection authorities.

The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary, which shows that the documents cover the action tasks in question and includes the conclusions of the committee or authority concerned (if any).

VALUES (— ARTICLE 14)

Gender mainstreaming

The beneficiaries must take all measures to promote equal opportunities between men and women in the implementation of the action and, where applicable, in line with the gender equality plan. They must aim, to the extent possible, for a gender balance at all levels of personnel assigned to the action, including at supervisory and managerial level.

INTELLECTUAL PROPERTY RIGHTS (IPR) — BACKGROUND AND RESULTS — ACCESS RIGHTS AND RIGHTS OF USE (— ARTICLE 16)

Definitions

Access rights — Rights to use results or background.

Dissemination — The public disclosure of the results by appropriate means, other than resulting from protecting or exploiting the results, including by scientific publications in any medium.

Exploit(ation) — The use of results in further research and innovation activities other than those covered by the action concerned, including among other things, commercial exploitation such as developing, creating, manufacturing and marketing a product or process, creating and providing a service, or in standardisation activities.

Fair and reasonable conditions — Appropriate conditions, including possible financial terms or royalty-free conditions, taking into account the specific circumstances of the request for access, for example the actual or potential value of the results or background to which access is requested and/or the scope, duration or other characteristics of the exploitation envisaged.

FAIR principles — ‘findability’, ‘accessibility’, ‘interoperability’ and ‘reusability’.

Open access — Online access to research outputs provided free of charge to the end-user.

Open science — An approach to the scientific process based on open cooperative work, tools and diffusing knowledge.

Research data management — The process within the research lifecycle that includes the organisation, storage, preservation, security, quality assurance, allocation of persistent identifiers (PIDs) and rules and procedures for sharing of data including licensing.

Research outputs — Results to which access can be given in the form of scientific publications, data or other engineered results and processes such as software, algorithms, protocols, models, workflows and electronic notebooks.

Scope of the obligations

For this section, references to ‘beneficiary’ or ‘beneficiaries’ do not include affiliated entities (if any).

Agreement on background

The beneficiaries must identify in a written agreement the background as needed for implementing the action or for exploiting its results.

Where the call conditions restrict control due to strategic interests reasons, background that is subject to control or other restrictions by a country (or entity from a country) which is not one of the eligible countries or target countries set out in the call conditions and that impact the exploitation of the results (i.e. would make the exploitation of the results subject to control or restrictions) must not be used and must be explicitly excluded from it in the agreement on background — unless otherwise agreed with the granting authority.

Ownership of results

Results are owned by the beneficiaries that generate them.

However, two or more beneficiaries own results jointly if:

- they have jointly generated them and
- it is not possible to:
 - establish the respective contribution of each beneficiary, or
 - separate them for the purpose of applying for, obtaining or maintaining their protection.

The joint owners must agree — in writing — on the allocation and terms of exercise of their joint ownership (**‘joint ownership agreement’**), to ensure compliance with their obligations under this Agreement.

Unless otherwise agreed in the joint ownership agreement or consortium agreement, each joint owner may grant non-exclusive licences to third parties to exploit the jointly-owned results (without any right to sub-license), if the other joint owners are given:

- at least 45 days advance notice and
- fair and reasonable compensation.

The joint owners may agree — in writing — to apply another regime than joint ownership.

If third parties (including employees and other personnel) may claim rights to the results, the beneficiary concerned must ensure that those rights can be exercised in a manner compatible with its obligations under the Agreement.

The beneficiaries must indicate the owner(s) of the results (results ownership list) in the final periodic report.

Protection of results

Beneficiaries which have received funding under the grant must adequately protect their results — for an appropriate period and with appropriate territorial coverage — if protection is possible and justified, taking into account all relevant considerations, including the prospects for commercial exploitation, the legitimate interests of the other beneficiaries and any other legitimate interests.

Exploitation of results

Beneficiaries which have received funding under the grant must — up to four years after the end of the action (see Data Sheet, Point 1) — use their best efforts to exploit their results directly or to have them exploited indirectly by another entity, in particular through transfer or licensing.

If, despite a beneficiary's best efforts, the results are not exploited within one year after the end of the action, the beneficiaries must (unless otherwise agreed in writing with the granting authority) use the Horizon Results Platform to find interested parties to exploit the results.

If results are incorporated in a standard, the beneficiaries must (unless otherwise agreed with the granting authority or unless it is impossible) ask the standardisation body to include the funding statement (see Article 17) in (information related to) the standard.

Additional exploitation obligations

Where the call conditions impose additional exploitation obligations (including obligations linked to the restriction of participation or control due to strategic assets, interests, autonomy or security reasons), the beneficiaries must comply with them — up to four years after the end of the action (see Data Sheet, Point 1).

Where the call conditions impose additional exploitation obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) grant for a limited period of time specified in the request, non-exclusive licences — under fair and reasonable conditions — to their results to legal entities that need the results to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Additional information obligation relating to standards

Where the call conditions impose additional information obligations relating to possible standardisation, the beneficiaries must — up to four years after the end of the action (see Data Sheet, Point 1) — inform the granting authority, if the results could reasonably be expected to contribute to European or international standards.

Transfer and licensing of results

Transfer of ownership

The beneficiaries may transfer ownership of their results, provided this does not affect compliance with their obligations under the Agreement.

The beneficiaries must ensure that their obligations under the Agreement regarding their results are passed on to the new owner and that this new owner has the obligation to pass them on in any subsequent transfer.

Moreover, they must inform the other beneficiaries with access rights of the transfer at least 45 days in advance (or less if agreed in writing), unless agreed otherwise in writing for specifically identified third parties including affiliated entities or unless impossible under the applicable law. This notification must include sufficient information on the new owner to enable the beneficiaries concerned to assess the effects on their access rights. The beneficiaries may object within 30 days of receiving notification (or less if agreed in writing), if they can show that the transfer would adversely affect their access rights. In this case, the transfer may not take place until agreement has been reached between the beneficiaries concerned.

Granting licences

The beneficiaries may grant licences to their results (or otherwise give the right to exploit them), including on an exclusive basis, provided this does not affect compliance with their obligations.

Exclusive licences for results may be granted only if all the other beneficiaries concerned have waived their access rights.

Granting authority right to object to transfers or licensing — Horizon Europe actions

Where the call conditions in Horizon Europe actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated with Horizon Europe, and
- the granting authority considers that the transfer or licence is not in line with EU interests.

Beneficiaries that intend to transfer ownership or grant an exclusive licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with ethical principles and security considerations.

The granting authority may request additional information.

If the granting authority decides to object to a transfer or exclusive licence, it must formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information it has requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Granting authority right to object to transfers or licensing — Euratom actions

Where the call conditions in Euratom actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive or non-exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated to the Euratom Research and Training Programme 2021-2025 and
- the granting authority considers that the transfer or licence is not in line with the EU interests.

Beneficiaries that intend to transfer ownership or grant a licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the results, the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with

ethical principles and security considerations (including the defence interests of the EU Member States under Article 24 of the Euratom Treaty).

The granting authority may request additional information.

If the granting authority decides to object to a transfer or licence, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Limitations to transfers and licensing due to strategic assets, interests, autonomy or security reasons of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security reasons, the beneficiaries may not transfer ownership of their results or grant licences to third parties which are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless they have requested and received prior approval by the granting authority.

The request must:

- identify the specific results concerned
- describe in detail the new owner and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or license on the strategic assets, interests, autonomy or security of the EU and its Member States.

The granting authority may request additional information.

Access rights to results and background

Exercise of access rights — Waiving of access rights — No sub-licensing

Requests to exercise access rights and the waiver of access rights must be in writing.

Unless agreed otherwise in writing with the beneficiary granting access, access rights do not include the right to sub-license.

If a beneficiary is no longer involved in the action, this does not affect its obligations to grant access.

If a beneficiary defaults on its obligations, the beneficiaries may agree that that beneficiary no longer has access rights.

Access rights for implementing the action

The beneficiaries must grant each other access — on a royalty-free basis — to background needed to implement their own tasks under the action, unless the beneficiary that holds the background has — before acceding to the Agreement —:

- informed the other beneficiaries that access to its background is subject to restrictions, or
- agreed with the other beneficiaries that access would not be on a royalty-free basis.

The beneficiaries must grant each other access — on a royalty-free basis — to results needed for implementing their own tasks under the action.

Access rights for exploiting the results

The beneficiaries must grant each other access — under fair and reasonable conditions — to results needed for exploiting their results.

The beneficiaries must grant each other access — under fair and reasonable conditions — to background needed for exploiting their results, unless the beneficiary that holds the background has — before acceding to the Agreement — informed the other beneficiaries that access to its background is subject to restrictions.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for entities under the same control

Unless agreed otherwise in writing by the beneficiaries, access to results and, subject to the restrictions referred to above (if any), background must also be granted — under fair and reasonable conditions — to entities that:

- are established in an EU Member State or Horizon Europe associated country
- are under the direct or indirect control of another beneficiary, or under the same direct or indirect control as that beneficiary, or directly or indirectly controlling that beneficiary and
- need the access to exploit the results of that beneficiary.

Unless agreed otherwise in writing, such requests for access must be made by the entity directly to the beneficiary concerned.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for the granting authority, EU institutions, bodies, offices or agencies and national authorities to results for policy purposes — Horizon Europe actions

In Horizon Europe actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, EU institutions, bodies, offices or agencies for developing, implementing and monitoring EU policies or programmes. Such access rights do not extend to beneficiaries' background.

Such access rights are limited to non-commercial and non-competitive use.

For actions under the cluster 'Civil Security for Society', such access rights also extend to national authorities of EU Member States for developing, implementing and monitoring their policies or programmes in this area. In this case, access is subject to a bilateral agreement to define specific conditions ensuring that:

- the access rights will be used only for the intended purpose and
- appropriate confidentiality obligations are in place.

Moreover, the requesting national authority or EU institution, body, office or agency (including the granting authority) must inform all other national authorities of such a request.

Access rights for the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy — Euratom actions

In Euratom actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy for developing, implementing and monitoring Euratom policies and programmes or for compliance with obligations assumed through international cooperation with non-EU countries and international organisations.

Such access rights include the right to authorise third parties to use the results in public procurement and the right to sub-license and are limited to non-commercial and non-competitive use.

Additional access rights

Where the call conditions impose additional access rights, the beneficiaries must comply with them.

COMMUNICATION, DISSEMINATION, OPEN SCIENCE AND VISIBILITY (— ARTICLE 17)

Dissemination

Dissemination of results

The beneficiaries must disseminate their results as soon as feasible, in a publicly available format, subject to any restrictions due to the protection of intellectual property, security rules or legitimate interests.

A beneficiary that intends to disseminate its results must give at least 15 days advance notice to the other beneficiaries (unless agreed otherwise), together with sufficient information on the results it will disseminate.

Any other beneficiary may object within (unless agreed otherwise) 15 days of receiving notification, if it can show that its legitimate interests in relation to the results or background would be significantly harmed. In such cases, the results may not be disseminated unless appropriate steps are taken to safeguard those interests.

Additional dissemination obligations

Where the call conditions impose additional dissemination obligations, the beneficiaries must also comply with those.

Open Science

Open science: open access to scientific publications

The beneficiaries must ensure open access to peer-reviewed scientific publications relating to their results. In particular, they must ensure that:

- at the latest at the time of publication, a machine-readable electronic copy of the published version or the final peer-reviewed manuscript accepted for publication, is deposited in a trusted repository for scientific publications
- immediate open access is provided to the deposited publication via the repository, under the latest available version of the Creative Commons Attribution International Public Licence (CC BY) or a licence with equivalent rights; for monographs and other long-text formats, the licence may exclude commercial uses and derivative works (e.g. CC BY-NC, CC BY-ND) and
- information is given via the repository about any research output or any other tools and instruments needed to validate the conclusions of the scientific publication.

Beneficiaries (or authors) must retain sufficient intellectual property rights to comply with the open access requirements.

Metadata of deposited publications must be open under a Creative Common Public Domain Dedication (CC 0) or equivalent, in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: publication (author(s), title, date of publication, publication venue); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the publication, the authors involved in the action and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for any research output or any other tools and instruments needed to validate the conclusions of the publication.

Only publication fees in full open access venues for peer-reviewed scientific publications are eligible for reimbursement.

Open science: research data management

The beneficiaries must manage the digital research data generated in the action ('data') responsibly, in line with the FAIR principles and by taking all of the following actions:

- establish a data management plan ('DMP') (and regularly update it)

- as soon as possible and within the deadlines set out in the DMP, deposit the data in a trusted repository; if required in the call conditions, this repository must be federated in the EOSC in compliance with EOSC requirements
- as soon as possible and within the deadlines set out in the DMP, ensure open access — via the repository — to the deposited data, under the latest available version of the Creative Commons Attribution International Public License (CC BY) or Creative Commons Public Domain Dedication (CC 0) or a licence with equivalent rights, following the principle ‘as open as possible as closed as necessary’, unless providing open access would in particular:
 - be against the beneficiary’s legitimate interests, including regarding commercial exploitation, or
 - be contrary to any other constraints, in particular the EU competitive interests or the beneficiary’s obligations under this Agreement; if open access is not provided (to some or all data), this must be justified in the DMP
- provide information via the repository about any research output or any other tools and instruments needed to re-use or validate the data.

Metadata of deposited data must be open under a Creative Commons Public Domain Dedication (CC 0) or equivalent (to the extent legitimate interests or constraints are safeguarded), in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: datasets (description, date of deposit, author(s), venue and embargo); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the dataset, the authors involved in the action, and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for related publications and other research outputs.

Open science: additional practices

Where the call conditions impose additional obligations regarding open science practices, the beneficiaries must also comply with those.

Where the call conditions impose additional obligations regarding the validation of scientific publications, the beneficiaries must provide (digital or physical) access to data or other results needed for validation of the conclusions of scientific publications, to the extent that their legitimate interests or constraints are safeguarded (and unless they already provided the (open) access at publication).

Where the call conditions impose additional open science obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) immediately deposit any research output in a repository and provide open access to it under a CC BY licence, a Public Domain Dedication (CC 0) or equivalent. As an exception, if the access would be against the beneficiaries’ legitimate interests, the beneficiaries must grant non-exclusive licenses — under fair and reasonable conditions — to legal entities that need the research output to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Plan for the exploitation and dissemination of results including communication activities

Unless excluded by the call conditions, the beneficiaries must provide and regularly update a plan for the exploitation and dissemination of results including communication activities.

SPECIFIC RULES FOR CARRYING OUT THE ACTION (— ARTICLE 18)

Implementation in case of restrictions due to strategic assets, interests, autonomy or security of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security, the beneficiaries must ensure that none of the entities that participate as affiliated entities, associated partners, subcontractors or recipients of financial support to third parties are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless otherwise agreed with the granting authority.

The beneficiaries must moreover ensure that any cooperation with entities established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) does not affect the strategic assets, interests, autonomy or security of the EU and its Member States.

Recruitment and working conditions for researchers

The beneficiaries must take all measures to implement the principles set out in the Commission Recommendation on the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers³, in particular regarding:

- working conditions
- transparent recruitment processes based on merit, and
- career development.

The beneficiaries must ensure that researchers and all participants involved in the action are aware of them.

³ Commission Recommendation 2005/251/EC of 11 March 2005 on the European Charter for Researchers and on a Code of Conduct for the Recruitment of Researchers (OJ L 75, 22.3.2005, p. 67).



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